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Table of Contents

List of Tables

1 Scope

This specification describes a transformation for a semantic translation from SysML v1 [SysMLv1] to SysML v2 [SysMLv2] in a precise way. (In this document, "SysML v1" refers to SysML v1.7, the last version of SysML prior to v2.0, and "SysML v2" refers to SysML v2.0, or whatever version corresponds to the current version of this specification.)

The main intent is to provide the rules on which automated conversions of SysML v1 models to the SysML v2 standard can be developed. In addition, this annex can be considered an educational document that provides useful information for people who would like to compare using SysML v2 and using SysML v1.

More sophisticated applications of this transformation can also be envisaged. For instance, a SysML v1 conformant tool could use this transformation to implement a limited subset of the SysML v2 API that will provide "SysMLv2-like" read-only access to its SysMLv1 models for external applications.

2 Conformance

A tool shall demonstrate *conformance* with this specification by meeting all of the following requirements.

1. The tool shall implement the UML4SysML abstract syntax and SysML v1 profile conformant with [SysMLv1]. The tool should, but is not required, to provide the ability to import a SysML v1 model using standard XMI Model Interchange format [XMI].
2. The tool shall implement the SysML v2 abstract syntax conformant with [SysML v2]. The tool should, but is not required, to provide the ability to export a SysML v2 model KerML-standard model interchange project (see [KerML], Clause 10; see also [SysML v2], Clause 2).
3. The tool shall implement a transformation from an abstract syntax representation of an input SysML v1 model to the abstract syntax representation of an output SysML v2, as specified in of this specification.

A tool may claim *partial conformance* with this specification by satisfying the first two requirements above, but only implementing an identified subset of the mappings specified in and . (Note that care must also be taken that certain mappings depend on other mappings, and so cannot reasonably be implemented separately.)

Note. A tool that conforms to [SysMLv2] is not required to necessarily implement a transformation conformant with this specification, or it may implement a SysML v1 to v2 transformation that is not claimed to conform with the transformation defined in this specification.

3 Normative References

The following normative documents contain provisions which, through reference in this text, constitute provisions of this specification.

[KerML] *Kernel Modeling Language (KerML),* Version 1.0
https://www.omg.org/spec/KerML/1.0

[MOF] *Meta Object Facility,* Version 2.5.1
https://www.omg.org/spec/MOF/2.5.1

[OCL] *Object Constraint Language,* Version 2.4
https://www.omg.org/spec/OCL/2.4

[SysML v1] *OMG Systems Modeling Language (SysML),* Version 1.7
https://www.omg.org/spec/SysML/1.7

[SysML v2] *OMG Systems Modeling Language (SysML),* Version 2.0
https://www.omg.org/spec/SysML/2.0

[UML] *Unified Modeling Language (UML),* Version 2.5.1
https://www.omg.org/spec/UML/2.5.1

[XMI] *XML Metadata Interchange,* Version 2.5.1
https://www.omg.org/spec/XMI/2.5.1

4 Terms and Definitions

Various terms and definitions are specified throughout the body of this specification.

5 Symbols

No special symbols are defined in this specification.

6 Introduction

6.1 Mapping Approach

The SysML v1 to v2 transformation is specified by directional mappings between UML metaclasses or stereotypes that are part of the SysML v1 specification [SysMLv1] (referenced below as the "SysML v1 scope") on the one hand, and the set of the metaclasses defined in the KerML [KerML] and SysMLv2 [SysMLv2] specifications (referenced below as "SysML v2") in the other hand. Some library classes are also involved.

Each mapping is a directed relationship that reifies a semantic link between a concept belonging to the SysML v1 scope on the source side and one concept belonging to SysML v2 (or one conforming library element) on the target side. As a set, those mappings constitute a declarative specification of a formal transformation that describes how the information encoded by the SysML v1 concepts can be reliably represented using constructs of SysML v2 metaclass instances.

In this approach, a mapping is represented by a UML class that has a pair of associations. One provides the `from` end that designates the source SysML v1 concept, while the other provides the `to` end that designates the target SysML v2 metaclass.

In addition to those associations, a mapping class provides a set of operations defining how the values of non-derived properties of the target metaclass instance have to be computed based on property values reachable from the source object. The computation algorithm is provided by the body condition of those operations and expressed using OCL code.

Note that the values assigned to the properties of the target object shall be instances of SysML v2 metaclasses, coming themselves from transformations of SysMLv1 objects to SysMLv2 objects. Since the specification is declarative, the order in which the individual transformations shall happen is not imposed. It is up to a conforming implementation to deal with this. Instead, the `getMapped` static operation is provided for referring to the result of a transformation from within an OCL rule. It returns a (possibly undefined) value, that is typed by the target metaclass of the mapping class from which it is invoked.

Each mapping class enables the transformation of any object that has the type specified by the `from` role to an object of the type specified by the `to` role, as long as it is not overloaded by a more specific mapping definition. In other words, assume a mapping is specified for the class `A` (i.e., it has `A` typing its `from` property), then it applies to any instance of a class `B` if `B` is a subclass of `A` and if there is no specialization of that mapping class specified for `B` (i.e., that has `B` typing its `from` property).

It is possible to restrict the applicability of a mapping specification to a specific subset of objects. This is achieved by the `filter` static operation that is evaluated against each candidate object. Only objects of the appropriate type for which this `filter` operation returns `true` shall be translated according to the specifications of that mapping class. The default `filter` operation always returns `true`.

Some mapping classes have one or more qualifiers for their `to` attribute. In such a case, each of those qualifiers reflects the specific property of the source type (i.e. the type of the `from` attribute) that has the same name and the same type. For those specific mappings, it is expected to get one instance of the target class (as specified by the type of the `to` attribute") for each actual combination of value of those properties for a given instance of object of the source type, assuming they pass the applicability filter as described above.

6.2 Acknowledgements

The primary authors of this specification document (and also developers of a proof-of-concept implementation of it) are:

- Yves Bernard, Airbus

- Tim Weilkiens, oose

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7 Mappings

7.1 Overview

This Clause is organized in order to match the packages that subdivide the model of the transformation. The `Foundations` package gathers the abstract classes that represent the concepts on top of which the mapping approach is built. The next subclause presents a utility class named `Helper` that provides reusable operations that simplify the OCL statements defining the computation rules of target properties and make them more readable. Libraries play an important role in SysML v2, and a specific one has been created in order to represent semantics equivalent to those of UML/SysML concepts, where needed. It is presented in this subclause as well.

The three next subclauses are dedicated to initializers, factories and generic mappings, respectively. They do not specify mappings, strictly speaking. Instead, they factorize more or less advanced OCL code that will be reused by the actual mapping specifications that are contained in the two last subclauses. The first of them is dedicated to UML metaclass from the UML4SYSML scope, while the second deals with SysML stereotypes more specifically.

7.2 Foundations

7.2.1 Overview

The concepts defined by KerML/SysML v2 are relatively similar to those of UML/SysML v1, but the ways they are built are different. This makes the specification of the global transformation quite complex. In order to keep it manageable, specific kinds of foundational classes are provided. They represent concepts on which classical "model to model" transformation technologies rely:

- The mappings built on top of the abstract class `Mapping` shall be executed only when they are explicitly called. Each call shall produce a new target element, whatever the source element. It specifies a `from` property typed by the `UML::CommonStructure::Element` metaclass that shall be redefined by any of its subclass for specifying the convenient type of source element. Also it specifies a default (neutral) filter and a set of `getMapped` operations for various purposes: regular mapping result, qualified mapping result and mapping result for a collection of elements.
- The mappings built on top of the abstract class `UniqueMapping`, specified as a specialization of the `Mapping` class, shall produce only one target element for a given source element, whatever the number of time they are called.
- The mappings built on top of the abstract class `MainMapping`, specified as a specialization of the `UniqueMapping` class, shall be systematically executed (i.e. implicitly called) for all the elements that match both theirs source type and filter. There can be at most one main mapping for a given source type and only one target element shall be produced for a given source element.

The corresponding classes are located the the `Foundations` package.

Sometimes, it is necessary to be able to generate elements in the target model without having to provide an explicit link with a source element. In such a case, a mapping class is not appropriate. Instead the mapping framework provides the concept of a `Factory`.

Last, the concept of an `Initializer` allows the factorization of the specification of properties' default values that can be inherited by mappings and factories, as convenient.

In the model of the transformation that is specified here, all of the abstract classes of this `Foundations` package are subject to direct or indirect subclassing. In other words, this specification is built as a set of interrelated initializers, factories, regular, unique and main mappings, where the initializers' operation factorizes the specification of default values for their target element, wherever possible. Those "default operations" are either used as-is or redefined by mappings or factories that can inherit for a specific initializer, as appropriate.

7.2.2 Foundational class specifications

7.2.2.1 UniqueMapping

Description

The mappings built on top of the abstract class UniqueMapping are a specific kind of Mappings that are intended to produce only one target element for a given source element, whatever the number of time they are called.If a getMapped is called several time with the same source element, the target element returned shall always be the same.

General Classes

- Mapping (from Foundations)

7.2.2.2 Factory

Description

Similarly to the well-known to the homonyms software design pattern, a Factory can be used for specifying the production of a target element without any link with a source element. Factories have in common with mapping classes the operations that specify how the properties of the target element shall be computed and the "to" property that specifies the type of the target element. However factories do not define source element. Instead, they can have parameters. Those parameters, if any, shall be specified by properties with appropriate types and multiplicities. Factories are expected to provide a "create" operation with parameters matching in type and multiplicity the properties that are intended to specify them.

General Classes

- Initializer (from Foundations)

7.2.2.3 Mapping

Description

This is the generic abstract class that provides the basic features of any mapping class mapping. The mappings built on top of the abstract class Mapping are intended to be executed only when explicitly called (e.g. by the rule of another mapping class). It specifies a "from" property typed by the UML::CommonStructure::Element metaclass that shall be redefined by any of its subclass for specifying the convenient type of source element. Also it specifies a default (neutral) filter and a set of getMapped operations for various purposes: regular mapping result, qualified mapping result and mapping result for a collection of elements. Each call to the getMapped operation shall produce a new target element, whatever the source element provided. Instances of Mapping class are represent a link between one source element and the target element produced by the transformation specified by that mapping class.

General Classes

- Initializer (from Foundations)

Association Ends

- from : Element [1]

Operations

- filter (in src : Element) : Boolean [1]
  returns "true" if the element provided as the actual parameter value can have a mapping to an instance of

the type specified by the "to" attribute (i.e. can be used as a value for the "from" attribute)

```
 true
```

- getMapped (in fromVar : Element) : Element [1]

  postConditions:

```
self.filter(fromVar) and
self.to.allFeatures()->selectByKind(UML::Property)->reject(isDerived)
->forAll(p | let ops: Operation = self.allFeatures()
    ->selectByKind(UML::Operation)->any(o | o.name = p.name) in
    p = ops()) and
result = self.to
```

- getMapped (in fromVar : Element, in qual : Element) : Element [1]

  postConditions:

```
self.filter(fromVar) and
self.to.allFeatures()->selectByKind(UML::Property)->reject(isDerived)
->forAll(p | let ops: Operation = self.allFeatures()
    ->selectByKind(UML::Operation)->any(o | o.name = p.name) in
    if ops.ownedParameter
        ->select(p | p.direction = UML::ParameterDirectionKind::_'in')
        ->size()=1 then
        p = ops(qual)
    else if ops.ownedParameter
        ->select(p | p.direction = UML::ParameterDirectionKind::_'in')
        ->size()=0 then
        p = ops()
    else
        invalid
    endif endif) and
result = self.to
```

- getMappedColl (in fromColl : Element) : Element [0..*]

  postConditions:

```
result = fromColl->collect(e | self.getMapped(e))
```

7.2.2.4 MainMapping

Description

The mappings built on top of the abstract class MainMapping are a specific kind of UniqueMappings class that are always implicitly called for any element in the source model that match both their source type (as specified by their "from" property) and their filter condition. If more than one main mapping is specified for a given source type, they shall have filters that specify mutually exclusive conditions. Also, as with any unique mapping, only one target element shall be produced for a given source element.

General Classes

- UniqueMapping (from Foundations)

7.2.2.5 Initializer

Description

The abstract class Initializer is the common ancestor of Mapping and Factory. It specifies a "to" property typed by the KerML::Root::Element metaclass that shall be redefined by any of its subclass for specifying the convenient type of target element. Initializers are intended to specify reusable properties' computation rules, mainly for initializing them with default values. Those rules will be inherited or redefined by the sub-classes, as appropriate.

Attributes

- /inputs (not typed) [0..*]

Association Ends

- to : Element [1]

7.3 Mapping Helper and Library

7.3.1 Helper

Description

The Helper class contains operations that are used by multiple mapping classes. The specification is in the bodyCondition.

Operations

- actionOwnedRelationship (in src : Element) : Relationship [0..*]
  Reusable mapping rule for owned relationships of a UML4SysML::Action mapping.

```
let actionInputPin: Set(UML::Element) =
    src.ownedElement->select(e | e.oclIsTypeOf(UML::ActionInputPin)) in
let triggers: Set(UML::Element) =
    src.ownedElement->select(e | e.oclIsKindOf(UML::Trigger)) in
let toElementFMS: Set(UML::Element) =
    src.ownedElement->select(e | e.oclIsKindOf(UML::Pin)) in
let toElementOMS: Set(UML::Element) =
    (((src.ownedElement - toElementFMS) - actionInputPin) - triggers) in
toElementOMS->collect(e | ElementOwningMembership_Mapping.getMapped(e))
->union(toElementFMS->collect(e | ElementFeatureMembership_Mapping.getMapped(e)))
```

- activityOwnedRelationship (in src : Element) : Relationship [0..*]
  Reusable mapping rule for owned relationships of a UML4SysML::Activity mapping.

```
let initialNodes: Set(UML!Element) = src.ownedElement
  ->select(e | e.oclIsKindOf(UML!InitialNode)) in
let flowFinalNodes: Set(UML!Element) = src.ownedElement
  ->select(e | e.oclIsKindOf(UML!FlowFinalNode)) in
let elementsFMS: Set(UML!Element) = (src.ownedElement
  ->select(e | e.oclIsKindOf(UML!ControlNode)
    or e.oclIsKindOf(UML!Action)
    or e.oclIsKindOf(UML!ActivityEdge)
    or e.oclIsKindOf(UML!Property))-initialNodes)-flowFinalNodes in
let parameters: Set(UML!Parameter) = src.ownedElement
```

```
  ->select(e | e.oclIsKindOf(UML!Parameter)) in
let ignoreParameterNodes: Set(UML!ActivityParameterNode) = src.ownedElement
  ->select(e | e.oclIsKindOf(UML!ActivityParameterNode)) in
let ignoreActivityPartition: Set(UML!ActivityPartition) = src.ownedElement
  ->select(e | e.oclIsKindOf(UML!ActivityPartition)) in
let ignoreInterruptibleActivityRegion: Set(UML!InterruptibleActivityRegion)
  = src.ownedElement
    ->select(e | e.oclIsKindOf(UML!InterruptibleActivityRegion)) in
let ownedClassifier: Sequence(UML!Classifier) = src.ownedElement
  ->select(e | e.oclIsKindOf(UML!Classifier)) in
let variables: Sequence(UML!Variable) = src.ownedElement
  ->select(e | e.oclIsKindOf(UML!Variable)) in
let parameterSets: Set(UML!ParameterSet) = src.ownedElement
  ->select(e | e.oclIsKindOf(UML!ParameterSet)) in
let elementsOMS: Set(UML!Element) = ((((((((((src.ownedElement-initialNodes)-
  flowFinalNodes)-elementsFMS)-parameters)-ignoreParameterNodes)-
  ignoreActivityPartition)-ignoreInterruptibleActivityRegion)-ownedClassifier)-
  variables)-parameterSets)-Set{src.classifierBehavior}) in
let memberships: Sequence(UML!Element) = elementsOMS
  ->collect(e | thisModule.ElementOwningMembership_Mapping((e)))
   ->union(initialNodes
     ->collect(e | thisModule.InitialNodeMembership_Mapping((e))))
   ->union(flowFinalNodes
     ->collect(e | thisModule.FlowFinalNodeMembership_Mapping((e))))
   ->union(elementsFMS
     ->collect(e | thisModule.ElementFeatureMembership_Mapping((e))))
   ->union(variables
     ->collect(e | thisModule.VariableMembership_Mapping((e))))
   ->union(parameterSets
     ->collect(e | thisModule.ParameterSetMembership_Mapping((e))))
   ->union(ownedClassifier
     ->collect(e | thisModule.ElementOwningMembership_Mapping((e)))) in
if src.classifierBehavior.oclIsUndefined() then
  memberships
else
  memberships
    ->append(thisModule.BehavioredClassifierFeatureMembership_Mapping((src)))
endif
```

- createUUID () : String [1]
  Creates a UUID. The specification is implementation-specific and therefore cannot provided here.

- excludedPin (in pin : Pin) : Boolean [1]
  Checks if a pin is excluded from the transformation, because it is already defined as a parameter in the SysMLv1Library.

```
if (pin.owner.oclIsTypeOf(UML::AddVariableValueAction) and
    (pin.name = 'value' or pin.name = 'insertAt')) then
    true
else if (pin.owner.oclIsTypeOf(UML::AddStructuralFeatureValueAction) and
    (pin.name = 'value' or pin.name = 'insertAt' or pin.name = 'object')) then
    true
else
    false
endif endif
```

- getAppliedStereotypes (in element : Element) : Stereotype [0..*]
  Returns the list of applied stereotypes. The specification is implementation-specific and therefore cannot provided here.

- getEnumerationType (in t : Enumeration) : EnumerationDefinition [1]
  Maps a given UML4SysM::Enumeration to the appropriate SysML v2 EnumerationDefinition.

```
 let enum: SYSML2::EnumerationDefinition =
    Enumeration_Mapping.getMapped(t) in
if enum.oclIsKindOf(SYSML2::EnumerationDefinition) then
    enum
else if t.name = 'VerdictKind' then
        SYSML2::EnumerationDefinition.allInstances()
        ->any(e | e.qualifiedName = 'VerificationCases::VerdictKind')

     else if t = UML::ParameterDirectionKind then
        KerML::FeatureDirectionKind

        else if t.qualifiedName =
            'SysML::Libraries::ControlValues::ControlValueKind' then
            SYSML2::EnumerationDefinition.allInstances()
            ->any(e | e.qualifiedName =
                'SysMLv1Library::Enumerations::ControlValueKind')

            else
                SYSML2::EnumerationDefinition.allInstances()
                ->any(e | e.qualifiedName =
                    'SysMLv1Library::Enumerations::' + t.name)
            endif
        endif
    endif
endif
```

- getFlowDirectionKind (in v : EnumerationLiteral) : FeatureDirectionKind [1]
  Maps a given SysMLv1 feature direction enumeration literal to a SysML v2 FeatureDirectionKind enumeration literal.

```
 if v.enumeration.qualifiedName =
    'SysML::Ports&Flows::FlowDirectionKind' then
    if v.name = 'out' then
        KerML::FeatureDirectionKind::_'out'
    else if v.name = 'in' then
        KerML::FeatureDirectionKind::_'in'
    else if v.name = 'inout' then
        KerML::FeatureDirectionKind::inout
    else
        invalid
    endif endif endif
else
    invalid
endif
```

- getID (in src : Element) : String [1]
  Returns the identifier of a UML4SysML::Element. The specification is implementation-specific and therefore cannot provided here.

- getKerMLFeatureDirectionKind (in v : EnumerationLiteral) : FeatureDirectionKind [1]
  Maps a given SysMLv1 feature direction enumeration literal to a SysML v2 FeatureDirectionKind enumeration literal.

```
 if v.enumeration.qualifiedName =
    'SysML::Ports&Flows::FeatureDirectionKind' or
```

```
    v.enumeration.qualifiedName = 'SysML::Ports&Flows::FeatureDirection' then
    if v = SysML::FeatureDirectionKind::provided then
        KerML::FeatureDirectionKind::_'out'
    else if (v = SysML::FeatureDirectionKind::required) then
        KerML::FeatureDirectionKind::_'in'
    else if (v = SysML::FeatureDirectionKind::providedRequired) then
        KerML::FeatureDirectionKind::inout
    else
        invalid
    endif endif endif
else
    invalid
endif
```

- getKerMLParameterDirectionKind (in v : ParameterDirectionKind) : FeatureDirectionKind [1]
  Maps a given SysMLv1 parameter direction enumeration literal to a SysML v2 FeatureDirectionKind enumeration literal.

```
 if v = UML::ParameterDirectionKind::_'in' then
    KerML::FeatureDirectionKind::_'in'
else if (v = UML::ParameterDirectionKind::return) then
    KerML::FeatureDirectionKind::out
else if (v = UML::ParameterDirectionKind::out) then
    KerML::FeatureDirectionKind::out
else if (v = UML::ParameterDirectionKind::inout) then
    KerML::FeatureDirectionKind::inout
else
    invalid
endif endif endif endif
```

- getKerMLVisibilityKind (in v : VisibilityKind) : VisibilityKind [1]
  Maps a given UML4SysML::VisibilityKind enumeration literal to a SysML v2 VisibilityKind enumeration literal.

```
 if (v = UML::VisibilityKind::public) then
    KerML::VisibilityKind::public
else if (v = UML::VisibilityKind::protected) then
    KerML::VisibilityKind::protected
else if (v = UML::VisibilityKind::private) then
    KerML::VisibilityKind::private
else if (v = UML::VisibilityKind::package) then
    KerML::VisibilityKind::public
else
    invalid
endif endif endif endif
```

- getMetadataByName (in mdName : String) : AttributeDefinition [1]
  Returns the metadata attribute definition element for a given metadata name.

```
 SYSML2::AttributeDefiniton.allInstances()->any(e | e.name = mdName)
```

- getMultiplicityRangeByName (in name : String) : MultiplicityRange [0..1]
  This operation retrieve a frequently used multiplicity range defiend in the KerML Base Library

```
 SYSML2::DataType.allInstances()
->any(e | e.qualifiedName = 'Base::' + name)
```

- getRequirementStereotype (in element : NamedElement) : Stereotype [0..1]
  Returns the requirement stereotype for a given element.

```
 let stereotypes: Set(UML::Stereotype) =
    Helper.getAppliedStereotypes(element) in
stereotypes->any(s | s.general->collect(g | g.qualifiedName)
->includes('SysML::Requirements::AbstractRequirement'))
```

- getScalarValueType (in t : DataType) : DataType [1]
  Maps a given SysMLv1 primitive type to a SysMLv2 scalar value type.

```
if t.oclIsUndefined()
  or t.name = ''
  or t.name.oclIsUndefined()
  or t.qualifiedName.oclIsUndefined() then
        OclUndefined
else if (t.qualifiedName = 'SysML::Libraries::PrimitiveValueTypes::UnlimitedNatural')
    or t.qualifiedName.indexOf('PrimitiveTypes::UnlimitedNatural') > 0 then
    SYSML2::DataType.allInstances()
    ->any(e | e.qualifiedName = 'ScalarValues::Natural')
else
    SYSML2::DataType.allInstances()
    ->any(e | e.qualifiedName = 'ScalarValues::' + t.name)
endif endif
```

- getScalarValueTypeByName (in ptName : String) : DataType [1]
  Maps a given SysMLv1 primitive type name string to a SysMLv2 scalar value type.

```
 SYSML2::DataType.allInstances()
->any(e | e.qualifiedName = 'ScalarValues::' + ptName)
```

- getTagValue (in element : Element, in stereotypeName : String, in tagValueName : String) [1]
  Returns the value of a stereotype property. The specification is implementation-specific and therefore cannot provided here.

- getTagValueAsElement (in element : Element, in stereotypeName : String, in tagValueName : String) : Element [1]
  Returns the value of a stereotype property. The specification is implementation-specific and therefore cannot provided here.

- getTagValueAsElementColl (in element : Element, in stereotypeName : String, in tagValueName : String) : Element [0..*]
  Returns the value of a stereotype property as a collection. The specification is implementation-specific and therefore cannot provided here.

- getTagValueAsString (in element : Element, in stereotypeName : String, in tagValueName : String) : String [1]
  Returns the value of a stereotype property as a string. The specification is implementation-specific and therefore cannot provided here.

- getTagValueAsStringColl (in element : Element, in stereotypeName : String, in tagValueName : String) : String [0..*]
  Returns the value of a stereotype property as a string collection. The specification is implementation-specific and therefore cannot provided here.

- globalNamespace () : Namespace [1]

```
 KerML::Package.allInstances()->any(p | p.owningNamespace->isEmpty())
```

- hasMainMapping (in element : Element) : Boolean [1]
- hasStereotypeApplied (in element : Element, in stereotypeName : String) : Boolean [1]
  Returns true if the given stereotype is applied to the element. The specification is implementation-specific and therefore cannot provided here.

- isConnectionDef (in association : Association) : Boolean [1]
  Checks if a UML4SysML::Association is mapped to a SysML v2 ConnectionDefinition.

```
-- Case 1: composite association with
-- multiplicity 1..1 on owner side
let case1: Boolean = association.memberEnd
->exists(e | not e.isComposite and e.lower=1) and
association.memberEnd->exists(e | e.isComposite) in

-- Case 2: association is not composite and
-- there is no owned end with multiplicity 0..*
let case2: Boolean = not association.memberEnd
->exists(e | e.isComposite) and
not association.ownedEnd
->exists(e | e.lower = 0 and e.upper = -1) in

association.oclIsTypeOf(UML::AssociationClass) or
case1 or
case2
```

- isInScope (in element : Element) : Boolean [1]
  The isInScope operation is intended to define the scope on which the transformation will apply. If the isInScope operation return "true" for a given model element, this element shall be consider by the transformation. Especially, main mappings - if any - will apply to it. It shall be ignored otherwise.
- isRequirement (in element : Element) : Boolean [1]
  Checks whether the stereotype AbstractRequirement is applied to the given element.

```
 let stereotypes: Set(UML::Stereotype) =
    Helper.getAppliedStereotypes(element) in
stereotypes->exists(s | s.general->collect(g | g.qualifiedName)
->includes('SysML::Requirements::AbstractRequirement'))
```

- packageOwnedRelationship (in src : Element) : Relationship [0..*]
  Reusable mapping rule for owned relationships of a UML4SysML::Package mapping.

```
let pkg: UML::Package = src.oclAsType(UML::Package) in
 if pkg.oclIsUndefined() then
   Set{}
 else
   let useCaseAssociations : Set(UML::Association) =
     pkg.ownedType->select(e | e.oclIsKindOf(UML::Association))
     ->select(a | a.memberEnd->exists(e | e.type.oclIsKindOf(UML::UseCase))) in
   let unmappedAssociations : Set(UML::Association) = pkg.ownedType
     ->select(e | e.oclIsKindOf(UML::Association))
     ->reject(a | Helper.isConnectionDef(a)) in
   let imports: Set(UML::PackageImport) = pkg.packageImport
```

```
      ->select(pi | Helper.isInScope(pi.importedPackage)) in
    let informationFlows: Set(UML::InformationFlow) = pkg.packagedElement
      ->select(e | e.oclIsKindOf(UML::InformationFlow))
      ->reject(i | i.realization->isEmpty() and i.realizingConnector->isEmpty()) in
    let fromIF: Set(SysMLv2::ConnectionUsage) = informationFlows
      ->collect(i | i.realization->collect(r | InformationFlow_Mapping.getMapped(i, r)))
      ->flatten()->asSet()
      ->union(informationFlows->collect(i | i.realizingConnector
      ->collect(r | InformationFlow_Mapping.getMapped(i, r)))
      ->flatten()->asSet())->asSet() in
    let relationships: Set(SysMLv2::Relationship) = pkg.ownedComment
      ->reject(c | c.annotatedElement->includes(pkg))
      ->collect(c| CommentOwnership_Mapping.getMapped(c))->asSet()
      ->union(((pkg.ownedType-useCaseAssociations)-unmappedAssociations)
      ->collect(e | ElementOwningMembership_Mapping.getMapped(e))->asSet())
      ->union(imports->collect(i | PackageImport_Mapping.getMapped(i))->asSet())
      ->union(pkg.ownedElement
      ->select(e | e.oclIsKindOf(UML::Dependency)
        or e.oclIsKindOf(UML::Package)
        or (e.oclIsKindOf(UML::InstanceSpecification)
        and e.oclAsType(UML::InstanceSpecification).classifier->notEmpty()))
      ->collect(e | ElementOwningMembership_Mapping.getMapped(e))->asSet())
      ->union(fromIF)->asSet() in
if pkg.URI.oclIsUndefined() or pkg.URI = '' then
 relationships
else
 relationships->including(PackageURIMetadataMembership_Mapping.getMapped(pkg))
endif endif
```

- stateOwnedRelationship (in src : Element) : Relationship [0..*]
  Reusable mapping rule for owned relationships of a UML4SysML::State mapping.

```
let initialState : Set(UML::Element) =
    from.ownedElement->select(e | e.oclIsKindOf(UML::Pseudostate) and
    e.oclAsType(UML::Pseudostate).kind = UML::PseudostateKind::initial) in
let toElementOMS : Set(UML::Element) = from.ownedElement - initialState in
toElementOMS->collect(e | ElementOwningMembership_Mapping.getMapped(e))
->union(initialState->collect(e | InitialStateMembership_Mapping.getMapped(e)))
```

7.3.2 SysML v1 Library

The SysML v1 library is a SysML v2 model library with metadata definitions for annotating some model elements resulting from a transformation from a SysML v1 model using the SysML v1 to SysML v2 transformation.

```
package SysMLv1Library {

        doc /*
         * The SysMLv1Library defines library elements and metadata for
         * SysML elements which cannot mapped to a SysML v2 element.
         */

        // Library elements

        action def AddValueAction {
                in insertAt : ScalarValues::Natural [0..1];
                in value : ScalarValues::Integer;
                in isReplaceAll : ScalarValues::Boolean = false;
                in target;

                if not isReplaceAll {
```

```
                    if insertAt == * {
                        assign target := SequenceFunctions::including(target, value);
                    }
                    else {
                        assign target :=
                            SequenceFunctions::includingAt(target, value, insertAt);
                    }
                } else {
                    target := value;
                }
            }

            action def AddStructuralFeatureValueAction :> AddValueAction {
                in object;
            }

            action def RemoveVariableValueAction :> Actions::AssignmentAction {
                in removeAt: ScalarValues::Integer [0..1];
                in value : ScalarValues::Integer;
                in isRemoveDuplicates : ScalarValues::Boolean = false;
                in variable;

                // isRemoveDuplicates not covered yet
                if isRemoveDuplicates {
                    if removeAt {
                        assign variable :=
                            SequenceFunctions::excludingAt(variable, value, removeAt);
                    } else {
                        assign variable := SequenceFunctions::excluding(variable, value);
                    }
                }
            }


            // Metadata

            metadata def ActivityEdgeData {
                doc /* Metadata definition for UML::ActivityEdge::weight property */
                attribute weight : ScalarValues::Natural;
            }

            metadata def AssociationData {
                doc /* Metadata definition for
                     * UML::StructuredClassifiers::Association::isDerived property mapping
                     */
            attribute isDerived : ScalarValues::Boolean;
        }

        metadata def BlockData {
                doc /* Metadata definition for
                     * SysML::Blocks::Block::isEncapsulated property
                     */
                attribute isEncapsulated : ScalarValues::Boolean;
        }

        metadata def ElementGroupData {
                doc /* Metadata definition for the criterion
                     * of a SysML::ModelElements::ElementGroup
                     */
            attribute criterion : ScalarValues::String;
        }

        metadata def ModelData :> PackageData {
```

```
        doc /* Metadata definition for the UML::Model::viewpoint property */
        :> annotatedElement : SysML::Package;
        attribute 'viewpoint' : ScalarValues::String;
}

metadata def PackageData {
        doc /* Metadata definition for the UML::Package::URI property */
        :> annotatedElement : SysML::Package;
        attribute URI : ScalarValues::String;
}

    metadata def ParameterSetData {
            doc /* Metadata definition for tagging parameters
             * mapped from a UML::ParameterSet
             */
            attribute isParameterSet : ScalarValues::Boolean;
    }

metadata def PortData {
        doc /* Metadata definition for tagging SysML v2 ports
         * mapped from a SysML::Ports&Flows::FullPort element
         */
        :> annotatedElement : SysML::PartUsage;
        attribute isFullPort : ScalarValues::Boolean;
}

metadata def ProbabilityData {
        doc /* Metadata definition for SysML::Activities::Probability stereotype */
        attribute probability : ScalarValues::Real;
}

metadata def RateData {
        doc /* Metadata definition for SysML::Activities::Rate and
         * specialized Discrete and Continuous stereotypes
         */
        :> annotatedElement : SysML::PartUsage;
        part rate;
        attribute isDiscrete : ScalarValues::Boolean;
        attribute isConcrete : ScalarValues::Boolean;
}

metadata def RefineData {
        doc /* Metadata definition for tagging SysML v2 dependencies
         * mapped from a SysML::Requirements::Refine relationship
         */
        :> annotatedElement : SysML::Dependency;
        attribute isRefine : ScalarValues::Boolean;
}

metadata def StakeholderData {
        doc /* Metadata definition for tagging SysML v2 item definitions
         * mapped from a SysML::ModelElements::Stakeholder element
         */
        :> annotatedElement : SysML::ItemDefinition;
        attribute isStakeholder : ScalarValues::Boolean;
}

metadata def traceData {
        doc /* Metadata definition for tagging SysML v2 dependencies
         * mapped from a SysML::Requirements::Trace relationship
         */
        :> annotatedElement : SysML::Dependency;
        attribute isTrace : ScalarValues::Boolean;
```

```
	}

	metadata def ViewpointData {
		doc /* Metadata definition for SysML::ModelElements::Viewpoint properties */
		attribute languages [0..*] : ScalarValues::String;
		attribute presentations [0..*] : ScalarValues::String;
	}

	package Enumerations {
		enum def ControlValueKind {
			doc /* The ControlValueKind enumeration is a type for
			 * treating control values as data and for UML control pins.
			 */
			enum disable;
			enum enable;
		}
	}
}
```

7.4 Initializers

7.4.1 Overview

The classes presented in this subclause provide set of rules that provide default values for all non-derived features of their target metaclasses. Intentionally, initializers do not specify any "source" element. This makes them easier to specialize but prevents them from being able to provide a computation algorithm for some target features. In such a case, the operation matching the feature will be specified as abstract.

7.4.2 Mapping Specifications

7.4.2.1 KerML Initializers

7.4.2.1.1 ToAnnotatingElement_Init

Description

Initializes the properties of the SysML v2 element AnnotatingElement.

General Classes

- ToElement_Init (from KerMLInitializers)

Association Ends

- to : AnnotatingElement [1]
  {redefines: ToElement_Init::to}

Operations

- annotation () : Annotation [0..*]

```
Set{}
```

7.4.2.1.2 ToAnnotation_Init

Description

Initializes the properties of the SysML v2 element Annotation.

General Classes

- ToRelationship_Init (from KerMLInitializers)

Association Ends

- to : Annotation [1]
  {redefines: ToRelationship_Init::to}

Operations

- annotatedElement () : Element [1] {redefines target, abstract}
- annotatingElement () : AnnotatingElement [1] {redefines source, abstract}
- owningAnnotatedElement () : Element [0..1]

```
null
```

7.4.2.1.3 ToAssociation_Init

Description

Initializes the properties of the SysML v2 element Association.

General Classes

- ToClassifier_Init (from KerMLInitializers)
- ToRelationship_Init (from KerMLInitializers)

Association Ends

- to : Association [1]
  {redefines: ToRelationship_Init::to}
  {redefines: ToClassifier_Init::to}

7.4.2.1.4 ToBehavior_Init

Description

Initializes the properties of the SysML v2 element Behavior.

General Classes

- ToClassifier_Init (from KerMLInitializers)

Association Ends

- to : Behavior [1]
  {redefines: ToClassifier_Init::to}

7.4.2.1.5 ToClassifier_Init

Description

Initializes the properties of the SysML v2 element Classifier.

General Classes

- ToType_Init (from KerMLInitializers)

Association Ends

- to : Classifier [1]
  {redefines: ToType_Init::to}

7.4.2.1.6 ToComment_Init

Description

Initializes the properties of the SysML v2 element Comment.

General Classes

- ToAnnotatingElement_Init (from KerMLInitializers)

Association Ends

- to : Comment [1]
  {redefines: ToAnnotatingElement_Init::to}

Operations

- body () : String [1]{abstract}
- locale () : String [1]

```
null
```

7.4.2.1.7 ToConjugation_Init

Description

Initializes the properties of the SysML v2 element Conjugation.

General Classes

- ToRelationship_Init (from KerMLInitializers)

Association Ends

- to : Conjugation [1]
  {redefines: ToRelationship_Init::to}

Operations

- conjugatedType () : Type [1] {redefines source, abstract}
- originalType () : Type [1] {redefines target, abstract}

7.4.2.1.8 ToConnector_Init

Description

Initializes the properties of the SysML v2 element Connector.

General Classes

- ToFeature_Init (from KerMLInitializers)
- ToRelationship_Init (from KerMLInitializers)

Association Ends

- to : Connector [1]
  {redefines: ToFeature_Init::to}
  {redefines: ToRelationship_Init::to}

Operations

- isDirected () : Boolean [1]

```
false
```

7.4.2.1.9 ToDocumentation_Init

Description

Initializes the properties of the SysML v2 element Documentation.

General Classes

- ToComment_Init (from KerMLInitializers)

Association Ends

- to : Documentation [1]
  {redefines: ToComment_Init::to}

7.4.2.1.10 ToElement_Init

Description

This is the general abstract class to be used as an ancestor for any class mapping specification.

General Classes

- Initializer (from Foundations)

Association Ends

- to : Element [1]
  {redefines: Initializer::to}

Operations

- aliasId () : String [0..*]

```
Set{}
```

- declaredName () : String [0..1]

```
null
```

- elementId () : String [1]

```
Helper.createUUID()
```

- ownedRelationship () : Relationship [0..*]

```
Set{}
```

- shortName () : String [0..1]

```
null
```

Constraints

- `from_and_to_types`
  `from.oclIsKindOf(factory.srcType) and to.oclIsKindOf(factory.tgtType)`

7.4.2.1.11 ToEndFeatureMembership_Init

Description

Initializes the properties of the SysML v2 element EndFeatureMembership.

General Classes

- ToFeatureMembership_Init (from KerMLInitializers)

Association Ends

- to : EndFeatureMembership [1]
  {redefines: ToFeatureMembership_Init::to}

7.4.2.1.12 ToExpression_Init

Description

Initializes the properties of the SysML v2 element Expression.

General Classes

- ToStep_Init (from KerMLInitializers)

Association Ends

- to : Expression [1]
  {redefines: ToStep_Init::to}

7.4.2.1.13 ToFeature_Init

Description

Initializes the properties of the SysML v2 element Feature.

General Classes

- ToType_Init (from KerMLInitializers)

Association Ends

- to : Feature [1]
  {redefines: ToType_Init::to}

Operations

- direction () : FeatureDirectionKind [0..1]

```
null
```

- isComposite () : Boolean [1]

```
false
```

- isDerived () : Boolean [1]

```
false
```

- isEnd () : Boolean [1]

```
false
```

- isOrdered () : Boolean [1]

```
false
```

- isPortion () : Boolean [1]

```
false
```

- isReadOnly () : Boolean [1]

```
false
```

- isUnique () : Boolean [1]

```
true
```

7.4.2.1.14 ToFeatureChainExpression_Init

Description

Initializes the properties of the SysML v2 element FeatureChainExpression.

General Classes

- ToOperatorExpression_Init (from KerMLInitializers)

Association Ends

- to : FeatureChainExpression [1]
  {redefines: ToOperatorExpression_Init::to}

7.4.2.1.15 ToFeatureChaining_Init

Description

Initializes the properties of the SysML v2 element FeatureChaining.

General Classes

- ToRelationship_Init (from KerMLInitializers)

Association Ends

- to : FeatureChaining [1]
  {redefines: ToRelationship_Init::to}

Operations

- chainingFeature () : Feature [1] {redefines target, abstract}

7.4.2.1.16 ToFeatureMembership_Init

Description

Initializes the properties of the SysML v2 element FeatureMembership.

General Classes

- ToOwningMembership_Init (from KerMLInitializers)
- ToTypeFeaturing_Init (from KerMLInitializers)

Association Ends

- to : FeatureMembership [1]
  {redefines: ToTypeFeaturing_Init::to}
  {redefines: ToOwningMembership_Init::to}

Operations

- ownedMemberFeature () : Feature [1] {redefines ownedMemberElement}

```
self.upperBound
```

- ownedRelatedElement () : Element [0..*] {redefines ownedRelatedElement}

```
Set{self.ownedMemberFeature()}
```

7.4.2.1.17 ToFeatureReferenceExpression_Init

Description

Initializes the properties of the SysML v2 element FeatureReferenceExpression.

General Classes

- ToExpression_Init (from KerMLInitializers)

Association Ends

- to : FeatureReferenceExpression [1]
  {redefines: ToExpression_Init::to}

7.4.2.1.18 ToFeatureTyping_Init

Description

Initializes the properties of the SysML v2 element FeatureTyping.

General Classes

- ToSpecialization_Init (from KerMLInitializers)

Association Ends

- to : FeatureTyping [1]
  {redefines: ToSpecialization_Init::to}

Operations

- type () : Type [1] {redefines general, abstract}
- typedFeature () : Feature [1] {redefines specific, abstract}

7.4.2.1.19 ToFeatureValue_Init

Description

Initializes the properties of the SysML v2 element FeatureValue.

General Classes

- ToOwningMembership_Init (from KerMLInitializers)

Association Ends

- to : FeatureValue [1]
  {redefines: ToOwningMembership_Init::to}

Operations

- featureWithValue () : Feature [1] {redefines ownedMemberElement, abstract}
- isDefault () : Boolean [1]

```
false
```

- isInitial () : Boolean [1]

```
false
```

- ownedRelatedElement () : Element [0..*] {redefines ownedRelatedElement}

```
Set{self.value()}
```

- value () : Expression [1] {redefines ownedMemberElement, abstract}

7.4.2.1.20 ToFlow_Init

Description

Initializes the properties of the SysML v2 element Flow.

General Classes

- ToConnector_Init (from KerMLInitializers)

Association Ends

- to : Flow [1]
  {redefines: ToConnector_Init::to}

7.4.2.1.21 ToFunction_Init

Description

Initializes the properties of the SysML v2 element Function.

General Classes

- ToBehavior_Init (from KerMLInitializers)

Association Ends

- to : Function [1]
  {redefines: ToBehavior_Init::to}

7.4.2.1.22 ToImport_Init

Description

Initializes the properties of the SysML v2 element Import.

General Classes

- ToRelationship_Init (from KerMLInitializers)

Association Ends

- to : Import [1]
  {redefines: ToRelationship_Init::to}

Operations

- importedMemberName () : String [0..1]

```
null
```

- isImportAll () : Boolean [1]

```
false
```

- isRecursive () : Boolean [1]

```
false
```

- visibility () : VisibilityKind [1]

```
KerML::VisibilityKind::public
```

7.4.2.1.23 ToInteraction_Init

Description

Initializes the properties of the SysML v2 element Interaction.

General Classes

- ToAssociation_Init (from KerMLInitializers)
- ToBehavior_Init (from KerMLInitializers)

Association Ends

- to : Interaction [1]
  {redefines: ToAssociation_Init::to}
  {redefines: ToBehavior_Init::to}

7.4.2.1.24 ToInvocationExpression_Init

Description

Initializes the properties of the SysML v2 element InvocationExpression.

General Classes

- ToExpression_Init (from KerMLInitializers)

Association Ends

- to : InvocationExpression [1]
  {redefines: ToExpression_Init::to}

7.4.2.1.25 ToMembership_Init

Description

Initializes the properties of the SysML v2 element Membership.

General Classes

- ToRelationship_Init (from KerMLInitializers)

Association Ends

- to : Membership [1]
  {redefines: ToRelationship_Init::to}

Operations

- memberElement () : Element [1] {redefines target, abstract}
- memberName () : String [0..1]

```
null
```

- memberShortName () : String [0..1]

```
null
```

- membershipOwningNamespace () : Element [0..*] {redefines source, abstract}
- visibility () : VisibilityKind [1]

```
KerML::VisibilityKind::public
```

7.4.2.1.26 ToMembershipImport_Init

Description

Initializes the properties of the SysML v2 element MembershipImport.

General Classes

- ToImport_Init (from KerMLInitializers)

Association Ends

- to : MembershipImport [1]
  {redefines: ToImport_Init::to}

Operations

- importedMembership () : Namespace [1] {redefines target, abstract}

7.4.2.1.27 ToNamespace_Init

Description

Initializes the properties of the SysML v2 element Namespace.

General Classes

- ToElement_Init (from KerMLInitializers)

Association Ends

- to : Namespace [1]
  {redefines: ToElement_Init::to}

7.4.2.1.28 ToNamespaceImport_Init

Description

Initializes the properties of the SysML v2 element NamespaceImport.

General Classes

- ToImport_Init (from KerMLInitializers)

Association Ends

- to : NamespaceImport [1]
  {redefines: ToImport_Init::to}

Operations

- importedNamespace () : Namespace [1] {redefines target, abstract}

7.4.2.1.29 ToOperatorExpression_Init

Description

Initializes the properties of the SysML v2 element OperatorExpression.

General Classes

- ToExpression_Init (from KerMLInitializers)

Association Ends

- to : OperatorExpression [1]
  {redefines: ToExpression_Init::to}

Operations

- operator () : String [1]{abstract}

7.4.2.1.30 ToOwningMembership_Init

Description

Initializes the properties of the SysML v2 element OwningMembership.

General Classes

- ToMembership_Init (from KerMLInitializers)

Association Ends

- to : OwningMembership [1]
  {redefines: ToMembership_Init::to}

Operations

- ownedMemberElement () : Element [1] {redefines memberElement, abstract}
- ownedRelatedElement () : Element [0..*] {redefines ownedRelatedElement}

```
Set{self.ownedMemberElement()}
```

7.4.2.1.31 ToPackage_Init

Description

Initializes the properties of the SysML v2 element Package.

General Classes

- ToNamespace_Init (from KerMLInitializers)

Association Ends

- to : Package [1]
  {redefines: ToNamespace_Init::to}

7.4.2.1.32 ToParameterMembership_Init

Description

Initializes the properties of the SysML v2 element ParameterMembership.

General Classes

- ToFeatureMembership_Init (from KerMLInitializers)

Association Ends

- to : ParameterMembership [1]
  {redefines: ToFeatureMembership_Init::to}
  {redefines: ElementOwningMembership_Mapping::to}

Operations

- ownedMemberParameter () : Feature [1] {redefines ownedMemberFeature}

```
null
```

- ownedRelatedElement () : Element [0..*] {redefines ownedRelatedElement}

```
Set{self.ownedMemberParameter()}
```

7.4.2.1.33 ToPredicate_Init

Description

Initializes the properties of the SysML v2 element Predicate.

General Classes

- ToFunction_Init (from KerMLInitializers)

Association Ends

- to : Predicate [1]
  {redefines: ToFunction_Init::to}

7.4.2.1.34 ToRedefinition_Init

Description

Initializes the properties of the SysML v2 element Redefinition.

General Classes

- ToSubsetting_Init (from KerMLInitializers)

Association Ends

- to : Redefinition [1]
  {redefines: ToSubsetting_Init::to}

Operations

- redefinedFeature () : Feature [1] {redefines subsettedFeature, abstract}
- redefiningFeature () : Feature [1]{abstract}

7.4.2.1.35 ToReferenceSubsetting_Init

Description

Initializes the properties of the SysML v2 element ReferenceSubsetting.

General Classes

- ToSubsetting_Init (from KerMLInitializers)

Association Ends

- to : ReferenceSubsetting [1]
  {redefines: ToSubsetting_Init::to}

Operations

- referencedFeature () : Feature [1] {redefines subsettedFeature, abstract}

7.4.2.1.36 ToRelationship_Init

Description

Initializes the properties of the SysML v2 element Relationship.

General Classes

- ToElement_Init (from KerMLInitializers)

Association Ends

- to : Relationship [1]
  {redefines: ToElement_Init::to}

Operations

- ownedRelatedElement () : Element [0..*]

```
Set{}
```

- source () : Element [0..*]

```
Set{}
```

- target () : Element [0..*]

```
Set{}
```

7.4.2.1.37 ToReturnParameterMembership_Init

Description

Initializes the properties of the SysML v2 element ReturnParameterMembership.

General Classes

- ToParameterMembership_Init (from KerMLInitializers)

Association Ends

- to : ReturnParameterMembership [1]
  {redefines: ToParameterMembership_Init::to}

Operations

- isComposite (in src : Element) : Boolean [1]
  returns "true" if the element provided as the actual parameter value can have a mapping to an instance of the type specified by the "to" attribute (i.e. can be used as a value for the "from" attribute)

```
false
```

7.4.2.1.38 ToSpecialization_Init

Description

Initializes the properties of the SysML v2 element Specialization.

General Classes

- ToRelationship_Init (from KerMLInitializers)

Association Ends

- to : Specialization [1]
  {redefines: ToRelationship_Init::to}

Operations

- general () : Type [1] {redefines target, abstract}
- specific () : Type [1] {redefines source, abstract}

7.4.2.1.39 ToStep_Init

Description

Initializes the properties of the SysML v2 element Step.

General Classes

- ToFeature_Init (from KerMLInitializers)

Association Ends

- to : Step [1]
  {redefines: ToFeature_Init::to}

7.4.2.1.40 ToSubclassification_Init

Description

Initializes the properties of the SysML v2 element Subclassification.

General Classes

- ToSpecialization_Init (from KerMLInitializers)

Association Ends

- to : Subclassification [1]
  {redefines: ToSpecialization_Init::to}

Operations

- subclassifier () : Classifier [1]

```
null
```

- superclassifier () : Classifier [1]

```
null
```

7.4.2.1.41 ToSubsetting_Init

Description

Initializes the properties of the SysML v2 element Subsetting.

General Classes

- ToSpecialization_Init (from KerMLInitializers)

Association Ends

- to : Subsetting [1]
  {redefines: ToSpecialization_Init::to}

Operations

- ownedRelatedElement () : Element [0..*] {redefines ownedRelatedElement}

```
Set{}
```

- subsettedFeature () : Feature [1] {redefines general, abstract}

7.4.2.1.42 ToSuccession_Init

Description

Initializes the properties of the SysML v2 element Succession.

General Classes

- ToConnector_Init (from KerMLInitializers)

Association Ends

- to : Succession [1]
  {redefines: ToConnector_Init::to}

7.4.2.1.43 ToSuccessionItemFlow_Init

Description

Initializes the properties of the SysML v2 element SuccessionFlow.

General Classes

- ToItemFlow_Init (from KerMLInitializers)
- ToSuccession_Init (from KerMLInitializers)

Association Ends

- to : SuccessionFlow [1]
  {redefines: ToSuccession_Init::to}
  {redefines: ToItemFlow_Init::to}

7.4.2.1.44 ToTextualRepresentation_Init

Description

Initializes the properties of the SysML v2 element TextualRepresentation.

General Classes

- ToAnnotatingElement_Init (from KerMLInitializers)

Association Ends

- to : TextualRepresentation [1]
  {redefines: ToAnnotatingElement_Init::to}

Operations

- body () : String [1]{abstract}
- language () : String [1]{abstract}

7.4.2.1.45 ToType_Init

Description

Initializes the properties of the SysML v2 element Type.

General Classes

- ToNamespace_Init (from KerMLInitializers)

Association Ends

- to : Type [1]
  {redefines: ToNamespace_Init::to}

Operations

- isAbstract () : Boolean [1]

```
false
```

- isSufficient () : Boolean [1]

```
false
```

7.4.2.1.46 ToTypeFeaturing_Init

Description

Initializes the properties of the SysML v2 element TypeFeaturing.

General Classes

- ToRelationship_Init (from KerMLInitializers)

Association Ends

- to : TypeFeaturing [1]
  {redefines: ToRelationship_Init::to}

Operations

- featureOfType () : Feature [1] {redefines source, abstract}

- featuringType () : Type [1] {redefines target, abstract}

7.4.2.2 System Initializers

7.4.2.2.1 ToActionUsage_Init

Description

Initializes the properties of the SysML v2 element ActionUsage.

General Classes

- ToStep_Init (from KerMLInitializers)
- ToUsage_Init (from SystemInitializers)

Association Ends

- to : ActionUsage [1]
  {redefines: ToStep_Init::to}
  {redefines: ToUsage_Init::to}

Operations

- isComposite () : Boolean [1] {redefines isComposite}

```
true
```

7.4.2.2.2 ToActorMembership_Init

Description

Initializes the properties of the SysML v2 element ActorMembership.

General Classes

- ToParameterMembership_Init (from KerMLInitializers)

Association Ends

- to : ActorMembership [1]
  {redefines: ToParameterMembership_Init::to}

7.4.2.2.3 ToAssignmentActionUsage_Init

Description

Initializes the properties of the SysML v2 element AssignmentActionUsage.

General Classes

- ToActionUsage_Init (from SystemInitializers)

Association Ends

- to : AssignmentActionUsage [1]
  {redefines: ToActionUsage_Init::to}

7.4.2.2.4 ToBindingConnectorAsUsage_Init

Description

Initializes the properties of the SysML v2 element BindingConnectorAsUsage.

General Classes

- ToConnectionUsage_Init (from SystemInitializers)

Association Ends

- to : BindingConnectorAsUsage [1]
  {redefines: ToConnectionUsage_Init::to}

7.4.2.2.5 ToCalculationUsage_Init

Description

Initializes the properties of the SysML v2 element CalculationUsage.

General Classes

- ToActionUsage_Init (from SystemInitializers)

Association Ends

- to : CalculationUsage [1]
  {redefines: ToActionUsage_Init::to}

7.4.2.2.6 ToConjugatedPortDefinition_Init

Description

Initializes the properties of the SysML v2 element ConjugatedPortDefinition.

General Classes

- ToPortDefinition_Init (from SystemInitializers)

Association Ends

- to : ConjugatedPortDefinition [1]
  {redefines: ToPortDefinition_Init::to}

7.4.2.2.7 ToConjugatedPortTyping_Init

Description

Initializes the properties of the SysML v2 element ConjugatedPortTyping.

General Classes

- ToFeatureTyping_Init (from KerMLInitializers)

Association Ends

- to : ConjugatedPortTyping [1]
  {redefines: ToFeatureTyping_Init::to}

Operations

- conjugatedPortDefinition () : ConjugatedPortDefinition [1] {redefines type, abstract}
- portDefinition () : PortDefinition [1]{abstract}

7.4.2.2.8 ToConnectionUsage_Init

Description

Initializes the properties of the SysML v2 element ConnectionUsage.

General Classes

- ToPartUsage_Init (from SystemInitializers)

Association Ends

- to : ConnectionUsage [1]
  {redefines: ToPartUsage_Init::to}

7.4.2.2.9 ToConstraintDefinition_Init

Description

Initializes the properties of the SysML v2 element ConstraintDefinition.

General Classes

- ToDefinition_Init (from SystemInitializers)

Association Ends

- to : ConstraintDefinition [1]
  {redefines: ToDefinition_Init::to}
  {redefines: ToFunction_Init::to}

7.4.2.2.10 ToConstraintUsage_Init

Description

Initializes the properties of the SysML v2 element ConstraintUsage.

General Classes

- ToUsage_Init (from SystemInitializers)

Association Ends

- to : ConstraintUsage [1]
  {redefines: ToUsage_Init::to}

7.4.2.2.11 ToDefinition_Init

Description

Initializes the properties of the SysML v2 element Definition.

General Classes

- ToClassifier_Init (from KerMLInitializers)

Association Ends

- to : Definition [1]
  {redefines: ToClassifier_Init::to}

Operations

- isVariation () : Boolean [1]

```
false
```

7.4.2.2.12 ToEventOccurerenceUsage_Init

Description

Initializes the properties of the SysML v2 element EventOccurrenceUsage.

General Classes

- ToOccurrenceUsage_Init (from SystemInitializers)

Association Ends

- to : EventOccurrenceUsage [1]
  {redefines: ToOccurrenceUsage_Init::to}

7.4.2.2.13 ToFlowUsage_Init

Description

Initializes the properties of the SysML v2 element FlowUsage.

General Classes

- ToActionUsage_Init (from SystemInitializers)
- ToConnector_Init (from KerMLInitializers)

Association Ends

- to : FlowUsage [1]
  {redefines: ToConnector_Init::to}
  {redefines: ToActionUsage_Init::to}

Operations

- isDirected () : Boolean [1] {redefines isDirected}

```
true
```

7.4.2.2.14 ToItemDefinition_Init

Description

Initializes the properties of the SysML v2 element ItemDefinition.

General Classes

- ToDefinition_Init (from SystemInitializers)

Association Ends

- to : ItemDefinition [1]
  {redefines: ToDefinition_Init::to}

7.4.2.2.15 ToItemFeature_Init

Description

Initializes the properties of the SysML v2 element ItemFeature.

General Classes

- ToFeature_Init (from KerMLInitializers)

Association Ends

- to : PayloadFeature [1]
  {redefines: ToFeature_Init::to}

7.4.2.2.16 ToItemUsage_Init

Description

Generic mapping class for mappings to the SysML v2 element ItemUsage.

General Classes

- ToOccurrenceUsage_Init (from SystemInitializers)

Association Ends

- to : ItemUsage [1]
  {redefines: ToOccurrenceUsage_Init::to}

7.4.2.2.17 ToMetadataUsage_Init

Description

Initializes the properties of the SysML v2 element MetadataUsage.

General Classes

- ToUsage_Init (from SystemInitializers)

Association Ends

- to : MetadataUsage [1]
  {redefines: ToUsage_Init::to}

7.4.2.2.18 ToObjectiveMembership_Init

Description

Initializes the properties of the SysML v2 element ObjectiveMembership.

General Classes

- ToFeatureMembership_Init (from KerMLInitializers)

Association Ends

- to : ObjectiveMembership [1]
  {redefines: ToFeatureMembership_Init::to}

7.4.2.2.19 ToOccurenceDefinition_Init

Description

Initializes the properties of the SysML v2 element OccurrenceDefinition.

General Classes

- ToDefinition_Init (from SystemInitializers)

Association Ends

- to : OccurrenceDefinition [1]
  {redefines: ToDefinition_Init::to}

Operations

- isIndividual () : Boolean [1]

```
false
```

7.4.2.2.20 ToOccurrenceUsage_Init

Description

Initializes the properties of the SysML v2 element OccurrenceUsage.

General Classes

- ToUsage_Init (from SystemInitializers)

Association Ends

- to : OccurrenceUsage [1]
  {redefines: ToUsage_Init::to}

Operations

- isIndividual () : Boolean [1]

```
false
```

- portionKind () : PortionKind [1]

```
invalid
```

7.4.2.2.21 ToPartUsage_Init

Description

Initializes the properties of the SysML v2 element PartUsage.

General Classes

- ToUsage_Init (from SystemInitializers)

Association Ends

- to : PartUsage [1]
  {redefines: ToUsage_Init::to}

7.4.2.2.22 ToPerformActionUsage_Init

Description

Initializes the properties of the SysML v2 element PerformActionUsage.

General Classes

- ToActionUsage_Init (from SystemInitializers)

Association Ends

- to : PerformActionUsage [1]
  {redefines: ToActionUsage_Init::to}

7.4.2.2.23 ToPortConjugation_Init

Description

Initializes the properties of the SysML v2 element PortConjugation.

General Classes

- ToConjugation_Init (from KerMLInitializers)

Association Ends

- to : PortConjugation [1]
  {redefines: ToConjugation_Init::to}

Operations

- originalPortDefinition () : PortDefinition [1] {redefines originalType, abstract}

7.4.2.2.24 ToPortDefinition_Init

Description

Initializes the properties of the SysML v2 element PortDefinition.

General Classes

- ToDefinition_Init (from SystemInitializers)

Association Ends

- to : PortDefinition [1]
  {redefines: ToDefinition_Init::to}

7.4.2.2.25 ToReferenceUsage_Init

Description

Provides the basic features to map to a ReferenceUsage element.

General Classes

- ToUsage_Init (from SystemInitializers)

Association Ends

- to : ReferenceUsage [1]
  {redefines: ToUsage_Init::to}

7.4.2.2.26 ToRequirementUsage_Init

Description

Initializes the properties of the SysML v2 element RequirementUsage.

General Classes

- ToUsage_Init (from SystemInitializers)

Association Ends

- to : RequirementUsage [1]
  {redefines: ToUsage_Init::to}

7.4.2.2.27 ToStateSubactionMembership_Init

Description

Initializes the properties of the SysML v2 element StateSubactionMembership.

General Classes

- ToFeatureMembership_Init (from KerMLInitializers)

Association Ends

- to : StateSubactionMembership [1]
  {redefines: ToFeatureMembership_Init::to}

7.4.2.2.28 ToStateUsage_Init

Description

Initializes the properties of the SysML v2 element StateUsage.

General Classes

- ToActionUsage_Init (from SystemInitializers)

Association Ends

- to : StateUsage [1]
  {redefines: ToActionUsage_Init::to}

7.4.2.2.29 ToSubjectMembership_Init

Description

Initializes the properties of the SysML v2 element SubjectMembership.

General Classes

- ToParameterMembership_Init (from KerMLInitializers)

Association Ends

- to : SubjectMembership [1]
  {redefines: ToParameterMembership_Init::to}

7.4.2.2.30 ToTransitionUsage_Init

Description

Initializes the properties of the SysML v2 element TransitionUsage.

General Classes

- ToActionUsage_Init (from SystemInitializers)

Association Ends

- to : TransitionUsage [1]
  {redefines: ToActionUsage_Init::to}

7.4.2.2.31 ToTriggerInvocationExpression_Init

Description

Initializes the properties of the SysML v2 element TriggerInvocationExpression.

General Classes

- ToInvocationExpression_Init (from KerMLInitializers)

Association Ends

- to : TriggerInvocationExpression [1]
  {redefines: ToInvocationExpression_Init::to}

Operations

- kind () : TriggerKind [0..1] {redefines direction, abstract}

7.4.2.2.32 ToUsage_Init

Description

Initializes the properties of the SysML v2 element Usage.

General Classes

- ToFeature_Init (from KerMLInitializers)

Association Ends

- to : Usage [1]
  {redefines: ToFeature_Init::to}

Operations

- isVariation () : Boolean [1]

```
false
```

7.5 Factories

7.5.1 Overview

The classes presented in this subclause specify facilities for creating elements in the target model form an arbitrary set of zero to many input parameters. After the target element is created, no link between it and an the value of inputs parameter (if any) will be preserved.

7.5.2 Mapping Specifications

7.5.2.1 LiteralString_Factory

Description

Factory class to create a LiteralString element.

General Classes

- Factory (from Foundations)
- ToExpression_Init (from KerMLInitializers)

Association Ends

- string : String [1]
- to : LiteralString [1]
  {redefines: ToExpression_Init::to}

Operations

- create (in string : String) : LiteralString [1]
- ownedRelationship () : Relationship [0..*] {redefines ownedRelationship}

```
Set{ReturnParameterFeatureMembership_Factory.create()}
```

7.5.2.2 StringParameterFeature_Factory

Description

Factory class to create a feature element representing a string.

General Classes

- Factory (from Foundations)
- ToFeature_Init (from KerMLInitializers)

Association Ends

- string : String [1]

Operations

- create (in string : String) : Feature [1]
- ownedRelationship () : Relationship [0..*] {redefines ownedRelationship}

```
Set{StringParameterFeatureValue_Factory.create(string)}
```

7.5.2.3 StringParameterFeatureValue_Factory

Description

Factory class to create a string feature value relationship for a feature element.

General Classes

- Factory (from Foundations)
- ToFeatureValue_Init (from KerMLInitializers)

Association Ends

- string : String [1]

Operations

- create (in string : String) : FeatureValue [1]
- value () : Expression [1] {redefines value}

```
LiteralString_Factory.create(string)
```

7.5.2.4 StringParameterMembership_Factory

Description

Factory class to create a parameter membership relationship for a feature element representing a string.

General Classes

- Factory (from Foundations)
- ToParameterMembership_Init (from KerMLInitializers)

Association Ends

- string : String [1]

Operations

- create (in string : String) : ParameterMembership [1]
- ownedMemberParameter () : Feature [1] {redefines ownedMemberParameter}

```
StringParameterFeature_Factory.create(string)
```

7.5.2.5 SubjectMembership_Factory

Description

Factory class to create a subject membership relationship for a given subject.

General Classes

- Factory (from Foundations)
- ToSubjectMembership_Init (from SystemInitializers)

Association Ends

- subject : Type [1]

Operations

- create (in subject : Type) : SubjectMembership [1]
- ownedMemberParameter () : Feature [1] {redefines ownedMemberParameter}

```
subject
```

7.5.2.6 AssignmentActionUsage_Factory

Description

Factory to create an assignment action usage.

General Classes

- Factory (from Foundations)
- ToAssignmentActionUsage_Init (from SystemInitializers)

Operations

- create () : AssignmentActionUsage [1]
- ownedRelationship () : Relationship [0..*] {redefines ownedRelationship}

```
Set{AssignmentActionUsageParameterMembership_Factory.create(),
DirectedReferenceUsageParameterMembership_Factory.create(
  KerML::FeatureDirectionKind::_'in')}
```

7.5.2.7 AssignmentActionUsageFeatureMembership2_Factory

Description

Factory class to create a feature membership relationship for a feature element created by the factory class AssignmentActionUsageTargetReferenceUsageIn2_Factory.

General Classes

- Factory (from Foundations)
- ToFeatureMembership_Init (from KerMLInitializers)

Operations

- create () : FeatureMembership [1]
- ownedMemberFeature () : Feature [1] {redefines ownedMemberFeature}

```
AssignmentActionUsageTargetReferenceUsageIn2_Factory.create()
```

7.5.2.8 AssignmentActionUsageFeatureMembership3_Factory

Description

Factory class to create a feature membership relationship for a feature element created by the factory class AssignmentActionUsageTargetReferenceUsageIn3_Factory.

General Classes

- Factory (from Foundations)
- ToFeatureMembership_Init (from KerMLInitializers)

Operations

- create () : FeatureMembership [1]
- ownedMemberFeature () : Feature [1] {redefines ownedMemberFeature}

```
AssignmentActionUsageTargetReferenceUsageIn3_Factory.create()
```

7.5.2.9 AssignmentActionUsageOwningMembership_Factory

Description

Factory class to create a owning membership relationship for an element created by the factory class AssignmentActionUsage_Factory.

General Classes

- Factory (from Foundations)
- ToOwningMembership_Init (from KerMLInitializers)

Operations

- create () : OwningMembership [1]
- ownedMemberElement () : Element [1] {redefines ownedMemberElement}

```
AssignmentActionUsage_Factory.create()
```

7.5.2.10 AssignmentActionUsageParameterMembership_Factory

Description

Factory class to create a parameter membership relationship for a feature element created by the factory class AssignmentActionUsageReferenceUsageIn1_Factory.

General Classes

- Factory (from Foundations)
- ToParameterMembership_Init (from KerMLInitializers)

Operations

- create () : ParameterMembership [1]
- ownedMemberParameter () : Feature [1] {redefines ownedMemberParameter}

```
AssignmentActionUsageReferenceUsageIn1_Factory.create()
```

7.5.2.11 AssignmentActionUsageReferenceUsageIn1_Factory

Description

Factory class creating a reference usage element with direction "in" as parameter of an assignment action usage.

General Classes

- Factory (from Foundations)
- ToReferenceUsage_Init (from SystemInitializers)

Operations

- create () : ReferenceUsage [1]
- direction () : FeatureDirectionKind [0..1] {redefines direction}

```
KerML::FeatureDirectionKind::_'in'
```

- ownedRelationship () : Relationship [0..*] {redefines ownedRelationship}

```
Set{AssignmentActionUsageFeatureMembership2_Factory.create()}
```

7.5.2.12 AssignmentActionUsageTargetReferenceUsageIn2_Factory

Description

Factory class creating a reference usage element as an owned feature of the reference usage of an assignment action usage.

General Classes

- Factory (from Foundations)
- ToReferenceUsage_Init (from SystemInitializers)

Operations

- create () : ReferenceUsage [1]
- ownedRelationship () : Relationship [0..*] {redefines ownedRelationship}

```
Set{AssignmentActionUsageFeatureMembership3_Factory.create()}
```

7.5.2.13 AssignmentActionUsageTargetReferenceUsageIn3_Factory

Description

Factory class creating a reference usage element as an owned feature of the reference usage of an assignment action usage.

General Classes

- Factory (from Foundations)
- ToReferenceUsage_Init (from SystemInitializers)

Operations

- create () : ReferenceUsage [1]

7.5.2.14 DirectedReferenceUsage_Factory

Description

Factory class creating a reference usage element with a given direction and without owned relationships.

General Classes

- Factory (from Foundations)
- ToReferenceUsage_Init (from SystemInitializers)

Association Ends

- featureDirectionKind : FeatureDirectionKind [1]

Operations

- create (in featureDirectionKind : FeatureDirectionKind) : ReferenceUsage [1]
- direction () : FeatureDirectionKind [0..1] {redefines direction}

```
featureDirectionKind
```

7.5.2.15 DirectedReferenceUsageParameterMembership_Factory

Description

Factory class to create a parameter membership relationship for a feature element created by the factory class DirectedReferenceUsage_Factory.

General Classes

- Factory (from Foundations)
- ToParameterMembership_Init (from KerMLInitializers)

Association Ends

- featureDirectionKind : FeatureDirectionKind [1]

Operations

- create (in featureDirectionKind : FeatureDirectionKind) : ParameterMembership [1]
- ownedMemberParameter () : Feature [1] {redefines ownedMemberParameter}

```
DirectedReferenceUsage_Factory.create(featureDirectionKind)
```

7.5.2.16 EmptyObjectiveMembership_Factory

Description

Factory class to create an objective membership without a source in the SysML v1 model.

General Classes

- Factory (from Foundations)
- ToObjectiveMembership_Init (from SystemInitializers)

Operations

- create () : ObjectiveMembership [1]
- ownedMemberFeature () : Feature [1] {redefines ownedMemberFeature}

```
EmptyRequirementUsage_Factory.create()
```

7.5.2.17 EmptyRequirementUsage_Factory

Description

Factory class to create a requirement usage without a source in the SysML v1 model.

General Classes

- Factory (from Foundations)
- ToRequirementUsage_Init (from SystemInitializers)

Operations

- create () : RequirementUsage [1]
- ownedRelationship () : Relationship [0..*] {redefines ownedRelationship}

```
 Set{
EmptySubjectMembership_Factory.create(),
ReturnParameterFeatureMembership_Factory.create()}
```

7.5.2.18 EmptySubject_Factory

Description

Factory class to create a reference usage representing a subject without a source in the SysML v1 model.

General Classes

- Factory (from Foundations)
- ToReferenceUsage_Init (from SystemInitializers)

Operations

- create () : ReferenceUsage [1]
- direction () : FeatureDirectionKind [0..1] {redefines direction}

```
KerML::FeatureDirectionKind::_'in'
```

7.5.2.19 EmptySubjectMembership_Factory

Description

Factory class to create a memberhsip relationship for a reference usage representing a subject without a source in the SysML v1 model.

General Classes

- Factory (from Foundations)
- ToSubjectMembership_Init (from SystemInitializers)

Operations

- create () : SubjectMembership [1]
- ownedMemberParameter () : Feature [1] {redefines ownedMemberParameter}

```
EmptySubject_Factory.create()
```

7.5.2.20 FeatureTyping_Factory

Description

Factory class to create a FeatureTyping relationship. The create parameter is set as the type.

General Classes

- Factory (from Foundations)
- ToFeatureTyping_Init (from KerMLInitializers)

Association Ends

- type : NamedElement [1]

Operations

- create (in type : NamedElement) : FeatureTyping [1]
- type () : Type [1] {redefines type}

```
type
```

7.5.2.21 FlowEndParameterMembership_Factory

Description

Factory class to create a ParameterMembership relationship for an end of a FlowUsage as a target element for a UML4SysML::InformationFlow that is realized by a UML4SysML::Connector.

General Classes

- Factory (from Foundations)
- ToParameterMembership_Init (from KerMLInitializers)

Association Ends

- end : NamedElement [1]
- informationFlow : InformationFlow [1]

Operations

- create (in informationFlow : InformationFlow, in end : NamedElement) : ParameterMembership [1]
- ownedMemberParameter () : Feature [1] {redefines ownedMemberParameter}

```
InformationFlowEventOccurrenceUsage_Factory.create(informationFlow, end)
```

7.5.2.22 FlowItem_Factory

Description

Factory class to create a ItemFeature element as a target element for the flowing entity specified by an UML4SysML::InformationFlow.

General Classes

- Factory (from Foundations)
- ToItemFeature_Init (from SystemInitializers)

Association Ends

- item : NamedElement [1]

Operations

- create (in item : NamedElement) : PayloadFeature [1]
- ownedRelationship () : Relationship [0..*] {redefines ownedRelationship}

```
if item.oclIsKindOf(UML::Classifier) then
    Set{FeatureTyping_Factory.create(item)}
else if item.oclIsKindOf(UML::Property) then
        Set{ReferenceSubsetting_Factory.create(item)}
     else
        Set{}
     endif
endif
```

7.5.2.23 FlowItemFeatureMembership_Factory

Description

Factory class to create a FeatureMembership relationship for an ItemFeature as a target element for the flowing entity specified by an UML4SysML::InformationFlow.

General Classes

- Factory (from Foundations)
- ToFeatureMembership_Init (from KerMLInitializers)

Association Ends

- item : NamedElement [1]

Operations

- create (in item : NamedElement) : FeatureMembership [1]
- ownedMemberFeature () : Feature [1] {redefines ownedMemberFeature}

```
FlowItem_Factory.create(item)
```

7.5.2.24 FlowUsage_Factory

Description

Factory class to create a FlowUsage as a target element for a UML4SysML::InformationFlow that is realized by a UML4SysML::Connector. The factory class only supports UML4SysML::InformationFlows which have exactly one source and one target element, which is implicitly assured since connectors in SysML may only ever have two ends.

General Classes

- Factory (from Foundations)
- ToFlowUsage_Init (from SystemInitializers)

Association Ends

- informationFlow : InformationFlow [1]

Operations

- create (in informationFlow : InformationFlow) : FlowUsage [1]
- ownedRelationship () : Relationship [0..*] {redefines ownedRelationship}

```
let relationships : Set(KerML::Relationship) =
  informationFlow.realizingConnector->collect(c|Subsetting_Factory.create(c))
  ->including(FeatureTyping_Factory.create(informationFlow))
  ->including(FlowEndParameterMembership_Factory.create(
              informationFlow,informationFlow.source.get(0)))
  ->including(FlowEndParameterMembership_Factory.create(
              informationFlow,informationFlow.target.get(0))) in
let itemProperty : UML::Property =
  if Helper.hasStereotypeApplied(informationFlow,
       'SysML::Ports&Flows::ItemFlow') then
    Helper.getTagValueAsElement(informationFlow,
       'SysML::Ports&Flows::ItemFlow', 'itemProperty')
  else
    invalid
  endif in

if itemProperty.oclIsUndefined() then
  relationships->union(informationFlow.conveyed->flatten()
    ->collect(i | FlowItemFeatureMembership_Factory.create(i)))
else
  relationships->including(
    FlowItemFeatureMembership_Factory.create(itemProperty))
endif
```

7.5.2.25 FlowUsageFeatureMembership_Factory

Description

Factory class to create a FeatureMembership relationship for a FlowUsage as a target element for a UML4SysML::InformationFlow that is realized by a UML4SysML::Connector.

General Classes

- Factory (from Foundations)
- ToFeatureMembership_Init (from KerMLInitializers)

Association Ends

- informationFlow : InformationFlow [1]

Operations

- create (in informationFlow : InformationFlow) : FeatureMembership [1]
- ownedMemberFeature () : Feature [1] {redefines ownedMemberFeature}

```
FlowUsage_Factory.create(informationFlow)
```

7.5.2.26 InformationFlowEventOccurrenceUsage_Factory

Description

General Classes

- Factory (from Foundations)
- ToEventOccurerenceUsage_Init (from SystemInitializers)

Association Ends

- end : NamedElement [1]
- informationFlow : InformationFlow [1]

Operations

- create (in informationFlow : InformationFlow, in end : NamedElement) : EventOccurrenceUsage [1]
- ownedRelationship () : Relationship [0..*] {redefines ownedRelationship}

```
Set{InformationFlowReferenceSubsetting_Factory.create(informationFlow, end)}
```

7.5.2.27 InformationFlowReferenceSubsetting_Factory

Description

Factory class to create a ReferenceSubsetting relationship for an end of a FlowUsage subsetting the target element of an end element of an UML4SysML::InformationFlow.

General Classes

- Factory (from Foundations)
- ToReferenceSubsetting_Init (from KerMLInitializers)

Association Ends

- end : NamedElement [1]
- informationFlow : InformationFlow [1]

Operations

- create (in informationFlow : InformationFlow, in end : NamedElement) : ReferenceSubsetting [1]
- referencedFeature () : Feature [1] {redefines referencedFeature}

```
InformationFlowEnd_Mapping.getMapped(informationFlow, end)
```

7.5.2.28 LiteralBoolean_Factory

Description

Factory class to create a LiteralBoolean element.

General Classes

- Factory (from Foundations)
- ToExpression_Init (from KerMLInitializers)

Association Ends

- boolean : Boolean [1]

- to : LiteralBoolean [1]
  {redefines: ToExpression_Init::to}

Operations

- create (in boolean : Boolean) : LiteralBoolean [1]
- ownedRelationship () : Relationship [0..*] {redefines ownedRelationship}

```
Set{ReturnParameterFeatureMembership_Factory.create()}
```

7.5.2.29 LiteralNull_Factory

Description

Factory class to create a LiteralNull element.

General Classes

- Factory (from Foundations)
- ToExpression_Init (from KerMLInitializers)

Association Ends

- to : NullExpression [1]
  {redefines: ToExpression_Init::to}

Operations

- create () : NullExpression [1]
- ownedRelationship () : Relationship [0..*] {redefines ownedRelationship}

```
Set{ReturnParameterFeatureMembership_Factory.create()}
```

7.5.2.30 LiteralRational_Factory

Description

Factory class to create a LiteralRational element.

General Classes

- Factory (from Foundations)
- ToExpression_Init (from KerMLInitializers)

Association Ends

- real : Real [1]
- to : LiteralRational [1]
  {redefines: ToExpression_Init::to}

Operations

- create (in real : Real) : LiteralReal [1]
- ownedRelationship () : Relationship [0..*] {redefines ownedRelationship}

```
Set{ReturnParameterFeatureMembership_Factory.create()}
```

7.5.2.31 LowerBound_Factory

Description

General Classes

- Factory (from Foundations)

Association Ends

- multiplicityLowerBoundMembership : MultiplicityLowerBoundMembership_Factory [1]
- to : LiteralInteger [1]
  {redefines: Initializer::to}

Operations

- create (in lowerValue : Integer) : LiteralInteger [1]
- ownedRelationship () : Relationship [0..*]

```
Set{ReturnParameterFeatureMembership_Factory.create()}
```

- value () : Integer [1]

```
lowerValue
```

7.5.2.32 MultiplicityElement_Factory

Description

General Classes

- Factory (from Foundations)
- ToFeature_Init (from KerMLInitializers)

Association Ends

- lowerValue : Integer [1]
- multiplicityLowerBoundMembership : MultiplicityLowerBoundMembership_Factory [1]
- multiplicityUpperBoundMembership : MultiplicityUpperBoundMembership_Factory [1]
- upperValue : Integer [1]

Operations

- create (in lowerValue : Integer, in upperValue : Integer) : Feature [1]
- ownedRelationship () : Relationship [0..*] {redefines ownedRelationship}

```
Set{self.multiplicityLowerBoundMembership, self.multiplicityUpperBoundMembership}
```

7.5.2.33 MultiplicityLowerBoundMembership_Factory

Description

General Classes

- Factory (from Foundations)
- ToFeatureMembership_Init (from KerMLInitializers)

Association Ends

- lowerBound : LowerBound_Factory [1]
- multiplicityElement : MultiplicityElement_Factory [1]

Operations

- create () : FeatureMembership [1]
- ownedMemberFeature () : Feature [1] {redefines ownedMemberFeature}

```
self.lowerBound
```

7.5.2.34 MultiplicityMembership_Factory

Description

General Classes

- Factory (from Foundations)
- ToFeatureMembership_Init (from KerMLInitializers)

Association Ends

- lowerValue : Integer [1]
- upperValue : Integer [1]

Operations

- create (in lowerValue : Integer, in upperValue : Integer) : FeatureMembership [1]
- ownedMemberFeature () : Feature [1] {redefines ownedMemberFeature}

```
 if upperValue = 1 then
    if lowerValue = 0 then
        Helper.getMultiplicityRangeByName('zeroOrOne')
    else if lowerValue = 1 then
        Helper.getMultiplicityRangeByName('exactlyOne')
    else
        MultiplicityElement_Factory.create(lowerValue, upperValue)
    endif endif
else if upperValue = -1 then
    if lowerValue = 0 then
        Helper.getMultiplicityRangeByName('zeroToMany')
    else if lowerValue = 1 then
        Helper.getMultiplicityRangeByName('oneToMany')
    else
        MultiplicityElement_Factory.create(lowerValue, upperValue)
    endif endif
else
        MultiplicityElement_Factory.create(lowerValue, upperValue)
endif endif
```

7.5.2.35 MultiplicityUpperBoundMembership_Factory

Description

General Classes

- Factory (from Foundations)
- ToFeatureMembership_Init (from KerMLInitializers)

Association Ends

- multiplicityElement : MultiplicityElement_Factory [1]
- upperBound : UpperBound_Factory [1]

Operations

- create (in upperValue : Integer) : FeatureMembership [1]

7.5.2.36 ObjectFlowItemFlowEndRedefinition_Factory

Description

General Classes

- Factory (from Foundations)
- ToRedefinition_Init (from KerMLInitializers)

Association Ends

- feature : Feature [1]

Operations

- create (in feature : Feature) : Redefinition [1]
- redefinedFeature () : Feature [1] {redefines redefinedFeature}

```
feature
```

7.5.2.37 ParameterMembership_Factory

Description

Factory class to create a ParameterMembership relationship.

General Classes

- Factory (from Foundations)
- ToParameterMembership_Init (from KerMLInitializers)

Operations

- create () : ParameterMembership [1]
- ownedMemberParameter () : Feature [1] {redefines ownedMemberParameter}

```
ReferenceUsage_Factory.create()
```

7.5.2.38 ReferenceSubsetting_Factory

Description

Factory class to create a ReferenceSubseeting relationship. The create parameter is set as the referenced feature.

General Classes

- Factory (from Foundations)
- ToReferenceSubsetting_Init (from KerMLInitializers)

Association Ends

- property : Property [1]

Operations

- create (in property : Property) : ReferenceSubsetting [1]
- referencedFeature () : Feature [1] {redefines referencedFeature}

```
property
```

7.5.2.39 ReferenceUsage_Factory

Description

Factory class to create a ReferenceUsage element with direction 'in'.

General Classes

- Factory (from Foundations)
- ToReferenceUsage_Init (from SystemInitializers)

Operations

- create () : ReferenceUsage [1]
- direction () : FeatureDirectionKind [0..1] {redefines direction}

```
KerML::FeatureDirectionKind::_'in'
```

7.5.2.40 ReturnParameterFeature_Factory

Description

Factory class to create a feature element with direction 'out' representing a return parameter.

General Classes

- Factory (from Foundations)
- ToFeature_Init (from KerMLInitializers)

Operations

- create () : Feature [1]
- direction () : FeatureDirectionKind [0..1] {redefines direction}

```
KerML::FeatureDirectionKind::_'out'
```

7.5.2.41 ReturnParameterFeatureMembership_Factory

Description

Factory class to create a feature membership relationship for a feature element with direction 'out' representing a return parameter.

General Classes

- Factory (from Foundations)
- ToReturnParameterMembership_Init (from KerMLInitializers)

Operations

- create () : ReturnParameterMembership [1]
- ownedMemberParameter () : Feature [1] {redefines ownedMemberParameter}

```
ReturnParameterFeature_Factory.create()
```

7.5.2.42 Subsetting_Factory

Description

Factory class to create a Subsetting relationship. The create parameter is set as the subsetted feature.

General Classes

- Factory (from Foundations)
- ToSubsetting_Init (from KerMLInitializers)

Association Ends

- subsetted : NamedElement [1]

Operations

- create (in subsetted : NamedElement) : Subsetting [1]
- subsettedFeature () : Feature [1] {redefines subsettedFeature}

```
subsetted
```

7.5.2.43 UpperBound_Factory

Description

General Classes

- Factory (from Foundations)

Association Ends

- multiplicityUpperBoundMembership : MultiplicityUpperBoundMembership_Factory [1]

- to : LiteralInteger [1]
  {redefines: Initializer::to}

Operations

- create (in upperValue : Integer) : LiteralInteger [1]
- ownedRelationship () : Relationship [0..*]

```
Set{ReturnParameterFeatureMembership_Factory.create()}
```

- value () : Integer [1]

```
upperValue
```

7.6 Generic Mappings

7.6.1 Overview

Generic mappings are partial definitions of transformation rules that are intended to factorize reusable algorithms for making the global specification more compact and easier to read and maintain. Basically, they provide a default value for all the non-derived attributes of their target metaclass wherever possible, or declare an abstract operation for them otherwise. They are similar to initializers, except that they have a source element defined. The operations provided by the generic mappings can be redefined by their specialization, as appropriate according to the source type specified by the redefinition of their `from` attribute.

All of these generic mappings are abstract.

7.6.2 Common Mappings

7.6.2.1 CommonFeatureReferenceExpression_Mapping

Description

Common mapping class for a feature reference expression.

General Mappings

ToFeatureReferenceExpression_Init
Mapping

Mapping Source

TypedElement

Mapping Target

FeatureReferenceExpression

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureReferenceExpression::ownedRelationship () : Relationship [0..*]

```
Set{CommonMembership_Mapping.getMapped(from),
CommonReturnParameterFeatureMembership_Mapping.getMapped(from)}
```

7.6.2.2 CommonMembership_Mapping

Description

Creates a membership relationship for *memberElement()*.

General Mappings

ToMembership_Init
Mapping

Mapping Source

TypedElement

Mapping Target

Membership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Membership::memberElement () : Element [1]

```
from
```

7.6.2.3 CommonParameterReferenceUsageInMembership_Mapping

Description

Creates a membership relationship for *memberElement()*.

General Mappings

ToParameterMembership_Init
Mapping

Mapping Source

Element

Mapping Target

ParameterMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ParameterMembership::ownedMemberParameter () : Feature [1]

```
if not from.oclIsKindOf(UML::TypedElement) then
    CommonParameterReferenceUsageIn_Mapping.getMapped(from)
else if from.oclAsType(UML::TypedElement).type.oclIsUndefined() then
        CommonParameterReferenceUsageIn_Mapping.getMapped(from)
    else
        CommonParameterReferenceUsageInUntyped_Mapping.getMapped(from)
    endif
endif
```

7.6.2.4 CommonParameterReferenceUsageIn_Mapping

Description

Common mapping class that creates a parameter reference usage element with direction 'in' and with a type.

General Mappings

CommonParameterReferenceUsageInUntyped_Mapping
Mapping

Mapping Source

Element

Mapping Target

ReferenceUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ReferenceUsage::ownedRelationship () : Relationship [0..*]

```
if from.oclIsKindOf(UML::TypedElement) then
Set{CommonParameterReferenceUsageInFeatureTyping_Mapping.getMapped(from)}
else Set{} endif
```

7.6.2.5 CommonParameterReferenceUsageInFeatureTyping_Mapping

Description

Creates a feature typing relationship owned by the element *typedFeature()*.

General Mappings

ToFeatureTyping_Init
Mapping

Mapping Source

Element

Mapping Target

FeatureTyping

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureTyping::type () : Type [1]

```
if from.oclIsKindOf(UML::TypedElement)
then
if from.oclAsType(UML::TypedElement).type.oclIsKindOf(UML::PrimitiveType) then
    Helper.getScalarValueType(from.oclAsType(UML::TypedElement).type)
else
    from.oclAsType(UML::TypedElement).type
endif
else invalid endif
```

7.6.2.6 CommonParameterReferenceUsageInUntyped_Mapping

Description

Common mapping class that creates a parameter reference usage element with direction 'in' and without a type.

General Mappings

ToReferenceUsage_Init
Mapping

Mapping Source

Element

Mapping Target

ReferenceUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ReferenceUsage::direction () : FeatureDirectionKind [0..1]

  ```
  KerML::FeatureDirectionKind::_'in'
  ```

7.6.2.7 CommonReturnParameterFeature_Mapping

Description

Common mapping class that creates a parameter feature element with a type.

General Mappings

CommonReturnParameterFeatureUntyped_Mapping
Mapping

Mapping Source

Element

Mapping Target

Feature

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Feature::ownedRelationship () : Relationship [0..*]

```
if from.oclIsKindOf(UML::Property) then
    Set{CommonReturnParameterFeatureTyping_Mapping.getMapped(from)}
else
    Set{}
endif
```

7.6.2.8 CommonReturnParameterFeatureTyping_Mapping

Description

Creates a feature typing relationship owned by the element *typedFeature()*.

General Mappings

ToFeatureTyping_Init
Mapping

Mapping Source

Element

Mapping Target

FeatureTyping

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureTyping::type () : Type [1]

```
if from.oclIsKindOf(UML::Property)
then
if from.oclAsType(UML::TypedElement).type.oclIsKindOf(UML::PrimitiveType) then
    Helper.getScalarValueType(from.oclAsType(UML::TypedElement).type)
else
    from.oclAsType(UML::TypedElement).type
endif
else invalid endif
```

7.6.2.9 CommonReturnParameterFeatureUntyped_Mapping

Description

Common mapping class that creates a parameter feature element without a type.

General Mappings

ToFeature_Init
Mapping

Mapping Source

Element

Mapping Target

Feature

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Feature::direction () : FeatureDirectionKind [0..1]

  ```
  KerML::FeatureDirectionKind::_'out'
  ```

7.6.2.10 CommonReturnParameterFeatureMembership_Mapping

Description

Creates a feature membership relationship for *ownedMemberFeature()*.

General Mappings

ToReturnParameterMembership_Init
Mapping

Mapping Source

Element

Mapping Target

ReturnParameterMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ReturnParameterMembership::ownedMemberParameter () : Feature [1]

```
if not from.oclIsKindOf(UML::TypedElement) then
    CommonReturnParameterFeatureUntyped_Mapping.getMapped(from)
else if from.oclAsType(UML::TypedElement).type.oclIsUndefined() then
        CommonReturnParameterFeatureUntyped_Mapping.getMapped(from)
     else
        CommonReturnParameterFeatureUntyped_Mapping.getMapped(from)
     endif
endif
```

7.6.2.11 CommonReturnParameterReferenceUsageMembership_Mapping

Description

Creates a membership relationship for *memberElement()*.

General Mappings

ToReturnParameterMembership_Init
Mapping

Mapping Source

Element

Mapping Target

ReturnParameterMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ReturnParameterMembership::ownedMemberParameter () : Feature [0..1]

```
if not from.oclIsKindOf(UML::TypedElement) then
    CommonReturnParameterReferenceUsageUntyped_Mapping.getMapped(from)
else if from.oclAsType(UML::TypedElement).type.oclIsUndefined() then
        CommonReturnParameterReferenceUsageUntyped_Mapping.getMapped(from)
     else
        CommonReturnParameterReferenceUsageUntyped_Mapping.getMapped(from)
```

```
        endif
    endif
```

7.6.2.12 CommonReturnParameterReferenceUsage_Mapping

Description

Creates a reference usage.

General Mappings

CommonReturnParameterReferenceUsageUntyped_Mapping
Mapping

Mapping Source

Element

Mapping Target

ReferenceUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ReferenceUsage::ownedRelationship () : Relationship [0..*]

```
if from.oclIsKindOf(UML::TypedElement) then
Set{CommonReturnParameterReferenceUsageFeatureTyping_Mapping.getMapped(from)}
else Set{} endif
```

7.6.2.13 CommonReturnParameterReferenceUsageFeatureTyping_Mapping

Description

Creates a feature typing relationship owned by the element *typedFeature()*.

General Mappings

ToFeatureTyping_Init
Mapping

Mapping Source

Element

Mapping Target

FeatureTyping

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureTyping::type () : Type [1]

```
if from.oclIsKindOf(UML::TypedElement)
then
if from.oclAsType(UML::TypedElement).type.oclIsKindOf(UML::PrimitiveType) then
    Helper.getScalarValueType(from.oclAsType(UML::TypedElement).type)
else
    from.oclAsType(UML::TypedElement).type
endif
else invalid endif
```

7.6.2.14 CommonReturnParameterReferenceUsageUntyped_Mapping

Description

Creates a reference usage.

General Mappings

ToReferenceUsage_Init
Mapping

Mapping Source

Element

Mapping Target

ReferenceUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ReferenceUsage::direction () : FeatureDirectionKind [0..1]

  ```
  KerML::FeatureDirectionKind::_'out'
  ```

7.6.2.15 CommonReferenceUsageIn_Mapping

Description

Common mapping class that creates a reference usage element with direction 'in'.

General Mappings

CommonReferenceUsageInUntyped_Mapping
Mapping

Mapping Source

TypedElement

Mapping Target

ReferenceUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ReferenceUsage::ownedRelationship () : Relationship [0..*]

  Common mapping class that creates a reference usage element with direction 'in'.

  ```
  Set{CommonReferenceUsageInFeatureTyping_Mapping.getMapped(from)}
  ```

7.6.2.16 CommonReferenceUsageInFeatureMembership_Mapping

Description

Creates a feature membership relationship for *ownedMemberFeature()*.

General Mappings

ToFeatureMembership_Init
Mapping

Mapping Source

TypedElement

Mapping Target

FeatureMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureMembership::ownedMemberFeature () : Feature [1]

```
if from.type.oclIsUndefined() then
    CommonReferenceUsageInUntyped_Mapping.getMapped(from)
else
    CommonReferenceUsageIn_Mapping.getMapped(from)
endif
```

7.6.2.17 CommonReferenceUsageInFeatureTyping_Mapping

Description

Creates a feature typing relationship owned by the element *typedFeature()*.

General Mappings

ToFeatureTyping_Init
Mapping

Mapping Source

TypedElement

Mapping Target

FeatureTyping

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureTyping::type () : Type [1]

```
if from.type.oclIsKindOf(UML::PrimitiveType) then
    Helper.getScalarValueType(from.type)
else
    from.type
endif
```

7.6.2.18 CommonReferenceUsageInUntyped_Mapping

Description

Common mapping class that creates an untyped reference usage element with direction 'in'.

General Mappings

ToReferenceUsage_Init
Mapping

Mapping Source

TypedElement

Mapping Target

ReferenceUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ReferenceUsage::declaredName () : String [0..1]

  ```
  from.name
  ```

- ReferenceUsage::direction () : FeatureDirectionKind [0..1]

  ```
  KerML::FeatureDirectionKind::_'in'
  ```

7.7 Mappings from UML4SysML metaclasses

7.7.1 Overview

`UML4SysML` is the subset of UML containing all model elements that are reused by SysML. The complete list of model elements is defined in [SysMLv1], subclause 4.1.

7.7.2 Actions

SYSML21-542: Transformation: overview tables shall be revised

7.7.2.1 Overview

SYSML21-542: Transformation: overview tables shall be revised

Table 1. List of High-Level Action Mappings

SysML v1 Abstract Syntax/Stereotype	SysML v2 Abstract Syntax
AcceptCallAction	Since the CallEvent is not supported by SysML v2, the AcceptCallAction is also not covered. It is mapped to an empty action usage to keep the connections within the activity respectively action definition.
AcceptEventAction	AcceptActionUsage
ActionInputPin	The UML4SysML::ActionInputPin concept is not covered by SysML v2. The model element is mapped as a input or output pin, but without the special action input pin semantics.
AddStructuralFeatureValueAction	ActionUsage
AddVariableValueAction	ActionUsage
BroadcastSignalAction	ActionUsage
CallBehaviorAction	ActionUsage
CallOperationAction	ActionUsage
Clause	No specific mapping specified yet.
ClearAssociationAction	ActionUsage
ClearStructuralFeatureAction	ActionUsage
ClearVariableAction	ActionUsage
ConditionalNode	No specific mapping specified yet.
CreateLinkAction	ActionUsage
CreateLinkObjectAction	ActionUsage
CreateObjectAction	ActionUsage
DestroyLinkAction	ActionUsage
DestroyObjectAction	ActionUsage
InputPin	ReferenceUsage
LinkEndCreationData	No specific mapping specified yet.
LinkEndData	No specific mapping specified yet.
LinkEndDestructionData	No specific mapping specified yet.
LoopNode	ActionUsage
OpaqueAction	ActionUsage
OutputPin	ReferenceUsage

SysML v1 Abstract Syntax/Stereotype	SysML v2 Abstract Syntax
RaiseExceptionAction	ActionUsage
ReadExtentAction	ActionUsage
ReadIsClassifiedObjectAction	ActionUsage
ReadLinkAction	ActionUsage
ReadLinkObjectEndAction	ActionUsage
ReadSelfAction	ActionUsage
ReadStructuralFeatureAction	ActionUsage
ReadVariableAction	ActionUsage
ReclassifyObjectAction	The UML4SysML::ReclassifyObjectAction is not supported by SysML v2. It is mapped to an empty action usage to keep the connections within the activity respectively action definition.
ReduceAction	ActionUsage
RemoveStructuralFeatureValueAction	ActionUsage
RemoveVariableValueAction	ActionUsage
ReplyAction	The UML4SysML::ReplyAction is only used with UML4SysML::AcceptCallAction. Since we have no mapping of AcceptCallAction to SysML v2, there is also no mapping for ReplyAction. However, it is mapped to an empty action usage to keep the connections within the activity respectively action definition.
SendObjectAction	ActionUsage
SendSignalAction	ActionUsage
SequenceNode	ActionUsage
StartClassifierBehaviorAction	The UML4SysML::StartClassifierBehaviorAction is not supported by SysML v2. It is mapped to an empty action usage to keep the connections within the activity respectively action definition.
StartObjectBehaviorAction	The UML4SysML::StartObjectBehaviorAction is not supported by SysML v2. It is mapped to an empty action usage to keep the connections within the activity respectively action definition.
StructuredActivityNode	ActionUsage
TestIdentityAction	CalculationUsage
UnmarshallAction	No specific mapping specified yet.
ValuePin	ReferenceUsage
ValueSpecificationAction	ActionUsage

7.7.2.2 Mapping Specifications

7.7.2.2.1 Accept Event Actions

7.7.2.2.1.1 AcceptCallAction_Mapping

Description

Since the CallEvent is not supported by SysML v2, the AcceptCallAction is also not covered. It is mapped to an empty action usage to keep the connections within the activity respectively action definition.

General Mappings

AcceptEventAction_Mapping

Mapping Source

AcceptCallAction

Mapping Target

AcceptActionUsage

Owned Mappings

(none)

Applicable filters

(none)

7.7.2.2.1.2 AcceptEventAction_Mapping

Description

The UML4SysML::AcceptEventAction is mapped to a AcceptActionUsage element.

If the trigger is a signal, it is mapped to an accept parameter typed by the signal.

SysMLv2 does not support more than one trigger. Therefore only the first specified trigger of the action is transformed. All further triggers are ignored.

The following shows an example of what the textual SysML v2 syntax of the result of the transformation may look like.

```
action acceptEventActionSignalEvent1 accept : SysMLv1Signal via sysMLv1Port;
action acceptEventActionChangeEvent1 accept when when changeExpression.result {
        calc changeExpression {
                return : ScalarValues::Boolean;
                language "OCL"
                        /*
                         * x > 0
                         */
        }
}
```

General Mappings

CommonAction_Mapping

Mapping Source

AcceptEventAction

Mapping Target

AcceptActionUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- AcceptActionUsage::ownedRelationship () : Relationship [0..*]

```
let relationships : Set(KerML::Relationship) = Helper.actionOwnedRelationship(from)
->including(AEAReceiverParameterMembership_Mapping.getMapped(from)) in
let relationshipsWithParameter : Set(KerML::Relationship) =
if (from.trigger.get(0).event.oclIsTypeOf(UML::SignalEvent) or
    from.trigger.get(0).event.oclIsTypeOf(UML::ChangeEvent)) then
    relationships->including(AEAParameterMembership_Mapping.getMapped(from))
else
    relationships
endif in
if from.trigger.get(0).event.oclIsTypeOf(UML::ChangeEvent) then
    relationshipsWithParameter
        ->including(ElementFeatureMembership_Mapping.getMapped(
            from.trigger.get(0).event.oclAsType(UML::ChangeEvent).changeExpression))
else relationshipsWithParameter
endif
```

7.7.2.2.1.3 AEAChangeExpressionMembership_Mapping

Description

Creates a membership relationship for *memberElement()*.

General Mappings

ToFeatureMembership_Init
Mapping

Mapping Source

AcceptEventAction

Mapping Target

FeatureMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureMembership::ownedMemberFeature () : Feature [1]

  ```
  from.trigger.get(0).event.oclAsType(UML::ChangeEvent).changeExpression
  ```

7.7.2.2.1.4 AEAChangeParameter_Mapping

Description

The mapping class transforms the change event specified at the AcceptEventAction.

General Mappings

ToReferenceUsage_Init
Mapping

Mapping Source

AcceptEventAction

Mapping Target

ReferenceUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ReferenceUsage::direction () : FeatureDirectionKind [0..1]

  ```
  KerML::FeatureDirectionKind::_'in'
  ```

- ReferenceUsage::ownedRelationship () : Relationship [0..*]

  ```
  Set{AEAChangeParameterFeatureValue_Mapping.getMapped(from)}
  ```

7.7.2.2.1.5 AEAChangeParameterFeatureValue_Mapping

Description

Creates a feature value relationship.

General Mappings

ToFeatureValue_Init
Mapping

Mapping Source

AcceptEventAction

Mapping Target

FeatureValue

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureValue::value () : Expression [1]

```
AEAChangeParameterTrigger_Mapping.getMapped(from)
```

7.7.2.2.1.6 AEAChangeParameterTrigger_Mapping

Description

The mapping class creates a TriggerInvocationExpression from the change event specified at the AcceptEventAction.

General Mappings

ToInvocationExpression_Init
Mapping

Mapping Source

AcceptEventAction

Mapping Target

TriggerInvocationExpression

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- TriggerInvocationExpression::ownedRelationship () : Relationship [0..*]

  ```
  Set{AEAChangeParameterFeatureMembership_Mapping.getMapped(from)}
  ```

7.7.2.2.1.7 AEAChangeParameterTriggerExpression_Mapping

Description

The mapping class creates the trigger expression element for the change parameter of the SysML v2 AcceptActionUsage element.

General Mappings

ToExpression_Init
Mapping

Mapping Source

AcceptEventAction

Mapping Target

Expression

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Expression::ownedRelationship () : Relationship [0..*]

  ```
  Set{AEAChangeParameterResultExpressionMembership_Mapping.getMapped(from)}
  ```

7.7.2.2.1.8 AEAChangeParameterResultExpressionMembership_Mapping

Description

Creates a membership relationship for *memberElement()*.

General Mappings

ToFeatureMembership_Init
Mapping

Mapping Source

AcceptEventAction

Mapping Target

ResultExpressionMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ResultExpressionMembership::ownedMemberFeature () : Feature [1]

  ```
  AEAChangeParameterFeatureChainExpression_Mapping.getMapped(from)
  ```

7.7.2.2.1.9 AEAChangeParameterFeatureChainExpression_Mapping

Description

The mapping class creates the feature chain expression element for the change parameter of the SysML v2 AcceptActionUsage element.

General Mappings

ToInvocationExpression_Init
Mapping

Mapping Source

AcceptEventAction

Mapping Target

FeatureChainExpression

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureChainExpression::ownedRelationship () : Relationship [0..*]

  `Set{AEAChangeParameterParameterMembership_Mapping.getMapped(from)}`

7.7.2.2.1.10 AEAChangeParameterFeatureMembership_Mapping

Description

Creates a feature membership relationship for *ownedMemberFeature()*.

General Mappings

ToFeatureMembership_Init
Mapping

Mapping Source

AcceptEventAction

Mapping Target

FeatureMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureMembership::ownedMemberFeature () : Feature [1]

  `AEAChangeParameterTriggerExpression_Mapping.getMapped(from)`

7.7.2.2.1.11 AEAChangeParameterFeature_Mapping

Description

The mapping class creates the feature for the feature chain expression element for the change parameter of the SysML v2 AcceptActionUsage element.

General Mappings

ToFeature_Init
Mapping

Mapping Source

AcceptEventAction

Mapping Target

Feature

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Feature::ownedRelationship () : Relationship [0..*]

  ```
  Set{AEAChangeParameterExpressionFeatureValue_Mapping.getMapped(from)}
  ```

7.7.2.2.1.12 AEAChangeParameterExpressionFeatureValue_Mapping

Description

Creates a feature value relationship.

General Mappings

ToFeatureValue_Init
Mapping

Mapping Source

AcceptEventAction

Mapping Target

FeatureValue

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureValue::value () : Expression [1]

  ```
  AEAChangeParameterFeatureReferenceExpression_Mapping.getMapped(from)
  ```

7.7.2.2.1.13 AEAChangeParameterFeatureReferenceExpression_Mapping

Description

The mapping class creates the feature reference expression for the feature chain expression element for the change parameter of the SysML v2 AcceptActionUsage element.

General Mappings

ToFeatureReferenceExpression_Init
Mapping

Mapping Source

AcceptEventAction

Mapping Target

FeatureReferenceExpression

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureReferenceExpression::ownedRelationship () : Relationship [0..*]

  `Set{AEAChangeParameterMembership_Mapping.getMapped(from)}`

7.7.2.2.1.14 AEAChangeParameterMembership_Mapping

Description

Creates a membership relationship for *memberElement()*.

General Mappings

ToMembership_Init
Mapping

Mapping Source

AcceptEventAction

Mapping Target

Membership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Membership::memberElement () : Element [1]

```
from.trigger.get(0).event.oclAsType(UML::ChangeEvent).changeExpression
```

7.7.2.2.1.15 AEAChangeParameterParameterMembership_Mapping

Description

Creates a membership relationship for *memberElement()*.

General Mappings

ToParameterMembership_Init
Mapping

Mapping Source

AcceptEventAction

Mapping Target

ParameterMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ParameterMembership::ownedMemberParameter () : Feature [1]

```
AEAChangeParameterFeature_Mapping.getMapped(from)
```

7.7.2.2.1.16 AEAReceiverParameter_Mapping

Description

The mapping class creates the reference usage element for the receiver parameter of the SysML v2 AcceptActionUsage element.

General Mappings

ToReferenceUsage_Init
Mapping

Mapping Source

AcceptEventAction

Mapping Target

ReferenceUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ReferenceUsage::ownedRelationship () : Relationship [0..*]

```
if from.trigger.get(0).port->size() > 0
then Set{AEAReceiverFeatureValue_Mapping.getMapped(from)}
else Set{}
endif
```

- ReferenceUsage::direction () : FeatureDirectionKind [0..1]

```
KerML::FeatureDirectionKind::_'in'
```

7.7.2.2.1.17 AEAReceiverParameterMembership_Mapping

Description

Creates a membership relationship for *memberElement()*.

General Mappings

ToParameterMembership_Init
Mapping

Mapping Source

AcceptEventAction

Mapping Target

ParameterMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ParameterMembership::ownedMemberParameter () : Feature [1]

  ```
  AEAReceiverParameter_Mapping.getMapped(from)
  ```

7.7.2.2.1.18 AEAReceiverFeatureValue_Mapping

Description

Creates a feature value relationship.

General Mappings

ToFeatureValue_Init
Mapping

Mapping Source

AcceptEventAction

Mapping Target

FeatureValue

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureValue::value () : Expression [1]

  ```
  AEAReceiverFeatureReferenceExpression_Mapping.getMapped(from)
  ```

7.7.2.2.1.19 AEASignalParameter_Mapping

Description

The mapping class creates the reference usage element for the signal parameter of the SysML v2 AcceptActionUsage element.

General Mappings

ToReferenceUsage_Init
Mapping

Mapping Source

AcceptEventAction

Mapping Target

ReferenceUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ReferenceUsage::ownedRelationship () : Relationship [0..*]

  ```
  Set{AEASignalParameterFeatureTyping_Mapping.getMapped(from)}
  ```

- ReferenceUsage::direction () : FeatureDirectionKind [0..1]

  ```
  KerML::FeatureDirectionKind::_'in'
  ```

7.7.2.2.1.20 AEASignalParameterFeatureTyping_Mapping

Description

Creates a feature typing relationship owned by the element *typedFeature()*.

General Mappings

ToFeatureTyping_Init
Mapping

Mapping Source

AcceptEventAction

Mapping Target

FeatureTyping

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureTyping::type () : Type [1]

```
let event : UML::Event = from.trigger.get(0).event in
if event.oclIsTypeOf(UML::SignalEvent) then
    event.oclAsType(UML::SignalEvent).signal
else invalid endif
```

7.7.2.2.1.21 AEAParameterMembership_Mapping

Description

The mapping class creates the parameter membership relationship for the element that can be received by the accept action. The source of the element is the trigger of the UML4SysML::AcceptEventAction.

Currently, more than one trigger is not supported by the transformation.

General Mappings

ToParameterMembership_Init
Mapping

Mapping Source

AcceptEventAction

Mapping Target

ParameterMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ParameterMembership::ownedMemberParameter () : Feature [1]

```
if from.trigger.get(0).event.oclIsTypeOf(UML::SignalEvent) then
    AEASignalParameter_Mapping.getMapped(from)
else if from.trigger.get(0).event.oclIsTypeOf(UML::ChangeEvent) then
    AEAChangeParameter_Mapping.getMapped(from)
```

```
else
    invalid
endif endif
```

7.7.2.2.1.22 AEAReceiverFeatureReferenceExpression_Mapping

Description

The mapping class creates the feature reference expression for the reference usage element for the receiver parameter of the SysML v2 AcceptActionUsage element.

General Mappings

ToFeatureReferenceExpression_Init
Mapping

Mapping Source

AcceptEventAction

Mapping Target

FeatureReferenceExpression

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureReferenceExpression::ownedRelationship () : Relationship [0..*]

  ```
  Set{AEAReceiverFeatureReferenceExpressionMembership_Mapping.getMapped(from),
  ReturnParameterFeatureMembership_Factory.create()}
  ```

7.7.2.2.1.23 AEAReceiverFeatureReferenceExpressionMembership_Mapping

Description

Creates a membership relationship for *memberElement()*.

General Mappings

ToMembership_Init
Mapping

Mapping Source

AcceptEventAction

Mapping Target

Membership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Membership::memberElement () : Element [1]

```
if from.trigger.get(0).port->size() > 0 then
    from.trigger.get(0).port.get(0)
else
    invalid
endif
```

7.7.2.2.1.24 ReplyAction_Mapping

Description

The UML4SysML::ReplyAction is only used with UML4SysML::AcceptCallAction. Since we have no mapping of AcceptCallAction to SysML v2, there is also no mapping for ReplyAction. However, it is mapped to an empty action usage to keep the connections within the activity respectively action definition.

General Mappings

CommonAction_Mapping

Mapping Source

ReplyAction

Mapping Target

ActionUsage

Owned Mappings

(none)

Applicable filters

(none)

7.7.2.2.1.25 UnmarshallAction_Mapping

Description

The mapping of UML4SysML::UnmarshallAction is not specified yet. It is currently mapped to an empty action usage to keep the connections within the activity respectively action definition.

General Mappings

CommonAction_Mapping

Mapping Source

UnmarshallAction

Mapping Target

ActionUsage

Owned Mappings

(none)

Applicable filters

(none)

7.7.2.2.2 Actions

7.7.2.2.2.1 CommonAction_Mapping

Description

Base mapping class for model elements of kind UML4SysML::Action. The target element is a SysML v2 ActionUsage.

General Mappings

ToActionUsage_Init
NamedElementMain_Mapping

Mapping Source

Action

Mapping Target

ActionUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ActionUsage::isComposite () : Boolean [1]

  ```
  true
  ```

- ActionUsage::ownedRelationship () : Relationship [0..*]

```
let actionInputPin: Set(UML::Element) =
    from.ownedElement->select(e | e.oclIsTypeOf(UML::ActionInputPin)) in
let triggers: Set(UML::Element) =
    from.ownedElement->select(e | e.oclIsKindOf(UML::Trigger)) in
let toElementFMS: Set(UML::Element) =
    from.ownedElement->select(e | e.oclIsKindOf(UML::Pin)) in
let toElementOMS: Set(UML::Element) =
    (((from.ownedElement-toElementFMS)-
    actionInputPin)-triggers)-from.ownedElement in
toElementOMS->collect(e | ElementOwningMembership_Mapping.getMapped(e))
  ->asSet()
->union(self.oclAsType(ElementMain_Mapping).ownedRelationship())
->union(toElementFMS
  ->collect(e | ElementFeatureMembership_Mapping.getMapped(e))->asSet())
```

7.7.2.2.2.2 OpaqueAction_Mapping

Description

The UML4SysML::OpaqueAction is mapped to a SysML v2 ActionUsage with a textual representation.

The following shows an example of the expected SysMLv2 textual syntax of a UML4SysML::OpaqueAction.

```
action thisIsAOpaqueAction {
  in x : ScalarValues::Integer;
  in y : ScalarValues::Integer;
  out result : ScalarValues::Boolean;

  language "OCL"
  /*
   * x = y + 1;
  */
}
```

General Mappings

CommonAction_Mapping

Mapping Source

OpaqueAction

Mapping Target

ActionUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ActionUsage::ownedRelationship () : Relationship [0..*]

```
if from.body->size() > 0 then
Helper.actionOwnedRelationship(from)->append(OABodyMembership_Mapping.getMapped(from))
else
Helper.actionOwnedRelationship(from)
endif
```

7.7.2.2.2.3 OABody_Mapping

Description

The languages and bodies of a UML4SysML::OpaqueAction are mapped to SysMLv2 TextualRepresentations.

General Mappings

ToAnnotatingElement_Init
Mapping

Mapping Source

OpaqueAction

Mapping Target

TextualRepresentation

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- TextualRepresentation::language () : String [1]

```
if from.language.notEmpty() then from.language.first() else invalid endif
```

- TextualRepresentation::body () : String [1]

```
if from.body.notEmpty() then from.body.first() else invalid endif
```

7.7.2.2.2.4 OABodyMembership_Mapping

Description

Creates a membership relationship for *memberElement()*.

General Mappings

ToOwningMembership_Init
Mapping

Mapping Source

OpaqueAction

Mapping Target

OwningMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- OwningMembership::ownedMemberElement () : Element [1]

  ```
  OABody_Mapping.getMapped(from)
  ```

7.7.2.2.2.5 Pin_Mapping

Description

Mapping class for model elements of kind UML4SysML::Pin. The operation ownedRelationship() makes a distinction between typed and untyped pins. The target element is a SysMLv2 ReferenceUsage.

The following shows an example of what the textual SysML v2 syntax of the result of the transformation may look like.

```
action def SysMLv1Activity {
        action sysMLv1Action {
                in sysMLv1InputPin : ScalarValues::Integer;
                out sysMLv1UntypedOutputPin;
        }
}
```

General Mappings

ToReferenceUsage_Init
NamedElementMain_Mapping

Mapping Source

Pin

Mapping Target

ReferenceUsage

Owned Mappings

(none)

Applicable filters

This mapping applies only if the following (OCL) condition implemented by the operation *filter(src : Element) : Boolean* is verified:

```
not Helper.excludedPin(src)
```

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ReferenceUsage::direction () : FeatureDirectionKind [0..1]

```
if from.oclIsTypeOf(UML::InputPin) then
    KerML::FeatureDirectionKind::_'in'
else if from.oclIsTypeOf(UML::OutputPin) then
    KerML::FeatureDirectionKind::_'out'
else
    invalid
endif endif
```

- ReferenceUsage::ownedRelationship () : Relationship [0..*]

```
self.oclAsType(ElementMain_Mapping).ownedRelationship()
->including(MultiplicityMembership_Mapping.getMapped(from))
```

7.7.2.2.2.6 ValuePin_Mapping

Description

A UML4SysML::ValuePin is mapped to a SysML v2 ReferenceUsage with assigned value.

The following shows an example of what the textual SysML v2 syntax of the result of the transformation may look like.

```
action sysMLv1Action {
        in sysMLv1ValuePin1 : ScalarValues::Integer = 42;
        in sysMLv1ValuePin2 = {
                return result;
                language "English"
                /*
                 * this is a opaque expression
                 */
                }.result;
}
```

General Mappings

Pin_Mapping

Mapping Source

ValuePin

Mapping Target

ReferenceUsage

Owned Mappings

(none)

Applicable filters

This mapping applies only if the following (OCL) condition implemented by the operation *filter(src : Element) : Boolean* is verified:

```
not Helper.excludedPin(src) and not src.type.oclIsUndefined()
```

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ReferenceUsage::ownedRelationship () : Relationship [0..*]

  ```
  Set{PinFeatureTyping_Mapping.getMapped(from),
  ValuePinFeatureValue_Mapping.getMapped(from),
  MultiplicityMembership_Mapping.getMapped(from)}
  ```

7.7.2.2.2.7 ValuePinFeatureValue_Mapping

Description

The mapping class creates the value expression for the reference usage element.

General Mappings

ToFeatureValue_Init
Mapping

Mapping Source

ValuePin

Mapping Target

FeatureValue

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureValue::value () : Expression [1]

  ```
  if from.value.oclIsUndefined() then invalid else from.value endif
  ```

7.7.2.2.2.8 ValuePinUntyped_Mapping

Description

Same as ValuePin_Mapping, but for UML4SysML::ValuePins without a specified type.

The following shows an example of what the textual SysML v2 syntax of the result of the transformation may look like.

```
action sysMLv1Action {
        in sysMLv1ValuePin1 = 42;
}
```

General Mappings

Pin_Mapping

Mapping Source

ValuePin

Mapping Target

ReferenceUsage

Owned Mappings

(none)

Applicable filters

This mapping applies only if the following (OCL) condition implemented by the operation *filter(src : Element) : Boolean* is verified:

```
not Helper.excludedPin(src) and src.type.oclIsUndefined()
```

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ReferenceUsage::ownedRelationship () : Relationship [0..*]

  ```
  self.oclAsType(Pin_Mapping).ownedRelationship()

    ->including(ValuePinFeatureValue_Mapping.getMapped(from))
  ```

7.7.2.2.3 Invocation Actions

7.7.2.2.3.1 BroadcastSignalAction_Mapping

Description

The UML4SysML::BroadcastSignalAction is mapped to a SysML v2 ActionUsage. The details of the mapping are not defined yet.

General Mappings

CommonAction_Mapping

Mapping Source

BroadcastSignalAction

Mapping Target

ActionUsage

Owned Mappings

(none)

Applicable filters

(none)

7.7.2.2.3.2 CallBehaviorAction_Mapping

Description

A UML4SysML::CallBehaviorAction is mapped to a SysML v2 ActionUsage.

The following shows an example of what the textual SysML v2 syntax of the result of the transformation may look like.

```
action def SysMLv1Activity1 {
        action sysMLv1CallBehaviorAction : SysMLv1Activity2;
}
action def SysMLv1Activity2;
```

General Mappings

CommonAction_Mapping

Mapping Source

CallBehaviorAction

Mapping Target

ActionUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ActionUsage::ownedRelationship () : Relationship [0..*]

  ```
  Helper.actionOwnedRelationship(from)
  ->append(CBAFeatureTyping_Mapping.getMapped(from))
  ```

7.7.2.2.3.3 CBAFeatureTyping_Mapping

Description

Creates a feature typing relationship owned by the element *typedFeature()*.

General Mappings

ToFeatureTyping_Init
Mapping

Mapping Source

CallBehaviorAction

Mapping Target

FeatureTyping

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureTyping::type () : Type [1]

  ```
  from.behavior
  ```

7.7.2.2.3.4 CallOperationAction_Mapping

Description

A UML4SysML::CallOperationAction is mapped to a SysML v2 ActionUsage which calls the operation.

The following shows an example of what the textual SysML v2 syntax of the result of the transformation may look like.

```
action sysMLv1CallOperationAction {
  in paramIn;
  in target : ThisIsABlock;
  out paramReturn = target.sysMLv1Operation;
}
```

General Mappings

CommonAction_Mapping

Mapping Source

CallOperationAction

Mapping Target

ActionUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ActionUsage::ownedRelationship () : Relationship [0..*]

```
Helper.actionOwnedRelationship(from)
->including(COAPerformActionFeatureMembership_Mapping.getMapped(from))
```

7.7.2.2.3.5 COAOutputPinFeature_Mapping

Description

The mapping class creates the feature element for the output parameter.

General Mappings

ToFeature_Init
Mapping

Mapping Source

OutputPin

Mapping Target

Feature

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Feature::direction () : FeatureDirectionKind [0..1]

```
KerML::FeatureDirectionKind::_'in'
```

- Feature::ownedRelationship () : Relationship [0..*]

```
Set{COAOutputPinFeatureFeatureValue_Mapping.getMapped(from),
COAOutputPinFeatureFeatureMembership_Mapping.getMapped(from)}
```

7.7.2.2.3.6 COAOutputPinFeatureChainExpression_Mapping

Description

The mapping class creates the feature chain expression for the output parameter feature value.

General Mappings

ToInvocationExpression_Init
Mapping

Mapping Source

OutputPin

Mapping Target

FeatureChainExpression

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureChainExpression::ownedRelationship () : Relationship [0..*]

```
Set{COAOutputPinParameterMembership_Mapping.getMapped(from),
COAOutputPinFeatureChainExpressionMembership_Mapping.getMapped(from),
ReturnParameterFeatureMembership_Factory.create()}
```

7.7.2.2.3.7 COAOutputPinFeatureChainExpressionMembership_Mapping

Description

Creates a membership relationship for *memberElement()*.

General Mappings

ToMembership_Init
Mapping

Mapping Source

OutputPin

Mapping Target

Membership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Membership::memberElement () : Element [1]

  ```
  from.owner.oclAsType(UML::CallOperationAction).operation
  ```

7.7.2.2.3.8 COAOutputPinFeatureFeature_Mapping

Description

Creates a feature element for the UML4SysML::CallOperationAction mapping.

General Mappings

ToFeature_Init
Mapping

Mapping Source

OutputPin

Mapping Target

Feature

Owned Mappings

(none)

Applicable filters

(none)

7.7.2.2.3.9 COAOutputPinFeatureFeatureMembership_Mapping

Description

Creates a feature membership relationship for *ownedMemberFeature()*.

General Mappings

ToFeatureMembership_Init
Mapping

Mapping Source

OutputPin

Mapping Target

FeatureMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureMembership::ownedMemberFeature () : Feature [1]

  ```
  COAOutputPinFeatureFeature_Mapping.getMapped(from)
  ```

7.7.2.2.3.10 COAOutputPinFeatureFeatureValue_Mapping

Description

Creates a feature value relationship.

General Mappings

ToFeatureValue_Init
Mapping

Mapping Source

OutputPin

Mapping Target

FeatureValue

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureValue::value () : Expression [1]

  `COAOutputPinFeatureReferenceExpression_Mapping.getMapped(from)`

7.7.2.2.3.11 COAOutputPinFeatureMembership_Mapping

Description

Creates a feature membership relationship for *ownedMemberFeature()*.

General Mappings

ToFeatureMembership_Init
Mapping

Mapping Source

OutputPin

Mapping Target

FeatureMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureMembership::ownedMemberFeature () : Feature [1]

  `COAOutputPinReferenceUsage_Mapping.getMapped(from)`

7.7.2.2.3.12 COAOutputPinFeatureReferenceExpression_Mapping

Description

The mapping class creates the feature reference expression for the output parameter.

General Mappings

ToFeatureReferenceExpression_Init
Mapping

Mapping Source

OutputPin

Mapping Target

FeatureReferenceExpression

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureReferenceExpression::ownedRelationship () : Relationship [0..*]

```
Set{COAOutputPinFeatureReferenceExpressionMembership_Mapping.getMapped(from),
ReturnParameterFeatureMembership_Factory.create()}
```

7.7.2.2.3.13 COAOutputPinFeatureReferenceExpressionMembership_Mapping

Description

Creates a membership relationship for *memberElement()*.

General Mappings

ToMembership_Init
Mapping

Mapping Source

OutputPin

Mapping Target

Membership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Membership::memberElement () : Element [1]

  ```
  from.owner.oclAsType(UML::CallOperationAction).target
  ```

7.7.2.2.3.14 COAOutputPinParameterMembership_Mapping

Description

Creates a membership relationship for *memberElement()*.

General Mappings

ToParameterMembership_Init
Mapping

Mapping Source

OutputPin

Mapping Target

ParameterMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ParameterMembership::visibility () : VisibilityKind [1]

  ```
  KerML::VisibilityKind::private
  ```

- ParameterMembership::ownedMemberParameter () : Feature [1]

  ```
  COAOutputPinFeature_Mapping.getMapped(from)
  ```

7.7.2.2.3.15 COAOutputPinReferenceUsage_Mapping

Description

Creates a reference usage.

General Mappings

ToReferenceUsage_Init
Mapping

Mapping Source

OutputPin

Mapping Target

ReferenceUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ReferenceUsage::ownedRelationship () : Relationship [0..*]

  ```
  Set{COAOutputPinReferenceUsageFeatureValue_Mapping.getMapped(from)}
  ```

7.7.2.2.3.16 COAOutputPinReferenceUsageFeatureValue_Mapping

Description

Creates a feature value relationship.

General Mappings

ToFeatureValue_Init
Mapping

Mapping Source

OutputPin

Mapping Target

FeatureValue

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureValue::value () : Expression [1]

  ```
  COAOutputPinFeatureChainExpression_Mapping.getMapped(from)
  ```

7.7.2.2.3.17 COAPerformAction_Mapping

Description

The mapping class creates the PerformActionUsage element.

General Mappings

ToPerformActionUsage_Init
Mapping

Mapping Source

CallOperationAction

Mapping Target

PerformActionUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- PerformActionUsage::ownedRelationship () : Relationship [0..*]

  ```
  Set{COAPerformActionReferenceSubsetting_Mapping.getMapped(from)}
  ```

7.7.2.2.3.18 COAPerformActionFeatureMembership_Mapping

Description

Creates a feature membership relationship for *ownedMemberFeature()*.

General Mappings

ToEndFeatureMembership_Init
Mapping

Mapping Source

CallOperationAction

Mapping Target

EndFeatureMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- EndFeatureMembership::ownedMemberFeature () : Feature [1]

```
COAPerformAction_Mapping.getMapped(from)
```

7.7.2.2.3.19 COAPerformActionReferenceSubsetting_Mapping

Description

Creates a subsetting relationship.

General Mappings

ToReferenceSubsetting_Init
Mapping

Mapping Source

CallOperationAction

Mapping Target

ReferenceSubsetting

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ReferenceSubsetting::ownedRelatedElement () : Element [0..*]

```
Set{COAPerformActionFeature_Mapping.getMapped(from)}
```

7.7.2.2.3.20 COAPerformActionFeature_Mapping

Description

The mapping class creates the feature element for the perform action usage.

General Mappings

ToFeature_Init
Mapping

Mapping Source

CallOperationAction

Mapping Target

Feature

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Feature::ownedRelationship () : Relationship [0..*]

```
Set{COAPerformActionFeatureChainingTarget_Mapping.getMapped(from),
COAPerformActionFeatureChainingOperation_Mapping.getMapped(from)}
```

7.7.2.2.3.21 COAPerformActionFeatureChainingOperation_Mapping

Description

The mapping class creates the feature chaining element for the operation of the perform action usage.

General Mappings

ToFeatureChaining_Init
Mapping

Mapping Source

CallOperationAction

Mapping Target

FeatureChaining

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureChaining::chainingFeature () : Feature [1]

  ```
  from.operation
  ```

7.7.2.2.3.22 COAPerformActionFeatureChainingTarget_Mapping

Description

The mapping class creates the feature chaining element for the target element of the perform action usage.

General Mappings

ToFeatureChaining_Init
Mapping

Mapping Source

CallOperationAction

Mapping Target

FeatureChaining

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureChaining::chainingFeature () : Feature [1]

  ```
  from.target
  ```

7.7.2.2.3.23 SendObjectAction_Mapping

Description

A UML4SysML::SendObjectAction is mapped to a SysMLv2 ActionUsage that includes a SendActionUsage.

The following shows an example of what the textual SysML v2 syntax of the result of the transformation may look like.

```
action sysMLv1SendObjectAction {
        in target : SysMLv1Block;
        send SysMLv1Object1() to target;
}
part def SysMLv1Block;
item def SysMLv1Object;
```

General Mappings

SendSignalAction_Mapping

Mapping Source

SendObjectAction

Mapping Target

ActionUsage

Owned Mappings

(none)

Applicable filters

(none)

7.7.2.2.3.24 SendSignalAction_Mapping

Description

A UML4SysML::SendSignalAction is mapped to a SysMLv2 ActionUsage that includes a SendActionUsage.

The following shows an example of what the textual SysML v2 syntax of the result of the transformation may look like.

```
action sysMLv1SendSignalAction {
        in target : SysMLv1Block;
        send SysMLv1Signal() to target;
}
part def SysMLv1Block;
item def SysMLv1Signal;
```

General Mappings

CommonAction_Mapping

Mapping Source

SendSignalAction

Mapping Target

ActionUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ActionUsage::ownedRelationship () : Relationship [0..*]

```
Helper.actionOwnedRelationship(from)
->including(SSAFeatureMembership_Mapping.getMapped(from))
```

7.7.2.2.3.25 SSAFeatureMembership_Mapping

Description

Creates a feature membership relationship for *ownedMemberFeature()*.

General Mappings

ToFeatureMembership_Init
Mapping

Mapping Source

InvocationAction

Mapping Target

FeatureMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureMembership::ownedMemberFeature () : Feature [1]

```
SSASendActionUsage_Mapping.getMapped(from)
```

7.7.2.2.3.26 SSAParameterMembership_Mapping

Description

Creates a membership relationship for *memberElement()*.

General Mappings

ToParameterMembership_Init
Mapping

Mapping Source

InvocationAction

Mapping Target

ParameterMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ParameterMembership::ownedMemberParameter () : Feature [1]

  ```
  SSAReferenceUsage_Mapping.getMapped(from)
  ```

7.7.2.2.3.27 SSAReferenceUsage_Mapping

Description

Creates a reference usage.

General Mappings

ToReferenceUsage_Init
Mapping

Mapping Source

InvocationAction

Mapping Target

ReferenceUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ReferenceUsage::direction () : FeatureDirectionKind [0..1]

  ```
  KerML::FeatureDirectionKind::_'in'
  ```

7.7.2.2.3.28 SSAItemParameterMembership_Mapping

Description

Creates a membership relationship for *memberElement()*.

General Mappings

ToParameterMembership_Init
Mapping

Mapping Source

InvocationAction

Mapping Target

ParameterMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ParameterMembership::ownedMemberParameter () : Feature [1]

  ```
  SSAItemReferenceUsage_Mapping.getMapped(from)
  ```

7.7.2.2.3.29 SSAItemReferenceUsage_Mapping

Description

Creates a reference usage.

General Mappings

ToReferenceUsage_Init
Mapping

Mapping Source

InvocationAction

Mapping Target

ReferenceUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ReferenceUsage::ownedRelationship () : Relationship [0..*]

  ```
  Set{SSAItemReferenceUsageFeatureValue_Mapping.getMapped(from)}
  ```

- ReferenceUsage::direction () : FeatureDirectionKind [0..1]

  ```
  KerML::FeatureDirectionKind::_'in'
  ```

7.7.2.2.3.30 SSAItemReferenceUsageFeatureValue_Mapping

Description

Creates a feature value relationship.

General Mappings

ToFeatureValue_Init
Mapping

Mapping Source

InvocationAction

Mapping Target

FeatureValue

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureValue::value () : Expression [1]

  ```
  SSAItemReferenceUsageInvocationExpression_Mapping.getMapped(from)
  ```

7.7.2.2.3.31 SSAItemReferenceUsageFeatureTyping_Mapping

Description

Creates a feature typing relationship owned by the element *typedFeature()*.

General Mappings

ToFeatureTyping_Init
Mapping

Mapping Source

InvocationAction

Mapping Target

FeatureTyping

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureTyping::type () : Type [1]

  ```
  if from.oclIsTypeOf(UML::SendSignalAction) then
      from.signal
  else if from.oclIsTypeOf(UML::SendObjectAction) then
      from.request
  else
      invalid
  endif endif
  ```

7.7.2.2.3.32 SSAItemReferenceUsageInvocationExpression_Mapping

Description

The mapping class creates the invocation expression for the SysML v2 SendActionUsage.

General Mappings

ToInvocationExpression_Init
Mapping

Mapping Source

InvocationAction

Mapping Target

InvocationExpression

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- InvocationExpression::ownedRelationship () : Relationship [0..*]

```
Set{SSAItemReferenceUsageFeatureTyping_Mapping.getMapped(from),
ReturnParameterFeatureMembership_Factory.create()}
```

7.7.2.2.3.33 SSATargetParameterMembership_Mapping

Description

Creates a membership relationship for *memberElement()*.

General Mappings

ToParameterMembership_Init
Mapping

Mapping Source

InvocationAction

Mapping Target

ParameterMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ParameterMembership::ownedMemberParameter () : Feature [1]

  `SSATargetReferenceUsage_Mapping.getMapped(from)`

7.7.2.2.3.34 SSATargetReferenceUsage_Mapping

Description

Creates a reference usage.

General Mappings

ToReferenceUsage_Init
Mapping

Mapping Source

InvocationAction

Mapping Target

ReferenceUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ReferenceUsage::ownedRelationship () : Relationship [0..*]

  `Set{SSATargetReferenceUsageFeatureValue_Mapping.getMapped(from)}`

- ReferenceUsage::direction () : FeatureDirectionKind [0..1]

  `KerML::FeatureDirectionKind::_'in'`

7.7.2.2.3.35 SSATargetReferenceUsageFeatureValue_Mapping

Description

Creates a feature value relationship.

General Mappings

ToFeatureValue_Init
Mapping

Mapping Source

InvocationAction

Mapping Target

FeatureValue

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureValue::value () : Expression [1]

  ```
  SSATargetReferenceUsageFeatureValueExpression_Mapping.getMapped(from)
  ```

7.7.2.2.3.36 SSATargetReferenceUsageFeatureValueMembership_Mapping

Description

Creates a membership relationship for *memberElement()*.

General Mappings

ToMembership_Init
Mapping

Mapping Source

InvocationAction

Mapping Target

Membership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Membership::memberElement () : Element [1]

```
from.target
```

7.7.2.2.3.37 SSATargetReferenceUsageFeatureValueExpression_Mapping

Description

The mapping class creates the feature reference expression for the target reference usage element of the SysML v2 SendActionUsage.

General Mappings

ToFeatureReferenceExpression_Init
Mapping

Mapping Source

InvocationAction

Mapping Target

FeatureReferenceExpression

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureReferenceExpression::ownedRelationship () : Relationship [0..*]

```
Set{SSATargetReferenceUsageFeatureValueMembership_Mapping.getMapped(from),
ReturnParameterFeatureMembership_Factory.create()}
```

7.7.2.2.3.38 SSASendActionUsage_Mapping

Description

The mapping class creates the SysML v2 element SendActionUsage for the UML4SysML::SendSignalAction mapping.

General Mappings

ToActionUsage_Init
Mapping

Mapping Source

InvocationAction

Mapping Target

SendActionUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- SendActionUsage::ownedRelationship () : Relationship [0..*]

```
Set{SSAItemParameterMembership_Mapping.getMapped(from),
SSAParameterMembership_Mapping.getMapped(from),
SSATargetParameterMembership_Mapping.getMapped(from)}
```

7.7.2.2.3.39 StartClassifierBehaviorAction_Mapping

Description

The UML4SysML::StartClassifierBehaviorAction is not supported by SysML v2. It is mapped to an empty action usage to keep the connections within the activity respectively action definition.

General Mappings

CommonAction_Mapping

Mapping Source

StartClassifierBehaviorAction

Mapping Target

ActionUsage

Owned Mappings

(none)

Applicable filters

(none)

7.7.2.2.3.40 StartObjectBehaviorAction_Mapping

Description

The UML4SysML::StartObjectBehaviorAction is not supported by SysML v2. It is mapped to an empty action usage to keep the connections within the activity respectively action definition.

General Mappings

CommonAction_Mapping

Mapping Source

StartObjectBehaviorAction

Mapping Target

ActionUsage

Owned Mappings

(none)

Applicable filters

(none)

7.7.2.2.4 Link Actions

7.7.2.2.4.1 ClearAssociationAction_Mapping

Description

The UML4SysML::ClearAssociationAction is mapped to a SysML v2 ActionUsage. The details of the mapping are not defined yet.

General Mappings

CommonAction_Mapping

Mapping Source

ClearAssociationAction

Mapping Target

ActionUsage

Owned Mappings

(none)

Applicable filters

(none)

7.7.2.2.4.2 CreateLinkAction_Mapping

Description

The UML4SysML::CreateLinkAction is mapped to a SysML v2 ActionUsage. The details of the mapping are not completely defined yet.

General Mappings

CommonAction_Mapping

Mapping Source

CreateLinkAction

Mapping Target

ActionUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ActionUsage::ownedRelationship () : Relationship [0..*]

```
let linkEndCreationData : Set(UML::Element) =
    from.ownedElement->select(e | e.oclIsTypeOf(UML::LinkEndCreationData)) in
let actionInputPin: Set(UML::Element) =
    from.ownedElement->select(e | e.oclIsTypeOf(UML::ActionInputPin)) in
let triggers: Set(UML::Element) =
    from.ownedElement->select(e | e.oclIsKindOf(UML::Trigger)) in
let toElementFMS: Set(UML::Element) =
    from.ownedElement->select(e | e.oclIsKindOf(UML::Pin)) in
let toElementOMS: Set(UML::Element) =
    ((((from.ownedElement - toElementFMS) - actionInputPin)
        - triggers) - linkEndCreationData) in
toElementOMS->collect(e | ElementOwningMembership_Mapping.getMapped(e))
->union(toElementFMS->collect(e | ElementFeatureMembership_Mapping.getMapped(e)))
```

7.7.2.2.4.3 CreateLinkObjectAction_Mapping

Description

A UML4SysML::CreateLinkObjectAction is mapped to a SysML v2 ActionUsage. The details of the mapping are not defined yet.

General Mappings

CreateLinkAction_Mapping

Mapping Source

CreateLinkObjectAction

Mapping Target

ActionUsage

Owned Mappings

(none)

Applicable filters

(none)

7.7.2.2.4.4 DestroyLinkAction_Mapping

Description

The UML4SysML::DestroyLinkAction is mapped to a SysML v2 ActionUsage. The details of the mapping are not completely defined yet.

General Mappings

CommonAction_Mapping

Mapping Source

DestroyLinkAction

Mapping Target

ActionUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ActionUsage::ownedRelationship () : Relationship [0..*]

```
let actionInputPin: Set(UML::Element) =
    from.ownedElement->select(e | e.oclIsTypeOf(UML::ActionInputPin)) in
let triggers: Set(UML::Element) =
    from.ownedElement->select(e | e.oclIsKindOf(UML::Trigger)) in
let linkData: Set(UML::Element) =
    from.ownedElement->select( e | e.oclIsKindOf(UML::LinkEndData) or
    e.oclIsKindOf(UML::LinkEndDestructionData)) in
let toElementFMS: Set(UML::Element) =
    from.ownedElement->select(e | e.oclIsKindOf(UML::Pin)) in
let toElementOMS: Set(UML::Element) =
    ((((from.ownedElement - toElementFMS) - actionInputPin)
         - triggers) - linkData) in
toElementOMS->collect(e | ElementOwningMembership_Mapping.getMapped(e))
->union(toElementFMS->collect(e | ElementFeatureMembership_Mapping.getMapped(e)))
```

7.7.2.2.4.5 ReadLinkAction_Mapping

Description

The UML4SysML::ReadLinkAction is mapped to a SysML v2 ActionUsage. The details of the mapping are not completely defined yet.

General Mappings

CommonAction_Mapping

Mapping Source

ReadLinkAction

Mapping Target

ActionUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ActionUsage::ownedRelationship () : Relationship [0..*]

```
let actionInputPin: Set(UML::Element) =
    from.ownedElement->select(e | e.oclIsTypeOf(UML::ActionInputPin)) in
let triggers: Set(UML::Element) =
    from.ownedElement->select(e | e.oclIsKindOf(UML::Trigger)) in
let linkData: Set(UML::Element) =
    from.ownedElement->select( e | e.oclIsKindOf(UML::LinkEndData)) in
let toElementFMS: Set(UML::Element) =
    from.ownedElement->select(e | e.oclIsKindOf(UML::Pin)) in
let toElementOMS: Set(UML::Element) =
    (((from.ownedElement - toElementFMS) - actionInputPin)
        - triggers) - linkData) in
toElementOMS->collect(e | ElementOwningMembership_Mapping.getMapped(e))
->union(toElementFMS->collect(e | ElementFeatureMembership_Mapping.getMapped(e)))
```

7.7.2.2.4.6 ReadLinkObjectEndAction_Mapping

Description

The UML4SysML::ReadLinkObjectEndAction is mapped to a SysML v2 ActionUsage. The details of the mapping are not defined yet.

General Mappings

CommonAction_Mapping

Mapping Source

ReadLinkObjectEndAction

Mapping Target

ActionUsage

Owned Mappings

(none)

Applicable filters

(none)

7.7.2.2.4.7 ReadLinkObjectEndQualifierAction_Mapping

Description

The UML4SysML::ReadLinkObjectEndQualifierAction is mapped to a SysML v2 ActionUsage. The details of the mapping are not defined yet.

General Mappings

CommonAction_Mapping

Mapping Source

ReadLinkObjectEndQualifierAction

Mapping Target

ActionUsage

Owned Mappings

(none)

Applicable filters

(none)

7.7.2.2.5 Object Actions

7.7.2.2.5.1 CreateObjectAction_Mapping

Description

A UML4SysML::CreateObjectAction is mapped to a SysML v2 ActionUsage.

The following shows an example of what the textual SysML v2 syntax of the result of the transformation may look like.

```
action def SysMLv1Activity {
        action sysMLv1CreateObjectAction {
                out result : SysMLv1Block = SysMLv1Block();
        }
}
part def SysMLv1Block;
```

General Mappings

CommonAction_Mapping

Mapping Source

CreateObjectAction

Mapping Target

ActionUsage

Owned Mappings

(none)

Applicable filters

(none)

7.7.2.2.5.2 COAInvocationExpessionFeatureTyping_Mapping

Description

Creates a feature typing relationship owned by the element *typedFeature()*.

General Mappings

ToFeatureTyping_Init
Mapping

Mapping Source

CreateObjectAction

Mapping Target

FeatureTyping

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureTyping::type () : Type [1]

  ```
  from.classifier
  ```

7.7.2.2.5.3 COAInvocationExpression_Mapping

Description

The mapping class creates the invocation expression to create the object.

General Mappings

ToInvocationExpression_Init
Mapping

Mapping Source

CreateObjectAction

Mapping Target

InvocationExpression

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- InvocationExpression::ownedRelationship () : Relationship [0..*]

```
Set{COAInvocationExpessionFeatureTyping_Mapping.getMapped(from),
CommonReturnParameterFeatureMembership_Mapping.getMapped(from.result)}
```

7.7.2.2.5.4 COAPin_Mapping

Description

The mapping class creates the output parameter of the ActionUsage for the mapping of UML4SysML::CreateObjectAction.

General Mappings

Pin_Mapping

Mapping Source

OutputPin

Mapping Target

ReferenceUsage

Owned Mappings

(none)

Applicable filters

This mapping applies only if the following (OCL) condition implemented by the operation *filter(src : Element) : Boolean* is verified:

```
src.owner.oclIsTypeOf(UML::CreateObjectAction)
```

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ReferenceUsage::ownedRelationship () : Relationship [0..*]

  ```
  Set{PinFeatureTyping_Mapping.getMapped(from),
  COAPinFeatureValue_Mapping.getMapped(from)}
  ```

7.7.2.2.5.5 COAPinFeatureValue_Mapping

Description

Creates a feature value relationship.

General Mappings

ToFeatureValue_Init
Mapping

Mapping Source

OutputPin

Mapping Target

FeatureValue

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureValue::value () : Expression [1]

  ```
  COAInvocationExpression_Mapping.getMapped(from.owner)
  ```

7.7.2.2.5.6 DestroyObjectAction_Mapping

Description

The UML4SysML::DestroyObjectAction is conceptually mapped to the SysML v2 library function OccurrenceFunctions::destroy.

The following shows an example of what the textual SysML v2 syntax of the result of the transformation may look like.

```
action def SysMLv1Activity {
        action sysMLv1DestroyObjectAction {
                in target : SysMLv1Block;
                  action : OccurrenceFunctions::destroy {
                        in occ = target;
                }
        }
}
part def SysMLv1Block;
```

General Mappings

CommonAction_Mapping

Mapping Source

DestroyObjectAction

Mapping Target

ActionUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ActionUsage::ownedRelationship () : Relationship [0..*]

  ```
  Helper.actionOwnedRelationship(from)
  ->including(DOADestroyFeatureMembership_Mapping.getMapped(from))
  ```

7.7.2.2.5.7 DOADestroyActionUsage_Mapping

Description

The mapping class creates the action usage for the destroy function.

General Mappings

ToActionUsage_Init
Mapping

Mapping Source

DestroyObjectAction

Mapping Target

ActionUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ActionUsage::ownedRelationship () : Relationship [0..*]

```
Set{DOADestroyActionUsageFeatureTyping_Mapping.getMapped(from),
DOADestroyActionUsageFeatureMembership_Mapping.getMapped(from)}
```

7.7.2.2.5.8 DOADestroyActionUsageFeatureMembership_Mapping

Description

Creates a feature membership relationship for *ownedMemberFeature()*.

General Mappings

ToFeatureMembership_Init
Mapping

Mapping Source

DestroyObjectAction

Mapping Target

FeatureMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureMembership::ownedMemberFeature () : Feature [1]

```
DOADestroyActionUsageReferenceUsage_Mapping.getMapped(from)
```

7.7.2.2.5.9 DOADestroyActionUsageFeatureReferenceExpression_Mapping

Description

The mapping class creates the feature reference expression for the UML4SysML::DestroyObjectAction mapping.

General Mappings

ToFeatureReferenceExpression_Init
Mapping

Mapping Source

DestroyObjectAction

Mapping Target

FeatureReferenceExpression

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureReferenceExpression::ownedRelationship () : Relationship [0..*]

```
Set{DOADestroyActionUsageMembership_Mapping.getMapped(from),
ReturnParameterFeatureMembership_Factory.create()}
```

7.7.2.2.5.10 DOADestroyActionUsageMembership_Mapping

Description

Creates a membership relationship for *memberElement()*.

General Mappings

ToMembership_Init
Mapping

Mapping Source

DestroyObjectAction

Mapping Target

Membership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Membership::memberElement () : Element [1]

```
from.target
```

7.7.2.2.5.11 DOADestroyActionUsageFeatureTyping_Mapping

Description

Creates a feature typing relationship owned by the element *typedFeature()*.

General Mappings

ToFeatureTyping_Init
Mapping

Mapping Source

DestroyObjectAction

Mapping Target

FeatureTyping

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureTyping::type () : Type [1]

```
SysMLv2::Function.allInstances(
)->any(e | e.qualifiedName =  'OccurrenceFunctions::destroy')
```

7.7.2.2.5.12 DOADestroyActionUsageFeatureValue_Mapping

Description

Creates a feature value relationship.

General Mappings

ToFeatureValue_Init
Mapping

Mapping Source

DestroyObjectAction

Mapping Target

FeatureValue

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureValue::value () : Expression [1]

  ```
  DOADestroyActionUsageFeatureReferenceExpression_Mapping.getMapped(from)
  ```

7.7.2.2.5.13 DOADestroyActionUsageReferenceUsage_Mapping

Description

Creates a reference usage.

General Mappings

ToReferenceUsage_Init
Mapping

Mapping Source

DestroyObjectAction

Mapping Target

ReferenceUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ReferenceUsage::ownedRelationship () : Relationship [0..*]

  ```
  Set{DOADestroyActionUsageFeatureValue_Mapping.getMapped(from)}
  ```

7.7.2.2.5.14 DOADestroyFeatureMembership_Mapping

Description

Creates a feature membership relationship for *ownedMemberFeature()*.

General Mappings

ToFeatureMembership_Init
Mapping

Mapping Source

DestroyObjectAction

Mapping Target

FeatureMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureMembership::ownedMemberFeature () : Feature [1]

  ```
  DOADestroyActionUsage_Mapping.getMapped(from)
  ```

7.7.2.2.5.15 ReadIsClassifiedObjectAction_Mapping

Description

The UML4SysML::ReadIsClassifiedObjectAction is conceptually mapped to a SysML v2 ActionUsage.

The following shows an example of what the textual SysML v2 syntax of the result of the transformation may look like.

```
action def SysMLv1Activity {
	action sysMLv1ReadIsClassifiedObjectActionDirect {
		in object;
		out result : ScalarValues::Boolean =
			object istype ThisIsABlock;
	}

	action sysMLv1ReadIsClassifiedObjectActionNonDirect {
		in object;
		out result : ScalarValues::Boolean =
			object hastype ThisIsABlock;
	}
}
```

General Mappings

CommonAction_Mapping

Mapping Source

ReadIsClassifiedObjectAction

Mapping Target

ActionUsage

Owned Mappings

(none)

Applicable filters

(none)

7.7.2.2.5.16 RICOAFeatureValue_Mapping

Description

Creates a feature value relationship.

General Mappings

ToFeatureValue_Init
Mapping

Mapping Source

ReadIsClassifiedObjectAction

Mapping Target

FeatureValue

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureValue::value () : Expression [1]

  ```
  RICOAFeatureValueOperatorExpression_Mapping.getMapped(from)
  ```

7.7.2.2.5.17 RICOAFeatureValueOperatorExpression_Mapping

Description

The mapping class creates the operator expression for the UML4SysML::ReadIsClassifiedObjectAction mapping.

General Mappings

ToOperatorExpression_Init
Mapping

Mapping Source

ReadIsClassifiedObjectAction

Mapping Target

OperatorExpression

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- OperatorExpression::operator () : String [1]

  ```
  if from.isDirect then 'istype' else 'hastype' endif
  ```

- OperatorExpression::ownedRelationship () : Relationship [0..*]

  ```
  Set{RICOAFeatureValueOperatorParameterMembership_Mapping.getMapped(from)}
  ```

7.7.2.2.5.18 RICOAFeatureValueOperatorExpressionFeature_Mapping

Description

The mapping class creates the feature for the operator expression of the UML4SysML::ReadIsClassifiedObjectAction mapping.

General Mappings

ToFeature_Init
Mapping

Mapping Source

ReadIsClassifiedObjectAction

Mapping Target

Feature

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Feature::ownedRelationship () : Relationship [0..*]

  `Set{RICOAFeatureValueOperatorExpressionFeatureValue_Mapping.getMapped(from)}`

- Feature::direction () : FeatureDirectionKind [0..1]

  `KerML::FeatureDirectionKind::_'in'`

7.7.2.2.5.19 RICOAFeatureValueOperatorExpressionFeatureValue_Mapping

Description

Creates a feature value relationship.

General Mappings

ToFeatureValue_Init
Mapping

Mapping Source

ReadIsClassifiedObjectAction

Mapping Target

FeatureValue

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureValue::value () : Expression [1]

  ```
  RICOAFeatureValueOperatorFeatureReferenceExpression_Mapping.getMapped(from)
  ```

7.7.2.2.5.20 RICOAFeatureValueOperatorFeatureReferenceExpression_Mapping

Description

The mapping class creates the feature reference expression for the UML4SysML::ReadIsClassifiedObjectAction mapping.

General Mappings

ToFeatureReferenceExpression_Init
Mapping

Mapping Source

ReadIsClassifiedObjectAction

Mapping Target

FeatureReferenceExpression

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureReferenceExpression::ownedRelationship () : Relationship [0..*]

  ```
  Set{RICOAFeatureValueOperatorMembership_Mapping.getMapped(from),
  CommonReturnParameterFeatureMembership_Mapping.getMapped(from)}
  ```

7.7.2.2.5.21 RICOAFeatureValueOperatorMembership_Mapping

Description

Creates a membership relationship for *memberElement()*.

General Mappings

ToMembership_Init
Mapping

Mapping Source

ReadIsClassifiedObjectAction

Mapping Target

Membership

Owned Mappings

(none)

Applicable filters

(none)

7.7.2.2.5.22 RICOAFeatureValueOperatorParameterMembership_Mapping

Description

Creates a membership relationship for *memberElement()*.

General Mappings

ToParameterMembership_Init
Mapping

Mapping Source

ReadIsClassifiedObjectAction

Mapping Target

ParameterMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ParameterMembership::visibility () : VisibilityKind [1]

  ```
  KerML::VisibilityKind::private
  ```

- ParameterMembership::ownedMemberParameter () : Feature [1]

```
RICOAFeatureValueOperatorExpressionFeature_Mapping.getMapped(from)
```

7.7.2.2.5.23 RICOAOutputPin_Mapping

Description

The mapping class creates the output parameter of the ActionUsage element for the UML4SysML::ReadIsClassifiedObjectAction mapping.

General Mappings

Pin_Mapping

Mapping Source

OutputPin

Mapping Target

ReferenceUsage

Owned Mappings

(none)

Applicable filters

This mapping applies only if the following (OCL) condition implemented by the operation *filter(src : Element) : Boolean* is verified:

```
src.owner.oclIsTypeOf(UML::ReadIsClassifiedObjectAction)
```

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ReferenceUsage::ownedRelationship () : Relationship [0..*]

```
Set{TypedElementFeatureTyping_Mapping.getMapped(from),
RICOAFeatureValue_Mapping.getMapped(from.owner),
MultiplicityMembership_Mapping.getMapped(from)}
```

7.7.2.2.5.24 ReadExtentAction_Mapping

Description

A UML4SysML::ReadExtentAction is mapped to a SysML v2 ActionUsage.

The following shows an example of what the textual SysML v2 syntax of the result of the transformation may look like.

```
action def SysMLv1Activity {
        action sysMLv1ReadExtentAction {
                out thisIsTheOutputPin : SysMLv1Block =
                        all SysMLv1Block;
        }
```

```
}
part def SysMLv1Block;
```

General Mappings

CommonAction_Mapping

Mapping Source

ReadExtentAction

Mapping Target

ActionUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ActionUsage::ownedRelationship () : Relationship [0..*]

  ```
  Helper.actionOwnedRelationship(from)
  ```

7.7.2.2.5.25 REAFeatureValue_Mapping

Description

Creates a feature value relationship.

General Mappings

ToFeatureValue_Init
Mapping

Mapping Source

OutputPin

Mapping Target

FeatureValue

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureValue::value () : Expression [1]

```
REAFeatureValueOperatorExpression_Mapping.getMapped(from)
```

7.7.2.2.5.26 REAFeatureValueOperatorExpression_Mapping

Description

The mapping class creates the operator expression for the UML4SysML::ReadExtentAction mapping.

General Mappings

ToOperatorExpression_Init
Mapping

Mapping Source

OutputPin

Mapping Target

OperatorExpression

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- OperatorExpression::operator () : String [1]

```
'all'
```

- OperatorExpression::ownedRelationship () : Relationship [0..*]

```
Set{REAFeatureValueOperatorExpressionMembership_Mapping.getMapped(from),
CommonReturnParameterFeatureMembership_Mapping.getMapped(from)}
```

7.7.2.2.5.27 REAFeatureValueOperatorExpressionFeature_Mapping

Description

The mapping class creates the feature for the operator expression for the UML4SysML::ReadExtentAction mapping.

General Mappings

ToFeature_Init
Mapping

Mapping Source

OutputPin

Mapping Target

Feature

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Feature::ownedRelationship () : Relationship [0..*]

  ```
  Set{REAFeatureValueOperatorExpressionFeatureTyping_Mapping.getMapped(from)}
  ```

7.7.2.2.5.28 REAFeatureValueOperatorExpressionFeatureTyping_Mapping

Description

Creates a feature typing relationship owned by the element *typedFeature()*.

General Mappings

ToFeatureTyping_Init
Mapping

Mapping Source

OutputPin

Mapping Target

FeatureTyping

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureTyping::type () : Type [1]

  ```
  from.owner.classifier
  ```

7.7.2.2.5.29 REAFeatureValueOperatorExpressionMembership_Mapping

Description

Creates a membership relationship for *memberElement()*.

General Mappings

ToFeatureMembership_Init
Mapping

Mapping Source

OutputPin

Mapping Target

FeatureMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureMembership::ownedMemberFeature () : Feature [1]

  ```
  REAFeatureValueOperatorExpressionFeature_Mapping.getMapped(from)
  ```

7.7.2.2.5.30 REAOutputPin_Mapping

Description

The mapping class creates the output parameter of the ActionUsage for the mapping of UML4SysML::ReadExtentAction.

General Mappings

Pin_Mapping

Mapping Source

OutputPin

Mapping Target

ReferenceUsage

Owned Mappings

(none)

Applicable filters

This mapping applies only if the following (OCL) condition implemented by the operation *filter(src : Element) : Boolean* is verified:

```
src.owner.oclIsTypeOf(UML::ReadExtentAction)
```

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ReferenceUsage::ownedRelationship () : Relationship [0..*]

```
Set {TypedElementFeatureTyping_Mapping.getMapped(from),
REAFeatureValue_Mapping.getMapped(from)}
->union(self.oclAsType(Pin_Mapping).ownedRelationship())
```

7.7.2.2.5.31 ReadSelfAction_Mapping

Description

A UML4SysML::ReadSelfAction is mapped to a SysML v2 ActionUsage.

The following shows an example of what the textual SysML v2 syntax of the result of the transformation may look like.

```
action def SysMLv1Activity {
        action sysMLv1ReadSelfAction {
                out : Base::Anything = this;
        }
}
```

General Mappings

CommonAction_Mapping

Mapping Source

ReadSelfAction

Mapping Target

ActionUsage

Owned Mappings

(none)

Applicable filters

(none)

7.7.2.2.5.32 RSAFeatureValue_Mapping

Description

Creates a feature value relationship.

General Mappings

ToFeatureValue_Init
Mapping

Mapping Source

OutputPin

Mapping Target

FeatureValue

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureValue::value () : Expression [1]

  `RSAFeatureValueFeatureReferenceExpression_Mapping.getMapped(from)`

7.7.2.2.5.33 RSAFeatureValueFeatureReferenceExpression_Mapping

Description

The mapping class creates the feature reference expression for the mapping of UML4SysML::ReadSelfAction.

General Mappings

ToFeatureReferenceExpression_Init
Mapping

Mapping Source

OutputPin

Mapping Target

FeatureReferenceExpression

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureReferenceExpression::ownedRelationship () : Relationship [0..*]

```
Set{RSAFeatureValueMembership_Mapping.getMapped(from),
CommonReturnParameterFeatureMembership_Mapping.getMapped(from)}
```

7.7.2.2.5.34 RSAFeatureValueMembership_Mapping

Description

Creates a membership relationship for *memberElement()*.

General Mappings

ToMembership_Init
Mapping

Mapping Source

OutputPin

Mapping Target

Membership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Membership::memberElement () : Element [1]

```
SYSML2::Feature.allInstances()
->any(e | e.qualifiedName = 'Occurrences::Occurrence::this')
```

7.7.2.2.5.35 RSAOutputPin_Mapping

Description

The mapping class creates the output parameter of the ActionUsage for the mapping of UML4SysML::ReadSelfAction.

General Mappings

Pin_Mapping

Mapping Source

OutputPin

Mapping Target

ReferenceUsage

Owned Mappings

(none)

Applicable filters

This mapping applies only if the following (OCL) condition implemented by the operation *filter(src : Element) : Boolean* is verified:

```
src.owner.oclIsKindOf(UML::ReadSelfAction)
```

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ReferenceUsage::ownedRelationship () : Relationship [0..*]

  ```
  Set{TypedElementFeatureTyping_Mapping.getMapped(from),
  RSAFeatureValue_Mapping.getMapped(from)}
  ->union(self.oclAsType(Pin_Mapping).ownedRelationship())
  ```

- ReferenceUsage::isAbstract () : Boolean [1]

  ```
  true
  ```

- ReferenceUsage::isUnique () : Boolean [1]

  ```
  false
  ```

7.7.2.2.5.36 ReclassifyObjectAction_Mapping

Description

The UML4SysML::ReclassifyObjectAction is not supported by SysML v2. It is mapped to an empty action usage to keep the connections within the activity respectively action definition.

General Mappings

CommonAction_Mapping

Mapping Source

ReclassifyObjectAction

Mapping Target

ActionUsage

Owned Mappings

(none)

Applicable filters

(none)

7.7.2.2.5.37 TestIdentityAction_Mapping

Description

A UML4SysML::TestIdentityAction is mapped to a SysML v2 ActionUsage.

The following shows an example of what the textual SysML v2 syntax of the result of the transformation may look like.

```
action def SysMLv1Activity {
        action sysMLv1TestIdentityAction {
                in firstParameter;
                in secondParameter;
                out result : ScalarValues::Boolean =
                        firstParameter == secondParameter;
        }
}
```

General Mappings

CommonAction_Mapping

Mapping Source

TestIdentityAction

Mapping Target

CalculationUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- CalculationUsage::ownedRelationship () : Relationship [0..*]

```
Helper.actionOwnedRelationship(from)
->including(TIAResultExpressionMembership_Mapping.getMapped(from))
```

7.7.2.2.5.38 TIAOperatorExpression_Mapping

Description

The mapping class creates the operator expression for the UML4SysML::TestIdentityAction mapping.

General Mappings

ToOperatorExpression_Init
Mapping

Mapping Source

TestIdentityAction

Mapping Target

OperatorExpression

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- OperatorExpression::operator () : String [1]

```
'=='
```

- OperatorExpression::ownedRelationship () : Relationship [0..*]

```
Set{EqualOperatorExpressionOperandParameterMembership_Mapping.getMapped(from.first),
EqualOperatorExpressionOperandParameterMembership_Mapping.getMapped(from.second),
CommonReturnParameterFeatureMembership_Mapping.getMapped(from.result)}
```

7.7.2.2.5.39 TIAResultExpressionMembership_Mapping

Description

Creates a membership relationship for *memberElement()*.

General Mappings

ToFeatureMembership_Init
Mapping

Mapping Source

TestIdentityAction

Mapping Target

ResultExpressionMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ResultExpressionMembership::ownedMemberFeature () : Feature [0..1]

  ```
  TIAOperatorExpression_Mapping.getMapped(from)
  ```

7.7.2.2.5.40 ValueSpecificationAction_Mapping

Description

A UML4SysML::ValueSpecificationAction is mapped to a SysML v2 ActionUsage.

The following shows an example of what the textual SysML v2 syntax of the result of the transformation may look like.

```
action def SysMLv1Acticity {
        action sysMLv1ValueSpecificationAction1 {
                out result : ScalarValues::Integer = 42;
        }

        action sysMLv1ValueSpecificationAction2 {
                out result = sysMLv1OpaqueExpression.result;
                        calc sysMLv1OpaqueExpression {
                            language "Math"
                            /*
                             * 42 + 23
                            */
                          }
        }
}
```

General Mappings

CommonAction_Mapping

Mapping Source

ValueSpecificationAction

Mapping Target

ActionUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ActionUsage::ownedRelationship () : Relationship [0..*]

```
let toElementFMS: Set(UML::Element) =
    from.ownedElement->select(e | e.oclIsKindOf(UML::Pin)) in
let toElementOMS: Set(UML::Element) =
    (from.ownedElement - toElementFMS) - Set{from.value} in
toElementFMS->collect(e | ElementFeatureMembership_Mapping.getMapped(e))
->union(toElementOMS->collect(e | ElementOwningMembership_Mapping.getMapped(e)))
```

7.7.2.2.5.41 VSAOutputPin_Mapping

Description

The mapping class creates the output parameter of the ActionUsage for the mapping of UML4SysML::ValueSpecificationAction.

General Mappings

Pin_Mapping

Mapping Source

OutputPin

Mapping Target

ReferenceUsage

Owned Mappings

(none)

Applicable filters

This mapping applies only if the following (OCL) condition implemented by the operation *filter(src : Element) : Boolean* is verified:

```
src.owner.oclIsKindOf(UML::ValueSpecificationAction)
```

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ReferenceUsage::ownedRelationship () : Relationship [0..*]

```
let relationships : Set(KerML::Relatiomship) =
  self.oclAsType(Pin_Mapping).ownedRelationship()
  ->including(VSAOutputPinFeatureValue_Mapping.getMapped(from)) in
if from.type.oclIsUndefined() then
  relationships
else
  relationships
    ->including(TypedElementFeatureTyping_Mapping.getMapped(from))
endif
```

7.7.2.2.5.42 VSAOutputPinFeatureValue_Mapping

Description

Creates a feature value relationship.

General Mappings

ToFeatureValue_Init
Mapping

Mapping Source

OutputPin

Mapping Target

FeatureValue

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureValue::value () : Expression [1]

```
if from.owner.value.oclIsTypeOf(UML::OpaqueExpression) then
    OpaqueExpressionAsValue_Mapping.getMapped(from.owner.value)
else
    from.owner.value
endif
```

7.7.2.2.6 Other Actions

7.7.2.2.6.1 RaiseExceptionAction_Mapping

Description

The UML4SysML::RaiseExceptionAction is mapped to a SysML v2 ActionUsage. The details of the mapping are not defined yet.

General Mappings

CommonAction_Mapping

Mapping Source

RaiseExceptionAction

Mapping Target

ActionUsage

Owned Mappings

(none)

Applicable filters

(none)

7.7.2.2.6.2 ReduceAction_Mapping

Description

The UML4SysML::ReduceAction is mapped to a SysML v2 ActionUsage. The details of the mapping are not defined yet.

General Mappings

CommonAction_Mapping

Mapping Source

ReduceAction

Mapping Target

ActionUsage

Owned Mappings

(none)

Applicable filters

(none)

7.7.2.2.7 Structural Feature Actions

7.7.2.2.7.1 AddStructuralFeatureValueAction_Mapping

Description

A UML4SysML::AddStructuralFeatureValueAction is mapped to a SysML v2 ActionUsage defined by the SysML v1 library action definition SysMLv1Library::AddStructuralFeatureValueAction.

The following shows an example of what the textual SysML v2 syntax of the result of the transformation may look like.

```
action thisIsAAddStructuralFeatureValueAction
  : SysMLv1Library::AddStructuralFeatureValueAction {
    :>> target := object.thisIsAnAttribute;
    :>> object : ThisIsABlock;
}
part def SysMLv1Block {
    attribute sysMLv1Property;
}
```

General Mappings

CommonAction_Mapping

Mapping Source

AddStructuralFeatureValueAction

Mapping Target

ActionUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ActionUsage::ownedRelationship () : Relationship [0..*]

  ```
  Set{ASFVAFeatureTyping_Mapping.getMapped(from),
  ASFVATargetFeatureMembership_Mapping.getMapped(from),
  ASFVAObjectFeatureMembership_Mapping.getMapped(from)}
  ```

7.7.2.2.7.2 ASFVAFeatureTyping_Mapping

Description

Creates a feature typing relationship owned by the element *typedFeature()*.

General Mappings

ToFeatureTyping_Init
Mapping

Mapping Source

AddStructuralFeatureValueAction

Mapping Target

FeatureTyping

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureTyping::type () : Type [1]

  ```
  SYSML2::ActionDefinition.allInstances()
  ->any(m | m.qualifiedName = 'SysMLv1Library::AddStructuralFeatureValueAction')
  ```

7.7.2.2.7.3 ASFVAObjectFeatureMembership_Mapping

Description

Creates a feature membership relationship for *ownedMemberFeature()*.

General Mappings

ToFeatureMembership_Init
Mapping

Mapping Source

AddStructuralFeatureValueAction

Mapping Target

FeatureMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureMembership::ownedMemberFeature () : Feature [1]

```
ASFVAObjectReferenceUsage_Mapping.getMapped(from)
```

7.7.2.2.7.4 ASFVAObjectReferenceUsage_Mapping

Description

Creates a reference usage.

General Mappings

UniqueMapping
ToReferenceUsage_Init

Mapping Source

AddStructuralFeatureValueAction

Mapping Target

ReferenceUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ReferenceUsage::ownedRelationship () : Relationship [0..*]

```
Set{ASFVAObjectReferenceUsageRedefinition_Mapping.getMapped(from),
ASFVAObjectReferenceUsageFeatureTyping_Mapping.getMapped(from)}
```

7.7.2.2.7.5 ASFVAObjectReferenceUsageFeatureTyping_Mapping

Description

Creates a feature typing relationship owned by the element *typedFeature()*.

General Mappings

ToFeatureTyping_Init
Mapping

Mapping Source

AddStructuralFeatureValueAction

Mapping Target

FeatureTyping

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureTyping::type () : Type [1]

```
from.structuralFeature.owner
```

7.7.2.2.7.6 ASFVAObjectReferenceUsageRedefinition_Mapping

Description

Creates a redefinition relationship for the *redefiningFeature()* and the *redefinedFeature()*.

General Mappings

ToRedefinition_Init
Mapping

Mapping Source

AddStructuralFeatureValueAction

Mapping Target

Redefinition

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Redefinition::redefinedFeature () : Feature [1]

```
SYSML2::ReferenceUsage.allInstances()
->any(m | m.qualifiedName = 'SysMLv1Library::AddStructuralFeatureValueAction::object')
```

7.7.2.2.7.7 ASFVATargetFeatureChainExpression_Mapping

Description

The mapping class creates the feature chain expression element for the target element of the UML4SysML::AddStructuralFeatureValueAction mapping.

General Mappings

ToFeatureChainExpression_Init
Mapping

Mapping Source

AddStructuralFeatureValueAction

Mapping Target

FeatureChainExpression

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureChainExpression::ownedRelationship () : Relationship [0..*]

```
Set{ASFVATargetParameterMembership_Mapping.getMapped(from),
ASFVATargetParameterFeatureExpressionMembership_Mapping.getMapped(from),
ReturnParameterFeatureMembership_Factory.create()}
```

7.7.2.2.7.8 ASFVATargetFeatureMembership_Mapping

Description

Creates a feature membership relationship for *ownedMemberFeature()*.

General Mappings

ToFeatureMembership_Init
Mapping

Mapping Source

AddStructuralFeatureValueAction

Mapping Target

FeatureMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureMembership::ownedMemberFeature () : Feature [1]

  `ASFVATargetReferenceUsage_Mapping.getMapped(from)`

7.7.2.2.7.9 ASFVATargetFeatureValue_Mapping

Description

Creates a feature value relationship.

General Mappings

ToFeatureValue_Init
Mapping

Mapping Source

AddStructuralFeatureValueAction

Mapping Target

FeatureValue

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureValue::isInitial () : Boolean [1]

  `true`

- FeatureValue::value () : Expression [1]

  `ASFVATargetFeatureChainExpression_Mapping.getMapped(from)`

7.7.2.2.7.10 ASFVATargetParameterExpressionFeature_Mapping

Description

The mapping class creates the feature element of the feature reference expression for the target element of the UML4SysML::AddStructuralFeatureValueAction mapping.

General Mappings

ToFeature_Init
Mapping

Mapping Source

AddStructuralFeatureValueAction

Mapping Target

Feature

Owned Mappings

(none)

Applicable filters

(none)

7.7.2.2.7.11 ASFVATargetParameterExpressionFeatureMembership_Mapping

Description

Creates a feature membership relationship for *ownedMemberFeature()*.

General Mappings

ToFeatureMembership_Init
Mapping

Mapping Source

AddStructuralFeatureValueAction

Mapping Target

FeatureMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureMembership::ownedMemberFeature () : Feature [1]

  `ASFVATargetParameterExpressionFeature_Mapping.getMapped(from)`

7.7.2.2.7.12 ASFVATargetParameterExpressionMembership_Mapping

Description

Creates a membership relationship for *memberElement()*.

General Mappings

ToMembership_Init
Mapping

Mapping Source

AddStructuralFeatureValueAction

Mapping Target

Membership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Membership::memberElement () : Element [1]

  `ASFVAObjectReferenceUsage_Mapping.getMapped(from)`

7.7.2.2.7.13 ASFVATargetParameterFeature_Mapping

Description

The mapping class creates the feature element for the target element of the UML4SysML::AddStructuralFeatureValueAction mapping.

General Mappings

ToFeature_Init
Mapping

Mapping Source

AddStructuralFeatureValueAction

Mapping Target

Feature

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Feature::direction () : FeatureDirectionKind [0..1]

  ```
  KerML::FeatureDirectionKind::_'in'
  ```

- Feature::ownedRelationship () : Relationship [0..*]

  ```
  Set{ASFVATargetParameterFeatureValue_Mapping.getMapped(from),
  ASFVATargetParameterExpressionFeatureMembership_Mapping.getMapped(from)}
  ```

7.7.2.2.7.14 ASFVATargetParameterFeatureExpressionMembership_Mapping

Description

Creates a membership relationship for *memberElement()*.

General Mappings

ToMembership_Init
Mapping

Mapping Source

AddStructuralFeatureValueAction

Mapping Target

Membership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Membership::memberElement () : Element [1]

```
from.structuralFeature
```

7.7.2.2.7.15 ASFVATargetParameterFeatureReferenceExpression_Mapping

Description

The mapping class creates the feature reference expression element for the target element of the UML4SysML::AddStructuralFeatureValueAction mapping.

General Mappings

ToFeatureReferenceExpression_Init
Mapping

Mapping Source

AddStructuralFeatureValueAction

Mapping Target

FeatureReferenceExpression

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureReferenceExpression::ownedRelationship () : Relationship [0..*]

```
Set{ASFVATargetParameterExpressionMembership_Mapping.getMapped(from),
ReturnParameterFeatureMembership_Factory.create()}
```

7.7.2.2.7.16 ASFVATargetParameterFeatureValue_Mapping

Description

Creates a feature value relationship.

General Mappings

ToFeatureValue_Init
Mapping

Mapping Source

AddStructuralFeatureValueAction

Mapping Target

FeatureValue

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureValue::value () : Expression [1]

  `ASFVATargetParameterFeatureReferenceExpression_Mapping.getMapped(from)`

7.7.2.2.7.17 ASFVATargetParameterMembership_Mapping

Description

Creates a membership relationship for *memberElement()*.

General Mappings

ToParameterMembership_Init
Mapping

Mapping Source

AddStructuralFeatureValueAction

Mapping Target

ParameterMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ParameterMembership::ownedMemberParameter () : Feature [1]

```
ASFVATargetParameterFeature_Mapping.getMapped(from)
```

- ParameterMembership::visibility () : VisibilityKind [1]

```
KerML::VisibilityKind::private
```

7.7.2.2.7.18 ASFVATargetReferenceUsage_Mapping

Description

Creates a reference usage.

General Mappings

ToReferenceUsage_Init
Mapping

Mapping Source

AddStructuralFeatureValueAction

Mapping Target

ReferenceUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ReferenceUsage::ownedRelationship () : Relationship [0..*]

```
Set{ASFVATargetReferenceUsageRedefinition_Mapping.getMapped(from),
ASFVATargetFeatureValue_Mapping.getMapped(from),
AssignmentActionUsageOwningMembership_Factory.create()}
```

7.7.2.2.7.19 ASFVATargetReferenceUsageRedefinition_Mapping

Description

Creates a redefinition relationship for the *redefiningFeature()* and the *redefinedFeature()*.

General Mappings

ToRedefinition_Init
Mapping

Mapping Source

AddStructuralFeatureValueAction

Mapping Target

Redefinition

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Redefinition::redefinedFeature () : Feature [1]

```
SYSML2::ReferenceUsage.allInstances()
->any(m | m.qualifiedName = 'SysMLv1Library::AddValueAction::target')
```

7.7.2.2.7.20 ClearStructuralFeatureAction_Mapping

Description

The UML4SysML::ClearStructuralFeatureAction is mapped to a SysML v2 ActionUsage. The details of the mapping are not defined yet.

General Mappings

CommonAction_Mapping

Mapping Source

ClearStructuralFeatureAction

Mapping Target

ActionUsage

Owned Mappings

(none)

Applicable filters

(none)

7.7.2.2.7.21 ReadStructuralFeatureAction_Mapping

Description

A UML4SysML::ReadStructuralFeatureAction is mapped to a SysML v2 ActionUsage that returns the value of the specified structural feature of the given object.

The following shows an example of what the textual SysML v2 syntax of the result of the transformation may look like.

```
action def SysMLv1Activity {
	action sysMLv1ReadStructuralFeatureAction {
		in object : SysMLv1Block;
		out result = object.sysMLv1Property;
	}
}
part def SysMLv1Block {
	attribute sysMLv1Property;
}
```

General Mappings

CommonAction_Mapping

Mapping Source

ReadStructuralFeatureAction

Mapping Target

ActionUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ActionUsage::ownedRelationship () : Relationship [0..*]

```
Helper.actionOwnedRelationship(from)
->including(RSFAReferenceUsageFeatureMembership_Mapping.getMapped(from))
```

7.7.2.2.7.22 RSFAReferenceUsage_Mapping

Description

Creates a reference usage.

General Mappings

ToReferenceUsage_Init
Mapping

Mapping Source

ReadStructuralFeatureAction

Mapping Target

ReferenceUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ReferenceUsage::ownedRelationship () : Relationship [0..*]

  ```
  Set{RSFAReferenceUsageFeatureValue_Mapping.getMapped(from)}
  ```

- ReferenceUsage::direction () : FeatureDirectionKind [0..1]

  ```
  KerML::FeatureDirectionKind::_'out'
  ```

7.7.2.2.7.23 RSFAReferenceUsageExpressionFeature_Mapping

Description

The mapping class creates the feature of the feature chain expression for the reference usage of the UML4SysML::ReadStructuralFeatureValueAction mapping.

General Mappings

ToFeature_Init
Mapping

Mapping Source

ReadStructuralFeatureAction

Mapping Target

Feature

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Feature::ownedRelationship () : Relationship [0..*]

```
Set{RSFAReferenceUsageExpressionFeatureValue_Mapping.getMapped(from),
RSFAReferenceUsageExpressionFeatureMembership_Mapping.getMapped(from)}
```

7.7.2.2.7.24 RSFAReferenceUsageExpressionFeatureMembership_Mapping

Description

Creates a feature membership relationship for *ownedMemberFeature()*.

General Mappings

ToFeatureMembership_Init
Mapping

Mapping Source

ReadStructuralFeatureAction

Mapping Target

FeatureMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureMembership::ownedMemberFeature () : Feature [1]

  ```
  RSFAReferenceUsageFeatureChainExpressionFeature_Mapping.getMapped(from)
  ```

7.7.2.2.7.25 RSFAReferenceUsageExpressionFeatureReferenceExpression_Mapping

Description

The mapping class creates the feature reference expression element for the UML4SysML::RemoveStructuralFeatureValueAction mapping.

General Mappings

ToFeatureReferenceExpression_Init
Mapping

Mapping Source

ReadStructuralFeatureAction

Mapping Target

FeatureReferenceExpression

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureReferenceExpression::ownedRelationship () : Relationship [0..*]

```
Set{RSFAReferenceUsageExpressionFeatureMembership_Mapping.getMapped(from),
ReturnParameterFeatureMembership_Factory.create()}
```

7.7.2.2.7.26 RSFAReferenceUsageExpressionFeatureValue_Mapping

Description

Creates a feature value relationship.

General Mappings

ToFeatureValue_Init
Mapping

Mapping Source

ReadStructuralFeatureAction

Mapping Target

FeatureValue

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureValue::value () : Expression [1]

```
RSFAReferenceUsageExpressionFeatureReferenceExpression_Mapping.getMapped(from)
```

7.7.2.2.7.27 RSFAReferenceUsageFeatureChainExpression_Mapping

Description

The mapping class creates the feature chain expression element for the reference usage of the UML4SysML::ReadStructuralFeatureValueAction mapping.

General Mappings

ToFeatureChainExpression_Init
Mapping

Mapping Source

ReadStructuralFeatureAction

Mapping Target

FeatureChainExpression

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureChainExpression::ownedRelationship () : Relationship [0..*]

```
Set{RSFAReferenceUsageParameterMembership_Mapping.getMapped(from),
RSFAReferenceUsageMembership_Mapping.getMapped(from),
ReturnParameterFeatureMembership_Factory.create()}
```

7.7.2.2.7.28 RSFAReferenceUsageFeatureChainExpressionFeature_Mapping

Description

The mapping class creates the feature element for the feature chain expression for the UML4SysML::RemoveStructuralFeatureValueAction mapping.

General Mappings

ToFeature_Init
Mapping

Mapping Source

ReadStructuralFeatureAction

Mapping Target

Feature

Owned Mappings

(none)

Applicable filters

(none)

7.7.2.2.7.29 RSFAReferenceUsageFeatureChainExpressionMembership_Mapping

Description

Creates a membership relationship for *memberElement()*.

General Mappings

ToMembership_Init
Mapping

Mapping Source

ReadStructuralFeatureAction

Mapping Target

Membership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Membership::memberElement () : Element [1]

  ```
  from.structuralFeature
  ```

7.7.2.2.7.30 RSFAReferenceUsageFeatureMembership_Mapping

Description

Creates a feature membership relationship for *ownedMemberFeature()*.

General Mappings

ToFeatureMembership_Init
Mapping

Mapping Source

ReadStructuralFeatureAction

Mapping Target

FeatureMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureMembership::ownedMemberFeature () : Feature [1]

  ```
  RSFAReferenceUsageFeatureValue_Mapping.getMapped(from)
  ```

7.7.2.2.7.31 RSFAReferenceUsageFeatureValue_Mapping

Description

Creates a feature value relationship.

General Mappings

ToFeatureValue_Init
Mapping

Mapping Source

ReadStructuralFeatureAction

Mapping Target

FeatureValue

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureValue::value () : Expression [1]

  ```
  RSFAReferenceUsageFeatureChainExpression_Mapping.getMapped(from)
  ```

7.7.2.2.7.32 RSFAReferenceUsageMembership_Mapping

Description

Creates a membership relationship for *memberElement()*.

General Mappings

ToMembership_Init
Mapping

Mapping Source

ReadStructuralFeatureAction

Mapping Target

Membership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Membership::memberElement () : Element [1]

  ```
  from.object
  ```

7.7.2.2.7.33 RSFAReferenceUsageParameterMembership_Mapping

Description

Creates a membership relationship for *memberElement()*.

General Mappings

ToParameterMembership_Init
Mapping

Mapping Source

ReadStructuralFeatureAction

Mapping Target

ParameterMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ParameterMembership::ownedMemberParameter () : Feature [1]

  `RSFAReferenceUsageExpressionFeature_Mapping.getMapped(from)`

7.7.2.2.7.34 RemoveStructuralFeatureValueAction_Mapping

Description

The UML4SysML::RemoveStructuralFeatureValueAction is mapped to a SysML v2 ActionUsage. The details of the mapping are not defined yet.

General Mappings

CommonAction_Mapping

Mapping Source

RemoveStructuralFeatureValueAction

Mapping Target

ActionUsage

Owned Mappings

(none)

Applicable filters

(none)

7.7.2.2.8 Structured Actions

7.7.2.2.8.1 LoopNode_Mapping

Description

The UML4SysML::LoopNode is mapped to a SysML v2 ActionUsage. The details of the mapping are not defined yet.

General Mappings

StructuredActivityNode_Mapping

Mapping Source

LoopNode

Mapping Target

ActionUsage

Owned Mappings

(none)

Applicable filters

(none)

7.7.2.2.8.2 SequenceNode_Mapping

Description

The UML4SysML::SequenceNode is mapped to a SysML v2 ActionUsage. The details of the mapping are not defined yet.

General Mappings

CommonAction_Mapping
StructuredActivityNode_Mapping

Mapping Source

SequenceNode

Mapping Target

ActionUsage

Owned Mappings

(none)

Applicable filters

(none)

7.7.2.2.8.3 StructuredActivityNode_Mapping

Description

The UML4SysML::StructuredActivityNode is mapped to a SysML v2 ActionUsage. The details of the mapping are not defined yet.

General Mappings

CommonAction_Mapping

Mapping Source

StructuredActivityNode

Mapping Target

ActionUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ActionUsage::ownedRelationship () : Relationship [0..*]

```
let initialNodes : Set(UML::Element) =
    from.ownedElement->select(e | e.oclIsKindOf(UML::InitialNode)) in
let finalNodes : Set(UML::Element) =
    from.ownedElement->select(e | e.oclIsKindOf(UML::FinalNode)) in
let objectFlowsWithGuard : Set(UML::ObjectFlow) =
    from.ownedElement->select(e | e.oclIsKindOf(UML::ObjectFlow)
        and not e.oclAsType(UML::ObjectFlow).guard.oclIsUndefined()) in
let objectFlows : Set(UML::ObjectFlow) =
    from.ownedElement->select(e | e.oclIsKindOf(UML::ObjectFlow))  in
let ignoreInterruptibleActivityRegion: Set(UML::InterruptibleActivityRegion) =
    from.ownedElement
      ->select(e | e.oclIsKindOf(UML::InterruptibleActivityRegion)) in
let elementsFMS : Set(UML::Element) =
    ((from.ownedElement->select(e | e.oclIsKindOf(UML::ControlNode) or
        e.oclIsKindOf(UML::Action) or (e.oclIsKindOf(UML::ControlFlow) or
        e.oclIsKindOf(UML::Pin))) - initialNodes) - finalNodes) in
let elementsOMS: Set(UML::Element) =
    (((((from.ownedElement-initialNodes)-finalNodes)-objectFlowsWithGuard)
        -objectFlows)-elementsFMS)-ignoreInterruptibleActivityRegion) in
elementsOMS->collect(e | ElementOwningMembership_Mapping.getMapped(e))
  ->union(elementsFMS->collect(e | ElementFeatureMembership_Mapping.getMapped(e)))
  ->union(initialNodes->collect(e | InitialNodeMembership_Mapping.getMapped(e)))
  ->union(finalNodes->collect(e | FlowFinalNodeMembership_Mapping.getMapped(e)))
  ->union(objectFlowsWithGuard
  ->collect(e | ObjectFlowGuardFeatureMembership_Mapping.getMapped(e)))
  ->union(objectFlows->collect(e | ObjectFlowFeatureMembership_Mapping.getMapped(e)))
```

7.7.2.2.9 Variable Actions

7.7.2.2.9.1 AddVariableValueAction_Mapping

Description

A UML4SysML::AddVariableValueAction is mapped to a SysML v2 ActionUsage defined by the SysML v1 library action definition SysMLv1Library::AddValueAction. The following shows an example of what the textual SysML v2 syntax of the result of the transformation may look like.

```
action def SysMLv1Activity {
        private attribute sysMLv1Variable1 : ScalarValues::Integer;
        private attribute sysMLv1Variable2 [0..*] : ScalarValues::Integer;

        action sysMLv1AddVariableValueAction1 : SysMLv1Library::AddValueAction {
                :>> target := sysMLv1Variable1;
        }

        action sysMLv1AddVariableValueAction1 : SysMLv1Library::AddValueAction {
```

```
                :>> target := thisIsAVariable;
                :>> isReplaceAll := true;
        }
}
```

General Mappings

CommonAction_Mapping

Mapping Source

AddVariableValueAction

Mapping Target

ActionUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ActionUsage::ownedRelationship () : Relationship [0..*]

```
let relationships : Set(KerML::Relationship) =
  Set{AVVAFeatureTyping_Mapping.getMapped(from)}
  ->including(AVVAVariableFeatureMembership_Mapping.getMapped(from)) in
if from.isReplaceAll then
  relationships
    ->including(AVVAIsReplaceAllFeatureMembership_Mapping.getMapped(from))
else
  relationships
endif
```

7.7.2.2.9.2 AVVAFeatureTyping_Mapping

Description

Creates a feature typing relationship owned by the element *typedFeature()*.

General Mappings

ToFeatureTyping_Init
Mapping

Mapping Source

AddVariableValueAction

Mapping Target

FeatureTyping

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureTyping::type () : Type [1]

```
SYSML2::ActionDefinition.allInstances()
->any(m | m.qualifiedName = 'SysMLv1Library::AddValueAction')
```

7.7.2.2.9.3 AVVAFeatureValue_Mapping

Description

Creates a feature value relationship.

General Mappings

ToFeatureValue_Init
Mapping

Mapping Source

AddVariableValueAction

Mapping Target

FeatureValue

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureValue::value () : Expression [1]

```
AVVAValueFeatureReferenceExpression_Mapping.getMapped(from)
```

7.7.2.2.9.4 AVVAIsReplaceAll_Mapping

Description

The mapping class creates a reference usage element as mapping target for the AddVariableValueAction::isReplaceAll property.

General Mappings

ToReferenceUsage_Init
Mapping

Mapping Source

AddVariableValueAction

Mapping Target

ReferenceUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ReferenceUsage::ownedRelationship () : Relationship [0..*]

```
Set{AVVAIsReplaceAllRedefinition_Mapping.getMapped(from),
AVVAIsReplaceAllValue_Mapping.getMapped(from),
AssignmentActionUsageOwningMembership_Factory.create()}
```

7.7.2.2.9.5 AVVAIsReplaceAllFeatureMembership_Mapping

Description

Creates a feature membership relationship for *ownedMemberFeature()*.

General Mappings

ToFeatureMembership_Init
Mapping

Mapping Source

AddVariableValueAction

Mapping Target

FeatureMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureMembership::ownedMemberFeature () : Feature [1]

```
AVVAIsReplaceAll_Mapping.getMapped(from)
```

7.7.2.2.9.6 AVVAIsReplaceAllRedefinition_Mapping

Description

Creates a redefinition relationship for the *redefiningFeature()* and the *redefinedFeature()*.

General Mappings

ToRedefinition_Init
Mapping

Mapping Source

AddVariableValueAction

Mapping Target

Redefinition

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Redefinition::redefinedFeature () : Feature [1]

```
SYSML2::ReferenceUsage.allInstances()
->any(m | m.qualifiedName = 'SysMLv1Library::AddValueAction::isReplaceAll')
```

7.7.2.2.9.7 AVVAIsReplaceAllValue_Mapping

Description

The mapping class maps the value of the AddVariableValueAction::isReplaceAll property.

General Mappings

ToFeatureValue_Init
Mapping

Mapping Source

AddVariableValueAction

Mapping Target

FeatureValue

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureValue::value () : Expression [1]

  ```
  LiteralBoolean_Factory.create(from.isReplaceAll)
  ```

7.7.2.2.9.8 AVVAValueExpressionMembership_Mapping

Description

Creates a membership relationship for *memberElement()*.

General Mappings

ToMembership_Init
Mapping

Mapping Source

AddVariableValueAction

Mapping Target

Membership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Membership::memberElement () : Element [1]

```
from.variable
```

7.7.2.2.9.9 AVVAValueFeatureReferenceExpression_Mapping

Description

The mapping class creates the feature reference expression element for the UML4SysML::AddStructuralFeatureValueAction mapping.

General Mappings

ToFeatureReferenceExpression_Init
Mapping

Mapping Source

AddVariableValueAction

Mapping Target

FeatureReferenceExpression

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureReferenceExpression::ownedRelationship () : Relationship [0..*]

```
Set{AVVAValueExpressionMembership_Mapping.getMapped(from),
ReturnParameterFeatureMembership_Factory.create()}
```

7.7.2.2.9.10 AVVAVariable_Mapping

Description

The mapping class creates a reference usage element for the UML4SysML::AddVariableValueAction mapping.

General Mappings

ToReferenceUsage_Init
Mapping

Mapping Source

AddVariableValueAction

Mapping Target

ReferenceUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ReferenceUsage::ownedRelationship () : Relationship [0..*]

  ```
  Set{AVVAVariableRedefinition_Mapping.getMapped(from),
  AVVAFeatureValue_Mapping.getMapped(from),
  AssignmentActionUsageOwningMembership_Factory.create()}
  ```

7.7.2.2.9.11 AVVAVariableFeatureMembership_Mapping

Description

Creates a feature membership relationship for *ownedMemberFeature()*.

General Mappings

ToFeatureMembership_Init
Mapping

Mapping Source

AddVariableValueAction

Mapping Target

FeatureMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureMembership::ownedMemberFeature () : Feature [1]

  ```
  AVVAVariable_Mapping.getMapped(from)
  ```

7.7.2.2.9.12 AVVAVariableRedefinition_Mapping

Description

Creates a redefinition relationship for the *redefiningFeature()* and the *redefinedFeature()*.

General Mappings

ToRedefinition_Init
Mapping

Mapping Source

AddVariableValueAction

Mapping Target

Redefinition

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Redefinition::redefinedFeature () : Feature [1]

  ```
  SYSML2::ReferenceUsage.allInstances()
  ->any(m | m.qualifiedName = 'SysMLv1Library::AddValueAction::target')
  ```

7.7.2.2.9.13 ClearVariableAction_Mapping

Description

The UML4SysML::ClearVariableAction is mapped to a SysML v2 ActionUsage that sets the attribute usage representing the variable to null.

The expected SysML v2 textual notation of a SysMLv1::ClearVariableAction is as follows

```
action def SysMLv1Activity {
        private attribute sysMLv1Variable : ScalarValues::Integer;
```

```
        action sysMLv1ClearVariableAction {
                sysMLv1Variable := null;
        }
}
```

General Mappings

CommonAction_Mapping

Mapping Source

ClearVariableAction

Mapping Target

ActionUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ActionUsage::ownedRelationship () : Relationship [0..*]

```
Helper.actionOwnedRelationship(from)
->including(CVAFeatureMembership_Mapping.getMapped(from))
```

7.7.2.2.9.14 CVAFeatureMembership_Mapping

Description

Creates a feature membership relationship for *ownedMemberFeature()*.

General Mappings

ToFeatureMembership_Init
Mapping

Mapping Source

ClearVariableAction

Mapping Target

FeatureMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureMembership::ownedMemberFeature () : Feature [1]

  ```
  CVAReferenceUsage_Mapping.getMapped(from)
  ```

7.7.2.2.9.15 CVAReferenceUsage_Mapping

Description

Creates a reference usage.

General Mappings

ToReferenceUsage_Init
Mapping

Mapping Source

ClearVariableAction

Mapping Target

ReferenceUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ReferenceUsage::declaredName () : String [0..1]

  ```
  from.variable.name
  ```

- ReferenceUsage::ownedRelationship () : Relationship [0..*]

  ```
  Set{CVAReferenceUsageFeatureValue_Mapping.getMapped(from),
  AssignmentActionUsageOwningMembership_Factory.create()}
  ```

7.7.2.2.9.16 CVAReferenceUsageFeatureValue_Mapping

Description

Creates a feature value relationship.

General Mappings

ToFeatureValue_Init
Mapping

Mapping Source

ClearVariableAction

Mapping Target

FeatureValue

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureValue::value () : Expression [1]

  ```
  LiteralNull_Factory.create()
  ```

7.7.2.2.9.17 ReadVariableAction_Mapping

Description

A UML4SysML::ReadVariableValueAction is mapped to a SysML v2 ActionUsage with an out parameter that returns the value of the attribute usage that is the transformation target of the UML4SysML::Variable.

The following shows an example of what the textual SysML v2 syntax of the result of the transformation may look like.

```
action def SysMLv1Activity {
        private attribute sysMLv1Variable : ScalarValues::Integer;

        action sysMLv1ReadVariableAction {
                out result : ScalarValues::Integer = sysMLv1Variable;
        }
}
```

General Mappings

CommonAction_Mapping

Mapping Source

ReadVariableAction

Mapping Target

ActionUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ActionUsage::ownedRelationship () : Relationship [0..*]

  ```
  Set{RVAFeatureMembership_Mapping.getMapped(from)}
  ```

7.7.2.2.9.18 RVAFeatureMembership_Mapping

Description

Creates a feature membership relationship for *ownedMemberFeature()*.

General Mappings

ToFeatureMembership_Init
Mapping

Mapping Source

ReadVariableAction

Mapping Target

FeatureMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureMembership::ownedMemberFeature () : Feature [1]

  ```
  RVAReferenceUsage_Mapping.getMapped(from.result)
  ```

7.7.2.2.9.19 RVAReferenceUsage_Mapping

Description

Creates a reference usage.

General Mappings

ToReferenceUsage_Init
Mapping

Mapping Source

Pin

Mapping Target

ReferenceUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ReferenceUsage::ownedRelationship () : Relationship [0..*]

```
let featureTyping : Set(KerML::FeatureTyping) =
    if from.type.oclIsUndefined() then
        Set{}
    else
        Set{RVAReferenceUsageFeatureTyping_Mapping.getMapped(from)}
    endif in
featureTyping
->including(RVAReferenceUsageFeatureValue_Mapping.getMapped(from))
```

7.7.2.2.9.20 RVAReferenceUsageFeatureReferenceExpression_Mapping

Description

The mapping class creates the feature reference expression element for the UML4SysML::ReadVariableAction mapping.

General Mappings

ToFeatureReferenceExpression_Init
Mapping

Mapping Source

Pin

Mapping Target

FeatureReferenceExpression

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureReferenceExpression::ownedRelationship () : Relationship [0..*]

```
Set{RVAReferenceUsageExpressionMembership_Mapping.getMapped(from),
ReturnParameterFeatureMembership_Factory.create()}
```

7.7.2.2.9.21 RVAReferenceUsageFeatureTyping_Mapping

Description

Creates a feature typing relationship owned by the element *typedFeature()*.

General Mappings

TypedElementFeatureTyping_Mapping

Mapping Source

Pin

Mapping Target

FeatureTyping

Owned Mappings

(none)

Applicable filters

(none)

7.7.2.2.9.22 RVAReferenceUsageFeatureValue_Mapping

Description

Creates a feature value relationship.

General Mappings

ToFeatureValue_Init
Mapping

Mapping Source

Pin

Mapping Target

FeatureValue

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureValue::value () : Expression [1]

  `RVAReferenceUsageFeatureReferenceExpression_Mapping.getMapped(from)`

7.7.2.2.9.23 RVAReferenceUsageExpressionMembership_Mapping

Description

Creates a membership relationship for *memberElement()*.

General Mappings

ToMembership_Init
Mapping

Mapping Source

Pin

Mapping Target

Membership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Membership::memberElement () : Element [1]

```
from.owner.oclAsType(UML::ReadVariableAction).variable
```

7.7.2.2.9.24 RemoveVariableValueAction_Mapping

Description

A UML4SysML::RemoveVariableValueAction is mapped to a SysML v2 ActionUsage defined by the SysML v1 library action definition SysMLv1Library::RemoveVariableValueAction.

The following shows an example of what the textual SysML v2 syntax of the result of the transformation may look like.

```
action def SysMLv1Activity {
        private sysMLv1Variable : ScalarValues::Integer;

        action sysMLv1RemoveVariableValueAction
            : SysMLv1Library::RemoveVariableValueAction {
                :>> variable := sysMLv1Variable;
        }
}
```

General Mappings

CommonAction_Mapping

Mapping Source

RemoveVariableValueAction

Mapping Target

ActionUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ActionUsage::ownedRelationship () : Relationship [0..*]

```
Helper.actionOwnedRelationship(from)
->including(RVVAFeatureTyping_Mapping.getMapped(from))
->including(RVVAVariableFeatureMembership_Mapping.getMapped(from))
```

7.7.2.2.9.25 RVVAFeatureTyping_Mapping

Description

Creates a feature typing relationship owned by the element *typedFeature()*.

General Mappings

ToFeatureTyping_Init
Mapping

Mapping Source

RemoveVariableValueAction

Mapping Target

FeatureTyping

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureTyping::type () : Type [1]

```
SYSML2::ActionDefinition.allInstances()
->any(m | m.qualifiedName = 'SysMLv1Library::RemoveVariableValueAction')
```

7.7.2.2.9.26 RVVAVariable_Mapping

Description

The mapping class creates a reference usage element for the UML4SysML::RemoveVariableValueAction mapping.

General Mappings

ToReferenceUsage_Init
Mapping

Mapping Source

RemoveVariableValueAction

Mapping Target

ReferenceUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ReferenceUsage::ownedRelationship () : Relationship [0..*]

```
Set{RVVAVariableRedefinition_Mapping.getMapped(from),
RVVAVariableFeatureValue_Mapping.getMapped(from),
AssignmentActionUsageOwningMembership_Factory.create()}
```

7.7.2.2.9.27 RVVAVariableExpressionMembership_Mapping

Description

Creates a membership relationship for *memberElement()*.

General Mappings

ToMembership_Init
Mapping

Mapping Source

RemoveVariableValueAction

Mapping Target

Membership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Membership::memberElement () : Element [1]

```
from.variable
```

7.7.2.2.9.28 RVVAVariableFeatureMembership_Mapping

Description

Creates a feature membership relationship for *ownedMemberFeature()*.

General Mappings

ToFeatureMembership_Init
Mapping

Mapping Source

RemoveVariableValueAction

Mapping Target

FeatureMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureMembership::ownedMemberFeature () : Feature [1]

  ```
  RVVAVariable_Mapping.getMapped(from)
  ```

7.7.2.2.9.29 RVVAVariableFeatureReferenceExpression_Mapping

Description

The mapping class creates the feature reference expression element for the UML4SysML::RemoveVariableValueAction mapping.

General Mappings

ToFeatureReferenceExpression_Init
Mapping

Mapping Source

RemoveVariableValueAction

Mapping Target

FeatureReferenceExpression

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureReferenceExpression::ownedRelationship () : Relationship [0..*]

```
Set{RVVAVariableExpressionMembership_Mapping.getMapped(from),
ReturnParameterFeatureMembership_Factory.create()}
```

7.7.2.2.9.30 RVVAVariableFeatureValue_Mapping

Description

Creates a feature value relationship.

General Mappings

ToFeatureValue_Init
Mapping

Mapping Source

RemoveVariableValueAction

Mapping Target

FeatureValue

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureValue::value () : Expression [1]

```
RVVAVariableFeatureReferenceExpression_Mapping.getMapped(from)
```

7.7.2.2.9.31 RVVAVariableRedefinition_Mapping

Description

Creates a redefinition relationship for the *redefiningFeature()* and the *redefinedFeature()*.

General Mappings

ToRedefinition_Init
Mapping

Mapping Source

RemoveVariableValueAction

Mapping Target

Redefinition

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Redefinition::redefinedFeature () : Feature [1]

```
SYSML2::ReferenceUsage.allInstances()
->any(m | m.qualifiedName = 'SysMLv1Library::RemoveVariableValueAction::variable')
```

7.7.3 Activities

SYSML21-542: Transformation: overview tables shall be revised

7.7.3.1 Overview

SYSML21-542: Transformation: overview tables shall be revised

Table 2. List of High-Level Activities Mappings

SysML v1 Abstract Syntax/Stereotype	SysML v2 Abstract Syntax
Activity	ActionDefinition
ActivityFinalNode	TerminateActionUsage
ActivityParameterNode	The parameter of the activity is mapped from SysML v1 to SysML v2. The additional concept of the activity parameter node is necessary for the token semantic of SysML v1 activities, which is not part of SysML v2. Therefore, the additional concept of the activity parameter node is not mapped to SysML v2.
ActivityPartition	No specific mapping specified yet.
CentralBufferNode	ActionUsage
ControlFlow	TransitionUsage, SuccessionAsUsage
DataStoreNode	ActionUsage
DecisionNode	DecisionNode
ExceptionHandler	No specific mapping specified yet.
FlowFinalNode	No specific mapping specified yet.
ForkNode	ForkNode
InitialNode	No specific mapping specified yet.

SysML v1 Abstract Syntax/Stereotype	SysML v2 Abstract Syntax
InterruptibleActivityRegion	No specific mapping specified yet.
JoinNode	JoinNode
MergeNode	MergeNode
ObjectFlow	SuccessionFlowUsage, TransitionUsage
Variable	AttributeUsage, ItemUsage

7.7.3.2 Mapping Specifications

7.7.3.2.1 ActivityAsDefinition_Mapping

Description

A UML4SysML::Activity is mapped to a SysMLv2 ActionDefinition.

The following shows an example of what the textual SysML v2 syntax of the result of the transformation may look like.

```
action def SysMLv1Activity {
  in parIn : SysMLv1Block;
  out parOut;
  out parReturn;
}
part def SysMLv1Block;
```

General Mappings

Behavior_Mapping

Mapping Source

Activity

Mapping Target

ActionDefinition

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ActionDefinition::ownedRelationship () : Relationship [0..*]

  ```
  let relationships : Set(KerML::Relationship) =
      Helper.activityOwnedRelationship(from) in
  ```

```
let parameters : Set(UML::Paramter) =
    from.ownedElement->select(e | e.oclIsKindOf(UML::Parameter)) in
relationships->union(parameters
    ->collect(p | ParameterMembership_Mapping.getMapped(p))
)
```

7.7.3.2.2 ActivityEdgeInitialNodeFeatureMembership_Mapping

Description

Creates a feature membership relationship for *ownedMemberFeature()*.

General Mappings

ToEndFeatureMembership_Init
Mapping

Mapping Source

InitialNode

Mapping Target

EndFeatureMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- EndFeatureMembership::ownedMemberFeature () : Feature [1]

  ```
  ActivityEdgeSourceInitialNode_Mapping.getMapped(from)
  ```

7.7.3.2.3 ActivityEdgeMetadata_Mapping

Description

Adds metadata to the transformation target elements of UML4SysML::ControlFlow and UML::ObjectFlow to map the UML4SysML::ActivityEdge::weight property which has no direct target in SysML v2.

General Mappings

ToMetadataUsage_Init
Mapping

Mapping Source

ActivityEdge

Mapping Target

MetadataUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- MetadataUsage::ownedRelationship () : Relationship [0..*]

  ```
  Set{ActivityEdgeMetadataFeatureTyping_Mapping.getMapped(from),
  ActivityEdgeMetadataFeatureMembership_Mapping.getMapped(from)}
  ```

- MetadataUsage::declaredName () : String [0..1]

  ```
  'weight'
  ```

7.7.3.2.4 ActivityEdgeMetadataFeatureMembership_Mapping

Description

Creates a feature membership relationship for *ownedMemberFeature()*.

General Mappings

ToFeatureMembership_Init
Mapping

Mapping Source

ActivityEdge

Mapping Target

FeatureMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureMembership::ownedMemberFeature () : Feature [1]

```
ActivityEdgeMetadataReferenceUsage_Mapping.getMapped(from)
```

7.7.3.2.5 ActivityEdgeMetadataFeatureTyping_Mapping

Description

Creates a feature typing relationship owned by the element *typedFeature()*.

General Mappings

ToFeatureTyping_Init
Mapping

Mapping Source

ActivityEdge

Mapping Target

FeatureTyping

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureTyping::type () : Type [1]

```
SYSML2::MetadataDefinition.allInstances()
->any(m | m.qualifiedName = 'SysMLv1Library::ActivityEdgeData')
```

7.7.3.2.6 ActivityEdgeMetadataFeatureValue_Mapping

Description

Creates a feature value relationship.

General Mappings

ToFeatureValue_Init
Mapping

Mapping Source

ActivityEdge

Mapping Target

FeatureValue

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureValue::value () : Expression [1]

  ```
  from.weight
  ```

7.7.3.2.7 ActivityEdgeMetadataOwningMembership_Mapping

Description

Creates a owning membership relationship for *ownedMemberElement()*.

General Mappings

ToOwningMembership_Init
Mapping

Mapping Source

ActivityEdge

Mapping Target

OwningMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- OwningMembership::ownedMemberElement () : Element [1]

  ```
  ActivityEdgeMetadata_Mapping.getMapped(from)
  ```

7.7.3.2.8 ActivityEdgeMetadataRedefinition_Mapping

Description

Creates a redefinition relationship for the *redefiningFeature()* and the *redefinedFeature()*.

General Mappings

ToRedefinition_Init
Mapping

Mapping Source

ActivityEdge

Mapping Target

Redefinition

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Redefinition::redefinedFeature () : Feature [1]

```
SYSML2::AttributeUsage.allInstances()
->any(m | m.qualifiedName = 'SysMLv1Library::ActivityEdgeData::weight')
```

7.7.3.2.9 ActivityEdgeMetadataReferenceUsage_Mapping

Description

Creates a reference usage.

General Mappings

ToReferenceUsage_Init
Mapping

Mapping Source

ActivityEdge

Mapping Target

ReferenceUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ReferenceUsage::ownedRelationship () : Relationship [0..*]

```
Set{ActivityEdgeMetadataRedefinition_Mapping.getMapped(from),
ActivityEdgeMetadataFeatureValue_Mapping.getMapped(from)}
```

7.7.3.2.10 ActivityEdgeSourceEndFeature_Mapping

Description

Creates a SysML v2 feature for the source activity node of the SysML v1 activity edge which subsets the SysML v2 target element of the source activity node.

General Mappings

ToFeature_Init
Mapping

Mapping Source

Element

Mapping Target

Feature

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Feature::isEnd () : Boolean [1]

```
true
```

- Feature::ownedRelationship () : Relationship [0..*]

```
Set{ActivityEdgeSourceEndSubsetting_Mapping.getMapped(from)}
```

7.7.3.2.11 ActivityEdgeSourceInitialNode_Mapping

Description

The UML4SysML::InitialNode is mapped to a subsetted feature of the SysML v2 library element Actions::start.

General Mappings

ToFeature_Init
Mapping

Mapping Source

InitialNode

Mapping Target

Feature

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Feature::isEnd () : Boolean [1]

  ```
  true
  ```

- Feature::ownedRelationship () : Relationship [0..*]

  ```
  Set{ActivityEdgeSourceInitialNodeSubsetting_Mapping.getMapped(from)}
  ```

7.7.3.2.12 ActivityEdgeSourceEndFeatureMembership_Mapping

Description

Creates a feature membership relationship for *ownedMemberFeature()*.

General Mappings

ToEndFeatureMembership_Init
Mapping

Mapping Source

Element

Mapping Target

EndFeatureMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- EndFeatureMembership::ownedMemberFeature () : Feature [1]

```
ActivityEdgeSourceEndFeature_Mapping.getMapped(from)
```

7.7.3.2.13 ActivityEdgeSourceInitialNodeSubsetting_Mapping

Description

Creates a subsetting relationship.

General Mappings

ToReferenceSubsetting_Init
Mapping

Mapping Source

InitialNode

Mapping Target

ReferenceSubsetting

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ReferenceSubsetting::referencedFeature () : Feature [1]

```
SYSML2::ActionUsage.allInstances()
->any(m | m.qualifiedName = 'Actions::Action::start')
```

7.7.3.2.14 ActivityEdgeSourceEndSubsetting_Mapping

Description

Creates a subsetting relationship.

General Mappings

ToReferenceSubsetting_Init
Mapping

Mapping Source

Element

Mapping Target

ReferenceSubsetting

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ReferenceSubsetting::referencedFeature () : Feature [1]

  ```
  from
  ```

7.7.3.2.15 ActivityEdgeTransitionUsageSourceMembership_Mapping

Description

Creates a membership relationship for *memberElement()*.

General Mappings

ToMembership_Init
Mapping

Mapping Source

ActivityNode

Mapping Target

Membership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Membership::memberElement () : Element [1]

```
if from.oclIsTypeOf(UML::ActivityParameterNode) then
    from.parameter
else
    from
endif
```

7.7.3.2.16 ActivityFinalNode_Mapping

Description

A UML4SysML::ActivityFinalNode is mapped to SysML v2 TerminateAction.

The following shows an example of what the textual SysML v2 syntax of the result of the transformation may look like.

```
action def SysMLv1Activity {
  first start;
  then action action1;
  then termine;
}
```

General Mappings

NamedElementMain_Mapping
ToActionUsage_Init

Mapping Source

ActivityFinalNode

Mapping Target

TerminateActionUsage

Owned Mappings

(none)

Applicable filters

(none)

7.7.3.2.17 CentralBufferNode_Mapping

Description

The mapping of the UML4SysML::CentralBufferNode is not defined in detail yet. It will be an action usage which contains the behavior of a central buffer node.

General Mappings

ToActionUsage_Init
NamedElementMain_Mapping

Mapping Source

CentralBufferNode

Mapping Target

ActionUsage

Owned Mappings

(none)

Applicable filters

(none)

7.7.3.2.18 CommonActivityEdgeSuccessionAsUsage_Mapping

Description

The mapping class provides a common mapping of a UML4SysML::ActivityEdge to a SysMLv2 SucessionAsUsage. The mapping is used for UML4SysML::ControlFlows and UML4SysML::ObjectFlows.

General Mappings

ToConnector_Init
Mapping

Mapping Source

ActivityEdge

Mapping Target

SuccessionAsUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- SuccessionAsUsage::ownedRelationship () : Relationship [0..*]

```
let relationships : Set(KerML::Relationship) = Set{
if from.source.oclIsKindOf(UML::InitialNode) then
  ActivityEdgeInitialNodeFeatureMembership_Mapping.getMapped(from.source)
else if from.source.oclIsKindOf(UML::ActivityParameterNode) then
    ActivityEdgeSourceEndFeatureMembership_Mapping.getMapped(
      from.source.parameter)
  else
    ActivityEdgeSourceEndFeatureMembership_Mapping.getMapped(from.source)
  endif
endif,
if from.oclIsKindOf(UML::ObjectFlow) then
  ObjectFlowGuardSuccessionTargetEndFeatureMembership_Mapping.getMapped(from)
else if from.target.oclIsKindOf(UML::FinalNode) then
    ControlFlowFinalNodeFeatureMembership_Mapping.getMapped(from.target)
  else
    ControlFlowTargetFeatureMembership_Mapping.getMapped(from.target)
  endif
endif} in
if from.guard.oclIsUndefined() then
  relationships
else
  relationships
  ->including(ElementFeatureMembership_Mapping.getMapped(from.guard))
endif
```

7.7.3.2.19 CommonVariable_Mapping

Description

Abstract mapping class for UML4SysML::Variable which is defined in the context of UML4SysML::Activity. A UML4SysML::Variable is mapped to a SysMLv2 AttributeUsage or SysMLv2 ItemUsage. See specialized mapping classes for the specific mapping rules.

General Mappings

PropertyCommon_Mapping

Mapping Source

Variable

Mapping Target

Feature

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Feature::isEnd () : Boolean [1]

```
false
```

- Feature::isComposite () : Boolean [1]

```
false
```

- Feature::ownedRelationship () : Relationship [0..*]

```
let typing: KerML::FeatureTyping =
    VariableFeatureTyping_Mapping.getMapped(from) in
if typing.oclIsUndefined() then
    Set{MultiplicityMembership_Mapping.getMapped(from)}
else
    Set{MultiplicityMembership_Mapping.getMapped(from), typing}
endif
```

- Feature::isDerived () : Boolean [1]

```
false
```

7.7.3.2.20 ControlFlowTransitionUsage_Mapping

Description

A UML4SysML::ControlFlow with a guard condition is mapped to a SysMLv2 TransitionUsage.

The following shows an example of what the textual SysML v2 syntax of the result of the transformation may look like.

```
action def SysMLv1Activity {
        action sysMLv1Action1;
        succession sysMLv1ControlFlow first sysMLv1Action1
                if guardCondition.result then sysMLv1Action2 {
                  calc guardCondition {
                    return : ScalarValues::Boolean;
                    language "English"
                    /*
                     * thisIsAGuard
                     */
                  }
                }
        action sysMLv1Action2;
}
```

General Mappings

ToTransitionUsage_Init
NamedElementMain_Mapping

Mapping Source

ControlFlow

Mapping Target

TransitionUsage

Owned Mappings

(none)

Applicable filters

This mapping applies only if the following (OCL) condition implemented by the operation *filter(src : Element) : Boolean* is verified:

```
not src.guard.oclIsUndefined()
```

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- TransitionUsage::ownedRelationship () : Relationship [0..*]

```
let relationships : Set(KerML::Relationship)=
  self.oclAsType(ElementMain_Mapping).ownedRelationship()
  ->union(Set{
    ActivityEdgeTransitionUsageSourceMembership_Mapping.getMapped(
      from.source),
    CommonParameterReferenceUsageInMembership_Mapping.getMapped(
      from.source),
    ControlFlowTransitionUsageFeatureMembership_Mapping.getMapped(
      from),
    CommonActivityEdgeSuccessionAsUsage_Mapping.getMapped(from),
    CommonReturnParameterReferenceUsageMembership_Mapping.getMapped(
      from)}) in
let relationshipsWithGuard : Set(KerML::Relationship) =
if from.guard.oclIsTypeOf(UML::OpaqueExpression) then
  relationships
  ->including(ElementFeatureMembership_Mapping.getMapped(from.guard))
else
  relationships
endif in
let relationshipsConsideringWeight : Set(KerML::Relationship) =
if from.weight.oclIsUndefined() then
  relationshipsWithGuard
else
  relationshipsWithGuard
  ->including(ActivityEdgeMetadataOwningMembership_Mapping.getMapped(from))
endif in
if Helper.hasStereotypeApplied(from, 'SysML::Activities::Probability') then
  relationshipsConsideringWeight
  ->including(ProbabilityOwningMembership_Mapping.getMapped(from))
else
  relationshipsConsideringWeight
endif
```

7.7.3.2.21 ControlFlowFinalNodeFeatureMembership_Mapping

Description

Creates a feature membership relationship for *ownedMemberFeature()*.

General Mappings

ToEndFeatureMembership_Init
Mapping

Mapping Source

ActivityNode

Mapping Target

EndFeatureMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- EndFeatureMembership::ownedMemberFeature () : Feature [1]

  ```
  ControlFlowTargetFinalNode_Mapping.getMapped(from)
  ```

7.7.3.2.22 ControlFlowTargetFinalNodeSubsetting_Mapping

Description

Creates a subsetting relationship.

General Mappings

ToReferenceSubsetting_Init
Mapping

Mapping Source

FinalNode

Mapping Target

ReferenceSubsetting

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ReferenceSubsetting::referencedFeature () : Feature [1]

```
SYSML2::ActionUsage.allInstances()
->any(m | m.qualifiedName = 'Actions::Action::done')
```

7.7.3.2.23 ControlFlowSuccessionAsUsage_Mapping

Description

A UML4SysML::ControlFlow without a guard condition is mapped to a SysMLv2 SuccessionAsUsage.

The following shows an example of what the textual SysML v2 syntax of the result of the transformation may look like.

```
action def SysMLv1Activity {
        action sysMLv1Action1;
        succession sysMLv1ControlFlow
                first sysMLv1Action1 then sysMLv1Action2;
        action sysMLv1Action2;
}
```

General Mappings

NamedElementMain_Mapping
CommonActivityEdgeSuccessionAsUsage_Mapping

Mapping Source

ControlFlow

Mapping Target

SuccessionAsUsage

Owned Mappings

(none)

Applicable filters

This mapping applies only if the following (OCL) condition implemented by the operation *filter(src : Element) : Boolean* is verified:

```
src.guard.oclIsUndefined()
```

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- SuccessionAsUsage::ownedRelationship () : Relationship [0..*]

```
let relationships : Set(KerML::Relationship) = Set{
if from.source.oclIsKindOf(UML::InitialNode) then
```

```
            ActivityEdgeInitialNodeFeatureMembership_Mapping.getMapped(from.source)
        else
            ActivityEdgeSourceEndFeatureMembership_Mapping.getMapped(from.source)
        endif,
        if from.oclIsKindOf(UML::ObjectFlow) then
            ObjectFlowGuardSuccessionTargetEndFeatureMembership_Mapping.getMapped(from)
        else if from.target.oclIsKindOf(UML::FinalNode) then
                ControlFlowFinalNodeFeatureMembership_Mapping.getMapped(from.target)
             else
                ControlFlowTargetFeatureMembership_Mapping.getMapped(from.target)
             endif
        endif} in
        let relationshipsWithGuard : Set(KerML::Relationship) =
        if from.guard.oclIsUndefined() then
            relationships
        else
            relationships
            ->including(ElementFeatureMembership_Mapping.getMapped(from.guard))
        endif in
        let relationshipsConsideringWeight : Set(KerML::Relationship) =
        if from.weight.oclIsUndefined() then
            relationshipsWithGuard
        else
            relationshipsWithGuard
            ->including(ActivityEdgeMetadataOwningMembership_Mapping.getMapped(from))
        endif in

        (if Helper.hasStereotypeApplied(from, 'SysML::Activities::Probability') then
            relationshipsConsideringWeight
            ->including(ProbabilityOwningMembership_Mapping.getMapped(from))
        else
            relationshipsConsideringWeight
        endif)->union(self.oclAsType(ElementMain_Mapping).ownedRelationship())
```

7.7.3.2.24 ControlFlowTargetFinalNode_Mapping

Description

The mapping class maps a UML4SysML::FinalNode to a Feature which will be subsetted by Actions::Action::done. The subsetting is created by the mapping class ControlFlowTargetFinalNodeSubsetting_Mapping.

General Mappings

ToFeature_Init
Mapping

Mapping Source

FinalNode

Mapping Target

Feature

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Feature::isEnd () : Boolean [1]

  ```
  true
  ```

- Feature::ownedRelationship () : Relationship [0..*]

  ```
  Set{ControlFlowTargetFinalNodeSubsetting_Mapping.getMapped(from)}
  ```

7.7.3.2.25 ControlFlowTargetEndFeature_Mapping

Description

The mapping class maps the UML4SysML::ActivityNode to a Feature which is subsetted by the mapping target of the UML4SysML::ActivityNode. The subsetting is created by the mapping class ControlFlowTargetEndSubsetting_Mapping.

General Mappings

ToFeature_Init
Mapping

Mapping Source

ActivityNode

Mapping Target

Feature

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Feature::ownedRelationship () : Relationship [0..*]

  ```
  Set{ControlFlowTargetEndSubsetting_Mapping.getMapped(from)}
  ```

- Feature::isEnd () : Boolean [1]

  ```
  true
  ```

7.7.3.2.26 ControlFlowTargetFeatureMembership_Mapping

Description

Creates a feature membership relationship for *ownedMemberFeature()*.

General Mappings

ToEndFeatureMembership_Init
Mapping

Mapping Source

ActivityNode

Mapping Target

EndFeatureMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- EndFeatureMembership::ownedMemberFeature () : Feature [1]

  ```
  ControlFlowTargetEndFeature_Mapping.getMapped(from)
  ```

7.7.3.2.27 ControlFlowTargetEndSubsetting_Mapping

Description

Creates a subsetting relationship.

General Mappings

ToReferenceSubsetting_Init
Mapping

Mapping Source

ActivityNode

Mapping Target

ReferenceSubsetting

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ReferenceSubsetting::referencedFeature () : Feature [1]

```
from
```

7.7.3.2.28 ControlFlowTransitionUsageFeatureMembership_Mapping

Description

Creates a feature membership relationship for *ownedMemberFeature()*.

General Mappings

ToFeatureMembership_Init
Mapping

Mapping Source

ControlFlow

Mapping Target

TransitionFeatureMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- TransitionFeatureMembership::ownedMemberFeature () : Feature [1]

```
if from.guard.oclIsKindOf(UML::OpaqueExpression) then
    OpaqueExpressionAsValue_Mapping.getMapped(from.guard)
else
    from.guard
endif
```

- TransitionFeatureMembership::kind () : TransitionFeatureKind [1]

```
KerML::TransitionFeatureKind::guard
```

7.7.3.2.29 ControlNodeObjectFlowFeatureMembership_Mapping

Description

Creates a feature membership relationship for *ownedMemberFeature()*.

General Mappings

ToFeatureMembership_Init
UniqueMapping

Mapping Source

ObjectFlow

Mapping Target

FeatureMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureMembership::ownedMemberFeature () : Feature [1]

  ```
  ControlNodeObjectFlowReferenceUsage_Mapping.getMapped(from)
  ```

7.7.3.2.30 ControlNodeObjectFlowFeatureValue_Mapping

Description

Creates a feature value relationship.

General Mappings

ToFeatureValue_Init
UniqueMapping

Mapping Source

ObjectFlow

Mapping Target

FeatureValue

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureValue::value () : Expression [1]

```
if from.source.oclIsTypeOf(UML::ForkNode) then
  ForkNodeObjectFlowFeatureReferenceExpression_Mapping.getMapped(from)
else if from.source.oclIsTypeOf(UML::JoinNode)
    or from.source.oclIsTypeOf(UML::MergeNode) then
  JoinMergeNodeObjectFlowOperatorExpression_Mapping.getMapped(from)
else
    OclUndefined
endif endif
```

7.7.3.2.31 ControlNodeObjectFlowReferenceUsage_Mapping

Description

Creates a reference usage.

General Mappings

ToReferenceUsage_Init
UniqueMapping

Mapping Source

ObjectFlow

Mapping Target

ReferenceUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ReferenceUsage::declaredName () : String [0..1]

```
if from.target.oclIsTypeOf(UML::ForkNode)
  or from.target.oclIsTypeOf(UML::JoinNode)
  or from.target.oclIsTypeOf(UML::MergeNode) then
  'inputObject' + from.target.incoming->indexOf(from).toString()
```

```
else if from.source.oclIsTypeOf(UML::ForkNode)
  or from.source.oclIsTypeOf(UML::JoinNode)
  or from.target.oclIsTypeOf(UML::MergeNode) then
  'outputObject' + from.source.outgoing->indexOf(from).toString()
else
  OclUndefined
endif endif
```

- ReferenceUsage::isUnique () : Boolean [1]

```
if from.source.oclIsTypeOf(UML::JoinNode) then
  if from.source.oclAsType(UML::JoinNode).isCombineDuplicate then
    true
  else
    false
  endif
else
  true
endif
```

- ReferenceUsage::ownedRelationship () : Relationship [0..*]

```
if from.source.oclIsTypeOf(UML::ForkNode)
  or from.source.oclIsTypeOf(UML::JoinNode)
  or from.source.oclIsTypeOf(UML::MergeNode) then
    Set{ControlNodeObjectFlowFeatureValue_Mapping.getMapped(from)}
else
  Set{}
endif
```

- ReferenceUsage::direction () : FeatureDirectionKind [0..1]

```
if from.target.oclIsTypeOf(UML::ForkNode)
  or from.target.oclIsTypeOf(UML::JoinNode)
  or from.target.oclIsTypeOf(UML::MergeNode) then
 KerML::FeatureDirectionKind::_'in'
else if from.source.oclIsTypeOf(UML::ForkNode)
  or from.target.oclIsTypeOf(UML::JoinNode)
  or from.target.oclIsTypeOf(UML::MergeNode) then
  KerML::FeatureDirectionKind::_'out'
else
  OclUndefined
endif endif
```

7.7.3.2.32 DataStoreNode_Mapping

Description

The mapping of the UML4SysML::DataStoreNode is not defined in detail yet. It will an action usage which contains the behavior of a data store node.

General Mappings

CentralBufferNode_Mapping

Mapping Source

DataStoreNode

Mapping Target

ActionUsage

Owned Mappings

(none)

Applicable filters

(none)

7.7.3.2.33 DecisionNode_Mapping

Description

The UML4SysML::DecisionNode is mapped to a SysMLv2 DecisionNode.

There is no suitable element in SysML v2 for the else condition of an outgoing UML4SysML::ActivityEdge. Therefore, it is mapped to a TextualRepresentation with language "SysML v1" and body "else" (see ExpressionElse_Mapping class).

The following shows an example of what the textual SysML v2 syntax of the result of the transformation may look like.

```
action def SysMLv1Activity {
        action sysMLv1Action1;
        succession sysMLv1ControlFlow1 first sysMLv1Action1 then sysMLv1DecisionNode;
        decide sysMLv1DecisionNode;
        succession sysMLv1ControlFlow2 first sysMLv1DecisionNode if {
                return : ScalarValues::Boolean;
                // guard expression, for example, opaque expression
        }.result then sysMLv1Action2;
        succession flow2 first sysMLv1DecisionNode if {
                return : ScalarValues::Boolean;
                language "SysMLv1"
                /*
                 * else
                  */
        }.result then sysMLv1Action2;
        action sysMLv1Action2;
}
```

General Mappings

ToUsage_Init
NamedElementMain_Mapping

Mapping Source

DecisionNode

Mapping Target

DecisionNode

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- DecisionNode::isComposite () : Boolean [1]

```
true
```

7.7.3.2.34 FlowFinalNodeMembership_Mapping

Description

The mapping class creates a membership relationship to the action usage library element Actions::Action::done.

General Mappings

ToMembership_Init
Mapping

Mapping Source

FlowFinalNode

Mapping Target

Membership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Membership::memberElement () : Element [1]

```
SysMLv2::ActionUsage.allInstances()
->any(e | e.qualifiedName =  'Actions::Action::done')
```

7.7.3.2.35 ForkNode_Mapping

Description

A UML4SysML::ForkNode is mapped to a SysMLv2 ForkNode. If object flows are connected with the UML4SYsML::ForkNode, corresponding input and output parameters are created to transfer the objects through the ForkNode.

The following shows an example of what the textual SysML v2 syntax of the result of the transformation may look like.

```
action def SysMLv1Activity {
  succession cf1 first sysMLv1Action1 then sysMLv1ForkNodeA;
  succession cf2 first sysMLv1Action2 then sysMLv1ForkNodeA;
  succession cf3 first sysMLv1ForkNodeA then sysMLv1Action4;

  succession flow of1 from sysMLv1Action1.result to sysMLv1ForkNodeB.inputObject1;
  succession flow of2 from sysMLv1ForkNodeB.outputObject1 to sysMLv1Action2.inputValue;
  succession flow of3 from sysMLv1ForkNodeB.outputObject2 to sysMLv1Action3.inputValue;

  fork sysMLv1ForkNodeA;
  fork sysMLv1ForkNodeB {
    in ref inputObject1;
    out ref outputObject1 = inputObject1;
    out ref outputObject2 = inputObject1;
  }
  action sysMLv1Action1 {
    out item result;
  }
  action sysMLv1Action2 {
    in item inputValue;
  }
  action sysMLv1Action3 {
    in item inputValue;
  }
  action sysMLv1Action4;
}
```

General Mappings

CommonAction_Mapping

Mapping Source

ForkNode

Mapping Target

ForkNode

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ForkNode::ownedRelationship () : Relationship [0..*]

```
if not (src.incoming->forAll(e | e.oclIsTypeOf(UML::ControlFlow))
```

```
    and src.outgoing->forAll(e | e.oclIsTypeOf(UML::ControlFlow))) then
      from.ownedElement->collect(e |
        ElementOwningMembership_Mapping.getMapped(e))->asSet()
      ->union(from.incoming
        ->collect(i |
          ControlNodeObjectFlowFeatureMembership_Mapping.getMapped(i))->asSet())
      ->union(from.outgoing
        ->collect(i |
          ControlNodeObjectFlowFeatureMembership_Mapping.getMapped(i))->asSet())
  else
    Set{}
  endif
```

7.7.3.2.36 ForkNodeObjectFlowFeatureReferenceExpression_Mapping

Description

Creates a feature reference expression.

General Mappings

UniqueMapping
ToFeatureReferenceExpression_Init

Mapping Source

ObjectFlow

Mapping Target

FeatureReferenceExpression

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureReferenceExpression::ownedRelationship () : Relationship [0..*]

```
Set{ForkNodeObjectFlowMembership_Mapping.getMapped(from)}
->including(ReturnParameterFeatureMembership_Factory.create())
```

7.7.3.2.37 ForkNodeObjectFlowMembership_Mapping

Description

Creates a membership relationship for *memberElement()*.

General Mappings

ToMembership_Init
UniqueMapping

Mapping Source

ObjectFlow

Mapping Target

Membership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Membership::memberElement () : Element [1]

```
ControlNodeObjectFlowReferenceUsage_Mapping.getMapped(
  from.source.oclAsType(UML::ForkNode).incoming
  ->asOrderedSet()->first())
```

7.7.3.2.38 JoinMergeNodeObjectFlowFeature_Mapping

Description

Creates a feature for the operator expression created by JoinMergeNodeObjectFlowOperatorExpression_Mapping.

General Mappings

ToFeature_Init
UniqueMapping

Mapping Source

ObjectFlow

Mapping Target

Feature

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Feature::ownedRelationship () : Relationship [0..*]

```
Set{JoinMergeNodeObjectFlowFeatureValue_Mapping.getMapped(from)}
```

7.7.3.2.39 JoinMergeNodeObjectFlowFeatureReferenceExpression_Mapping

Description

Creates a feature reference expression.

General Mappings

ToFeatureReferenceExpression_Init
UniqueMapping

Mapping Source

ObjectFlow

Mapping Target

FeatureReferenceExpression

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureReferenceExpression::ownedRelationship () : Relationship [0..*]

```
Set{JoinMergeNodeObjectFlowMembership_Mapping.getMapped(from)}
->including(ReturnParameterFeatureMembership_Factory.create())
```

7.7.3.2.40 JoinMergeNodeObjectFlowFeatureValue_Mapping

Description

Creates a feature value relationship.

General Mappings

UniqueMapping
ToFeatureValue_Init

Mapping Source

ObjectFlow

Mapping Target

FeatureValue

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureValue::value () : Expression [1]

  `JoinMergeNodeObjectFlowFeatureReferenceExpression_Mapping.getMapped(from)`

7.7.3.2.41 JoinMergeNodeObjectFlowMembership_Mapping

Description

Creates a membership relationship for *memberElement()*.

General Mappings

ToMembership_Init
UniqueMapping

Mapping Source

ObjectFlow

Mapping Target

Membership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Membership::memberElement () : Element [1]

```
ControlNodeObjectFlowReferenceUsage_Mapping.getMapped(from)
```

7.7.3.2.42 JoinMergeNodeObjectFlowOperatorExpression_Mapping

Description

Creates an operator expression to combine the input objects.

General Mappings

ToOperatorExpression_Init
UniqueMapping

Mapping Source

ObjectFlow

Mapping Target

OperatorExpression

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- OperatorExpression::operator () : String [1]

  ```
  ','
  ```

- OperatorExpression::ownedRelationship () : Relationship [0..*]

  ```
  if from.source.oclIsKindOf(UML::ControlNode) then
    from.source.oclAsType(UML::ControlNode).incoming
    ->collect(o | JoinMergeNodeObjectFlowParameterMembership_Mapping.getMapped(o))
    ->including(ReturnParameterFeatureMembership_Factory.create())
  else
    Set{}
  endif
  ```

7.7.3.2.43 JoinMergeNodeObjectFlowParameterMembership_Mapping

Description

Creates a membership relationship for *memberElement()*.

General Mappings

ToParameterMembership_Init
UniqueMapping

Mapping Source

ObjectFlow

Mapping Target

ParameterMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ParameterMembership::ownedMemberParameter () : Feature [1]

  `JoinMergeNodeObjectFlowFeature_Mapping.getMapped(from)`

7.7.3.2.44 InitialNodeMembership_Mapping

Description

The mapping class creates a membership relationship to the action usage library element Actions::Action::start.

General Mappings

ToMembership_Init
Mapping

Mapping Source

InitialNode

Mapping Target

Membership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Membership::memberElement () : Element [1]

```
SysMLv2::ActionUsage.allInstances()
->any(e | e.qualifiedName =  'Actions::Action::start')
```

- Membership::memberName () : String [0..1]

```
if from.name = '' then null else from.name endif
```

7.7.3.2.45 JoinNode_Mapping

Description

A UML4SysML::JoinNode is mapped to a SysMLv2 JoinNode. If object flows are connected with the UML4SYsML::JoinNode, corresponding input and output parameters are created to transfer the objects through the JoinNode.

The output object is specified as follows if UML4SysML::JoinNode::isCombineDuplicate is false:

```
out ref outputObject1 nonunique = (inputObject1, inputObject2)
```

The following shows an example of what the textual SysML v2 syntax of the result of the transformation may look like.

```
action def SysMLv1Activity {
  succession cf1 first sysMLv1Action1 then sysMLv1JoinNodeA;
  succession cf2 first sysMLv1Action2 then sysMLv1JoinNodeA;

  succession flow of1 from sysMLv1Action2.result to sysMLv1JoinNodeB.inputObject1;
  succession flow of2 from sysMLv1Action3.result to sysMLv1JoinNodeB.inputObject2;
  succession flow of3 from sysMLv1JoinNodeB.outputObject1 to sysMLv1Action4.inputValue;

  join sysMLv1JoinNodeA;
  join sysMLv1JoinNodeB {
    in ref inputObject1;
    in ref inputObject2;
    out ref outputObject1 = (inputObject1, inputObject2);
  }
  action sysMLv1Action1;
  action sysMLv1Action2 {
    out item result;
  }
  action sysMLv1Action3 {
    out item result;
  }
  action sysMLv1Action4 {
    in item inputValue;
  }
}
```

General Mappings

CommonAction_Mapping

Mapping Source

JoinNode

Mapping Target

JoinNode

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- JoinNode::ownedRelationship () : Relationship [0..*]

```
if not (src.incoming->forAll(e | e.oclIsTypeOf(UML::ControlFlow))
  and src.outgoing->forAll(e | e.oclIsTypeOf(UML::ControlFlow))) then
    from.ownedElement
      ->collect(e | ElementOwningMembership_Mapping.getMapped(e))->asSet()
      ->union(from.incoming
        ->collect(i |
          ControlNodeObjectFlowFeatureMembership_Mapping.getMapped(i))->asSet())
      ->union(from.outgoing
        ->collect(i |
          ControlNodeObjectFlowFeatureMembership_Mapping.getMapped(i))->asSet())
else
  Set{}
endif
```

7.7.3.2.46 MergeNode_Mapping

Description

A UML4SysML::MergeNode is mapped to a SysMLv2 MergeNode. If object flows are connected with the UML4SYsML::MergeNode, corresponding input and output parameters are created to transfer the objects through the MergeNode.

The following shows an example of what the textual SysML v2 syntax of the result of the transformation may look like.

```
action def SysMLv1Activity {
  succession cf1 first sysMLv1Action1 then sysMLv1MergeNodeA;
  succession cf2 first sysMLv1Action2 then sysMLv1MergeNodeA;

  succession flow of1 from sysMLv1Action2.result to sysMLv1MergeNodeB.inputObject1;
  succession flow of2 from sysMLv1Action3.result to sysMLv1MergeNodeB.inputObject2;
  succession flow of3 from sysMLv1MergeNodeB.outputObject1 to sysMLv1Action4.inputValue;

  merge sysMLv1MergeNodeA;
  merge sysMLv1MergeNodeB {
    in ref inputObject1;
    in ref inputObject2;
    out ref outputObject1 = (inputObject1, inputObject2);
```

```
  }
  action sysMLv1Action1;
  action sysMLv1Action2 {
    out item result;
  }
  action sysMLv1Action3 {
    out item result;
  }
  action sysMLv1Action4 {
    in item inputValue;
  }
}
```

General Mappings

CommonAction_Mapping

Mapping Source

MergeNode

Mapping Target

MergeNode

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- MergeNode::ownedRelationship () : Relationship [0..*]

```
if not (src.incoming->forAll(e | e.oclIsTypeOf(UML::ControlFlow))
  and src.outgoing->forAll(e | e.oclIsTypeOf(UML::ControlFlow))) then
    from.ownedElement
      ->collect(e | ElementOwningMembership_Mapping.getMapped(e))->asSet()
      ->union(from.incoming
        ->collect(i |
          ControlNodeObjectFlowFeatureMembership_Mapping.getMapped(i))->asSet())
      ->union(from.outgoing
        ->collect(i |
          ControlNodeObjectFlowFeatureMembership_Mapping.getMapped(i))->asSet())
else
  Set{}
endif
```

7.7.3.2.47 ObjectFlow_Mapping

Description

A UML4SysML::ObjectFlowFlow without a guard condition is mapped to a SysMLv2 SuccessionFlowUsage.

The following shows an example of what the textual SysML v2 syntax of the result of the transformation may look like.

```
action def SysMLv1Acticity {
        action sysMLv1Action1 {
                out outputValue;
        }
        succession flow sysMLv1ObjectFlow of ScalarValues::String
                from sysMLv1Action1.outputValue to sysMLv1Action1.inputValue;
        action sysMLv1Action2 {
                out inputValue;
        }
}
```

General Mappings

ToConnector_Init
NamedElementMain_Mapping
ToFlowUsage_Init

Mapping Source

ObjectFlow

Mapping Target

SuccessionFlowUsage

Owned Mappings

(none)

Applicable filters

This mapping applies only if the following (OCL) condition implemented by the operation *filter(src : Element) : Boolean* is verified:

```
src.guard.oclIsUndefined()
and (not src.target.oclIsTypeOf(UML::ActivityFinalNode))
```

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- SuccessionFlowUsage::ownedRelationship () : Relationship [0..*]

```
let relationships : Set(KerML::Relationship) =
  let sourceFeatureMembership : KerML::FeatureMembership
    = ObjectFlowEndFeatureMembership_Mapping.getMapped(from.source) in
  let targetFeatureMembership : KerML::FeatureMembership
    = ObjectFlowEndFeatureMembership_Mapping.getMapped(from.target) in
  if from.source.oclIsKindOf(UML::ObjectNode) then
    Set{ObjectFlowItemFeatureMembership_Mapping.getMapped(from),
    sourceFeatureMembership, targetFeatureMembership}
```

```
  else
    Set{sourceFeatureMembership, targetFeatureMembership}
  endif in

let relationshipsConsideringWeight : Set(KerML::Relationship) =
if from.weight.oclIsUndefined() then
  relationships
else
  relationships
    ->including(ActivityEdgeMetadataOwningMembership_Mapping.getMapped(from))
endif in

let relationshipsConsideringRate : Set(KerML::Relationship) =
if (Helper.hasStereotypeApplied(from, 'SysML::Activities::Rate') or
  Helper.hasStereotypeApplied(from, 'SysML::Activities::Discrete') or
  Helper.hasStereotypeApplied(from, 'SysML::Activities::Continuous')) then

  relationshipsConsideringWeight
    ->including(RateOwningMembership_Mapping.getMapped(from))
else
  relationshipsConsideringWeight
endif in

self.oclAsType(ElementMain_Mapping).ownedRelationship()->union(
  if Helper.hasStereotypeApplied(from, 'SysML::Activities::Probability') then
    relationshipsConsideringRate
      ->including(ProbabilityOwningMembership_Mapping.getMapped(from))
   else
    relationshipsConsideringRate
   endif
)
```

7.7.3.2.48 ObjectFlowFeatureMembership_Mapping

Description

Creates a feature membership relationship for *ownedMemberFeature()*.

General Mappings

ToFeatureMembership_Init
Mapping

Mapping Source

ObjectFlow

Mapping Target

FeatureMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureMembership::ownedMemberFeature () : Feature [1]

  ```
  ObjectFlow_Mapping.getMapped(from)
  ```

7.7.3.2.49 ObjectFlowGuardFeatureMembership_Mapping

Description

Creates a feature membership relationship for *ownedMemberFeature()*.

General Mappings

ToFeatureMembership_Init
Mapping

Mapping Source

ObjectFlow

Mapping Target

FeatureMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureMembership::ownedMemberFeature () : Feature [1]

  ```
  ObjectFlowGuard_Mapping.getMapped(from)
  ```

7.7.3.2.50 ObjectFlowGuard_Mapping

Description

A UML4SysML::ObjectFlowFlow with a guard condition is mapped to a combined SysMLv2 TransitionUsage and SysMLv2 SuccessionFlowUsage.

The following shows an example of what the textual SysML v2 syntax of the result of the transformation may look like.

```
action def SysMLv1Activity {
        action sysMLv1Action1 {
                out outputValue;
        }
```

```
        first sysMLv1Action1 if guardCondition.result then sysMLv1ObjectFlow {
          calc guardCondition {
            return : ScalarValues::Boolean;
            language "English"
            /*
             * guard says ok
             */
          }
        }
        succession flow sysMLv1ObjectFlow of SysMLv1Block from
                sysMLv1Action1.outputValue to sysMLv1Action2.inputValue;

        action sysMLv1Action2 {
                out inputValue;
        }
}
```

General Mappings

ToTransitionUsage_Init
NamedElementMain_Mapping

Mapping Source

ObjectFlow

Mapping Target

TransitionUsage

Owned Mappings

(none)

Applicable filters

This mapping applies only if the following (OCL) condition implemented by the operation *filter(src : Element) : Boolean* is verified:

```
(not src.guard.oclIsUndefined())
and (not src.target.oclIsTypeOf(UML::ActivityFinalNode))
```

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- TransitionUsage::ownedRelationship () : Relationship [0..*]

```
Set{
ActivityEdgeTransitionUsageSourceMembership_Mapping.getMapped(from.source),
CommonParameterReferenceUsageInMembership_Mapping.getMapped(from.source),
ObjectFlowTransitionUsageFeatureMembership_Mapping.getMapped(from),
ObjectFlowGuardSuccessionTargetEndFeatureMembership_Mapping.getMapped(from),
CommonActivityEdgeSuccessionAsUsage_Mapping.getMapped(from),
CommonReturnParameterReferenceUsageMembership_Mapping.getMapped(from)
}->union(self.oclAsType(ElementMain_Mapping).ownedRelationship())
```

7.7.3.2.51 ObjectFlowGuardSuccessionTargetEndFeature_Mapping

Description

Creates a feature element for the UML4SysML::ObjectFlow mapping.

General Mappings

ToFeature_Init
Mapping

Mapping Source

ObjectFlow

Mapping Target

Feature

Owned Mappings

- objectFlowGuardSuccessionTargetEndSubsetting : ObjectFlowGuardSuccessionTargetEndSubsetting_Mapping

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Feature::ownedRelationship () : Relationship [0..*]

  ```
  Set{objectFlowGuardSuccessionTargetEndSubsetting.to}
  ```

- Feature::isEnd () : Boolean [1]

  ```
  true
  ```

7.7.3.2.52 ObjectFlowGuardSuccessionTargetEndFeatureMembership_Mapping

Description

Creates a feature membership relationship for *ownedMemberFeature()*.

General Mappings

ToEndFeatureMembership_Init
Mapping

Mapping Source

ObjectFlow

Mapping Target

EndFeatureMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- EndFeatureMembership::ownedMemberFeature () : Feature [1]

  `ObjectFlowGuardSuccessionTargetEndFeature_Mapping.getMapped(from)`

7.7.3.2.53 ObjectFlowGuardSuccessionTargetEndSubsetting_Mapping

Description

Creates a subsetting relationship.

General Mappings

ToSubsetting_Init
Mapping

Mapping Source

ObjectFlow

Mapping Target

Subsetting

Owned Mappings

- objectFlowGuardSuccessionTargetEndFeature : ObjectFlowGuardSuccessionTargetEndFeature_Mapping

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Subsetting::subsettingFeature () : Feature [1]

  `objectFlowGuardSuccessionTargetEndFeature.to`

- Subsetting::subsettedFeature () : Feature [1]

  `ObjectFlow_Mapping.getMapped(from)`

7.7.3.2.54 ObjectFlowItemFeature_Mapping

Description

The mapping class maps the source UML4SysML::ObjectNode to a ItemFeature which is typed by the UML4SysML::ObjectNode type.

General Mappings

ObjectFlowItemFeatureUntyped_Mapping

Mapping Source

ObjectNode

Mapping Target

PayloadFeature

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- PayloadFeature::ownedRelationship () : Relationship [0..*]

  ```
  Set{ObjectFlowItemFeatureTyping_Mapping.getMapped(from)}
  ```

7.7.3.2.55 ObjectFlowItemFeatureMembership_Mapping

Description

Creates a feature membership relationship for *ownedMemberFeature()*.

General Mappings

ToFeatureMembership_Init
Mapping

Mapping Source

ObjectFlow

Mapping Target

FeatureMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureMembership::ownedMemberFeature () : Feature [1]

```
if from.source.type.oclIsUndefined() then
    ObjectFlowItemFeatureUntyped_Mapping.getMapped(from.source)
else
    ObjectFlowItemFeature_Mapping.getMapped(from.source)
endif
```

7.7.3.2.56 ObjectFlowItemFeatureTyping_Mapping

Description

Creates a feature typing relationship owned by the element *typedFeature()*.

General Mappings

TypedElementFeatureTyping_Mapping

Mapping Source

ObjectNode

Mapping Target

FeatureTyping

Owned Mappings

(none)

Applicable filters

(none)

7.7.3.2.57 ObjectFlowItemFeatureUntyped_Mapping

Description

The mapping class maps the source UML4SysML::ObjectNode to a ItemFeature without a type.

General Mappings

ToFeature_Init

Mapping Source

ObjectNode

Mapping Target

PayloadFeature

Owned Mappings

(none)

Applicable filters

(none)

7.7.3.2.58 ObjectFlowEndFeatureMembership_Mapping

Description

Creates a feature membership relationship for *ownedMemberFeature()*.

General Mappings

ToEndFeatureMembership_Init
Mapping

Mapping Source

ActivityNode

Mapping Target

EndFeatureMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- EndFeatureMembership::ownedMemberFeature () : Feature [1]

  ```
  ObjectFlowItemFlowEnd_Mapping.getMapped(from)
  ```

7.7.3.2.59 ObjectFlowItemFlowEnd_Mapping

Description

The mapping class maps a UML4SysML::ActivityNode to a ItemFlowEnd which is subsetted by the transformation target of the UML4SysML::ActivityNode.

General Mappings

ToFeature_Init
Mapping

Mapping Source

ActivityNode

Mapping Target

FlowEnd

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FlowEnd::isEnd () : Boolean [1]

  ```
  true
  ```

- FlowEnd::ownedRelationship () : Relationship [0..*]

  ```
  Set{ObjectFlowItemFlowEndSubsetting_Mapping.getMapped(from),
  ObjectFlowItemFlowEndFeatureMembership_Mapping.getMapped(from)}
  ```

7.7.3.2.60 ObjectFlowItemFlowEndReferenceUsage_Mapping

Description

Creates a feature element for the UML4SysML::ObjectFlow mapping.

General Mappings

ToReferenceUsage_Init
Mapping

Mapping Source

ActivityNode

Mapping Target

ReferenceUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ReferenceUsage::ownedRelationship () : Relationship [0..*]

```
let redefinition : KerML::Redefinition =
  if from.owner.oclIsTypeOf(UML::AddVariableValueAction) or
    from.owner.oclIsTypeOf(UML::AddStructuralFeatureValueAction) then
      if from.name = 'value' then
        ObjectFlowItemFlowEndRedefinition_Factory.create(
          SYSML2::ReferenceUsage.allInstances()
          ->any(m | m.qualifiedName = 'SysMLv1Library::AddValueAction::value'))
      else if from.name = 'insertAt' then
        ObjectFlowItemFlowEndRedefinition_Factory.create(
          SYSML2::ReferenceUsage.allInstances()
          ->any(m | m.qualifiedName = 'SysMLv1Library::AddValueAction::insertAt'))
      else if from.owner.oclIsTypeOf(UML::AddStructuralFeatureValueAction)
        and (from.name = 'object') then
         ObjectFlowItemFlowEndRedefinition_Factory.create(
           SYSML2::ReferenceUsage.allInstances()
           ->any(m | m.qualifiedName =
            'SysMLv1Library::AddStructuralFeatureValueAction::object'))
      else
        ObjectFlowItemFlowEndRedefinition_Factory.create(
          ElementMain_Mapping.getMapped(from))
      endif endif endif
else
  if from.oclIsTypeOf(UML::ActivityParameterNode) then
    ObjectFlowItemFlowEndRedefinition_Factory.create(
      ElementMain_Mapping.getMapped(from.oclAsType(
        UML::ActivityParameterNode).parameter))
    else if from.oclIsTypeOf(UML::FlowFinalNode) then
      ObjectFlowItemFlowEndRedefinition_Factory.create(
        ElementMain_Mapping.getMapped(
        SysMLv2::ActionUsage.allInstances()
        ->any(e | e.qualifiedName =  'Actions::Action::done')))
    else
        ObjectFlowItemFlowEndRedefinition_Factory.create(
          ElementMain_Mapping.getMapped(from))
    endif endif
endif in
Set{redefinition}
```

7.7.3.2.61 ObjectFlowItemFlowEndFeatureMembership_Mapping

Description

Creates a feature membership relationship for *ownedMemberFeature()*.

General Mappings

ToFeatureMembership_Init
Mapping

Mapping Source

ActivityNode

Mapping Target

FeatureMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureMembership::ownedMemberFeature () : Feature [1]

  `ObjectFlowItemFlowEndReferenceUsage_Mapping.getMapped(from)`

7.7.3.2.62 ObjectFlowItemFlowEndRedefinition_Mapping

Description

Creates a redefinition relationship for the *redefiningFeature()* and the *redefinedFeature()*.

General Mappings

ToRedefinition_Init
Mapping

Mapping Source

ActivityNode

Mapping Target

Redefinition

Owned Mappings

(none)

Applicable filters

(none)

7.7.3.2.63 ObjectFlowItemFlowEndSubsetting_Mapping

Description

Creates a subsetting relationship.

General Mappings

ToReferenceSubsetting_Init
Mapping

Mapping Source

ActivityNode

Mapping Target

ReferenceSubsetting

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ReferenceSubsetting::referencedFeature () : Feature [1]

```
if from.oclIsKindOf(UML::ActivityParameterNode) then
    Parameter_Mapping.getMapped(from.parameter)
else if from.oclIsKindOf(UML::Pin) then
        CommonAction_Mapping.getMapped(from.owner)
    else if from.oclIsKindOf(UML::InitialNode) then
            SysMLv2::ActionUsage.allInstances()
            ->any(e | e.qualifiedName =  'Actions::Action::start')
        else if from.oclIsKindOf(UML::FinalNode) then
                SysMLv2::ActionUsage.allInstances()
                ->any(e | e.qualifiedName =  'Actions::Action::done')
            else
                from
            endif
        endif
    endif
endif
```

7.7.3.2.64 ObjectFlowTransitionUsageFeatureMembership_Mapping

Description

Creates a feature membership relationship for *ownedMemberFeature()*.

General Mappings

ToFeatureMembership_Init
Mapping

Mapping Source

ObjectFlow

Mapping Target

TransitionFeatureMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- TransitionFeatureMembership::ownedMemberFeature () : Feature [1]

```
if from.guard.oclIsKindOf(UML::OpaqueExpression) then
    OpaqueExpressionAsValue_Mapping.getMapped(from.guard)
else
    from.guard
endif
```

- TransitionFeatureMembership::kind () : TransitionFeatureKind [1]

```
KerML::TransitionFeatureKind::guard
```

7.7.3.2.65 VariableAttribute_Mapping

Description

A UML4SysML::Variable is mapped to a SysML v2 AttributeUsage if the type of the variable is of kind UML4SysML::DataType.

The following shows an example of what the textual SysML v2 syntax of the result of the transformation may look like.

```
action def SysMLv1Activity {
  private attribute sysmlv1Variable : ScalarValues::Integer;
}
```

General Mappings

NamedElementMain_Mapping
CommonVariable_Mapping

Mapping Source

Variable

Mapping Target

AttributeUsage

Owned Mappings

(none)

Applicable filters

This mapping applies only if the following (OCL) condition implemented by the operation *filter(src : Element) : Boolean* is verified:

```
src.type.oclIsKindOf(UML::DataType)
```

Mapping rules

The mapping class only has inherited rules. See the mapping classes in the general mapping section for details.

7.7.3.2.66 VariableFeatureTyping_Mapping

Description

Creates a feature typing relationship owned by the element *typedFeature()*.

General Mappings

TypedElementFeatureTyping_Mapping

Mapping Source

Variable

Mapping Target

FeatureTyping

Owned Mappings

(none)

Applicable filters

(none)

7.7.3.2.67 VariableItem_Mapping

Description

A UML4SysML::Variable is mapped to a SysML v2 ItemUsage if the type of the variable is not of kind UML4SysML::DataType.

The following shows an example of what the textual SysML v2 syntax of the result of the transformation may look like.

```
action def SysMLv1Activity {
  private item sysmlv1Variable : SysMLv1Block;
}
part def SysMLv1Block;
```

General Mappings

NamedElementMain_Mapping
CommonVariable_Mapping

Mapping Source

Variable

Mapping Target

ItemUsage

Owned Mappings

(none)

Applicable filters

This mapping applies only if the following (OCL) condition implemented by the operation *filter(src : Element) : Boolean* is verified:

```
not src.type.oclIsKindOf(UML::DataType)
```

Mapping rules

The mapping class only has inherited rules. See the mapping classes in the general mapping section for details.

7.7.3.2.68 VariableMembership_Mapping

Description

Creates a membership relationship for *memberElement()*.

General Mappings

ElementFeatureMembership_Mapping

Mapping Source

Variable

Mapping Target

FeatureMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureMembership::visibility () : VisibilityKind [1]

  ```
  KerML::VisibilityKind::private
  ```

7.7.4 Classification

7.7.4.1 Overview

SYSML21-542: Transformation: overview tables shall be revised

Table 3. List of High-Level Classification Mappings

SysML v1 Abstract Syntax/Stereotype	SysML v2 Abstract Syntax
Generalization	Subclassification
GeneralizationSet	No specific mapping specified yet.
InstanceSpecification	ConnectionUsage, PartUsage
InstanceValue	FeatureReferenceExpression
Operation	PerformActionUsage
Parameter	ReferenceUsage
ParameterSet	No specific mapping specified yet.
Property	AttributeUsage, OccurrenceUsage, ReferenceUsage, Feature
Slot	Feature
Substitution	Dependency

7.7.4.2 Mapping Specifications

7.7.4.2.1 BehavioralFeature_Mapping

Description

The mapping class is the abstract base class for UML4SysML::BehavioralFeature mappings.

General Mappings

ToUsage_Init
Namespace_Mapping

Mapping Source

BehavioralFeature

Mapping Target

Usage

Owned Mappings

(none)

Applicable filters

(none)

7.7.4.2.2 Classifier_Mapping

Description

The mapping class is the abstract base class for all mapping classes that map specializations of UML4SysML::Classifier elements.

General Mappings

ToClassifier_Init
Namespace_Mapping

Mapping Source

Classifier

Mapping Target

Classifier

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Classifier::isAbstract () : Boolean [1]

```
from.isAbstract
```

- Classifier::ownedRelationship () : Relationship [0..*]

```
let generalizations : Set(UML::Generalization) =
    from.ownedElement->select(e | e.oclIsKindOf(UML::Generalization))->asSet() in
let toElementFMS: Set(UML::Element) =
    from.ownedElement->select(e | e.oclIsKindOf(UML::Feature))->asSet() in
let toElementOMS: Set(UML::Element) =
    ((from.ownedElement - toElementFMS) - generalizations) - from.ownedComment  in
toElementOMS->collect(e | ElementOwningMembership_Mapping.getMapped(e))
  ->asSet()
->union(toElementFMS->collect(e | ElementFeatureMembership_Mapping.getMapped(e))
  ->asSet())
->union(generalizations->collect(e | Generalization_Mapping.getMapped(e))
  ->asSet())
->union(self.oclAsType(ElementMain_Mapping).ownedRelationship())
```

7.7.4.2.3 DefaultLowerBound_Mapping

Description

The mapping class creates the default lower bound of a multiplicity element.

General Mappings

ToExpression_Init
Mapping

Mapping Source

Element

Mapping Target

LiteralInteger

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- LiteralInteger::ownedRelationship () : Relationship [0..*]

  ```
  Set{CommonReturnParameterFeatureMembership_Mapping.getMapped(from)}
  ```

- LiteralInteger::value () : Integer [1]

  ```
  1
  ```

7.7.4.2.4 DefaultMultiplicityBoundFeatureMembership_Mapping

Description

Creates a feature membership relationship for *ownedMemberFeature()*.

General Mappings

ToFeatureMembership_Init
Mapping

Mapping Source

Element

Mapping Target

FeatureMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureMembership::isComposite () : Boolean [1]

  `true`

7.7.4.2.5 DefaultMultiplicityElement_Mapping

Description

The mapping class creates a feature element representing the default multiplicity.

General Mappings

ToFeature_Init
Mapping

Mapping Source

Element

Mapping Target

MultiplicityRange

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- MultiplicityRange::isUnique () : Boolean [1]

  `true`

- MultiplicityRange::ownedRelationship () : Relationship [0..*]

  ```
  OrderedSet{DefaultMultiplicityLowerBoundFeatureMembership_Mapping.getMapped(from),
  DefaultMultiplicityUpperBoundFeatureMembership_Mapping.getMapped(from)}
  ```

- MultiplicityRange::declaredName () : String [0..1]

  `'defaultMultiplicity'`

7.7.4.2.6 DefaultMultiplicityLowerBoundFeatureMembership_Mapping

Description

Creates a feature membership relationship for *ownedMemberFeature()*.

General Mappings

DefaultMultiplicityBoundFeatureMembership_Mapping

Mapping Source

Element

Mapping Target

FeatureMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureMembership::ownedMemberFeature () : MultiplicityRange [1]

  ```
  DefaultLowerBound_Mapping.getMapped(from)
  ```

7.7.4.2.7 DefaultMultiplicityMembership_Mapping

Description

Creates a membership relationship for *memberElement()*.

General Mappings

Mapping
ToOwningMembership_Init

Mapping Source

Element

Mapping Target

OwningMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- OwningMembership::ownedMemberElement () : Element [1]

  `DefaultMultiplicityElement_Mapping.getMapped(from)`

7.7.4.2.8 DefaultMultiplicityUpperBoundFeatureMembership_Mapping

Description

Creates a feature membership relationship for *ownedMemberFeature()*.

General Mappings

DefaultMultiplicityBoundFeatureMembership_Mapping

Mapping Source

Element

Mapping Target

FeatureMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureMembership::ownedMemberFeature () : MultiplicityRange [1]

  `DefaultUpperBound_Mapping.getMapped(from)`

7.7.4.2.9 DefaultUpperBound_Mapping

Description

The mapping class creates the default upper bound of a multiplicity element.

General Mappings

ToExpression_Init
Mapping

Mapping Source

Element

Mapping Target

LiteralInteger

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- LiteralInteger::ownedRelationship () : Relationship [0..*]

  ```
  Set{CommonReturnParameterFeatureMembership_Mapping.getMapped(from)}
  ```

- LiteralInteger::value () : Integer [1]

  ```
  1
  ```

7.7.4.2.10 DefaultValue_Mapping

Description

The expected SysML v2 textual syntax of a mapped SysML v2 default value is as follows:

```
attribute sysMLv1Property : ScalarValues::String default := "default value";
```

General Mappings

ToFeatureValue_Init
Mapping

Mapping Source

Property

Mapping Target

FeatureValue

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureValue::value () : Expression [1]

```
from.defaultValue
```

- FeatureValue::isDefault () : Boolean [1]

```
true
```

7.7.4.2.11 ElementFeatureMembership_Mapping

Description

Creates a feature membership relationship for *ownedMemberFeature()*.

General Mappings

ToFeatureMembership_Init
Mapping

Mapping Source

Element

Mapping Target

FeatureMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureMembership::visibility () : VisibilityKind [1]

```
if from.oclIsKindOf(UML::NamedElement) then
Helper.getKerMLVisibilityKind(from.oclAsType(UML::NamedElement).visibility)
else KerML::VisibilityKind::public endif
```

- FeatureMembership::ownedMemberFeature () : Feature [1]

```
NamedElementMain_Mapping.getMapped(from)
```

7.7.4.2.12 Generalization_Mapping

Description

A UML4SysML::Generalization relationship is mapped to a SysML v2 Subclassification.

The following shows an example of what the textual SysML v2 syntax of the result of the transformation may look like.

```
part def SysMLv1BlockGeneral;
part def SysMLv1BlockSpecial :> SysMLv1BlockGeneral;
```

General Mappings

ToSpecialization_Init
ElementMain_Mapping

Mapping Source

Generalization

Mapping Target

Subclassification

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Subclassification::superclassifier () : Classifier [1]

```
if from.general.oclIsTypeOf(UML::PrimitiveType)
            and not (Helper.getScalarValueType(from.general)
                = invalid) then
    Helper.getScalarValueType(from.general)
else
    Classifier_Mapping.getMapped(from.general)
endif
```

- Subclassification::subclassifier () : Classifier [1]

```
Classifier_Mapping.getMapped(from.specific)
```

7.7.4.2.13 InstanceSpecificationLink_Mapping

Description

The UML4SysML::InstanceSpecification that is a link is mapped to a SysMLv2 ConnectionUsage.

The following shows an example of what the textual SysML v2 syntax of the result of the transformation may look like.

```
part def SysMLv1Block1;
part def SysMLv1Block2;
connection def SysMLv1Association {
```

```
        end : SysMLv1Block1[1];
        end : SysMLv1Block2[1];
}
part sysMLv1InstanceSpecification1 : SysMLv1Block1;
part sysMLv1InstanceSpecification2 : SysMLv1Block2;
connection sysMLv1Link : SysMLv1Association
        connect sysMLv1InstanceSpecification1 to sysMLv1InstanceSpecification2;
```

General Mappings

NamedElementMain_Mapping
ToConnectionUsage_Init

Mapping Source

InstanceSpecification

Mapping Target

ConnectionUsage

Owned Mappings

(none)

Applicable filters

This mapping applies only if the following (OCL) condition implemented by the operation *filter(src : Element) : Boolean* is verified:

```
src.classifier->select( c | c.oclIsTypeOf(UML::Association))->size() > 0
```

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ConnectionUsage::ownedRelationship () : Relationship [0..*]

```
self.oclAsType(ElementMain_Mapping).ownedRelationship()
->union(SlotMembership_Mapping.getMappedColl(from.slot)->asSet())
->union(from.classifier
  ->collect(g |
    InstanceSpecificationFeatureTyping_Mapping.getMapped(from, g))
    ->asSet())
  ->asSet()
```

7.7.4.2.14 InstanceSpecification_Mapping

Description

The UML4SysML::InstanceSpecification that is not a link is mapped to a SysMLv2 PartDefinition.

The following shows an example of what the textual SysML v2 syntax of the result of the transformation may look like.

```
part def SysMLv1Block {
        attribute sysMLv1ValueProperty : ScalarValues::String;
```

```
}

part sysMLv1InstanceSpecification : SysMLv1Block {
        redefines sysMLv1ValueProperty = "Hello InstanceSpecification";
}
```

General Mappings

NamedElementMain_Mapping
ToPartUsage_Init

Mapping Source

InstanceSpecification

Mapping Target

PartUsage

Owned Mappings

(none)

Applicable filters

This mapping applies only if the following (OCL) condition implemented by the operation *filter(src : Element) : Boolean* is verified:

```
src.classifier->select( c | c.oclIsTypeOf(UML::Association))->size() = 0
```

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- PartUsage::ownedFeatureMembership () : FeatureMembership [0..*]

  ```
  from.classifier
  ->collect(c | InstanceSpecificationToGeneralization_Mapping.getMapped(from, c))
  ```

- PartUsage::ownedRelationship () : Relationship [0..*]

  ```
  SlotMembership_Mapping.getMappedColl(from.slot)->asSet()
  ->union(from.classifier
    ->collect(g |
      InstanceSpecificationFeatureTyping_Mapping.getMapped(from, g))
      ->asSet())
  ->union(self.oclAsType(ElementMain_Mapping).ownedRelationship())
  ->asSet()
  ```

7.7.4.2.15 InstanceSpecificationFeatureTyping_Mapping

Description

Creates a feature typing relationship owned by the element *typedFeature()*.

General Mappings

ToFeatureTyping_Init
Mapping

Mapping Source

InstanceSpecification

Mapping Target

FeatureTyping with qualifier: classifier:Classifier

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureTyping::type (in classifier : Classifier) : Type [1]

  ```
  Classifier_Mapping.getMapped(classifier)
  ```

7.7.4.2.16 InstanceValue_Mapping

Description

The UML4SysML::InstanceValue is mapped to a SysMLv2 FeatureReferenceExpression.

The following shows an example of what the textual SysML v2 syntax of the result of the transformation may look like.

```
part def SysMLv1Block1;
part sysMLv1InstanceSpecification : SysMLv1Block1;
part def SysMLv1Block2 {
        part sysMLv1PartProperty : SysMLv1Block1
                = sysMLv1InstanceSpecification;
}
```

General Mappings

ValueSpecification_Mapping

Mapping Source

InstanceValue

Mapping Target

FeatureReferenceExpression

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureReferenceExpression::ownedRelationship () : Relationship [0..*]

```
self.oclAsType(ElementMain_Mapping).ownedRelationship()
->including(InstanceValueMembership_Mapping.getMapped(from.instance))
->including(ReturnParameterFeatureMembership_Factory.create())
```

7.7.4.2.17 InstanceValueMembership_Mapping

Description

Creates a membership relationship for *memberElement()*.

General Mappings

ToMembership_Init
Mapping

Mapping Source

InstanceSpecification

Mapping Target

Membership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Membership::memberElement () : Element [1]

```
from
```

7.7.4.2.18 LowerBoundValueFeatureMembership_Mapping

Description

Creates a feature membership relationship for *ownedMemberFeature()*.

General Mappings

ToFeatureMembership_Init

Mapping Source

MultiplicityElement

Mapping Target

FeatureMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureMembership::ownedMemberFeature () : Feature [0..1]

  ```
  LiteralInteger_Mapping.getMapped(from.lowerValue)
  ```

7.7.4.2.19 MultiplicityElement_Mapping

Description

A UML4SysML::MultiplicityElement is mapped to a SysML v2 MultiplicityRange.

General Mappings

ToFeature_Init
Mapping

Mapping Source

MultiplicityElement

Mapping Target

MultiplicityRange

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- MultiplicityRange::isUnique () : Boolean [1]

```
from.isUnique
```

- MultiplicityRange::declaredName () : String [0..1]

```
'multiplicity'
```

- MultiplicityRange::ownedRelationship () : Relationship [0..*]

```
OrderedSet{MultiplicityLowerBoundOwningMembership_Mapping.getMapped(from),
MultiplicityUpperBoundOwningMembership_Mapping.getMapped(from)}
```

7.7.4.2.20 MultiplicityLowerBoundOwningMembership_Mapping

Description

Creates a owning membership relationship for *ownedMemberElement()*.

General Mappings

ToOwningMembership_Init
Mapping

Mapping Source

MultiplicityElement

Mapping Target

OwningMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- OwningMembership::ownedMemberElement () : Element [1]

```
if from.lowerValue.oclIsUndefined() then
    DefaultLowerBound_Mapping.getMapped(from)
else
    from.lowerValue
endif
```

- OwningMembership::memberName () : String [0..1]

```
'lowerBound'
```

7.7.4.2.21 MultiplicityMembership_Mapping

Description

Creates a membership relationship for *memberElement()*.

General Mappings

ToOwningMembership_Init
Mapping

Mapping Source

MultiplicityElement

Mapping Target

OwningMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- OwningMembership::ownedMemberElement () : Element [1]

  ```
  MultiplicityElement_Mapping.getMapped(from)
  ```

7.7.4.2.22 MultiplicityUpperBoundOwningMembership_Mapping

Description

Creates a owning membership relationship for *ownedMemberElement()*.

General Mappings

ToOwningMembership_Init
Mapping

Mapping Source

MultiplicityElement

Mapping Target

OwningMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- OwningMembership::ownedMemberElement () : Element [1]

  ```
  if from.upperValue.oclIsUndefined() then
      DefaultUpperBound_Mapping.getMapped(from)
  else
      from.upperValue
  endif
  ```

- OwningMembership::memberName () : String [0..1]

  ```
  'upperBound'
  ```

7.7.4.2.23 Operation_Mapping

Description

A UML4SysML::Operation is mapped to a SysML v2 PerformActionUsage.

The following shows an example of what the textual SysML v2 syntax of the result of the transformation may look like.

```
part def SysMLv1Block {
          perform action sysMLv1Operation {
                  in parIn : ScalarValues::Boolean;
                 out result : ScalarValues::Integer;
          }
}
```

General Mappings

BehavioralFeature_Mapping
ToPerformActionUsage_Init

Mapping Source

Operation

Mapping Target

PerformActionUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- PerformActionUsage::ownedRelationship () : Relationship [0..*]

```
let parameters: Set(UML::Element) =
  from.ownedElement->select(e | e.oclIsKindOf(UML::Parameter)) in
let parameterSets: Set(UML::Element) =
  from.ownedElement->select(e | e.oclIsKindOf(UML::ParameterSet)) in
self.oclAsType(ElementMain_Mapping).ownedRelationship()
->union(parameters->collect(e |
  ParameterMembership_Mapping.getMapped(e))->asSet())
->union(parameterSets->collect(e |
  ParameterSetMembership_Mapping.getMapped(e))->asSet())
```

7.7.4.2.24 Parameter_Mapping

Description

A UML4SysML::Parameter is mapped to a SysML v2 ReferenceUsage.

The following shows an example of what the textual SysML v2 syntax of the result of the transformation may look like.

```
action def SysMLv1Activity {
        in parIn : ScalarValues::Boolean;
}
```

General Mappings

ToReferenceUsage_Init
NamedElementMain_Mapping

Mapping Source

Parameter

Mapping Target

ReferenceUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ReferenceUsage::declaredName () : String [0..1]

```
if from.direction = UML::ParameterDirectionKind::return then

  'result'

else

  from.name

endif
```

- ReferenceUsage::direction () : FeatureDirectionKind [0..1]

```
Helper.getKerMLParameterDirectionKind(from.direction)
```

- ReferenceUsage::ownedRelationship () : Relationship [0..*]

```
let typings: Set(KerML::FeatureTyping) =
    if from.type.oclIsUndefined() then
        Set{}
    else
        Set{ParameterToFeatureTyping_Mapping.getMapped(from)}
    endif in
let multiplicities: Set(KerML::Relationship) =
    Set{MultiplicityMembership_Mapping.getMapped(from)} in
let defaultValues: Set(KerML::Relationship) =
    if from.defaultValue.oclIsUndefined() then
        Set{}
    else
        Set{ParameterDefaultValue_Mapping.getMapped(from)}
    endif in
self.oclAsType(ElementMain_Mapping).ownedRelationship()
->union(typings)
->union(multiplicities)
->union(defaultValues)
```

7.7.4.2.25 ParameterDefaultValue_Mapping

Description

The following shows an example of what the textual SysML v2 syntax of the result of the transformation may look like.

```
attribute value : ScalarValues::String default := "default value";
```

General Mappings

ToFeatureValue_Init
Mapping

Mapping Source

Parameter

Mapping Target

FeatureValue

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureValue::value () : Expression [1]

  ```
  from.defaultValue
  ```

- FeatureValue::isDefault () : Boolean [1]

  ```
  true
  ```

7.7.4.2.26 ParameterMembership_Mapping

Description

Creates a membership relationship for *memberElement()*.

General Mappings

ToParameterMembership_Init
Mapping

Mapping Source

Parameter

Mapping Target

ParameterMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ParameterMembership::ownedMemberParameter () : Feature [1]

  ```
  Parameter_Mapping.getMapped(from)
  ```

7.7.4.2.27 ParameterSet_Mapping

Description

A UML4SysML::ParameterSet is mapped to a SysML v2 ReferenceUsage.

The following shows an example of what the textual SysML v2 syntax of the result of the transformation may look like.

```
action def SysMLv1Activity {
        in parIn [0..1];
        inout parInOut [0..1];
        out parOut [0..1];
        out parReturn [0..1];

        sysMLv1ParameterSet1 [1] {
                ref parIn = SysMLv1Activity::parIn;
                assert constraint sysMLv1ParameterSet1Condition {
                        language "English"
                        /*
                         * opaque expression parameter set 1
                        */
                }
        }
        sysMLv1ParameterSet2  [1]  {
                ref parInOut = SysMLv1Activity::parInOut;
                ref parOut = SysMLv1Activity::parOut;
                ref parReturn = SysMLv1Activity::parReturn;
        }
}
```

General Mappings

ToReferenceUsage_Init
Mapping

Mapping Source

ParameterSet

Mapping Target

ReferenceUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ReferenceUsage::ownedRelationship () : Relationship [0..*]

```
from.parameter
->collect(p | ParameterSetParameterFeatureMembership_Mapping.getMapped(from, p))
->asSet()
```

- ReferenceUsage::declaredName () : String [0..1]

```
from.name
```

7.7.4.2.28 ParameterSetMembership_Mapping

Description

Creates a membership relationship for *memberElement()*.

General Mappings

ToFeatureMembership_Init
Mapping

Mapping Source

ParameterSet

Mapping Target

FeatureMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureMembership::ownedMemberFeature () : Feature [1]

```
ParameterSet_Mapping.getMapped(from)
```

7.7.4.2.29 ParameterSetParameterFeatureMembership_Mapping

Description

Creates a feature membership relationship for *ownedMemberFeature()*.

General Mappings

ToFeatureMembership_Init
Mapping

Mapping Source

ParameterSet

Mapping Target

FeatureMembership with qualifier: parameter:Parameter

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureMembership::ownedMemberFeature (in parameter : Parameter) : Feature [1]

  ```
  ParameterSetParameterReferenceUsage_Mapping.getMapped(parameter)
  ```

7.7.4.2.30 ParameterSetParameterReferenceUsage_Mapping

Description

The mapping class creates the reference usage element for the UML4SysML::ParameterSet mapping.

General Mappings

ToReferenceUsage_Init
Mapping

Mapping Source

Parameter

Mapping Target

ReferenceUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ReferenceUsage::ownedRelationship () : Relationship [0..*]

```
Set{ParameterSetParameterReferenceUsageFeatureValue_Mapping.getMapped(from),
MultiplicityMembership_Mapping.getMapped(from)}
```

7.7.4.2.31 ParameterSetParameterReferenceUsageFeatureValue_Mapping

Description

The mapping class creates the feature reference expression for the reference usage element of the UML4SysML::ParameterSet mapping.

General Mappings

ToFeatureValue_Init
Mapping

Mapping Source

Parameter

Mapping Target

FeatureValue

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureValue::value () : Expression [1]

```
ParameterSetParameterReferenceUsageFeatureValueExpression_Mapping.getMapped(from)
```

7.7.4.2.32 ParameterSetParameterReferenceUsageFeatureValueExpression_Mapping

Description

The mapping class creates the feature reference expression for the UML4SysML::ParameterSet mapping.

General Mappings

ToFeatureReferenceExpression_Init
Mapping

Mapping Source

Parameter

Mapping Target

FeatureReferenceExpression

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureReferenceExpression::ownedRelationship () : Relationship [0..*]

```
Set{ParameterSetParameterReferenceUsageMembership_Mapping.getMapped(from),
CommonReturnParameterFeatureMembership_Mapping.getMapped(from)}
```

7.7.4.2.33 ParameterSetParameterReferenceUsageMembership_Mapping

Description

Creates a membership relationship for *memberElement()*.

General Mappings

ToMembership_Init
Mapping

Mapping Source

Parameter

Mapping Target

Membership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Membership::memberElement () : Element [1]

```
from
```

7.7.4.2.34 ParameterToFeatureTyping_Mapping

Description

Creates a feature typing relationship owned by the element *typedFeature()*.

General Mappings

TypedElementFeatureTyping_Mapping
Mapping

Mapping Source

Parameter

Mapping Target

FeatureTyping

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureTyping::typedFeature () : Feature [1]

```
parameter.to
```

7.7.4.2.35 PropertyCommon_Mapping

Description

The mapping class is the abstract base class for UML4SysML::Property mappings.

General Mappings

StructuralFeature_Mapping
Mapping

Mapping Source

Property

Mapping Target

Feature

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Feature::isEnd () : Boolean [1]

```
if from.association.oclIsUndefined() then
    false
else
    from.association.ownedEnd->includes(from)
endif
```

- Feature::ownedRelationship () : Relationship [0..*]

```
let typings: Set(KerML::FeatureTyping) = if from.type.oclIsUndefined() then
    Set{}
else
    Set{StructuralFeatureToFeatureTyping_Mapping.getMapped(from)}
endif in
let subsettings: Set(KerML::Subsetting) = from.subsettedProperty
    ->collect(p | PropertySubsetting_Mapping.getMapped(from, p))->asSet() in
let defaultValue: Set(KerML::OwningMembership) =
    if from.defaultValue.oclIsUndefined() then
        Set{}
    else
        Set{DefaultValue_Mapping.getMapped(from)}
    endif in
typings->union(subsettings)->union(defaultValue)
->including(MultiplicityMembership_Mapping.getMapped(from))->asSet()
```

- Feature::isDerived () : Boolean [1]

```
from.isDerived
```

- Feature::isComposite () : Boolean [1]

```
from.isComposite
```

7.7.4.2.36 PropertySubsetting_Mapping

Description

Creates a subsetting relationship.

General Mappings

ToSubsetting_Init
Mapping

Mapping Source

Property

Mapping Target

Subsetting with qualifier: subsettedProperty:Property

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Subsetting::subsettingFeature () : Feature [1]

  ```
  Property_Mapping.getMapped(from)
  ```

- Subsetting::subsettedFeature (in subsettedProperty : Property) : Feature [1]

  ```
  Property_Mapping.getMapped(subsettedProperty)
  ```

7.7.4.2.37 PropertyTypedByClassInterface_Mapping

Description

A UML4SysML::Property typed by a UML4SysML::Class or UML4SysML::Interface is mapped to a SysML v2 OccurrenceUsage.

The following shows an example of what the textual SysML v2 syntax of the result of the transformation may look like.

```
part def SysMLv1Block {
        occurrence sysMLv1Property1 [0..1] : SysMLv1Class;
        ref occurrence sysMLv1ReferencedProperty [0..1] : SysMLv1Class;
        occurrence sysMLv1Property2 [0..1] : SysMLv1Interface;
}
```

General Mappings

PropertyCommon_Mapping
NamedElementMain_Mapping

Mapping Source

Property

Mapping Target

OccurrenceUsage

Owned Mappings

(none)

Applicable filters

This mapping applies only if the following (OCL) condition implemented by the operation *filter(src : Element) : Boolean* is verified:

```
if src.oclIsTypeOf(UML::Property) then
    let p: UML::Property = src.oclAsType(UML::Property) in
    if p.type.oclIsUndefined() then
        false
    else
        (p.type.oclIsTypeOf(UML::Class) or
        p.type.oclIsTypeOf(UML::Interface)) and
        not (p.name.indexOf('base_') > 0) and
        (p.association.oclIsUndefined() or p.association.ownedEnd->excludes(p))
    endif
else
    false
endif
```

Mapping rules

The mapping class only has inherited rules. See the mapping classes in the general mapping section for details.

7.7.4.2.38 PropertyUntyped_Mapping

Description

A UML4SysML::Property is mapped to a SysML v2 Feature. The mapping class maps properties without a type.

The following shows an example of what the textual SysML v2 syntax of the result of the transformation may look like.

```
part def SysMLv1Block {
        attribute sysMLv1Property;
}
```

General Mappings

PropertyCommon_Mapping
ToReferenceUsage_Init
NamedElementMain_Mapping

Mapping Source

Property

Mapping Target

ReferenceUsage

Owned Mappings

(none)

Applicable filters

This mapping applies only if the following (OCL) condition implemented by the operation *filter(src : Element) : Boolean* is verified:

```
src.type.oclIsUndefined() and not
Helper.hasStereotypeApplied(src.owner, 'SysML::ConstraintBlocks::ConstraintBlock')
```

Mapping rules

The mapping class only has inherited rules. See the mapping classes in the general mapping section for details.

7.7.4.2.39 Realization_Mapping

Description

A UML4SysML::Realization relationship is mapped to a SysML v2 Dependency.

General Mappings

Abstraction_Mapping

Mapping Source

Realization

Mapping Target

Dependency

Owned Mappings

(none)

Applicable filters

(none)

7.7.4.2.40 Slot_Mapping

Description

A UML4SysML::Slot is mapped to a SysML v2 Feature.

General Mappings

ToFeature_Init
ElementMain_Mapping

Mapping Source

Slot

Mapping Target

Feature

Owned Mappings

(none)

Applicable filters

(none)

7.7.4.2.41 SlotMembership_Mapping

Description

Creates a membership relationship for *memberElement()*.

General Mappings

ToFeatureMembership_Init
Mapping

Mapping Source

Slot

Mapping Target

FeatureMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureMembership::ownedMemberFeature () : Feature [1]

  ```
  from
  ```

- FeatureMembership::memberName () : String [0..1]

  ```
  from.definingFeature.name
  ```

- FeatureMembership::isReadOnly () : Boolean [1]

  ```
  from.isReadOnly
  ```

7.7.4.2.42 SlotFeatureTyping_Mapping

Description

Creates a feature typing relationship owned by the element *typedFeature()*.

General Mappings

ToFeatureTyping_Init
Mapping

Mapping Source

Slot

Mapping Target

FeatureTyping

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureTyping::type () : Type [1]

  ```
  ElementMain_Mapping.getMapped(from)
  ```

7.7.4.2.43 SlotValue_Mapping

Description

Issue here since a KerML feature cannot have more than one FeatureValue while a UML4SysML::Slot can. How to manage collection of values?

General Mappings

ToFeatureValue_Init
Mapping

Mapping Source

ValueSpecification

Mapping Target

FeatureValue

Owned Mappings

(none)

Applicable filters

This mapping applies only if the following (OCL) condition implemented by the operation *filter(src : Element) : Boolean* is verified:

```
src.owner.oclIsKindOf(UML::Slot)
```

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureValue::featureWithValue () : Feature [1]

```
Slot_Mapping.getMapped(from.owner)
```

- FeatureValue::value () : Expression [1]

```
from
```

7.7.4.2.44 StructuralFeature_Mapping

Description

The mapping class is the abstract base class for all UML4SysML::StructuralFeature mappings.

General Mappings

ToFeature_Init
Mapping

Mapping Source

StructuralFeature

Mapping Target

Feature

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Feature::ownedRelationship () : Relationship [0..*]

```
let typing: KerML::FeatureTyping =
    StructuralFeatureToFeatureTyping_Mapping.getMapped(from) in
if typing.oclIsUndefined() then
    Set{MultiplicityMembership_Mapping.getMapped(from)}
else
    Set{MultiplicityMembership_Mapping.getMapped(from), typing}
endif
```

- Feature::isUnique () : Boolean [1]

```
from.isUnique
```

- Feature::isOrdered () : Boolean [1]

```
from.isOrdered
```

- Feature::isAbstract () : Boolean [1]

```
false
```

- Feature::isReadOnly () : Boolean [1]
  abstract rule

7.7.4.2.45 StructuralFeatureMembership_Mapping

Description

Creates a feature membership relationship for *ownedMemberFeature()*.

General Mappings

ToFeatureMembership_Init
Mapping

Mapping Source

StructuralFeature

Mapping Target

FeatureMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureMembership::ownedMemberFeature () : Feature [0..1]

```
NamedElementMain_Mapping.getMapped(from)
```

- FeatureMembership::visibility () : VisibilityKind [1]

```
if (from.oclIsKindOf(UML::NamedElement)) then
    Helper.getKerMLVisibilityKind(from.oclAsType(UML::NamedElement).visibility)
else
    KerML::VisibilityKind::public
endif
```

7.7.4.2.46 StructuralFeatureToFeatureTyping_Mapping

Description

Creates a feature typing relationship owned by the element *typedFeature()*.

General Mappings

TypedElementFeatureTyping_Mapping

Mapping Source

StructuralFeature

Mapping Target

FeatureTyping

Owned Mappings

(none)

Applicable filters

(none)

7.7.4.2.47 TypedElementFeatureTyping_Mapping

Description

Creates a feature typing relationship owned by the element *typedFeature()*.

General Mappings

ToFeatureTyping_Init
Mapping

Mapping Source

TypedElement

Mapping Target

FeatureTyping

Owned Mappings

(none)

Applicable filters

This mapping applies only if the following (OCL) condition implemented by the operation *filter(src : Element) : Boolean* is verified:

```
not src.type.oclIsUndefined()
```

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureTyping::type () : Type [1]

```
if from.type.oclIsKindOf(UML::PrimitiveType) then
    Helper.getScalarValueType(from.type)
else if from.type.oclIsKindOf(UML::Enumeration) then
    Helper.getEnumerationType(from.type)
else
    Classifier_Mapping.getMapped(from.type)
endif endif
```

7.7.4.2.48 UpperBoundValueFeatureMembership_Mapping

Description

Creates a feature membership relationship for *ownedMemberFeature()*.

General Mappings

ToFeatureMembership_Init

Mapping Source

MultiplicityElement

Mapping Target

FeatureMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureMembership::ownedMemberFeature () : Feature [0..1]

```
if from.upper <> -1 then
    LiteralUnlimitedToInteger_Mapping.getMapped(from.upperValue)
else
    LiteralUnlimitedToUnbounded_Mapping.getMapped(from.upperValue)
endif
```

7.7.5 CommonBehavior

SYSML21-542: Transformation: overview tables shall be revised

7.7.5.1 Overview

SYSML21-542: Transformation: overview tables shall be revised

Table 4. List of High-Level CommonBehavior Mappings

SysML v1 Abstract Syntax/Stereotype	SysML v2 Abstract Syntax
AnyReceiveEvent	No specific mapping specified yet.
CallEvent	The concept of a CallEvent is not supported by SysML v2.
ChangeEvent	CalculationUsage
FunctionBehavior	ActionDefinition
OpaqueBehavior	ActionDefinition
SignalEvent	Mapping is not specified yet.
TimeEvent	CalculationUsage
Trigger	AcceptActionUsage

7.7.5.2 Mapping Specifications

7.7.5.2.1 Behavior_Mapping

Description

The mapping class is the abstract base class for all UML4SysML::Behavior mappings.

General Mappings

ToBehavior_Init
Class_Mapping

Mapping Source

Behavior

Mapping Target

Behavior

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Behavior::ownedRelationship () : Relationship [0..*]

```
let parameters: Set(UML::Element) =
    from.ownedElement->select(e | e.oclIsKindOf(UML::Parameter)) in
let parameterSets: Set(UML::Element) =
```

```
        from.ownedElement->select(e | e.oclIsKindOf(UML::ParameterSet)) in
    let features: Set(UML::Element) =
        from.ownedElement->select(e | e.oclIsKindOf(UML::Property)) in
    let elementsOMS: Set(UML::Element) =
        (((from.ownedElement - parameters) parameterSets) - features) in
    elementsOMS->collect(e | ElementOwningMembership_Mapping.getMapped(e))
    ->union(features->collect(e | PropertyMembership_Mapping.getMapped(e)))
    ->union(parameters->collect(e | ParameterMembership_Mapping.getMapped(e)))
    ->union(parameterSets->collect(e | ParameterSetMembership_Mapping.getMapped(e)))
```

7.7.5.2.2 ChangeEvent_Mapping

Description

Main mapping class for the mapping of UML4SysML::ChangeEvent.

```
calc sysMLv1ChangeEvent1 {
  language "language"
  /* change expression */
}
```

General Mappings

NamedElementMain_Mapping
ToCalculationUsage_Init

Mapping Source

ChangeEvent

Mapping Target

CalculationUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- CalculationUsage::ownedRelationship () : Relationship [0..*]

```
from.ownedComment->reject(c | c.annotatedElement->includes(from))
  ->collect(c| CommentOwnership_Mapping.getMapped(c))->asSet()
  ->including(ElementOwningMembership_Mapping.getMapped(from.changeExpression))
```

7.7.5.2.3 ChangeEventReturnParameter_Mapping

Description

Creates the reference usage for the return parameter of the calculation usage which is the target of the UML4SysML::ChangeEvent mapping.

General Mappings

UniqueMapping
ToReferenceUsage_Init

Mapping Source

ChangeEvent

Mapping Target

ReferenceUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ReferenceUsage::direction () : FeatureDirectionKind [0..1]

  ```
  KerML::FeatureDirectionKind::_'out'
  ```

7.7.5.2.4 ChangeEventReturnParameterMembership_Mapping

Description

Creates a membership relationship for *memberElement()*.

General Mappings

UniqueMapping
ToReturnParameterMembership_Init

Mapping Source

ChangeEvent

Mapping Target

ReturnParameterMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ReturnParameterMembership::ownedMemberParameter () : Feature [1]

  ```
  ChangeEventReturnParameter_Mapping.getMapped(from)
  ```

7.7.5.2.5 ChangeTriggerBindingConnector_Mapping

Description

Creates the binding connector between the result of the trigger calculation usage and the result of the time event calculation usage.

General Mappings

ToBindingConnectorAsUsage_Init
UniqueMapping

Mapping Source

Trigger

Mapping Target

BindingConnectorAsUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- BindingConnectorAsUsage::ownedRelationship () : Relationship [0..*]

  ```
  Set{ChangeTriggerReturnEndFeatureMembership_Mapping.getMapped(from)}

  ->including(ChangeTriggerEndFeatureMembership_Mapping.getMapped(from))
  ```

7.7.5.2.6 ChangeTriggerConstraintUsage_Mapping

Description

Creates the constraint usage of the target of the mapping of a UML4SysML::Trigger referencing a UML4SysML::ChangeEvent.

General Mappings

ToConstraintUsage_Init
UniqueMapping

Mapping Source

Trigger

Mapping Target

ConstraintUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ConstraintUsage::ownedRelationship () : Relationship [0..*]

  ```
  Set{ChangeTriggerFeatureMembership_Mapping.getMapped(from)}
  ->including(ChangeTriggerReturnParameterMembership_Mapping.getMapped(from))
  ```

7.7.5.2.7 ChangeTriggerEndFeatureMembership_Mapping

Description

Creates a feature membership relationship for *ownedMemberFeature()*.

General Mappings

ToEndFeatureMembership_Init
UniqueMapping

Mapping Source

Trigger

Mapping Target

EndFeatureMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- EndFeatureMembership::ownedMemberFeature () : Feature [1]

  ```
  ChangeTriggerReferenceUsage_Mapping.getMapped(from)
  ```

7.7.5.2.8 ChangeTriggerEventChainingFeature_Mapping

Description

Creates the chaining feature for the event for the mapping of a UML4SysML::Trigger referencing a UML4SysML::ChangeEvent.

General Mappings

UniqueMapping
ToFeatureChaining_Init

Mapping Source

Trigger

Mapping Target

FeatureChaining

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureChaining::chainingFeature () : Feature [1]

  ```
  from.event
  ```

7.7.5.2.9 ChangeTriggerEventReturnParameterChainingFeature_Mapping

Description

Creates the chaining feature for the return parameter for the mapping of a UML4SysML::Trigger referencing a UML4SysML::ChangeEvent.

General Mappings

ToFeatureChaining_Init
UniqueMapping

Mapping Source

Trigger

Mapping Target

FeatureChaining

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureChaining::chainingFeature () : Feature [1]

  ```
  ChangeEventReturnParameter_Mapping.getMapped(from.event)
  ```

7.7.5.2.10 ChangeTriggerExpressionFeature_Mapping

Description

Creates the feature for the trigger invocation expression of the target of the mapping of a UML4SysML::Trigger referencing a UML4SysML::ChangeEvent.

General Mappings

ToFeature_Init
UniqueMapping

Mapping Source

Trigger

Mapping Target

Feature

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Feature::ownedRelationship () : Relationship [0..*]

  ```
  Set{ChangeTriggerExpressionFeatureValue_Mapping.getMapped(from)}
  ```

7.7.5.2.11 ChangeTriggerExpressionFeatureMembership_Mapping

Description

Creates a feature membership relationship for *ownedMemberFeature()*.

General Mappings

UniqueMapping
ToFeatureMembership_Init

Mapping Source

Trigger

Mapping Target

FeatureMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureMembership::ownedMemberFeature () : Feature [1]

  ```
  ChangeTriggerExpressionInvocationExpression_Mapping.getMapped(from)
  ```

7.7.5.2.12 ChangeTriggerExpressionFeatureReferenceExpression_Mapping

Description

Creates the feature reference expression for the feature value in the trigger invocation expression of the target of the mapping of a UML4SysML::Trigger referencing a UML4SysML::ChangeEvent.

General Mappings

UniqueMapping
ToFeatureReferenceExpression_Init

Mapping Source

Trigger

Mapping Target

FeatureReferenceExpression

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureReferenceExpression::ownedRelationship () : Relationship [0..*]

```
Set{ChangeTriggerExpressionFeatureMembership_Mapping.getMapped(from)}
->including(ReturnParameterFeatureMembership_Factory.create())
```

7.7.5.2.13 ChangeTriggerExpressionFeatureTyping_Mapping

Description

Creates a feature typing relationship owned by the element *typedFeature()*.

General Mappings

ToFeatureTyping_Init
UniqueMapping

Mapping Source

Trigger

Mapping Target

FeatureTyping

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureTyping::type () : Type [1]

```
TransitionChangeTriggerConstraintUsage_Mapping.getMapped(from)
```

7.7.5.2.14 ChangeTriggerExpressionFeatureValue_Mapping

Description

Creates a feature value relationship.

General Mappings

UniqueMapping
ToFeatureValue_Init

Mapping Source

Trigger

Mapping Target

FeatureValue

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureValue::value () : Expression [1]

  ```
  ChangeTriggerExpressionFeatureReferenceExpression_Mapping.getMapped(from)
  ```

7.7.5.2.15 ChangeTriggerExpressionInvocationExpression_Mapping

Description

Creates the invocation expression for the trigger invocation expression of the target of the mapping of a UML4SysML::Trigger referencing a UML4SysML::ChangeEvent.

General Mappings

ToInvocationExpression_Init
UniqueMapping

Mapping Source

Trigger

Mapping Target

InvocationExpression

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- InvocationExpression::ownedRelationship () : Relationship [0..*]

  ```
  Set{ChangeTriggerExpressionFeatureTyping_Mapping.getMapped(from)}
  ```

  ```
  ->including(ReturnParameterFeatureMembership_Factory.create())
  ```

7.7.5.2.16 ChangeTriggerExpressionParameterMembership_Mapping

Description

Creates a membership relationship for *memberElement()*.

General Mappings

ToParameterMembership_Init
UniqueMapping

Mapping Source

Trigger

Mapping Target

ParameterMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ParameterMembership::ownedMemberParameter () : Feature [1]

  ```
  ChangeTriggerExpressionFeature_Mapping.getMapped(from)
  ```

7.7.5.2.17 ChangeTriggerFeature_Mapping

Description

Creates the feature for the mapping of a UML4SysML::Trigger referencing a UML4SysML::ChangeEvent.

General Mappings

ToFeature_Init
UniqueMapping

Mapping Source

Trigger

Mapping Target

Feature

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Feature::ownedRelationship () : Relationship [0..*]

```
Set{ChangeTriggerEventChainingFeature_Mapping.getMapped(from)}
```

```
->including(ChangeTriggerEventReturnParameterChainingFeature_Mapping.getMapped(from))
```

7.7.5.2.18 ChangeTriggerFeatureMembership_Mapping

Description

Creates a feature membership relationship for *ownedMemberFeature()*.

General Mappings

UniqueMapping
ToFeatureMembership_Init

Mapping Source

Trigger

Mapping Target

FeatureMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureMembership::ownedMemberFeature () : Feature [1]

  ```
  ChangeTriggerBindingConnector_Mapping.getMapped(from)
  ```

7.7.5.2.19 ChangeTriggerFeatureValue_Mapping

Description

Creates a feature value relationship.

General Mappings

ToFeatureValue_Init
UniqueMapping

Mapping Source

Trigger

Mapping Target

FeatureValue

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureValue::value () : Expression [1]

  ```
  ChangeTriggerInvocationExpression_Mapping.getMapped(from)
  ```

7.7.5.2.20 ChangeTriggerInvocationExpression_Mapping

Description

Creates the trigger invocation expression of the target of the mapping of a UML4SysML::Trigger referencing a UML4SysML::ChangeEvent.

General Mappings

ToTriggerInvocationExpression_Init
UniqueMapping

Mapping Source

Trigger

Mapping Target

TriggerInvocationExpression

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- TriggerInvocationExpression::kind () : TriggerKind [0..1]

  ```
  SysML::Systems::Actions::TriggerKind::when
  ```

- TriggerInvocationExpression::ownedRelationship () : Relationship [0..*]

  ```
  Set{ChangeTriggerExpressionParameterMembership_Mapping.getMapped(from)}
  ->including(ReturnParameterFeatureMembership_Factory.create())
  ```

7.7.5.2.21 ChangeTriggerReferenceSubsetting_Mapping

Description

Creates a subsetting relationship.

General Mappings

UniqueMapping
ToReferenceSubsetting_Init

Mapping Source

Trigger

Mapping Target

ReferenceSubsetting

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ReferenceSubsetting::ownedRelatedElement () : Element [0..*]

  `Set{ChangeTriggerFeature_Mapping.getMapped(from)}`

- ReferenceSubsetting::referencedFeature () : Feature [1]

  `ChangeTriggerFeature_Mapping.getMapped(from)`

7.7.5.2.22 ChangeTriggerReferenceUsage_Mapping

Description

Creates a reference usage.

General Mappings

UniqueMapping
ToReferenceUsage_Init

Mapping Source

Trigger

Mapping Target

ReferenceUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ReferenceUsage::ownedRelationship () : Relationship [0..*]

  `Set{ChangeTriggerFeatureValue_Mapping.getMapped(from)}`

- ReferenceUsage::direction () : FeatureDirectionKind [0..1]

  `KerML::FeatureDirectionKind::_'in'`

7.7.5.2.23 ChangeTriggerReturnEndFeatureMembership_Mapping

Description

Creates a feature membership relationship for *ownedMemberFeature()*.

General Mappings

UniqueMapping
ToEndFeatureMembership_Init

Mapping Source

Trigger

Mapping Target

EndFeatureMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- EndFeatureMembership::ownedMemberFeature () : Feature [1]

  ```
  ChangeTriggerReturnReferenceUsage_Mapping.getMapped(from)
  ```

7.7.5.2.24 ChangeTriggerReturnParameter_Mapping

Description

Creates the return parameter feature for the mapping of a UML4SysML::Trigger referencing a UML4SysML::ChangeEvent.

General Mappings

UniqueMapping
ToReferenceUsage_Init

Mapping Source

Trigger

Mapping Target

ReferenceUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ReferenceUsage::direction () : FeatureDirectionKind [0..1]

  ```
  KerML::FeatureDirectionKind::_'out'
  ```

7.7.5.2.25 ChangeTriggerReturnParameterMembership_Mapping

Description

Creates a membership relationship for *memberElement()*.

General Mappings

ToReturnParameterMembership_Init
UniqueMapping

Mapping Source

Trigger

Mapping Target

ReturnParameterMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ReturnParameterMembership::ownedMemberParameter () : Feature [1]

  ```
  ChangeTriggerReturnParameter_Mapping.getMapped(from)
  ```

7.7.5.2.26 ChangeTriggerReturnReferenceSubsetting_Mapping

Description

Creates a subsetting relationship.

General Mappings

ToReferenceSubsetting_Init
UniqueMapping

Mapping Source

Trigger

Mapping Target

ReferenceSubsetting

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ReferenceSubsetting::referencedFeature () : Feature [1]

  ```
  ChangeTriggerReturnParameter_Mapping.getMapped(from)
  ```

7.7.5.2.27 ChangeTriggerReturnReferenceUsage_Mapping

Description

Creates a reference usage.

General Mappings

UniqueMapping
ToReferenceUsage_Init

Mapping Source

Trigger

Mapping Target

ReferenceUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ReferenceUsage::isEnd () : Boolean [1]

  ```
  true
  ```

- ReferenceUsage::ownedRelationship () : Relationship [0..*]

```
Set{ChangeTriggerReturnReferenceSubsetting_Mapping.getMapped(from)}
```

7.7.5.2.28 OpaqueBehavior_Mapping

Description

A UML4SysML::OpaqueBehavior is mapped to a SysML v2 ActionDefinition.

The following shows an example of what the textual SysML v2 syntax of the result of the transformation may look like.

```
action def SysMLv1OpaqueBehavior {
        language "Built-in Math"
        /*
         * result = 42 + 23;
        */
}
```

General Mappings

Behavior_Mapping

Mapping Source

OpaqueBehavior

Mapping Target

ActionDefinition

Owned Mappings

(none)

Applicable filters

This mapping applies only if the following (OCL) condition implemented by the operation *filter(src : Element) : Boolean* is verified:

```
src.owner.oclIsKindOf(UML::Package)
```

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ActionDefinition::ownedRelationship () : Relationship [0..*]

```
let parameters : Set(UML::Parameter) =
    from.ownedElement->select(e | e.oclIsKindOf(UML::Parameter)) in
let parameterSets : Set(UML::ParameterSet) =
    from.ownedElement->select(e | e.oclIsKindOf(UML::ParameterSet)) in
let features : Set(UML::Property) =
    from.ownedElement->select(e | e.oclIsKindOf(UML::Property)) in
let elementsOMS: Set(UML::Element) =
```

```
    (((from.ownedElement - parameters) - parameterSets) - features) in
elementsOMS->collect(e | ElementOwningMembership_Mapping.getMapped(e))
->union(features->collect(e | PropertyMembership_Mapping.getMapped(e)))
->union(parameters->collect(e | ParameterMembership_Mapping.getMapped(e)))
->union(parameterSets->collect(e | ParameterSetMembership_Mapping.getMapped(e)))
->union(from.language
    ->collect(l | OpaqueBehaviorMembership_Mapping.getMapped(from, l)))
```

7.7.5.2.29 OpaqueBehaviorMembership_Mapping

Description

Creates a membership relationship for *memberElement()*.

General Mappings

ToOwningMembership_Init
Mapping

Mapping Source

OpaqueBehavior

Mapping Target

OwningMembership with qualifier: language:String

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- OwningMembership::ownedMemberElement (in language : String) : Element [1]

```
OpaqueBehaviorSpecification_Mapping.getMapped(from, language)
```

7.7.5.2.30 OpaqueBehaviorSpecification_Mapping

Description

The mapping class creates the SysML v2 TextualRepresentation elements from the languages and bodies properties of the given UML4SysML::OpaqueBehavior.

General Mappings

ToTextualRepresentation_Init
Mapping

Mapping Source

OpaqueBehavior

Mapping Target

TextualRepresentation with qualifier: language:String

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- TextualRepresentation::body () : String [1]

```
let index:Integer = from.language->indexOf(language) in
from._'body'->at(index)
```

- TextualRepresentation::language () : String [1]

```
language
```

7.7.5.2.31 SignalTriggerReferenceUsage_Mapping

Description

Creates a reference usage.

General Mappings

UniqueMapping
ToReferenceUsage_Init

Mapping Source

Trigger

Mapping Target

ReferenceUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ReferenceUsage::direction () : FeatureDirectionKind [0..1]

  ```
  KerML::FeatureDirectionKind::_'in'
  ```

- ReferenceUsage::ownedRelationship () : Relationship [0..*]

  ```
  Set{SignalTriggerReferenceUsageFeatureTyping_Mapping.getMapped(from)}
  ```

7.7.5.2.32 SignalTriggerReferenceUsageFeatureTyping_Mapping

Description

Creates a feature typing relationship owned by the element *typedFeature()*.

General Mappings

UniqueMapping
ToFeatureTyping_Init

Mapping Source

Trigger

Mapping Target

FeatureTyping

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureTyping::type () : Type [1]

  ```
  from.event.oclAsType(UML::SignalEvent).signal
  ```

7.7.5.2.33 TimeEvent_Mapping

Description

Main mapping class for the mapping of UML4SysML::TimeEvent.

```
calc sysMLv1TimeEvent1 {
  language "language"
  /* duration */
}
```

General Mappings

NamedElementMain_Mapping
ToCalculationUsage_Init

Mapping Source

TimeEvent

Mapping Target

CalculationUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- CalculationUsage::ownedRelationship () : Relationship [0..*]

```
from.ownedComment
  ->reject(c | c.annotatedElement->includes(from))
  ->collect(c| CommentOwnership_Mapping.getMapped(c))->asSet()
  ->including(OpaqueExpressionMembership_Mapping.getMapped(from.when.expr))
```

7.7.5.2.34 TimeTriggerBindingConnector_Mapping

Description

Creates the binding connector between the result of the trigger calculation usage and the result of the time event calculation usage.

General Mappings

UniqueMapping
ToBindingConnectorAsUsage_Init

Mapping Source

Trigger

Mapping Target

BindingConnectorAsUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- BindingConnectorAsUsage::ownedRelationship () : Relationship [0..*]

```
Set{TimeTriggerReturnEndFeatureMembership_Mapping.getMapped(from)}
->including(TimeTriggerEndFeatureMembership_Mapping.getMapped(from))
```

7.7.5.2.35 TimeTriggerCalculationUsage_Mapping

Description

Creates the calculation usage of the target of the mapping of a UML4SysML::Trigger referencing a UML4SysML::TimeEvent.

General Mappings

UniqueMapping
ToCalculationUsage_Init

Mapping Source

Trigger

Mapping Target

CalculationUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- CalculationUsage::ownedRelationship () : Relationship [0..*]

```
Set{TimeTriggerReturnParameterMembership_Mapping.getMapped(from)}

->including(TimeTriggerFeatureMembership_Mapping.getMapped(from))
```

- CalculationUsage::declaredName () : String [0..1]

```
from.name
```

7.7.5.2.36 TimeTriggerEndFeatureMembership_Mapping

Description

Creates a feature membership relationship for *ownedMemberFeature()*.

General Mappings

ToEndFeatureMembership_Init
UniqueMapping

Mapping Source

Trigger

Mapping Target

EndFeatureMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- EndFeatureMembership::ownedMemberFeature () : Feature [1]

```
TimeTriggerReferenceUsage_Mapping.getMapped(from)
```

7.7.5.2.37 TimeTriggerEventChainingFeature_Mapping

Description

Creates the chaining feature for the event for the mapping of a UML4SysML::Trigger referencing a UML4SysML::TimeEvent.

General Mappings

ToFeatureChaining_Init
UniqueMapping

Mapping Source

Trigger

Mapping Target

FeatureChaining

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureChaining::chainingFeature () : Feature [1]

  ```
  from.event
  ```

7.7.5.2.38 TimeTriggerEventReturnParameterChainingFeature_Mapping

Description

Creates the chaining feature for the return parameter for the mapping of a UML4SysML::Trigger referencing a UML4SysML::TimeEvent.

General Mappings

UniqueMapping
ToFeatureChaining_Init

Mapping Source

Trigger

Mapping Target

FeatureChaining

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureChaining::chainingFeature () : Feature [1]

  ```
  TimeTriggerReturnParameter_Mapping.getMapped(from)
  ```

7.7.5.2.39 TimeTriggerExpressionFeature_Mapping

Description

Creates the feature for the trigger invocation expression of the target of the mapping of a UML4SysML::Trigger referencing a UML4SysML::TimeEvent.

General Mappings

UniqueMapping
ToFeature_Init

Mapping Source

Trigger

Mapping Target

Feature

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Feature::ownedRelationship () : Relationship [0..*]

  ```
  Set{TimeTriggerExpressionFeatureValue_Mapping.getMapped(from)}
  ```

7.7.5.2.40 TimeTriggerExpressionFeatureTyping_Mapping

Description

Creates a feature typing relationship owned by the element *typedFeature()*.

General Mappings

ToFeatureTyping_Init
UniqueMapping

Mapping Source

Trigger

Mapping Target

FeatureTyping

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureTyping::type () : Type [1]

  ```
  TransitionTimeTriggerCalculationUsage_Mapping.getMapped(from)
  ```

7.7.5.2.41 TimeTriggerExpressionFeatureValue_Mapping

Description

Creates a feature value relationship.

General Mappings

ToFeatureValue_Init
UniqueMapping

Mapping Source

Trigger

Mapping Target

FeatureValue

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureValue::value () : Expression [1]

  ```
  TimeTriggerExpressionInvocationExpression_Mapping.getMapped(from)
  ```

7.7.5.2.42 TimeTriggerExpressionInvocationExpression_Mapping

Description

Creates the invocation expression for the trigger invocation expression of the target of the mapping of a UML4SysML::Trigger referencing a UML4SysML::TimeEvent.

General Mappings

UniqueMapping
ToInvocationExpression_Init

Mapping Source

Trigger

Mapping Target

InvocationExpression

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- InvocationExpression::ownedRelationship () : Relationship [0..*]

```
Set{TimeTriggerExpressionFeatureTyping_Mapping.getMapped(from)}
->including(ReturnParameterFeatureMembership_Factory.create())
```

7.7.5.2.43 TimeTriggerExpressionParameterMembership_Mapping

Description

Creates a membership relationship for *memberElement()*.

General Mappings

UniqueMapping
ToParameterMembership_Init

Mapping Source

Trigger

Mapping Target

ParameterMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ParameterMembership::ownedMemberParameter () : Feature [1]

  ```
  TimeTriggerExpressionFeature_Mapping.getMapped(from)
  ```

7.7.5.2.44 TimeTriggerFeature_Mapping

Description

Creates the feature for the mapping of a UML4SysML::Trigger referencing a UML4SysML::TimeEvent.

General Mappings

UniqueMapping
ToFeature_Init

Mapping Source

Trigger

Mapping Target

Feature

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Feature::ownedRelationship () : Relationship [0..*]

  ```
  Set{TimeTriggerEventChainingFeature_Mapping.getMapped(from)}

  ->including(TimeTriggerEventReturnParameterChainingFeature_Mapping.getMapped(from))
  ```

7.7.5.2.45 TimeTriggerFeatureMembership_Mapping

Description

Creates a feature membership relationship for *ownedMemberFeature()*.

General Mappings

ToFeatureMembership_Init
UniqueMapping

Mapping Source

Trigger

Mapping Target

FeatureMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureMembership::ownedMemberFeature () : Feature [1]

  ```
  TimeTriggerBindingConnector_Mapping.getMapped(from)
  ```

7.7.5.2.46 TimeTriggerFeatureTyping_Mapping

Description

Creates a feature typing relationship owned by the element *typedFeature()*.

General Mappings

UniqueMapping
ToFeatureTyping_Init

Mapping Source

Trigger

Mapping Target

FeatureTyping

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureTyping::type () : Type [1]

```
if from.event.oclAsType(UML::TimeEvent).isRelative then
  SYSML2::AttributeDefinition.allInstances()
  ->any(m | m.qualifiedName = 'ISQ::DurationValue')
else
  SYSML2::AttributeDefinition.allInstances()
  ->any(m | m.qualifiedName = 'Time:TimeInstantValue')
endif
```

7.7.5.2.47 TimeTriggerFeatureValue_Mapping

Description

Creates a feature value relationship.

General Mappings

ToFeatureValue_Init
UniqueMapping

Mapping Source

Trigger

Mapping Target

FeatureValue

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureValue::value () : Expression [1]

  ```
  TimeTriggerInvocationExpression_Mapping.getMapped(from)
  ```

7.7.5.2.48 TimeTriggerInvocationExpression_Mapping

Description

Creates the trigger invocation expression of the target of the mapping of a UML4SysML::Trigger referencing a UML4SysML::TimeEvent.

General Mappings

ToTriggerInvocationExpression_Init
UniqueMapping

Mapping Source

Trigger

Mapping Target

TriggerInvocationExpression

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- TriggerInvocationExpression::kind () : TriggerKind [0..1]

```
if from.event.oclAsType(UML::TimeEvent).isRelative then
  SysML::Systems::Actions::TriggerKind::after
else
  SysML::Systems::Actions::TriggerKind::at
endif
```

- TriggerInvocationExpression::ownedRelationship () : Relationship [0..*]

```
Set{TimeTriggerExpressionParameterMembership_Mapping.getMapped(from)}
```

7.7.5.2.49 TimeTriggerReferenceSubsetting_Mapping

Description

Creates a subsetting relationship.

General Mappings

ToReferenceSubsetting_Init
UniqueMapping

Mapping Source

Trigger

Mapping Target

ReferenceSubsetting

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ReferenceSubsetting::referencedFeature () : Feature [1]

  ```
  TimeTriggerFeature_Mapping.getMapped(from)
  ```

- ReferenceSubsetting::ownedRelatedElement () : Element [0..*]

  ```
  Set{TimeTriggerFeature_Mapping.getMapped(from)}
  ```

7.7.5.2.50 TimeTriggerReferenceUsage_Mapping

Description

Creates a reference usage.

General Mappings

ToReferenceUsage_Init
UniqueMapping

Mapping Source

Trigger

Mapping Target

ReferenceUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ReferenceUsage::direction () : FeatureDirectionKind [0..1]

  ```
  KerML::FeatureDirectionKind::_'in'
  ```

- ReferenceUsage::ownedRelationship () : Relationship [0..*]

  ```
  Set{TimeTriggerFeatureValue_Mapping.getMapped(from)}
  ```

7.7.5.2.51 TimeTriggerReturnEndFeatureMembership_Mapping

Description

Creates a feature membership relationship for *ownedMemberFeature()*.

General Mappings

ToEndFeatureMembership_Init
UniqueMapping

Mapping Source

Trigger

Mapping Target

EndFeatureMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- EndFeatureMembership::ownedMemberFeature () : Feature [1]

  ```
  TimeTriggerReturnReferenceUsage_Mapping.getMapped(from)
  ```

7.7.5.2.52 TimeTriggerReturnParameter_Mapping

Description

Creates the return parameter feature for the mapping of a UML4SysML::Trigger referencing a UML4SysML::TimeEvent.

General Mappings

UniqueMapping
ToReferenceUsage_Init

Mapping Source

Trigger

Mapping Target

ReferenceUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ReferenceUsage::ownedRelationship () : Relationship [0..*]

  ```
  Set{TimeTriggerFeatureTyping_Mapping.getMapped(from)}
  ```

7.7.5.2.53 TimeTriggerReturnParameterMembership_Mapping

Description

Creates a membership relationship for *memberElement()*.

General Mappings

UniqueMapping
ToReturnParameterMembership_Init

Mapping Source

Trigger

Mapping Target

ReturnParameterMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ReturnParameterMembership::ownedMemberParameter () : Feature [1]

  ```
  TimeTriggerReturnParameter_Mapping.getMapped(from)
  ```

7.7.5.2.54 TimeTriggerReturnReferenceSubsetting_Mapping

Description

Creates a subsetting relationship.

General Mappings

UniqueMapping
ToReferenceSubsetting_Init

Mapping Source

Trigger

Mapping Target

ReferenceSubsetting

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ReferenceSubsetting::referencedFeature () : Feature [1]

  ```
  TimeTriggerReturnParameter_Mapping.getMapped(from)
  ```

7.7.5.2.55 TimeTriggerReturnReferenceUsage_Mapping

Description

Creates a reference usage.

General Mappings

UniqueMapping
ToReferenceUsage_Init

Mapping Source

Trigger

Mapping Target

ReferenceUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ReferenceUsage::isEnd () : Boolean [1]

```
true
```

- ReferenceUsage::ownedRelationship () : Relationship [0..*]

```
Set{TimeTriggerReturnReferenceSubsetting_Mapping.getMapped(from)}
```

7.7.5.2.56 Trigger_Mapping

Description

A UML4SysML::Trigger is mapped to a SysML v2 AcceptActionUsage.

General Mappings

NamedElementMain_Mapping
ToActionUsage_Init

Mapping Source

Trigger

Mapping Target

AcceptActionUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- AcceptActionUsage::ownedRelationship () : Relationship [0..*]

```
from.ownedComment

->reject(c | c.annotatedElement->includes(from))

  ->collect(c| CommentOwnership_Mapping.getMapped(c))->asSet()

->including(TriggerParameterMembership_Mapping.getMapped(from))

->including(ParameterMembership_Factory.create())
```

7.7.5.2.57 TriggerParameterMembership_Mapping

Description

Creates a membership relationship for *memberElement()*.

General Mappings

UniqueMapping
ToParameterMembership_Init

Mapping Source

Trigger

Mapping Target

ParameterMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ParameterMembership::ownedMemberParameter () : Feature [1]

```
if from.event.oclIsKindOf(UML::SignalEvent) then
  SignalTriggerReferenceUsage_Mapping.getMapped(from)
else if from.event.oclIsKindOf(UML::TimeEvent) then
  TimeTriggerReferenceUsage_Mapping.getMapped(from)
else if from.event.oclIsKindOf(UML::ChangeEvent) then
  ChangeTriggerReferenceUsage_Mapping.getMapped(from)
else
  OclUndefined
endif endif endif
```

7.7.6 CommonStructure

7.7.6.1 Overview

SYSML21-542: Transformation: overview tables shall be revised

Table 5. List of High-Level CommonStructure Mappings

SysML v1 Abstract Syntax/Stereotype	SysML v2 Abstract Syntax
Abstraction	Dependency
Comment	Comment
Constraint	ConstraintDefinition
Dependency	Dependency
ElementImport	MembershipImport
PackageImport	NamespaceImport
Realization	Dependency
Usage	Dependency

7.7.6.2 Mapping Specifications

7.7.6.2.1 Abstraction_Mapping

Description

A UML4SysML::Abstraction relationship is mapped to a SysML v2 Dependency relationship.

General Mappings

Dependency_Mapping

Mapping Source

Abstraction

Mapping Target

Dependency

Owned Mappings

(none)

Applicable filters

(none)

7.7.6.2.2 Comment_Mapping

Description

A UML4SysML::Comment is mapped to a SysML v2 Comment.

The following shows an example of what the textual SysML v2 syntax of the result of the transformation may look like.

```
part def SysMLv1Block1;
part def SysMLv1Block2;
action def SysMLv1Activitiy {
        comment about SysMLv1Activity, SysMLv1Block1
                /* comment body */
}
comment about SysMLv1Block1, SysMLv1Block /* comment body */
```

General Mappings

ElementMain_Mapping
ToAnnotatingElement_Init

Mapping Source

Comment

Mapping Target

Comment

Owned Mappings

(none)

Applicable filters

This mapping applies only if the following (OCL) condition implemented by the operation *filter(src : Element) : Boolean* is verified:

```
not Helper.hasStereotypeApplied(src, 'SysML::ModelElements::ElementGroup')
```

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Comment::body () : String [1]

  ```
  if from.body->isEmpty() then '' else from.body endif
  ```

- Comment::ownedRelationship () : Relationship [0..*]

  ```
  self.oclAsType(ElementMain_Mapping).ownedRelationship()
  ->union(self.annotation()->asSet())
  ```

- Comment::annotation () : Annotation [0..*]

  ```
  from.annotatedElement
  ->collect(e | CommentAnnotation_Mapping.getMapped(from, e))
  ```

7.7.6.2.3 CommentAnnotation_Mapping

Description

The mapping class creates the annotation relationship for the UML4SysML::Comment mapping.

General Mappings

ToAnnotation_Init
Mapping

Mapping Source

Comment

Mapping Target

Annotation with qualifier: annotatedElement:Element

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Annotation::annotatedElement (in annotatedElement : Element) : Element [1]

  ```
  ElementMain_Mapping.getMapped(annotatedElement)
  ```

- Annotation::annotatingElement () : AnnotatingElement [1]

  ```
  Comment_Mapping.getMapped(from)
  ```

- Annotation::owningAnnotatedElement () : Element [0..1]

  ```
  null
  ```

7.7.6.2.4 CommentOwnership_Mapping

Description

That mapping class creates an ownership relation that is convenient for a Comment. In SysMLv1/UML can be owned by any kind of element, including some that are not translated to SysMLv2 Namespaces.

General Mappings

ToAnnotation_Init
UniqueMapping

Mapping Source

Comment

Mapping Target

Annotation

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Annotation::ownedRelatedElement () : Element [0..*]

  ```
  Set{self.annotatingElement()}
  ```

- Annotation::annotatedElement () : Element [1]

  ```
  ElementMain_Mapping.getMapped(from.owner)
  ```

- Annotation::annotatingElement () : AnnotatingElement [1]

```
Comment_Mapping.getMapped(from)
```

7.7.6.2.5 Constraint_Mapping

Description

A UML4SysML::Constraint is mapped to a SysML v2 ConstraintDefinition and AssertConstraintUsages for the constrained elements.

The following shows an example of what the textual SysML v2 syntax of the result of the transformation may look like.

```
part def SysMLv1Block {
        constraint def SysMLv1Constraint {
                calc sysMLv1Constraint {
                         language "English"
                        /*
                         * constraint specification
                         */
                 }
        }
        assert constraint assert_sysMLv1Constraint : SysMLv1Constraint;
}
```

General Mappings

ToConstraintDefinition_Init
NamedElementMain_Mapping

Mapping Source

Constraint

Mapping Target

ConstraintDefinition

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ConstraintDefinition::ownedRelationship () : Relationship [0..*]

```
self.oclAsType(ElementMain_Mapping).ownedRelationship()
->union(Set{ElementFeatureMembership_Mapping.getMapped(from.specification),
CommonReturnParameterReferenceUsageMembership_Mapping.getMapped(from.specification)})
```

7.7.6.2.6 ConstrainedElementFeatureMembership_Mapping

Description

Creates a feature membership relationship for *ownedMemberFeature()*.

General Mappings

ToFeatureMembership_Init
Mapping

Mapping Source

Constraint

Mapping Target

FeatureMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureMembership::ownedMemberFeature () : Feature [1]

  ```
  ConstraintUsage_Mapping.getMapped(from)
  ```

7.7.6.2.7 ConstraintUsageFeatureTyping_Mapping

Description

Creates a feature typing relationship owned by the element *typedFeature()*.

General Mappings

ToFeatureTyping_Init
Mapping

Mapping Source

Constraint

Mapping Target

FeatureTyping

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureTyping::type () : Type [1]

```
from
```

7.7.6.2.8 ConstraintUsage_Mapping

Description

The mapping class creates the SysML v2 AssertConstraintUsage elements for the constrained elements of the UML4SysML::Constraint mapping.

General Mappings

ToUsage_Init
Mapping

Mapping Source

Constraint

Mapping Target

AssertConstraintUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- AssertConstraintUsage::declaredName () : String [0..1]

```
'assert_' + from.name
```

- AssertConstraintUsage::ownedRelationship () : Relationship [0..*]

```
from.ownedComment->reject(c | c.annotatedElement->includes(from))
  ->collect(c| CommentOwnership_Mapping.getMapped(c))->asSet()
```

```
->union(Set{ConstraintUsageFeatureTyping_Mapping.getMapped(from),
CommonReturnParameterReferenceUsageMembership_Mapping.getMapped(from)})
```

7.7.6.2.9 Dependency_Mapping

Description

A UML4SysML::Dependency relationship is mapped to a SysML v2 Dependency relationship.

General Mappings

DirectedRelationship_Mapping

Mapping Source

Dependency

Mapping Target

Dependency

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Dependency::client () : Element [0..*]

  ```
  from.source->collect(e | ElementMain_Mapping.getMapped(e))
  ```

- Dependency::supplier () : Element [0..*]

  ```
  from.target->collect(e | ElementMain_Mapping.getMapped(e))
  ```

- Dependency::declaredName () : String [0..1]

  ```
  from.name
  ```

7.7.6.2.10 DirectedRelationship_Mapping

Description

The mapping class is the abstract base class for all UML4SysML::DirectedRelationship mappings.

General Mappings

Relationship_Mapping

Mapping Source

DirectedRelationship

Mapping Target

Relationship

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Relationship::target () : Element [0..*]

  ```
  from.target->collect(e | ElementMain_Mapping.getMapped(e))
  ```

- Relationship::source () : Element [0..*]

  ```
  from.source->collect(e | ElementMain_Mapping.getMapped(e))
  ```

7.7.6.2.11 ElementMain_Mapping

Description

This is the general abstract class to be used as an ancestor for any class mapping specification.

General Mappings

ToElement_Init
MainMapping

Mapping Source

Element

Mapping Target

Element

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Element::ownedRelationship () : Relationship [0..*]

```
from.ownedComment->reject(c | c.annotatedElement->includes(from))

->collect(c| CommentOwnership_Mapping.getMapped(c))->asSet()
```

- Element::elementId () : String [1]

```
Helper.getID(from)
```

7.7.6.2.12 ElementMembership_Mapping

Description

Creates a membership relationship for *memberElement()*.

General Mappings

ToMembership_Init
Mapping

Mapping Source

Element

Mapping Target

Membership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Membership::memberElement () : Element [1]

```
ElementMain_Mapping.getMapped(from)
```

- Membership::membershipOwningNamespace () : Element [0..*]

```
Set{ElementMain_Mapping(from)}
-- will not be used since corresponding attribute is derived,
-- but required for redefinition
```

- Membership::visibility () : VisibilityKind [1]

```
if (from.oclIsKindOf(UML::NamedElement)) then
    from.oclAsType(UML::NamedElement).visibility
else
    KerML::VisibilityKind::public
endif
```

7.7.6.2.13 ElementOwnership_Mapping

Description

The mapping class is the abstract base class for mappings that target ownership relationships.

General Mappings

ToRelationship_Init
UniqueMapping

Mapping Source

Element

Mapping Target

Relationship

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Relationship::ownedRelatedElement () : Element [0..*]

  ```
  self.target()
  ```

- Relationship::source () : Element [0..*]

  ```
  OrderedSet{ElementMain_Mapping.getMapped(from.owner)}
  ```

- Relationship::target () : Element [0..*]

  ```
  OrderedSet{ElementMain_Mapping.getMapped(from)}
  ```

7.7.6.2.14 ElementOwningMembership_Mapping

Description

Creates a owning membership relationship for *ownedMemberElement()*.

General Mappings

ElementMembership_Mapping
ElementOwnership_Mapping

Mapping Source

Element

Mapping Target

OwningMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- OwningMembership::ownedMemberElement () : Element [1]

  ```
  ElementMain_Mapping.getMapped(from)
  ```

- OwningMembership::membershipOwningNamespace () : Element [0..*]

  ```
  Set{ElementMain_Mapping(from)}
  -- will not be used since corresponding attribute is derived,
  -- but required for redefinition
  ```

- OwningMembership::ownedRelatedElement () : Element [0..*]

  ```
  Set{self.ownedMemberElement()}
  ```

7.7.6.2.15 NamedElementMain_Mapping

Description

The mapping class is the abstract base class for mappings of UML4SysML::NamedElements.

General Mappings

ElementMain_Mapping

Mapping Source

NamedElement

Mapping Target

Element

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Element::declaredName () : String [0..1]

  ```
  from.name
  ```

7.7.6.2.16 Namespace_Mapping

Description

The mapping class is the abstract base class for UML4SysML::Namespace mappings.

General Mappings

ToNamespace_Init
NamedElementMain_Mapping

Mapping Source

Namespace

Mapping Target

Namespace

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Namespace::ownedImport () : Import [0..*]

  ```
  Set{}
  ```

7.7.6.2.17 Relationship_Mapping

Description

Th mapping class is the abstract base class for UML4SysML::Relationship mappings.

General Mappings

ToRelationship_Init
ElementMain_Mapping

Mapping Source

Relationship

Mapping Target

Relationship

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Relationship::ownedRelatedElement () : Element [0..*]

```
from.relatedElement->select(e | from.ownedElement->includes(e))
->collect(e | ElementMain_Mapping.getMapped(e))
```

- Relationship::owningRelatedElement () : Element [0..1]

```
ElementMain_Mapping.getMapped(from.owner)
```

7.7.6.2.18 Usage_Mapping

Description

A UML4SysML::Usage relationship is mapped to a SysML v2 Dependency relationship.

General Mappings

Dependency_Mapping

Mapping Source

Usage

Mapping Target

Dependency

Owned Mappings

(none)

Applicable filters

(none)

7.7.7 InformationFlows

7.7.7.1 Overview

SYSML21-542: Transformation: overview tables shall be revised

Table 6. List of High-Level InformationFlows Mappings

SysML v1 Abstract Syntax/Stereotype	SysML v2 Abstract Syntax
InformationFlow	No specific mapping specified yet.
InformationItem	ItemDefinition

7.7.7.2 Mapping Specifications

7.7.7.2.1 InformationFlow_Mapping

Description

A UML4SysML::InformationFlow is mapped to a FlowDefinition, if the UML4SysML::InformationFlow has defined realizing connectors or if it is realized by an association. If the information flow has more that one realizing connector, a FlowDefinition element is created for each of them.

General Mappings

ToConnectionUsage_Init
UniqueMapping

Mapping Source

InformationFlow

Mapping Target

FlowDefinition with qualifier: realization:NamedElement

Owned Mappings

(none)

Applicable filters

This mapping applies only if the following (OCL) condition implemented by the operation *filter(src : Element) : Boolean* is verified:

```
src.oclIsKindOf(UML::InformationFlow) and
(src.oclAsType(UML::InformationFlow).realizingConnector->notEmpty()
or src.oclAsType(UML::InformationFlow).realization->notEmpty())
```

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FlowDefinition::ownedRelationship () : Relationship [0..*]

```
from.informationSource
  ->collect(s | InformationFlowEndFeatureMembership_Mapping.getMapped(from, s))
    ->asSet()
->union(from.informationTarget
  ->collect(t | InformationFlowEndFeatureMembership_Mapping.getMapped(from, t))
    ->asSet())
->union(from.conveyed
  ->collect(i | InformationFlowConveyedFeatureMembership_Mapping.getMapped(i))
    ->asSet())
->union(from.realization->select( a | a.oclIsKindOf(UML::Association))
  ->collect(r | InformationFlowSubclassification_Mapping.getMapped(from, r))
    ->asSet())
->asOrderedSet()
```

7.7.7.2.2 InformationFlowConveyedFeatureMembership_Mapping

Description

Creates a feature membership relationship for *ownedMemberFeature()*.

General Mappings

ToFeatureMembership_Init
Mapping

Mapping Source

Classifier

Mapping Target

FeatureMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureMembership::ownedMemberFeature () : Feature [1]

```
InformationItemFlowConveyedItemUsage_Mapping.getMapped(from)
```

7.7.7.2.3 InformationFlowEnd_Mapping

Description

The mapping class creates the source feature of the FlowDefinition for the mapping of UML4SysML::InformationFlow.

General Mappings

ToFeature_Init
UniqueMapping

Mapping Source

InformationFlow

Mapping Target

Feature with qualifier: end:NamedElement

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Feature::ownedRelationship () : Relationship [0..*]

  ```
  Set{InformationFlowFeatureTyping_Mapping.getMapped(from, end)}
  ```

- Feature::isEnd () : Boolean [1]

  ```
  true
  ```

7.7.7.2.4 InformationFlowEndFeatureMembership_Mapping

Description

The mapping class creates the source and the target membership relationships of the FlowDefinition for the UML4SysML::InformationFlow mapping.

General Mappings

ToFeatureMembership_Init
UniqueMapping

Mapping Source

InformationFlow

Mapping Target

FeatureMembership with qualifier: end:NamedElement

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureMembership::ownedMemberFeature (in end : NamedElement) : Feature [1]

```
InformationFlowEnd_Mapping.getMapped(from, end)
```

7.7.7.2.5 InformationFlowFeatureTyping_Mapping

Description

Creates a feature typing relationship owned by the element *typedFeature()*.

General Mappings

ToFeatureTyping_Init
UniqueMapping

Mapping Source

InformationFlow

Mapping Target

FeatureTyping with qualifier: element:NamedElement

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureTyping::type (in source : NamedElement) : Type [1]

```
ElementMain_Mapping.getMapped(element)
```

7.7.7.2.6 InformationFlowSubclassification_Mapping

Description

Creates a Subclassification relationship between the target element of the UML4SysML::InformationFlow mapping and the target element of the UML4SysML::Association which realizes the flow.

General Mappings

ToSubclassification_Init
Mapping

Mapping Source

InformationFlow

Mapping Target

Subclassification with qualifier: element:Relationship

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Subclassification::subclassifier () : Classifier [1]

  `from`

- Subclassification::superclassifier () : Classifier [1]

  `element`

7.7.7.2.7 InformationItem_Mapping

Description

A UML4SysML::InformationItem is mapped to a SysML v2 ItemDefinition.

General Mappings

Classifier_Mapping

Mapping Source

InformationItem

Mapping Target

ItemDefinition

Owned Mappings

(none)

Applicable filters

(none)

7.7.7.2.8 InformationItemFlowConveyedItemUsage_Mapping

Description

Creates an ItemUsage element representing the conveyed classifier of an UML4SysML::InformationFlow.

General Mappings

ToItemUsage_Init
Mapping

Mapping Source

Classifier

Mapping Target

ItemUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ItemUsage::ownedRelationship () : Relationship [0..*]

  ```
  Set{InformationItemFlowConveyedItemUsageFeatureTyping_Mapping.getMapped(from)}
  ```

7.7.7.2.9 InformationItemFlowConveyedItemUsageFeatureTyping_Mapping

Description

Creates a feature typing relationship owned by the element *typedFeature()*.

General Mappings

ToFeatureTyping_Init
Mapping

Mapping Source

Classifier

Mapping Target

FeatureTyping

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureTyping::type () : Type [1]

  `from`

7.7.8 Interactions

SYSML21-542: Transformation: overview tables shall be revised

7.7.8.1 Overview

SYSML21-542: Transformation: overview tables shall be revised

Table 7. List of High-Level Interactions Mappings

SysML v1 Abstract Syntax/Stereotype	SysML v2 Abstract Syntax
ActionExecutionSpecification	ActionUsage
BehaviorExecutionSpecification	ActionUsage
CombinedFragment	Interaction
ConsiderIgnoreFragment	Interaction
Continuation	No specific mapping specified yet.
DestructionOccurrenceSpecification	No specific mapping specified yet.
ExecutionOccurrenceSpecification	No specific mapping specified yet.
Gate	No specific mapping specified yet.
GeneralOrdering	No specific mapping specified yet.
Interaction	Interaction, Behavior
InteractionConstraint	ConstraintDefinition
InteractionOperand	Interaction, Namespace
InteractionUse	Step
Lifeline	PartUsage
Message	Flow
MessageOccurrenceSpecification	No specific mapping specified yet.
OccurrenceSpecification	No specific mapping specified yet.
PartDecomposition	Step
StateInvariant	Invariant

7.7.8.2 Mapping Specifications

7.7.8.2.1 ActionExecutionSpecification_Mapping

Description

A UML4SysML::ActionExecutionSpecification is mapped to a SysML v2 ActionUsage.

General Mappings

ToActionUsage_Init
NamedElementMain_Mapping

Mapping Source

ActionExecutionSpecification

Mapping Target

ActionUsage

Owned Mappings

(none)

Applicable filters

(none)

7.7.8.2.2 BehaviorExecutionSpecification_Mapping

Description

A UML4SysML::BehaviorExecutionSpecification is mapped to a SysML v2 ActionUsage.

General Mappings

ToActionUsage_Init
NamedElementMain_Mapping

Mapping Source

BehaviorExecutionSpecification

Mapping Target

ActionUsage

Owned Mappings

(none)

Applicable filters

(none)

7.7.8.2.3 CombinedFragment_Mapping

Description

A UML4SysML::CombinedFragment is mapped to a SysMLv2 Interaction.

General Mappings

NamedElementMain_Mapping
ToInteraction_Init

Mapping Source

CombinedFragment

Mapping Target

Interaction

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Interaction::ownedRelationship () : Relationship [0..*]

```
let operands: Set(UML::Element) =
  from.ownedElement->select(e | e.oclIsKindOf(UML::InteractionOperand)) in
let occurrencesSpecs: Set(UML::Element) =
  from.ownedElement->select(e | e.oclIsKindOf(UML::OccurrenceSpecification)) in
let elements: Set(UML::Element) =
  (from.ownedElement - operands) - occurrencesSpecs in
    elements->collect(e |
      ElementOwningMembership_Mapping.getMapped(e))->asSet()
    ->union(operands->collect(e |
      InteractionOperandMembership_Mapping.getMapped(e))->asSet())
    ->union(self.oclAsType(ElementMain_Mapping).ownedRelationship())
```

7.7.8.2.4 CombinedFragmentMembership_Mapping

Description

Creates a membership relationship for *memberElement()*.

General Mappings

ToFeatureMembership_Init
Mapping

Mapping Source

CombinedFragment

Mapping Target

FeatureMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureMembership::ownedMemberFeature () : Feature [0..1]

  ```
  self.memberFeature()
  ```

- FeatureMembership::memberFeature () : Feature [1]

  ```
  ElementMain_Mapping.getMapped(from)
  ```

7.7.8.2.5 ExecutionSpecificationMembership_Mapping

Description

Creates a membership relationship for *memberElement()*.

General Mappings

ToEndFeatureMembership_Init
Mapping

Mapping Source

ExecutionSpecification

Mapping Target

EndFeatureMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- EndFeatureMembership::memberFeature () : Feature [1]

  ```
  ElementMain_Mapping.getMapped(from)
  ```

- EndFeatureMembership::ownedMemberFeature () : Feature [0..1]

```
self.memberFeature()
```

7.7.8.2.6 Interaction_Mapping

Description

A UML4SysML::Interaction is mapped to a SysMLv2 Interaction.

General Mappings

Namespace_Mapping
ToInteraction_Init

Mapping Source

Interaction

Mapping Target

Interaction

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Interaction::ownedRelationship () : Relationship [0..*]

```
let lifelines: Set(UML::Element) = from.lifeline in
let messageOccurrences: Set(UML::Element) =
  from.ownedElement->select(e |
    e.oclIsKindOf(UML::MessageOccurrenceSpecification)) in
let executionOccurrences: Set(UML::Element) =
  from.fragment->select(e | e.oclIsKindOf(UML::ExecutionSpecification)) in
let occurrencesSpecs: Set(UML::Element) =
  from.fragment->select(e | e.oclIsKindOf(UML::OccurrenceSpecification)) in
let messages: Set(UML::Element) = from.message in
let invariants: Set(UML::Element) =
  from.fragment->select(e | e.oclIsKindOf(UML::StateInvariant)) in
let interactionUsages: Set(UML::Element) =
  from.fragment->select(e | e.oclIsKindOf(UML::InteractionUse)) in
let combinedFragments: Set(UML::Element) =
  from.ownedElement->select( e | e.oclIsKindOf(UML::CombinedFragment)) in
let continuations: Set(UML::Element) =
  from.ownedElement->select(e | e.oclIsKindOf(UML::Continuation)) in
let elements: Set(UML::Element) =
  ((((((((from.ownedElement - lifelines) - messageOccurrences)
  - executionOccurrences) - occurrencesSpecs) - messages) -
```

```
  combinedFragments) - invariants) -
  interactionUsages) - continuations) - from.ownedComment in

elements->collect(e | ElementOwningMembership_Mapping.getMapped(e))->asSet()
->union(lifelines->collect(e | LifelineMembership_Mapping.getMapped(e))->asSet())
->union(executionOccurrences
  ->collect(e | ExecutionSpecificationMembership_Mapping.getMapped(e))->asSet())
->union(messages->collect(e | MessageMembership_Mapping.getMapped(e))->asSet())
->union(combinedFragments
  ->collect(e | CombinedFragmentMembership_Mapping.getMapped(e))->asSet())
->union(invariants
  ->collect(e | StateInvariantMembership_Mapping.getMapped(e))->asSet())
->union(interactionUsages
  ->collect(e | InteractionUseMembership_Mapping.getMapped(e))->asSet())
->union(self.oclAsType(ElementMain_Mapping).ownedRelationship())
```

7.7.8.2.7 InteractionOperand_Mapping

Description

A UML4SysML::InteractionOperand is mapped to a SysML v2 Interaction.

General Mappings

NamedElementMain_Mapping
ToInteraction_Init

Mapping Source

InteractionOperand

Mapping Target

Interaction

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Interaction::ownedRelationship () : Relationship [0..*]

```
let executionOccurrences: Set(UML::Element) =
    from.ownedElement->select(e | e.oclIsKindOf(UML::ExecutionSpecification)) in
let occurrencesSpecs: Set(UML::Element) =
    from.ownedElement->select(e | e.oclIsKindOf(UML::OccurrenceSpecification)) in
let continuations: Set(UML::Element) =
    from.ownedElement->select(e | e.oclIsKindOf(UML::Continuation)) in
let elements: Set(UML::Element) =
    (((from.ownedElement - executionOccurrences) - occurrencesSpecs) -
    continuations) - from.ownedComment in
```

```
elements->collect(e | ElementOwningMembership_Mapping.getMapped(e))->asSet()
->union(self.oclAsType(ElementMain_Mapping).ownedRelationship())
->union(executionOccurrences
    ->collect(e |ExecutionSpecificationMembership_Mapping.getMapped(e))->asSet())
```

7.7.8.2.8 InteractionOperandMembership_Mapping

Description

Creates a membership relationship for *memberElement()*.

General Mappings

ToFeatureMembership_Init
Mapping

Mapping Source

InteractionOperand

Mapping Target

FeatureMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureMembership::memberFeature () : Feature [1]

  ```
  ElementMain_Mapping.getMapped(from)
  ```

- FeatureMembership::ownedMemberFeature () : Feature [0..1]

  ```
  self.memberFeature()
  ```

7.7.8.2.9 InteractionUse_Mapping

Description

A UML4SysML::InteractionUse is mapped to a SysML v2 Step.

General Mappings

ToStep_Init
Namespace_Mapping

Mapping Source

InteractionUse

Mapping Target

Step

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Step::ownedRelationship () : Relationship [0..*]

```
self.oclAsType(ElementMain_Mapping).ownedRelationship()

->including(InteractionUseFeatureTyping_Mapping.getMapped(from))
```

7.7.8.2.10 InteractionUseMembership_Mapping

Description

Creates a membership relationship for *memberElement()*.

General Mappings

ToFeatureMembership_Init
Mapping

Mapping Source

InteractionUse

Mapping Target

FeatureMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureMembership::memberFeature () : Feature [1]

  ```
  ElementMain_Mapping.getMapped(from)
  ```

- FeatureMembership::ownedMemberFeature () : Feature [0..1]

  ```
  self.memberFeature()
  ```

7.7.8.2.11 InteractionUseFeatureTyping_Mapping

Description

Creates a feature typing relationship owned by the element *typedFeature()*.

General Mappings

ToFeatureTyping_Init
Mapping

Mapping Source

InteractionUse

Mapping Target

FeatureTyping

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureTyping::type () : Type [1]

  ```
  ElementMain_Mapping.getMapped(from.refersTo)
  ```

7.7.8.2.12 LifelineMembership_Mapping

Description

Creates a membership relationship for *memberElement()*.

General Mappings

ToFeatureMembership_Init
Mapping

Mapping Source

Lifeline

Mapping Target

FeatureMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureMembership::ownedMemberFeature () : Feature [0..1]

  ```
  self.memberFeature()
  ```

- FeatureMembership::memberFeature () : Feature [1]

  ```
  ElementMain_Mapping.getMapped(from)
  ```

7.7.8.2.13 LifelinePartUsage_Mapping

Description

A UML4SysML::Lifeline is mapped to a SysML v2 PartUsage.

General Mappings

ToPartUsage_Init
NamedElementMain_Mapping

Mapping Source

Lifeline

Mapping Target

PartUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- PartUsage::ownedRelationship () : Relationship [0..*]

```
self.oclAsType(ElementMain_Mapping).ownedRelationship()
->including(LifelineFeatureTyping_Mapping.getMapped(from))
```

7.7.8.2.14 LifelineFeatureTyping_Mapping

Description

Creates a feature typing relationship owned by the element *typedFeature()*.

General Mappings

ToFeatureTyping_Init
Mapping

Mapping Source

Lifeline

Mapping Target

FeatureTyping

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureTyping::type () : Type [1]

```
ElementMain_Mapping.getMapped(from.represents.type)
```

7.7.8.2.15 Message_Mapping

Description

A UML4SysML::Message is mapped to a SysML v2 ItemFlow.

General Mappings

ToItemFlow_Init
NamedElementMain_Mapping

Mapping Source

Message

Mapping Target

Flow

Owned Mappings

(none)

Applicable filters

(none)

7.7.8.2.16 MessageMembership_Mapping

Description

Creates a membership relationship for *memberElement()*.

General Mappings

ToFeatureMembership_Init

Mapping Source

Message

Mapping Target

FeatureMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureMembership::memberFeature () : Feature [1]

  ```
  ElementMain_Mapping.getMapped(from)
  ```

- FeatureMembership::ownedMemberFeature () : Feature [0..1]

  ```
  self.memberFeature()
  ```

7.7.8.2.17 StateInvariant_Mapping

Description

A UML4SysML::StateInvariant is mapped to a SysML v2 Invariant.

General Mappings

ToExpression_Init
Namespace_Mapping

Mapping Source

StateInvariant

Mapping Target

Invariant

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Invariant::ownedRelationship () : Relationship [0..*]

```
self.oclAsType(ElementMain_Mapping).ownedRelationship()
->including(StateInvariantFeatureTyping_Mapping.getMapped(from))
```

7.7.8.2.18 StateInvariantMembership_Mapping

Description

Creates a membership relationship for *memberElement()*.

General Mappings

ToFeatureMembership_Init
Mapping

Mapping Source

StateInvariant

Mapping Target

FeatureMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureMembership::memberFeature () : Feature [1]

  ```
  ElementMain_Mapping.getMapped(from)
  ```

- FeatureMembership::ownedMemberFeature () : Feature [0..1]

  ```
  self.memberFeature()
  ```

7.7.8.2.19 StateInvariantFeatureTyping_Mapping

Description

Creates a feature typing relationship owned by the element *typedFeature()*.

General Mappings

ToFeatureTyping_Init
Mapping

Mapping Source

StateInvariant

Mapping Target

FeatureTyping

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureTyping::type () : Type [1]

  ```
  ElementMain_Mapping.getMapped(from.invariant)
  ```

7.7.9 Packages

SYSML21-542: Transformation: overview tables shall be revised

7.7.9.1 Overview

SYSML21-542: Transformation: overview tables shall be revised

Table 8. List of High-Level Packages Mappings

SysML v1 Abstract Syntax/Stereotype	SysML v2 Abstract Syntax
Extension	ConnectionDefinition
ExtensionEnd	The mapping of the extension relationship is performed in the context of Stereotype_Mapping.
Image	No specific mapping specified yet.
Model	Package
Package	Package
PackageMerge	The concept of the PackageMerge relationship is not supported by SysML v2.
Profile	Package
ProfileApplication	No specific mapping specified yet.
Stereotype	MetadataDefinition

7.7.9.2 Mapping Specifications

7.7.9.2.1 ElementImport_Mapping

Description

A UML4SysML::ElementImport is mapped to a SysMLv2 MembershipImport. The following shows an example of what the textual SysML v2 syntax of the result of the transformation may look like.

```
package SysMLv1Package1 {
    import SysMLv1Package2::SysMLv1Block;
    import SysMLv1Package2::SysMLv1ValueType;
}
package SysMLv1Package2 {
   part def SysMLv1Block;
   attribute def SysMLv1ValueType;
}
```

General Mappings

ToMembershipImport_Init
NamedElementMain_Mapping

Mapping Source

ElementImport

Mapping Target

MembershipImport

Owned Mappings

(none)

Applicable filters

This mapping applies only if the following (OCL) condition implemented by the operation *filter(src : Element) : Boolean* is verified:

```
if src.oclIsKindOf(UML::ElementImport) then
    Helper.hasMainMapping(src.oclAsType(UML::ElementImport).importedElement)
else
    false
endif
```

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- MembershipImport::importedMembership () : Namespace [1]

  ```
  ElementOwningMembership_Mapping.getMapped(from.importedElement)
  ```

- MembershipImport::visibility () : VisibilityKind [1]

  ```
  Helper.getKerMLVisibilityKind(from.visibility)
  ```

- MembershipImport::importedMemberName () : String [0..1]

  ```
  from.alias
  ```

7.7.9.2.2 Model_Mapping

Description

SysMLv2 has no explicit model element for a model. The UML4SysML::Model element is mapped to a SysMLv2 Package. The property "viewpoint" is mapped to a metadata defined in the SysML v1 library. The expected SysML v2 textual notation of a UML4SysML::Model with URI and viewpoint is as follows. If URI or viewpoint are not set in the source model, the metadata is not generated.

```
package SysMLv1Model {
  @SysMLv1Library::PackageData {URI="https://omg.org";}
  @SysMLv1Library::ModelData {'viewpoint'="The viewpoint of the model element.";}
}
```

General Mappings

Package_Mapping

Mapping Source

Model

Mapping Target

Package

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Package::ownedRelationship () : Relationship [0..*]

```
let relationships : Set(KerML::Relationship) =
    self.oclAsType(Package_Mapping).ownedRelationship() in
if from.viewpoint.oclIsUndefined() or from.viewpoint = '' then
    relationships
else
    relationships
    ->including(ModelViewpointMetadataMembership_Mapping.getMapped(from))
endif
```

7.7.9.2.3 ModelViewpointMetadataUsage_Mapping

Description

The mapping class creates the annotating feature to annotate the generated Package element with metadata to store the UML4SysML::Model::viewpoint property.

General Mappings

ToMetadataUsage_Init
Mapping

Mapping Source

Model

Mapping Target

MetadataUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- MetadataUsage::declaredName () : String [0..1]

```
'viewpoint'
```

- MetadataUsage::ownedRelationship () : Relationship [0..*]

```
Set{ModelViewpointMetadataFeatureTyping_Mapping.getMapped(from), ModelViewpointMetadataFeatur
```

7.7.9.2.4 ModelViewpointMetadataFeatureMembership_Mapping

Description

The mapping class creates the feature membership relationship for the metadata feature to store the UML4SysML::Model::viewpoint property.

General Mappings

ToFeatureMembership_Init
Mapping

Mapping Source

Model

Mapping Target

FeatureMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureMembership::ownedMemberFeature () : Feature [0..1]

```
ModelViewpointMetadataReferenceUsage_Mapping.getMapped(from)
```

7.7.9.2.5 ModelViewpointMetadataReferenceUsage_Mapping

Description

The mapping class creates the MetadataFeature for the mapping of the property UML4SysML::Model::viewpoint.

General Mappings

ToReferenceUsage_Init
Mapping

Mapping Source

Model

Mapping Target

ReferenceUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ReferenceUsage::ownedRelationship () : Relationship [0..*]

```
Set{ModelViewpointMetadataRedefinition_Mapping.getMapped(from),
ModelViewpointMetadataFeatureValue_Mapping.getMapped(from)}
```

7.7.9.2.6 ModelViewpointMetadataFeatureTyping_Mapping

Description

The mapping class creates the FeatureTyping relationship for the AnnotatingFeature for the metadata to store the UML4SysML::Model::viewpoint property.

General Mappings

ToFeatureTyping_Init
Mapping

Mapping Source

Model

Mapping Target

FeatureTyping

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureTyping::type () : Type [1]

```
SysMLv2::MetadataDefinition.allInstances()
->any(m | m.qualifiedName = 'SysMLv1Library::ModelData')
```

7.7.9.2.7 ModelViewpointMetadataMembership_Mapping

Description

The mapping class creates a membership relationship for the metadata feature value for the UML4SysML::Model::viewpoint property.

General Mappings

ToOwningMembership_Init
Mapping

Mapping Source

Model

Mapping Target

OwningMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- OwningMembership::ownedMemberElement () : Element [1]

  ```
  ModelViewpointMetadataUsage_Mapping.getMapped(from)
  ```

7.7.9.2.8 ModelViewpointMetadataFeatureValue_Mapping

Description

The mapping class maps the value of the property UML4SysML::Model::viewpoint.

General Mappings

ToFeatureValue_Init
Mapping

Mapping Source

Model

Mapping Target

FeatureValue

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureValue::value () : Expression [1]

```
ModelViewpointValue_Mapping.getMapped(from)
```

7.7.9.2.9 ModelViewpointMetadataRedefinition_Mapping

Description

The mapping class creates the redefinition of the attribute for the metadata UML4SysML::Model::viewpoint.

General Mappings

ToRedefinition_Init
Mapping

Mapping Source

Model

Mapping Target

Redefinition

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Redefinition::redefinedFeature () : Feature [1]

```
let m : SYSML2::Membership =
    SYSML2::AttributeUsage.allInstances()
    ->collect(dt | dt.owningRelationship)
    ->select(r | r.oclIsKindOf(SYSML2::Membership))
    ->any(m | m.memberName = 'viewpoint') in
if (m.oclIsUndefined()) then
    invalid
else
```

```
        m.memberElement
    endif
```

7.7.9.2.10 ModelViewpointValue_Mapping

Description

The mapping class maps the value expression of the property UML4SysML::Model::viewpoint.

General Mappings

ToExpression_Init
Mapping

Mapping Source

Model

Mapping Target

LiteralString

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- LiteralString::value () : String [1]

```
LiteralString_Factory.create(from.viewpoint)
```

7.7.9.2.11 Package_Mapping

Description

A UML4SysML::Package is mapped to a SysML v2 Package. The property "URI" is mapped to a metadata if it has a value. The expected SysML v2 textual notation of a UML4SysML::Package is as follows:

```
package ThisIsAPackageWithURI {
  metadata SysMLv1Library::PackageData {URI="https://omg.org";}
}
```

General Mappings

Namespace_Mapping

Mapping Source

Package

Mapping Target

Package

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Package::ownedRelationship () : Relationship [0..*]

  ```
  Helper.packageOwnedRelationship(from)
  ```

7.7.9.2.12 PackageImport_Mapping

Description

A UML4SysML::PackageImport is mapped to a SysML v2 NamespaceImport. The following shows an example of what the textual SysML v2 syntax of the result of the transformation may look like.

```
import SysMLv1Package::*;
```

General Mappings

ToNamespaceImport_Init
ElementMain_Mapping

Mapping Source

PackageImport

Mapping Target

NamespaceImport

Owned Mappings

(none)

Applicable filters

This mapping applies only if the following (OCL) condition implemented by the operation *filter(src : Element) : Boolean* is verified:

```
if src.oclIsKindOf(UML::PackageImport) then
    Helper.isInScope(src.oclAsType(UML::PackageImport).importedPackage)
else
    false
endif
```

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- NamespaceImport::visibility () : VisibilityKind [0..1]

  ```
  Helper.getKerMLVisibilityKind(from.visibility)
  ```

- NamespaceImport::importedNamespace () : Namespace [1]

  ```
  Namespace_Mapping.getMapped(from.importedPackage)
  ```

7.7.9.2.13 PackageURIMetadataUsage_Mapping

Description

The mapping class creates the annotating feature to annotate the generated Package element with metadata to store the UML4SysML::Package::URI property.

General Mappings

ToMetadataUsage_Init
Mapping

Mapping Source

Package

Mapping Target

MetadataUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- MetadataUsage::ownedRelationship () : Relationship [0..*]

  ```
  Set{PackageURIFeatureTyping_Mapping.getMapped(from),
  PackageURIFeatureMembership_Mapping.getMapped(from)}
  ```

- MetadataUsage::declaredName () : String [0..1]

  ```
  'URI'
  ```

7.7.9.2.14 PackageURIFeatureMembership_Mapping

Description

The mapping class creates the feature membership relationship for the metadata feature to store the UML4SysML::Package::URI property.

General Mappings

ToFeatureMembership_Init
Mapping

Mapping Source

Package

Mapping Target

FeatureMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureMembership::ownedMemberFeature () : Feature [1]

  ```
  PackageURIMetadataReferenceUsage_Mapping.getMapped(from)
  ```

7.7.9.2.15 PackageURIFeatureTyping_Mapping

Description

The mapping class creates the FeatureTyping relationship for the AnnotatingFeature for the metadata to store the UML4SysML::Package::URI property.

General Mappings

ToFeatureTyping_Init
Mapping

Mapping Source

Package

Mapping Target

FeatureTyping

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureTyping::type () : Type [1]

```
let m: SysMLv2::Membership = SysMLv2::AttributeDefinition.allInstances()
->collect(dt | dt.owningRelationship)
->select(r | r.oclIsKindOf(SysMLv2::Membership))
->any(m | m.memberName =  'PackageData' ) in

if (m.oclIsUndefined()) then
          invalid
else
    m.memberElement
endif
```

7.7.9.2.16 PackageURIMetadataReferenceUsage_Mapping

Description

The mapping class creates the MetadataFeature for the mapping of the property UML4SysML::Package::URI.

General Mappings

ToReferenceUsage_Init
Mapping

Mapping Source

Package

Mapping Target

ReferenceUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ReferenceUsage::ownedRelationship () : Relationship [0..*]

```
Set{PackageURIRedefinition_Mapping.getMapped(from),
PackageURIMetadataFeatureValue_Mapping.getMapped(from)}
```

7.7.9.2.17 PackageURIMetadataFeatureValue_Mapping

Description

The mapping class maps the value of the property UML4SysML::Package::URI.

General Mappings

ToFeatureValue_Init
Mapping

Mapping Source

Package

Mapping Target

FeatureValue

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureValue::value () : Expression [1]

  ```
  PackageURIValue_Mapping.getMapped(from)
  ```

- FeatureValue::featureWithValue () : Feature [1]

  ```
  packageURIMetadataReferenceUsage.to
  ```

7.7.9.2.18 PackageURIMetadataMembership_Mapping

Description

The mapping class creates a membership relationship for the metadata feature value for the UML4SysML::Package::URI property.

General Mappings

ToOwningMembership_Init
Mapping

Mapping Source

Package

Mapping Target

OwningMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- OwningMembership::ownedMemberElement () : Element [1]

```
PackageURIMetadataUsage_Mapping.getMapped(from)
```

7.7.9.2.19 PackageURIRedefinition_Mapping

Description

The mapping class creates the redefinition of the attribute for the metadata UML4SysML::Package::URI.

General Mappings

ToRedefinition_Init
Mapping

Mapping Source

Package

Mapping Target

Redefinition

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Redefinition::redefinedFeature () : Feature [1]

```
let m : SysMLv2::Membership =
    SysMLv2::AttributeUsage.allInstances()
    ->collect(dt | dt.owningRelationship)
    ->select(r | r.oclIsKindOf(SYSML2::Membership))
    ->any(m | m.memberName = 'URI') in
```

```
if (m.oclIsUndefined()) then
    invalid
else
    m.memberElement
endif
```

7.7.9.2.20 PackageURIValue_Mapping

Description

The mapping class maps the value expression of the property UML4SysML::Package::URI.

General Mappings

ToExpression_Init
Mapping

Mapping Source

Package

Mapping Target

LiteralString

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- LiteralString::value () : String [1]

```
from.URI
```

7.7.9.2.21 Profile_Mapping

Description

A UML4SysML::Profile is mapped to a SysML v2 Package.

General Mappings

Package_Mapping

Mapping Source

Profile

Mapping Target

Package

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Package::ownedRelationship () : Relationship [0..*]

```
self.oclAsType(Package_Mapping).ownedRelationship()
->including(ProfileMetadataMembership_Mapping.getMapped(from))
```

7.7.9.2.22 ProfileMetadataMembership_Mapping

Description

The mapping class creates a membership relationship for the metadata feature value for the UML4SysML::Model::viewpoint property.

General Mappings

ToOwningMembership_Init
Mapping

Mapping Source

Profile

Mapping Target

OwningMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- OwningMembership::ownedMemberElement () : Element [1]

```
ProfileMetadataUsage_Mapping.getMapped(from)
```

7.7.9.2.23 ProfileMetadataUsage_Mapping

Description

The mapping class creates the annotating feature to annotate the generated Package element with metadata to store the UML4SysML::Model::viewpoint property.

General Mappings

ToMetadataUsage_Init
Mapping

Mapping Source

Profile

Mapping Target

MetadataUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- MetadataUsage::declaredName () : String [0..1]

  ```
  'Profile'
  ```

7.7.9.2.24 StereotypeMetadataDefinition_Mapping

Description

A UML4SysML::Stereotype is mapped to a SysML v2 MetadataDefinition.

General Mappings

Class_Mapping

Mapping Source

Stereotype

Mapping Target

MetadataDefinition

Owned Mappings

(none)

Applicable filters

(none)

7.7.9.2.25 StereotypeMetadataDefinitionMembership_Mapping

Description

Creates a membership relationship for *memberElement()*.

General Mappings

ElementOwningMembership_Mapping

Mapping Source

Stereotype

Mapping Target

OwningMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- OwningMembership::ownedMemberElement () : Element [0..1]

  ```
  ElementMain_Mapping.getMapped(from)
  ```

7.7.9.2.26 StereotypeOccurenceUsage_Mapping

Description

The mapping class maps the usage of a stereotype to a SysML v2 OccurrenceUsage.

General Mappings

ToOccurrenceUsage_Init
Mapping

Mapping Source

Stereotype

Mapping Target

OccurrenceUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- OccurrenceUsage::ownedRelationship () : Relationship [0..*]

```
Set{StereotypeOccurenceUsageFeatureTyping_Mapping.getMapped(from),
StereotypeOccurenceUsageMultiplicityMembership_Mapping.getMapped(from)}
```

7.7.9.2.27 StereotypeOccurenceUsageFeatureTyping_Mapping

Description

Creates a feature typing relationship owned by the element *typedFeature()*.

General Mappings

ToFeatureTyping_Init
Mapping

Mapping Source

Stereotype

Mapping Target

FeatureTyping

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureTyping::type () : Type [1]

```
StereotypeOccurenceDefinition_Mapping.getMapped(from)
```

7.7.9.2.28 StereotypeOccurenceUsageMembership_Mapping

Description

Creates a membership relationship for *memberElement()*.

General Mappings

ToMembership_Init
Mapping

Mapping Source

Stereotype

Mapping Target

Membership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Membership::memberElement () : Element [1]

  ```
  StereotypeOccurenceUsage_Mapping.getMapped(from)
  ```

7.7.9.2.29 StereotypeOccurenceUsageMultiplicityMembership_Mapping

Description

Creates a membership relationship for *memberElement()*.

General Mappings

ToMembership_Init
Mapping

Mapping Source

Stereotype

Mapping Target

Membership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Membership::memberElement () : Element [1]

  ```
  self.ownedMemberElement()
  ```

- Membership::ownedMemberElement () : Element [0..1]

  ```
  StereotypeOccurenceUsageMultiplicityRange_Mapping.getMapped(from)
  ```

7.7.9.2.30 StereotypeOccurenceUsageMultiplicityRange_Mapping

Description

The mapping class creates the multiplicity range element for the UML4SysML::Stereotype mapping.

General Mappings

ToFeature_Init
Mapping

Mapping Source

Stereotype

Mapping Target

MultiplicityRange

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- MultiplicityRange::ownedRelationship () : Relationship [0..*]

  ```
  Set{StereotypeOccurenceUsageMultiplicityRangeMembership_Mapping.getMapped(from)}
  ```

7.7.9.2.31 StereotypeOccurenceUsageMultiplicityRangeInfinity_Mapping

Description

The mapping class creates the literal infinity element for the multiplicity range element for the UML4SysML::Stereotype mapping.

General Mappings

ToExpression_Init
Mapping

Mapping Source

Stereotype

Mapping Target

LiteralInfinity

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- LiteralInfinity::ownedRelationship () : Relationship [0..*]

  ```
  Set{StereotypeOccurenceUsageInfinityReturnParameterMembership_Mapping.getMapped(from)}
  ```

7.7.9.2.32 StereotypeOccurenceUsageInfinityReturnParameter_Mapping

Description

The mapping class creates the return parameter relationship for the literal infinity element for the multiplicity range element for the UML4SysML::Stereotype mapping.

General Mappings

ToFeature_Init
Mapping

Mapping Source

Stereotype

Mapping Target

Feature

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Feature::direction () : FeatureDirectionKind [0..1]

```
SysMLv2::FeatureDirectionKind::out
```

7.7.9.2.33 StereotypeOccurenceUsageInfinityReturnParameterMembership_Mapping

Description

General Mappings

ToReturnParameterMembership_Init
Mapping

Mapping Source

Stereotype

Mapping Target

ReturnParameterMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ReturnParameterMembership::ownedMemberParameter () : Feature [0..1]

```
StereotypeOccurenceUsageInfinityReturnParameter_Mapping.getMapped(from)
```

- ReturnParameterMembership::ownedRelatedElement () : Element [0..*]

```
let member: KerML::Element = self.ownedMemberParameter() in
if member.oclIsUndefined() then
    Set{}
else
    Set{self.ownedMemberParameter()}
endif
```

- ReturnParameterMembership::memberParameter () : Feature [1]

```
self.ownedMemberParameter()
```

7.7.9.2.34 StereotypeOccurenceUsageMultiplicityRangeMembership_Mapping

Description

Creates a membership relationship for *memberElement()*.

General Mappings

ToMembership_Init
Mapping

Mapping Source

Stereotype

Mapping Target

Membership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Membership::ownedMemberElement () : Element [0..1]

```
StereotypeOccurenceUsageMultiplicityRangeInfinity_Mapping.getMapped(from)
```

- Membership::memberElement () : Element [1]

```
self.ownedMemberElement()
```

7.7.10 SimpleClassifiers

7.7.10.1 Overview

SYSML21-542: Transformation: overview tables shall be revised

Table 9. List of High-Level SimpleClassifiers Mappings

SysML v1 Abstract Syntax/Stereotype	SysML v2 Abstract Syntax
DataType	AttributeDefinition
Enumeration	EnumerationDefinition
EnumerationLiteral	ConnectionUsage, EnumerationUsage

SysML v1 Abstract Syntax/Stereotype	SysML v2 Abstract Syntax
Interface	PortDefinition
InterfaceRealization	Dependency
PrimitiveType	AttributeDefinition
Reception	ItemUsage
Signal	ItemDefinition

7.7.10.2 Mapping Specifications

7.7.10.2.1 Attribute_Mapping

Description

An UML4SysML::Property is mapped to a SysMLv2 AttributeUsage.

General Mappings

PropertyCommon_Mapping
NamedElementMain_Mapping

Mapping Source

Property

Mapping Target

AttributeUsage

Owned Mappings

(none)

Applicable filters

This mapping applies only if the following (OCL) condition implemented by the operation *filter(src : Element) : Boolean* is verified:

```
if src.oclIsKindOf(UML::Property) and not
    Helper.hasStereotypeApplied(src.owner,
        'SysML::ConstraintBlocks::ConstraintBlock') then
        let p: UML::Property = src.oclAsType(UML::Property) in
        if p.type.oclIsUndefined() then
            false
        else
         p.type.oclIsKindOf(UML::DataType) and
         (p.association.oclIsUndefined() or p.association.ownedEnd->excludes(p))
        endif
else
    false
endif
```

Mapping rules

The mapping class only has inherited rules. See the mapping classes in the general mapping section for details.

7.7.10.2.2 AttributeRedefined_Mapping

Description

An UML4SysML::SimpleClassifiers::Property is mapped to a SysML v2 AttributeUsage.

General Mappings

PropertyCommon_Mapping

Mapping Source

Property

Mapping Target

ReferenceUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ReferenceUsage::ownedRelationship () : Relationship [0..*]

```
let typing: KerML::FeatureTyping =
    AssociationToFeatureTyping_Mapping.getMapped(from) in
let subsetting: Set(KerML::Subsetting) =
    from.subsettedProperty
    ->collect(p | PropertySubsetting_Mapping.getMapped(from, p))->asSet() in
let subsettingMultiplicityTyping: Set(KerML::Relationship) =
    subsetting
    ->union(Set{AttributeRedefinedRedefinition_Mapping.getMapped(from)})->union(
        if typing.oclIsUndefined() then
            Set{MultiplicityMembership_Mapping.getMapped(from)}
        else
            Set{MultiplicityMembership_Mapping.getMapped(from), typing}
        endif)->asSet() in
if from.defaultValue.oclIsUndefined() then
    subsettingMultiplicityTyping
else
    subsettingMultiplicityTyping
    ->including(PropertyDefaultValue_Mapping.getMapped(from))
endif
```

7.7.10.2.3 AttributeRedefinedRedefinition_Mapping

Description

Creates a redefinition relationship for the *redefiningFeature()* and the *redefinedFeature()*.

General Mappings

ToRedefinition_Init
Mapping

Mapping Source

Property

Mapping Target

Redefinition

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Redefinition::redefinedFeature () : Feature [1]

```
from.redefinedProperty.get(0)
```

7.7.10.2.4 AttributeRedefinedMembership_Mapping

Description

Creates a membership relationship for *memberElement()*.

General Mappings

ElementFeatureMembership_Mapping

Mapping Source

Element

Mapping Target

FeatureMembership

Owned Mappings

(none)

Applicable filters

This mapping applies only if the following (OCL) condition implemented by the operation *filter(src : Element) : Boolean* is verified:

```
src.oclIsKindOf(UML::Property)
and (src.oclAsType(UML::Property).redefinedElement->size() > 0)
```

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureMembership::ownedMemberFeature () : Feature [0..1]

  ```
  AttributeRedefined_Mapping.getMapped(from)
  ```

7.7.10.2.5 AttributeRedefinedFeatureTyping_Mapping

Description

Creates a feature typing relationship owned by the element *typedFeature()*.

General Mappings

StructuralFeatureToFeatureTyping_Mapping

Mapping Source

StructuralFeature

Mapping Target

FeatureTyping

Owned Mappings

(none)

Applicable filters

(none)

7.7.10.2.6 BehavioredClassifier_Mapping

Description

The abstract mapping class maps the abstract metaclass UML4SysML::BehavioredClassifiers to a SysMLv2 Classifier. The mapping class is used by concrete mapping classes, for example, Block_Mapping.

General Mappings

Classifier_Mapping

Mapping Source

BehavioredClassifier

Mapping Target

Classifier

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Classifier::ownedRelationship () : Relationship [0..*]

```
let toElementFMS: Set(UML::Element) =
    from.ownedElement->select(e | (e.oclIsKindOf(UML::Property) and
        (e.oclAsType(UML::Property).redefinedProperty->size() = 0)) or
         e.oclIsKindOf(UML::Operation) or e.oclIsKindOf(UML::Connector)) in
let redefinedAttributes: Set(UML::Element) =
    from.ownedElement->select(e | from.oclIsKindOf(UML::DataType) and
        (e.oclAsType(UML::Property).redefinedProperty->size() > 0)) in
let generalizations : Set(UML::Generalization) =
    from.ownedElement
    ->select(e | e.oclIsKindOf(UML::Generalization)) in
let constraints : Set(UML::Constraint) =
    UML::Constraint.allInstances()
    ->select( c | c.constrainedElement->includes(from)) in
let toElementOMS: Set(UML::Element) =
    (((from.ownedElement - toElementFMS) - redefinedAttributes) -
    generalizations) - from.ownedComment  in
let relationships: Sequence(KerML::Relationship) =
toElementOMS->collect(e | ElementOwningMembership_Mapping.getMapped(e))->asSet()
->union(toElementFMS->collect(e |
    ElementFeatureMembership_Mapping.getMapped(e))->asSet())
->union(constraints->collect(e |
    ConstrainedElementFeatureMembership_Mapping.getMapped(e))->asSet())
->union(redefinedAttributes->collect(e |
    AttributeRedefinedMembership_Mapping.getMapped(e))->asSet())
->union(generalizations->collect(e |
    Generalization_Mapping.getMapped(e))->asSet())
->union(self.oclAsType(ElementMain_Mapping).ownedRelationship()) in
if from.classifierBehavior.oclIsUndefined() then
    relationships
else
    relationships
    ->including(BehavioredClassifierFeatureMembership_Mapping.getMapped(from))
endif
```

7.7.10.2.7 BehavioredClassifierFeatureMembership_Mapping

Description

General Mappings

ToFeatureMembership_Init
Mapping

Mapping Source

BehavioredClassifier

Mapping Target

FeatureMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureMembership::ownedMemberFeature () : Feature [0..1]

  ```
  BehavioredClassifierActionUsage_Mapping.getMapped(from)
  ```

7.7.10.2.8 BehavioredClassifierFeatureTyping_Mapping

Description

Creates a feature typing relationship owned by the element *typedFeature()*.

General Mappings

ToFeatureTyping_Init
Mapping

Mapping Source

BehavioredClassifier

Mapping Target

FeatureTyping

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureTyping::type () : Type [1]

```
from
```

7.7.10.2.9 BehavioredClassifierActionUsage_Mapping

Description

The BehavioredClassifierToPerformActionUsage_Mapping class creates a PerformActionUsage element to call the transformed SysML v1 classifier behavior.

General Mappings

ToActionUsage_Init
Mapping

Mapping Source

BehavioredClassifier

Mapping Target

ActionUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ActionUsage::ownedRelationship () : Relationship [0..*]

  ```
  Set{BehavioredClassifierFeatureTyping_Mapping.getMapped(from)}
  ```

- ActionUsage::declaredName () : String [0..1]

  ```
  'classifierBehavior'
  ```

7.7.10.2.10 DataType_Mapping

Description

A UML4SysML::SimpleClassifiers::DataType is mapped to a SysML v2 AttributeDefinition. The mapping also cover the transformation of UML4SysML::PrimitiveType elements.

The following shows an example of what the textual SysML v2 syntax of the result of the transformation may look like.

```
part def SysMLv1Block {
        attribute sysMLv1Property : ScalarValues::Integer;
}
```

General Mappings

Classifier_Mapping

Mapping Source

DataType

Mapping Target

AttributeDefinition

Owned Mappings

(none)

Applicable filters

(none)

7.7.10.2.11 Enumeration_Mapping

Description

A UML4SysML::Enumeration is mapped to a SysML v2 EnumerationDefinition.

The following shows an example of what the textual SysML v2 syntax of the result of the transformation may look like.

```
enum def SysMLv1Enumeration {
        enum sysMLv1Literal1;
        enum sysMLv1Literal2;
}
```

General Mappings

DataType_Mapping

Mapping Source

Enumeration

Mapping Target

EnumerationDefinition

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- EnumerationDefinition::isVariation () : Boolean [1]

  ```
  true
  ```

- EnumerationDefinition::ownedRelationship () : Relationship [0..*]

  ```
  self.oclAsType(Classifier_Mapping).ownedRelationship()
  ->union(from.ownedLiteral->collect(e |
    EnumerationVariantMembership_Mapping.getMapped(e))->asSet())
  ```

7.7.10.2.12 EnumerationLiteral_Mapping

Description

A UML4SysML::EnumerationLiteral is mapped to a SysML v2 EnumerationUsage.

General Mappings

ToFeature_Init
InstanceSpecification_Mapping

Mapping Source

EnumerationLiteral

Mapping Target

EnumerationUsage

Owned Mappings

(none)

Applicable filters

(none)

7.7.10.2.13 EnumerationVariantMembership_Mapping

Description

The EnumerationVariantMembership_Mapping class creates the variant membership relationship between the enumeration definition and a enumeration usage.

General Mappings

ToOwningMembership_Init
Mapping

Mapping Source

EnumerationLiteral

Mapping Target

VariantMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- VariantMembership::ownedMemberElement () : Element [1]

  ```
  from
  ```

7.7.10.2.14 Interface_Mapping

Description

A UML4SysML::Interface is mapped to a SysMLv2 PortDefinition. The mapping also includes the generation of an appropriate ConjugatedPortDefinition. That mappings is performed by the mapping classes InterfaceConjugatedPortDefinitionMembership_Mapping, InterfacePortConjugation_Mapping, and InterfaceConjugatedPortDefinition_Mapping.

The following shows an example of what the textual SysML v2 syntax of the result of the transformation may look like.

```
port def SysMLv1Interface {
        attribute sysMLv1Property;
}
```

General Mappings

ToPortDefinition_Init
Classifier_Mapping

Mapping Source

Interface

Mapping Target

PortDefinition

Owned Mappings

- conjugatedPortDefinitionMembership : InterfaceConjugatedPortDefinitionMembership_Mapping

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- PortDefinition::ownedRelationship () : Relationship [0..*]

```
self.oclAsType(Classifier_Mapping).ownedRelationship()
->including(conjugatedPortDefinitionMembership)
```

7.7.10.2.15 InterfaceConjugatedPortDefinition_Mapping

Description

As part of the mapping from a UML4SysML::Interface to a SysMLv2 PortDefinition, this mapping class is used to create the appropriate ConjugatedPortDefinition.

General Mappings

ToPortDefinition_Init
Mapping

Mapping Source

Interface

Mapping Target

ConjugatedPortDefinition

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ConjugatedPortDefinition::declaredName () : String [0..1]

```
'~'+from.name
```

- ConjugatedPortDefinition::ownedRelationship () : Relationship [0..*]

```
Set{InterfacePortConjugation_Mapping.getMapped(from)}
```

7.7.10.2.16 InterfaceConjugatedPortDefinitionMembership_Mapping

Description

As part of the mapping from a UML4SysML::Interface to a SysML v2 PortDefinition, this mapping class is used to create the membership relationship for the ConjugatedPortDefinition.

General Mappings

ToOwningMembership_Init
Mapping

Mapping Source

Interface

Mapping Target

OwningMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- OwningMembership::ownedMemberElement () : Element [1]

  ```
  InterfaceConjugatedPortDefinition_Mapping.getMapped(from)
  ```

7.7.10.2.17 InterfacePortConjugation_Mapping

Description

As part of the mapping from a UML4SysML::Interface to a SysML v2 PortDefinition, this mapping class is used to create the appropriate PortConjugation relationship.

General Mappings

ToRelationship_Init
Mapping

Mapping Source

Interface

Mapping Target

PortConjugation

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- PortConjugation::conjugatedType () : Type [1]

```
SysMLv2::ConjugatedPortDefinition.allInstances()
->collect(cpd | cpd.owningRelationship)
->select(r | r.oclIsKindOf(SysMLv2::Membership))
->any(m | m.memberName = from.name)
```

- PortConjugation::originalPortDefinition () : PortDefinition [1]

```
from
```

7.7.10.2.18 InterfaceRealization_Mapping

Description

A UML4SysML::InterfaceRealization is mapped to a SysMLv2 Subclassification relationship.

General Mappings

ToSpecialization_Init
Mapping

Mapping Source

InterfaceRealization

Mapping Target

Subclassification

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Subclassification::subclassifier () : Type [1]

```
Classifier_Mapping.getMapped(from.specific)
```

- Subclassification::superclassifier () : Type [1]

```
Classifier_Mapping.getMapped(from.general)
```

7.7.10.2.19 PrimitiveType_Mapping

Description

The PrimitiveType_Mapping class maps a UML4SysML::PrimitiveType to a SysML v2 AttributeDefinition.

General Mappings

DataType_Mapping

Mapping Source

PrimitiveType

Mapping Target

AttributeDefinition

Owned Mappings

(none)

Applicable filters

(none)

7.7.10.2.20 Reception_Mapping

Description

A UML4SysML::Reception is mapped to a SysML v2 AttributeUsage with feature direction "in".

General Mappings

BehavioralFeature_Mapping

Mapping Source

Reception

Mapping Target

ItemUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ItemUsage::ownedRelationship () : Relationship [0..*]

```
self.oclAsType(ElementMain_Mapping).ownedRelationship()
```

```
->including(ReceptionFeatureTyping_Mapping.getMapped(from))
```

- ItemUsage::direction () : FeatureDirectionKind [0..1]

```
SysMLv2::FeatureDirectionKind::in
```

7.7.10.2.21 ReceptionFeatureTyping_Mapping

Description

A UML4SysML::Reception is mapped to SysML v2 AttributeUsage. The ReceptionToFeatureTyping_Mapping class creates the type of the AttributeUsage which is the Signal of the Reception.

General Mappings

TypedElementFeatureTyping_Mapping

Mapping Source

Reception

Mapping Target

FeatureTyping

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureTyping::type () : Type [1]

```
Classifier_Mapping.getMapped(from.signal)
```

7.7.10.2.22 Signal_Mapping

Description

A UML4SysML::Signal is mapped to a SysML v2 ItemDefinition.

General Mappings

Classifier_Mapping

Mapping Source

Signal

Mapping Target

ItemDefinition

Owned Mappings

(none)

Applicable filters

(none)

7.7.11 StateMachines

7.7.11.1 Overview

SYSML21-542: Transformation: overview tables shall be revised

Table 10. List of High-Level StateMachines Mappings

SysML v1 Abstract Syntax/Stereotype	SysML v2 Abstract Syntax
ConnectionPointReference	StateUsage
FinalState	StateUsage
Pseudostate	ActionUsage, StateUsage
Region	StateUsage
State	StateUsage
StateMachine	StateDefinition
Transition	TransitionUsage

7.7.11.2 Mapping Specifications

7.7.11.2.1 ChangeTriggerReferenceUsage_Mapping

Description

*** not specified yet ***

General Mappings

UniqueMapping
ToReferenceUsage_Init

Mapping Source

Trigger

Mapping Target

ReferenceUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ReferenceUsage::ownedRelationship () : Relationship [0..*]

  ```
  Set{ChangeTriggerReferenceSubsetting_Mapping.getMapped(from)}
  ```

- ReferenceUsage::isEnd () : Boolean [1]

  ```
  true
  ```

7.7.11.2.2 CommonPseudostate_Mapping

Description

Abstract mapping class for common rules for pseudostates mappings.

General Mappings

Namespace_Mapping

Mapping Source

Pseudostate

Mapping Target

Namespace

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Namespace::ownedRelationship () : Relationship [0..*]

  ```
  let toFeatureMS : Set(UML::Element) =
      from.ownedElement->select(e | e.oclIsKindOf(UML::Region))->asSet() in
  let toElementOMS : Set(UML::Element) =
      from.ownedElement - toFeatureMS in
  toElementOMS
  ->collect(e | ElementOwningMembership_Mapping.getMapped(e))->asSet()
  ->union(toFeatureMS
  ->collect(e | ElementFeatureMembership_Mapping.getMapped(e))->asSet())
  ->union(self.oclAsType(ElementMain_Mapping).ownedRelationship())
  ```

7.7.11.2.3 ConnectionPointReference_Mapping

Description

A UML4SysML::ConnectionPointReference element is mapped to a SysML v2 StateUsage.

General Mappings

Namespace_Mapping
ToStateUsage_Init

Mapping Source

ConnectionPointReference

Mapping Target

StateUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- StateUsage::ownedRelationship () : Relationship [0..*]

```
let toFeatureMS : Set(UML::Element) =
  from.ownedElement->select(e | e.oclIsKindOf(UML::Region)) in
let toElementOMS : Set(UML::Element) =
  (from.ownedElement - toFeatureMS) - from.ownedComment in
toElementOMS->collect(e | ElementOwningMembership_Mapping.getMapped(e))
  ->asSet()
->union(toFeatureMS
  ->collect(e | ElementFeatureMembership_Mapping.getMapped(e))->asSet())
->union(self.oclAsType(ElementMain_Mapping).ownedRelationship())
```

- StateUsage::isComposite () : Boolean [1]

```
false
```

7.7.11.2.4 DoBehaviorStateSubactionMembership_Mapping

Description

Creates a state subaction membership relationship for *memberFeature()*.

General Mappings

StateBehaviorStateSubactionMembership_Mapping

Mapping Source

Behavior

Mapping Target

StateSubactionMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- StateSubactionMembership::kind () : StateSubactionKind [1]

  ```
  SysMLv2::SubactionKind::do
  ```

7.7.11.2.5 EntryBehaviorStateSubactionMembership_Mapping

Description

Creates a state subaction membership relationship for *memberFeature()*.

General Mappings

StateBehaviorStateSubactionMembership_Mapping

Mapping Source

Behavior

Mapping Target

StateSubactionMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- StateSubactionMembership::kind () : StateSubactionKind [1]

```
SysMLv2::SubactionKind::entry
```

7.7.11.2.6 ExitBehaviorStateSubactionMembership_Mapping

Description

Creates a state subaction membership relationship for *memberFeature()*.

General Mappings

StateBehaviorStateSubactionMembership_Mapping

Mapping Source

Behavior

Mapping Target

StateSubactionMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- StateSubactionMembership::kind () : StateSubactionKind [1]

  ```
  SysMLv2::SubactionKind::exit
  ```

7.7.11.2.7 FinalState_Mapping

Description

A UML4SysML::FinalState is mapped to a SysML v2 StateUsage. The details of the mapping are not defined yet.

General Mappings

State_Mapping

Mapping Source

FinalState

Mapping Target

StateUsage

Owned Mappings

(none)

Applicable filters

This mapping applies only if the following (OCL) condition implemented by the operation *filter(src : Element) : Boolean* is verified:

```
src.oclIsTypeOf(UML::FinalState)
```

Mapping rules

The mapping class only has inherited rules. See the mapping classes in the general mapping section for details.

7.7.11.2.8 InitialState_Mapping

Description

The mapping class maps a Pseudostate with kind = initial to a SysML v2 ActionUsage.

General Mappings

CommonPseudostate_Mapping

Mapping Source

Pseudostate

Mapping Target

ActionUsage

Owned Mappings

(none)

Applicable filters

This mapping applies only if the following (OCL) condition implemented by the operation *filter(src : Element) : Boolean* is verified:

```
(src.kind = PseudostateKind::initial)
```

Mapping rules

The mapping class only has inherited rules. See the mapping classes in the general mapping section for details.

7.7.11.2.9 InitialStateSubactionMembership_Mapping

Description

Creates a StateSubactionMembership relationship.

General Mappings

ToStateSubactionMembership_Init
Mapping

Mapping Source

Pseudostate

Mapping Target

StateSubactionMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- StateSubactionMembership::kind () : StateSubactionKind [1]

  ```
  SysMLv2::SubactionKind::entry
  ```

- StateSubactionMembership::ownedMemberFeature () : Feature [1]

  ```
  InitialState_Mapping.getMapped(from)
  ```

7.7.11.2.10 PseudoState_Mapping

Description

A UML4SysML::PseudoState is mapped to a SysML v2 StateUsage.

General Mappings

CommonPseudostate_Mapping
ToStateUsage_Init

Mapping Source

Pseudostate

Mapping Target

StateUsage

Owned Mappings

(none)

Applicable filters

This mapping applies only if the following (OCL) condition implemented by the operation *filter(src : Element) : Boolean* is verified:

```
(src.kind <> PseudostateKind::initial)
```

Mapping rules

The mapping class only has inherited rules. See the mapping classes in the general mapping section for details.

7.7.11.2.11 Region_Mapping

Description

A UML4SysML::Region is mapped to SysML v2 StateUsage.

General Mappings

Namespace_Mapping
ToStateUsage_Init

Mapping Source

Region

Mapping Target

StateUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- StateUsage::ownedRelationship () : Relationship [0..*]

```
let initialState : Set(UML::Pseudostate) =
  from.ownedElement->select(e | e.oclIsKindOf(UML::Pseudostate)
    and e.oclAsType(UML::Pseudostate).kind = PseudostateKind::initial)->asSet() in
let toFeatureMS : Set(UML::Element) =
  from.ownedElement->select(e | e.oclIsKindOf(UML::Region))->asSet() in
let toElementOMS : Set(UML::Element) =
  ((from.ownedElement - initialState) - toFeatureMS) - from.ownedComment in
toElementOMS->collect(e | ElementOwningMembership_Mapping.getMapped(e))->asSet()
->union(toFeatureMS->collect(e |
  ElementFeatureMembership_Mapping.getMapped(e))->asSet())
->union(initialState->collect(e |
  InitialStateMembership_Mapping.getMapped(e))->asSet())
->union(self.oclAsType(ElementMain_Mapping).ownedRelationship())
```

7.7.11.2.12 State_Mapping

Description

A UML4SysML::State is mapped to a SysMLv2 StateUsage. If it is a composite state, it is mapped to a parallel state.

The following shows an example of what the textual SysML v2 syntax of the result of the transformation may look like.

```
state SysMLv1State parallel {
  entry; then SysMLv1StateA;
  state SysMLv1StateA;
}
```

General Mappings

Namespace_Mapping
ToStateUsage_Init

Mapping Source

State

Mapping Target

StateUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- StateUsage::ownedRelationship () : Relationship [0..*]

```
let toFeatureMS : Set(UML::Element) =
  from.ownedElement->select(e | e.oclIsKindOf(UML::Region))->asSet() in
let toElementOMS : Set(UML::Element) =
  (from.ownedElement - toFeatureMS) - from.ownedComment in
let relationships : Set(KerML::Relationship) =
toElementOMS
  ->collect(e | ElementOwningMembership_Mapping.getMapped(e))->asSet()
->union(toFeatureMS
  ->collect(e | ElementFeatureMembership_Mapping.getMapped(e))->asSet())
->union(self.oclAsType(ElementMain_Mapping).ownedRelationship()) in

let consideredEntry : Set(KerML::Relationship) =
if (from.entry.oclIsUndefined()) then
  relationships
else
  relationships
  ->including(
    EntryBehaviorStateSubactionMembership_Mapping.getMapped(
     from.entry))
endif in
```

```
let consideredDo : Set(KerML::Relationship) =
if (from.doActivity.oclIsUndefined()) then
  consideredEntry
else
  consideredEntry
    ->including(DoBehaviorStateSubactionMembership_Mapping.getMapped(
      from.doActivity))
endif in
if (from.exit.oclIsUndefined()) then
  consideredDo
else
  consideredDo
    ->including(ExitBehaviorStateSubactionMembership_Mapping.getMapped(
      from.exit))
endif
```

- StateUsage::isParallel () : Boolean [1]

```
from.isComposite
```

7.7.11.2.13 StateBehaviorPerformActionUsage_Mapping

Description

The mapping class creates a perform action usage typed by the target element of the mapping of the source behavior element.

General Mappings

ToPerformActionUsage_Init
Mapping

Mapping Source

Behavior

Mapping Target

PerformActionUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- PerformActionUsage::ownedRelationship () : Relationship [0..*]

```
Set{StateBehaviorPerformActionUsageFeatureTyping_Mapping.getMapped(from)}
```

7.7.11.2.14 StateBehaviorPerformActionUsageFeatureTyping_Mapping

Description

Creates a feature typing relationship owned by the element *typedFeature()*.

General Mappings

ToFeatureTyping_Init
Mapping

Mapping Source

Behavior

Mapping Target

FeatureTyping

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureTyping::type () : Type [1]

  ```
  from
  ```

7.7.11.2.15 StateBehaviorStateSubactionMembership_Mapping

Description

Abstract mapping class for mapping classes for state behavior mappings (enty, do and exit).

General Mappings

ToStateSubactionMembership_Init
Mapping

Mapping Source

Behavior

Mapping Target

StateSubactionMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- StateSubactionMembership::ownedMemberFeature () : Feature [1]

```
StateBehaviorPerformActionUsage_Mapping.getMapped(from)
```

7.7.11.2.16 StateDefinition_Mapping

Description

A UML4SysML::StateMachine is mapped to a SysML v2 StateDefinition.

General Mappings

Behavior_Mapping

Mapping Source

StateMachine

Mapping Target

StateDefinition

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- StateDefinition::isParallel () : Boolean [1]

```
from.region->size() > 1
```

- StateDefinition::ownedRelationship () : Relationship [0..*]

```
let initialState : Set(UML::Element) =
  from.ownedElement
  ->select(e | e.oclIsKindOf(UML::Pseudostate) and
  e.oclAsType(UML::Pseudostate).kind = UML::PseudostateKind::initial) in
let toParameterMS : Set(UML::Element) =
  from.ownedElement->select(e | e.oclIsKindOf(UML::Parameter)) in
let parameterSets: Set(UML::Element) =
```

```
  from.ownedElement->select(e | e.oclIsKindOf(UML::ParameterSet)) in
let toFeatureMS : Set(UML::Element) =
  from.ownedElement->select(e | e.oclIsKindOf(UML::Region)
    or e.oclIsKindOf(UML::ChangeEvent) or e.oclIsKindOf(UML::TimeEvent)) in
let rejectedElements : Set(UML::Element) =
  from.ownedElement->select(e | e.oclIsKindOf(UML::SignalEvent)) in
let toElementOMS : Set(UML::Element) =
  ((from.ownedElement - toFeatureMS) - toParameterMS) - initialState in
toElementOMS->collect(e | ElementOwningMembership_Mapping.getMapped(e))
->union(toFeatureMS->collect(e | ElementFeatureMembership_Mapping.getMapped(e)))
->union(toParameterMS->collect(e | ParameterMembership_Mapping.getMapped(e)))
->union(parameterSets->collect(e | ParameterSetMembership_Mapping.getMapped(e)))
->union(initialState->collect(e | InitialStateMembership_Mapping.getMapped(e)))
```

7.7.11.2.17 TimeTriggerReferenceUsage_Mapping

Description

*** not specified yet ***

General Mappings

ToReferenceUsage_Init
UniqueMapping

Mapping Source

Trigger

Mapping Target

ReferenceUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ReferenceUsage::ownedRelationship () : Relationship [0..*]

  ```
  Set{TimeTriggerReferenceSubsetting_Mapping.getMapped(from)}
  ```

- ReferenceUsage::isEnd () : Boolean [1]

  ```
  true
  ```

7.7.11.2.18 Transition_Mapping

Description

A UML4SysML::Transition is mapped to a SysML v2 TransitionUsage.

General Mappings

Namespace_Mapping
ToTransitionUsage_Init

Mapping Source

Transition

Mapping Target

TransitionUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- TransitionUsage::ownedRelationship () : Relationship [0..*]

```
self.oclAsType(ElementMain_Mapping).ownedRelationship()
->union((from.ownedElement - from.ownedComment)
  ->collect(e | ElementOwningMembership_Mapping.getMapped(e))->asSet())
->union(from.trigger->select(t | t.event.oclIsKindOf(UML::ChangeEvent)
  or t.event.oclIsKindOf(UML::TimeEvent))
  ->collect(e | TransitionTriggerFeatureMembership_Mapping.getMapped(e))
    ->asSet())
->including(TransitionSuccession_Mapping.getMapped(from))
```

- TransitionUsage::source () : ActionUsage [1]

```
from.source
```

- TransitionUsage::target () : ActionUsage [1]

```
from.target
```

7.7.11.2.19 TransitionSuccession_Mapping

Description

The mapping class creates the source Feature element of the Succession that is part of the TransitionUsage that is the target element of the UML4SysML::Transition mapping.

General Mappings

ToConnector_Init
ToMembership_Init
Mapping

Mapping Source

Transition

Mapping Target

Succession

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Succession::ownedRelationship () : Relationship [0..*]

```
OrderedSet{TransitionSuccessionSourceMembership_Mapping.getMapped(from),
TransitionSuccessionTargetMembership_Mapping.getMapped(from)}
```

7.7.11.2.20 TransitionSourceToSubsetting_Mapping

Description

Creates a subsetting relationship.

General Mappings

ToSubsetting_Init
Mapping

Mapping Source

Transition

Mapping Target

Subsetting

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Subsetting::subsettedFeature () : Feature [1]

  ```
  ElementMain_Mapping.getMapped(from.source)
  ```

- Subsetting::subsettingFeature () : Feature [1]

  ```
  TransitionSuccessionSource_Mapping.getMapped(from)
  ```

7.7.11.2.21 TransitionSuccessionSource_Mapping

Description

The mapping class creates the Succession element that is part of the TransitionUsage that is the target element of the UML4SysML::Transition mapping.

General Mappings

ToFeature_Init
Mapping

Mapping Source

Transition

Mapping Target

Feature

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Feature::declaredName () : String [0..1]

  ```
  'source'
  ```

- Feature::ownedRelationship () : Relationship [0..*]

  ```
  Set{TransitionSourceToSubsetting_Mapping.getMapped(from)}
  ```

- Feature::isEnd () : Boolean [1]

  ```
  true
  ```

7.7.11.2.22 TransitionSuccessionSourceMembership_Mapping

Description

Creates a membership relationship for *memberElement()*.

General Mappings

ToEndFeatureMembership_Init
Mapping

Mapping Source

Transition

Mapping Target

EndFeatureMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- EndFeatureMembership::ownedMemberFeature () : Feature [1]

  ```
  TransitionSuccessionSource_Mapping.getMapped(from)
  ```

7.7.11.2.23 TransitionSuccessionTarget_Mapping

Description

The mapping class creates the target Feature element of the Succession that is part of the TransitionUsage that is the target element of the UML4SysML::Transition mapping.

General Mappings

ToFeature_Init
Mapping

Mapping Source

Transition

Mapping Target

Feature

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Feature::ownedRelationship () : Relationship [0..*]

  `Set{TransitionTargetToSubsetting_Mapping.getMapped(from)}`

- Feature::declaredName () : String [0..1]

  `'target'`

- Feature::isEnd () : Boolean [1]

  `true`

7.7.11.2.24 TransitionSuccessionTargetMembership_Mapping

Description

Creates a membership relationship for *memberElement()*.

General Mappings

ToEndFeatureMembership_Init
Mapping

Mapping Source

Transition

Mapping Target

EndFeatureMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- EndFeatureMembership::ownedMemberFeature () : Feature [1]

  `TransitionSuccessionTarget_Mapping.getMapped(from)`

7.7.11.2.25 TransitionTargetToSubsetting_Mapping

Description

Creates a subsetting relationship.

General Mappings

ToSubsetting_Init
Mapping

Mapping Source

Transition

Mapping Target

Subsetting

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Subsetting::subsettedFeature () : Feature [1]

  ```
  ElementMain_Mapping.getMapped(from.target)
  ```

- Subsetting::subsettingFeature () : Feature [1]

  ```
  TransitionSuccessionTarget_Mapping.getMapped(from)
  ```

7.7.11.2.26 TransitionTriggerFeatureMembership_Mapping

Description

Creates a feature membership relationship for *ownedMemberFeature()*.

General Mappings

ToFeatureMembership_Init
UniqueMapping

Mapping Source

Trigger

Mapping Target

FeatureMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureMembership::ownedMemberFeature () : Feature [1]

```
if from.event.oclIsKindOf(UML::TimeEvent) then
  TimeTriggerCalculationUsage_Mapping.getMapped(from)
else if from.event.oclIsKindOf(UML::ChangeEvent) then
  ChangeTriggerConstraintUsage_Mapping.getMapped(from)
else
  OclUndefined
endif endif
```

7.7.12 StructuredClassifiers

7.7.12.1 Overview

SYSML21-542: Transformation: overview tables shall be revised

Table 11. List of High-Level StructuredClassifiers Mappings

SysML v1 Abstract Syntax/Stereotype	SysML v2 Abstract Syntax
Association	ConnectionDefinition
AssociationClass	OccurrenceDefinition, ConnectionDefinition
Class	OccurrenceDefinition
Connector	ConnectionUsage
ConnectorEnd	Feature
Port	OccurrenceUsage, PortUsage, Feature, AttributeUsage

7.7.12.2 Mapping Specifications

7.7.12.2.1 AssociationClass_Mapping

Description

A UML4SysML::AssociationClass is mapped to a SysML v2 ConnectionDefinition. The following shows an example of what the textual SysML v2 syntax of the result of the transformation may look like.

```
part def SysMLv1Block1;
part def SysMLv1Block2;
connection def SysMLv1AssociationBlock {
```

```
        end : SysMLv1Block1;
        end : SysMLv1Block2;
}
```

General Mappings

AssociationCommon_Mapping

Mapping Source

AssociationClass

Mapping Target

ConnectionDefinition

Owned Mappings

(none)

Applicable filters

This mapping applies only if the following (OCL) condition implemented by the operation *filter(src : Element) : Boolean* is verified:

```
not Helper.hasStereotypeApplied(src, 'SysML::Blocks::Block')
```

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ConnectionDefinition::ownedRelationship () : Relationship [0..*]

```
let nonOwnedEnds: OrderedSet(UML::Property) =
    (from.memberEnd-from.ownedEnd)->asOrderedSet() in
let generalizations : Set(UML::Generalization) =
    from.ownedElement->select(e | e.oclIsKindOf(UML::Generalization)) in
let others: OrderedSet(UML::Element) =
    ((from.ownedElement-from.memberEnd)-generalizations)->asOrderedSet() in
nonOwnedEnds->collect(e | NonOwnedEndMembership_Mapping.getMapped(e))
->union(from.ownedEnd->collect(e | OwnedEndMembership_Mapping.getMapped(e)))
->union(generalizations->collect(e | Generalization_Mapping.getMapped(e)))
->union(others->collect(e | ElementOwningMembership_Mapping.getMapped(e)))
->asOrderedSet()
```

7.7.12.2.2 AssociationCommon_Mapping

Description

A UML4SysML::Association is mapped to a SysML v2 ConnectionDefinition. This is the abstract base class of all concrete association mapping classes.

General Mappings

Classifier_Mapping
Relationship_Mapping

Mapping Source

Association

Mapping Target

Association

Owned Mappings

(none)

Applicable filters

This mapping applies only if the following (OCL) condition implemented by the operation *filter(src : Element) : Boolean* is verified:

```
src.memberEnd->select( m | m.type.oclIsKindOf(UML::UseCase))->isEmpty()
```

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Association::ownedRelationship () : Relationship [0..*]

```
let nonOwnedEnds: OrderedSet(UML::Property) =
    (from.memberEnd-from.ownedEnd)->asOrderedSet() in
nonOwnedEnds->collect(e | NonOwnedEndMembership_Mapping.getMapped(e))->asOrderedSet()
->union(self.oclAsType(Classifier_Mapping).ownedRelationship()->asOrderedSet())
->asOrderedSet()
```

7.7.12.2.3 AssociationMetadataUsage_Mapping

Description

The mapping class creates the MetadataUsage element to annotate a ConnectionDefinition that its mapping source element is a derived association.

General Mappings

ToMetadataUsage_Init
Mapping

Mapping Source

Association

Mapping Target

MetadataUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- MetadataUsage::ownedRelationship () : Relationship [0..*]

  ```
  Set{AssociationToFeatureTyping_Mapping.getMapped(from),
  AssociationMetadataUsageFeatureMembership_Mapping.getMapped(from)}
  ```

7.7.12.2.4 AssociationMetadataUsageFeatureMembership_Mapping

Description

Creates a feature membership relationship for *ownedMemberFeature()*.

General Mappings

ToFeatureMembership_Init
Mapping

Mapping Source

Association

Mapping Target

FeatureMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureMembership::ownedMemberFeature () : Feature [1]

  ```
  AssociationMetadataUsageFeature_Mapping.getMapped(from)
  ```

7.7.12.2.5 AssociationMetadataUsageFeatureTyping_Mapping

Description

Creates a feature typing relationship owned by the element *typedFeature()*.

General Mappings

ToFeatureTyping_Init
Mapping

Mapping Source

Association

Mapping Target

FeatureTyping

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureTyping::type () : Type [1]

```
SYSML2::MetadataDefinition.allInstances()
->any(m | m.qualifiedName = 'SysMLv1Library::AssociationData')
```

7.7.12.2.6 AssociationMetadataUsageFeature_Mapping

Description

The mapping class creates the feature of the MetadataUsage.

General Mappings

ToFeature_Init
Mapping

Mapping Source

Association

Mapping Target

Feature

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Feature::ownedRelationship () : Relationship [0..*]

```
Set{AssociationMetadataUsageRedefinition_Mapping.getMapped(from),
AssociationMetadataUsageFeatureValue_Mapping.getMapped(from)}
```

7.7.12.2.7 AssociationMetadataUsageFeatureValue_Mapping

Description

Creates a feature value relationship.

General Mappings

ToFeatureValue_Init
Mapping

Mapping Source

Association

Mapping Target

FeatureValue

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureValue::value () : Expression [1]

```
LiteralBoolean_Factory.create(from.isDerived)
```

7.7.12.2.8 AssociationMetadataUsageMembership_Mapping

Description

Creates a membership relationship for *memberElement()*.

General Mappings

ToOwningMembership_Init
Mapping

Mapping Source

Association

Mapping Target

OwningMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- OwningMembership::ownedMemberElement () : Element [1]

```
AssociationMetadataUsage_Mapping.getMapped(from)
```

7.7.12.2.9 AssociationMetadataUsageRedefinition_Mapping

Description

Creates a redefinition relationship for the *redefiningFeature()* and the *redefinedFeature()*.

General Mappings

ToRedefinition_Init
Mapping

Mapping Source

Association

Mapping Target

Redefinition

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Redefinition::redefinedFeature () : Feature [1]

```
SYSML2::AttributeUsage.allInstances()
->any(m | m.qualifiedName = 'SysMLv1Library::AssociationData::isDerived')
```

7.7.12.2.10 Class_Mapping

Description

A UML4SysML::Class is mapped to a SysML v2 OccurrenceDefinition. The following shows an example of what the textual SysML v2 syntax of the result of the transformation may look like.

```
occurrence def UML4SysMLClass;
```

General Mappings

BehavioredClassifier_Mapping

Mapping Source

Class

Mapping Target

OccurrenceDefinition

Owned Mappings

(none)

Applicable filters

This mapping applies only if the following (OCL) condition implemented by the operation *filter(src : Element) : Boolean* is verified:

```
not Helper.isRequirement(src) and not src.oclIsTypeOf(UML::AssociationClass)
```

Mapping rules

The mapping class only has inherited rules. See the mapping classes in the general mapping section for details.

7.7.12.2.11 ConnectionDefEnd_Mapping

Description

*** not specified yet ***

General Mappings

UniqueMapping
End_Mapping

Mapping Source

Property

Mapping Target

Feature

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Feature::ownedRelationship () : Relationship [0..*]

```
let crossFMultiplicity: Set(SysML2::ReferenceUsage) =
    if from.association.ownedEnd->includes(from) and
        not ((from.opposite.isComposite and from.lower = 0) or
        (from.lower = 0 and from.upper = -1)) then
        Set {MultiplicityReferenceUsage_Mapping.getMapped(from)}
    else
        Set{}
    endif in
let typings: Set(KerML::FeatureTyping) = if from.type.oclIsUndefined() then
    Set{}
else
    Set{StructuralFeatureToFeatureTyping_Mapping.getMapped(from)}
endif in
let subsettings: Set(KerML::CrossSubsetting) =
    if from.association.ownedEnd->excludes(from) and from.opposite.lower = 0 and
        not (from.isComposite or from.opposite.upper = -1) then
        Set{CrossSubsetting_Mapping.getMapped(from)}
    else
        Set{}
    endif in
let defaultValue: Set(KerML::OwningMembership) =
    if from.defaultValue.oclIsUndefined() then
        Set{}
    else
        Set{DefaultValue_Mapping.getMapped(from)}
    endif in
crossFMultiplicity->union(typings)
    ->union(subsettings)->union(defaultValue)
    ->including(MultiplicityMembership_Factory.create(1,1))->asSet()
```

7.7.12.2.12 ConnectionDefEndMembership_Mapping

Description

Creates a membership relationship for *memberElement()*.

General Mappings

UniqueMapping
ToFeatureMembership_Init

Mapping Source

Property

Mapping Target

FeatureMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureMembership::ownedMemberFeature () : Feature [1]

```
ConnectionDefEnd_Mapping.getMapped(from)
```

7.7.12.2.13 ConnectionEndToSubsetting_Mapping

Description

Creates a subsetting relationship.

General Mappings

ToSubsetting_Init
Mapping

Mapping Source

ConnectorEnd

Mapping Target

Subsetting

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Subsetting::subsettedFeature () : Feature [1]

```
let propertyPath: OrderedSet(UML::Property) =
    Helper.getTagValueAsElementColl
    (src, 'SysML::Blocks::NestedConnectorEnd','propertyPath')
    ->asOrderedSet() in
if propertyPath->isEmpty() then
```

```
        ElementMain_Mapping.getMapped(from.role)
    else
        ConnectorEndToSubsettedFeature_Mapping.getMapped(from)
    endif
```

- Subsetting::ownedRelationship () : Relationship [0..*]

```
let propertyPath: OrderedSet(UML::Property) =
    Helper.getTagValueAsElementColl
        (from, 'SysML::Blocks::NestedConnectorEnd','propertyPath')
    ->asOrderedSet() in
if propertyPath->notEmpty() then
        OrderedSet{ConnectorEndToSubsettedFeatureMembership_Mapping.getMapped(from)}
else
        OrderedSet{}
endif
```

- Subsetting::subsettingFeature () : Feature [1]

```
ConnectorEndToOwnedFeature_Mapping.getMapped(from)
```

7.7.12.2.14 Connector_Mapping

Description

A UML4SysML::Connector is mapped to a SysMLv2 ConnectionUsage. The following shows an example of what the textual SysML v2 syntax of the result of the transformation may look like.

```
part def SysMLv1Block3 {
        part sysMLv1PartProperty1 : SysMLv1Block1;
        part sysMLv1PartProperty2 : SysMLv1Block2;
        connection sysMLv1Connector connect sysMLv1PartProperty1 to sysMLv1PartProperty2;
}
part def SysMLv1Block1;
part def SysMLv1Block2;
```

General Mappings

NamedElementMain_Mapping
ToConnector_Init

Mapping Source

Connector

Mapping Target

ConnectionUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ConnectionUsage::ownedRelationship () : Relationship [0..*]

```
from.end->collect(e | ConnectorEndToMembership_Mapping.getMapped(e))->asSet()
    ->union(self.oclAsType(ElementMain_Mapping).ownedRelationship())
```

7.7.12.2.15 ConnectorEndToFeatureCommon_Mapping

Description

The mapping class is the abstract base class for UML4SysML::ConnectorEnd mapping classes.

General Mappings

ToFeature_Init
Mapping

Mapping Source

ConnectorEnd

Mapping Target

Feature

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Feature::isOrdered () : Boolean [1]

```
from.isOrdered
```

7.7.12.2.16 ConnectorEndToMembership_Mapping

Description

Creates a membership relationship for *memberElement()*.

General Mappings

ToFeatureMembership_Init
Mapping

Mapping Source

ConnectorEnd

Mapping Target

EndFeatureMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- EndFeatureMembership::ownedMemberFeature () : Feature [1]

```
ConnectorEndToOwnedFeature_Mapping.getMapped(from)
```

7.7.12.2.17 ConnectorEndToOwnedFeature_Mapping

Description

The mapping class creates the SysML v2 Feature element for the UML4SysML::ConnectorEnd mapping.

General Mappings

ConnectorEndToFeatureCommon_Mapping
ElementMain_Mapping

Mapping Source

ConnectorEnd

Mapping Target

Feature

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Feature::ownedRelationship () : Relationship [0..*]

```
let subsetting: KerML::Subsetting =
    ConnectionEndToSubsetting_Mapping.getMapped(from) in
```

```
if subsetting.oclIsUndefined() then
    OrderedSet{MultiplicityMembership_Mapping.getMapped(from)}
else
    OrderedSet{MultiplicityMembership_Mapping.getMapped(from), subsetting}
endif
```

7.7.12.2.18 ConnectorEndToSubsettedFeature_Mapping

Description

The mapping class maps UML4SysML::ConnectorEnd that are part of a SysML::Ports&Flows::NestedConnectorEnd.

General Mappings

ConnectorEndToFeatureCommon_Mapping

Mapping Source

ConnectorEnd

Mapping Target

Feature

Owned Mappings

(none)

Applicable filters

This mapping applies only if the following (OCL) condition implemented by the operation *filter(src : Element) : Boolean* is verified:

```
let propertyPath: OrderedSet(UML::Property) =
Helper.getTagValueAsElementColl(src, 'SysML::Blocks::NestedConnectorEnd','propertyPath')
->asOrderedSet() in
propertyPath->notEmpty()
```

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Feature::ownedRelationship () : Relationship [0..*]

```
let propertyPath: OrderedSet(UML::Property) =
    Helper.getTagValueAsElementColl
    (from, 'SysML::Blocks::NestedConnectorEnd','propertyPath')
    ->asOrderedSet() in
let chain: OrderedSet(KerML::FeatureChaining) =
    propertyPath->collect(p | PropertyToFeatureChaining_Mapping.getMapped(p))
    ->asOrderedSet()
    ->including(PropertyToFeatureChaining_Mapping.getMapped(from.role)) in
chain->union(OrderedSet{MultiplicityMembership_Mapping.getMapped(from)})
```

- Feature::declaredName () : String [0..1]

```
'featureChain'
```

7.7.12.2.19 ConnectorEndToSubsettedFeatureMembership_Mapping

Description

Creates a feature membership relationship for *ownedMemberFeature()*.

General Mappings

ToFeatureMembership_Init
Mapping

Mapping Source

ConnectorEnd

Mapping Target

EndFeatureMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- EndFeatureMembership::ownedMemberFeature () : Feature [1]

  `ConnectorEndToSubsettedFeature_Mapping.getMapped(from)`

7.7.12.2.20 ConnectorType_Mapping

Description

A UML4SysML::Association is mapped to a SysML v2 ConnectionDefinition.

General Mappings

AssociationCommon_Mapping

Mapping Source

Association

Mapping Target

ConnectionDefinition

Owned Mappings

(none)

Applicable filters

This mapping applies only if the following (OCL) condition implemented by the operation *filter(src : Element) : Boolean* is verified:

```
let this: UML::Association = src.oclAsType(UML::Association) in
if this.oclIsUndefined() then
    false
else
    not src.memberEnd->exists( m | m.type.oclIsKindOf(UML::UseCase)) and
    not src.isDerived and
    not src.oclIsTypeOf(UML::AssociationClass) and
    Helper.isConnectionDef(src)
endif
```

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ConnectionDefinition::ownedRelationship () : Relationship [0..*]

```
from.memberEnd
  ->collect(e | ConnectionDefEndMembership_Mapping.getMapped(e))
    ->asOrderedSet()
```

7.7.12.2.21 ConnectorTypeDerived_Mapping

Description

The mapping class is a concrete mapping class of the abstract AssociationCommon_Mapping class for mappings of derived associations. The UML4SysML::Association::isDerived property is not supported in SysML v2. To preserve the information, it is stored in a metadata annotation.

General Mappings

AssociationCommon_Mapping

Mapping Source

Association

Mapping Target

ConnectionDefinition

Owned Mappings

(none)

Applicable filters

This mapping applies only if the following (OCL) condition implemented by the operation *filter(src : Element) : Boolean* is verified:

```
(src.memberEnd->select( m | m.type.oclIsKindOf(UML::UseCase))->isEmpty()) and
(let this: UML::Association = src.oclAsType(UML::Association) in
if this.oclIsUndefined() then
```

```
    false
else
    this.isDerived and
    not this.oclIsTypeOf(UML::AssociationClass) and
    Helper.isConnectionDef(this)
endif)
```

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ConnectionDefinition::ownedRelationship () : Relationship [0..*]

  ```
  self.oclAsType(AssociationCommon_Mapping).ownedRelationship()
  ->including(AssociationMetadataUsageMembership_Mapping.getMapped(from))
  ```

7.7.12.2.22 CrossSubsetting_Mapping

Description

Creates a subsetting relationship.

General Mappings

UniqueMapping
ToSubsetting_Init

Mapping Source

Property

Mapping Target

CrossSubsetting

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- CrossSubsetting::subsettedFeature () : Feature [1]

  ```
  NonOwnedEnd_Mapping.getMapped(from)
  ```

7.7.12.2.23 End_Mapping

Description

The mapping class is the abstract base class of mapping classes for properties that are defined by association ends.

General Mappings

PropertyCommon_Mapping

Mapping Source

Property

Mapping Target

Feature

Owned Mappings

(none)

Applicable filters

This mapping applies only if the following (OCL) condition implemented by the operation *filter(src : Element) : Boolean* is verified:

```
src.oclIsKindOf(UML::Property) and
not src.oclAsType(UML::Property).association.oclIsUndefined()
```

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Feature::isEnd () : Boolean [1]

  ```
  true
  ```

7.7.12.2.24 EndMembership_Mapping

Description

Creates a membership relationship for *memberElement()*.

General Mappings

StructuralFeatureMembership_Mapping

Mapping Source

Property

Mapping Target

EndFeatureMembership

Owned Mappings

(none)

Applicable filters

(none)

7.7.12.2.25 EndToSubsettedFeature_Mapping

Description

The mapping class creates a feature element for the UML4SysML::ConnectorEnd mapping.

General Mappings

PropertyCommon_Mapping

Mapping Source

Property

Mapping Target

Feature

Owned Mappings

(none)

Applicable filters

This mapping applies only if the following (OCL) condition implemented by the operation *filter(src : Element) : Boolean* is verified:

```
let property: UML::Property = src.oclAsType(UML::Property) in
not property.association.oclIsUndefined()
and property.association.ownedEnd->excludes(property)
```

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Feature::ownedRelationship () : Relationship [0..*]

```
let chain: OrderedSet(KerML::FeatureChaining) =
    OrderedSet{EndToSubsettedFeatureChaining_Mapping.getMapped(from)} in
chain->including(MultiplicityMembership_Mapping.getMapped(from))
```

7.7.12.2.26 EndToSubsettedFeatureChaining_Mapping

Description

The mapping class creates a feature chaining element for the UML4SysML::ConnectorEnd mapping.

General Mappings

ToRelationship_Init
Mapping

Mapping Source

Property

Mapping Target

FeatureChaining

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureChaining::chainingFeature () : Feature [1]

  ```
  from
  ```

- FeatureChaining::declaredName () : String [0..1]

  ```
  'featureChain'
  ```

7.7.12.2.27 MultiplicityReferenceUsage_Mapping

Description

Creates a reference usage.

General Mappings

UniqueMapping
ToReferenceUsage_Init

Mapping Source

Property

Mapping Target

ReferenceUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ReferenceUsage::ownedRelationship () : Relationship [0..*]

  ```
  Set{MultiplicityMembership_Factory.create(from.lower,from.upper)}
  ```

7.7.12.2.28 NonOwnedEndSubsetting_Mapping

Description

Creates a subsetting relationship.

General Mappings

ToSubsetting_Init
Mapping

Mapping Source

Property

Mapping Target

Subsetting

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Subsetting::subsettedFeature () : Feature [1]

  ```
  from
  ```

7.7.12.2.29 NonOwnedEndToSubsettedFeatureMembership_Mapping

Description

Creates a feature membership relationship for *ownedMemberFeature()*.

General Mappings

ToFeatureMembership_Init
Mapping

Mapping Source

Property

Mapping Target

FeatureMembership

Owned Mappings

(none)

Applicable filters

This mapping applies only if the following (OCL) condition implemented by the operation *filter(src : Element) : Boolean* is verified:

```
src.oclIsKindOf(UML::Property) and
not src.oclAsType(UML::Property).association.oclIsUndefined()
```

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureMembership::ownedMemberFeature () : Feature [1]

  ```
  EndToSubsettedFeature_Mapping.getMapped(from)
  ```

7.7.12.2.30 NonOwnedEnd_Mapping

Description

The mapping class maps UML4SysML::Property elements that are not owned by an association to a SysML v2 Feature element.

General Mappings

End_Mapping
UniqueMapping

Mapping Source

Property

Mapping Target

Feature

Owned Mappings

- nonOwnedEndTyping : NonOwnedEndFeatureTyping_Mapping

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Feature::ownedRelationship () : Relationship [0..*]

  ```
  Set{MultiplicityMembership_Mapping.getMapped(from),
  nonOwnedEndTyping.to,
  NonOwnedEndSubsettingMembership_Mapping.getMapped(from),
  NonOwnedEndToSubsettedFeatureMembership_Mapping.getMapped(from)}
  ->union(from.qualifier
  ->collect(q | ElementFeatureMembership_Mapping.getMapped(q))->asSet())
  ```

- Feature::declaredName () : String [0..1]

  ```
  'nonOwnedEnd'
  ```

7.7.12.2.31 NonOwnedEndMembership_Mapping

Description

Creates a membership relationship for *memberElement()*.

General Mappings

EndMembership_Mapping

Mapping Source

Property

Mapping Target

EndFeatureMembership

Owned Mappings

(none)

Applicable filters

This mapping applies only if the following (OCL) condition implemented by the operation *filter(src : Element) : Boolean* is verified:

```
src.oclIsKindOf(UML::Property)
    and not src.oclAsType(UML::Property).association.oclIsUndefined()
    and src.oclAsType(UML::Property).association.ownedEnd->excludes(src)
```

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- EndFeatureMembership::ownedMemberFeature () : Feature [1]

  ```
  NonOwnedEnd_Mapping.getMapped(from)
  ```

7.7.12.2.32 NonOwnedEndSubsettingMembership_Mapping

Description

Creates a membership relationship for *memberElement()*.

General Mappings

ToOwningMembership_Init
Mapping

Mapping Source

Property

Mapping Target

OwningMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- OwningMembership::ownedMemberElement () : Element [1]

  ```
  NonOwnedEndSubsetting_Mapping.getMapped(from)
  ```

7.7.12.2.33 NonOwnedEndFeatureTyping_Mapping

Description

Creates a feature typing relationship owned by the element *typedFeature()*.

General Mappings

StructuralFeatureToFeatureTyping_Mapping
UniqueMapping

Mapping Source

Property

Mapping Target

FeatureTyping

Owned Mappings

- nonOwnedEnd : NonOwnedEnd_Mapping

Applicable filters

(none)

7.7.12.2.34 OwnedEnd_Mapping

Description

The mapping class maps UML4SysML::Property elements that are owned by an association to a SysML v2 Feature element.

General Mappings

End_Mapping
NamedElementMain_Mapping

Mapping Source

Property

Mapping Target

Feature

Owned Mappings

(none)

Applicable filters

This mapping applies only if the following (OCL) condition implemented by the operation *filter(src : Element) : Boolean* is verified:

```
let p: UML::Property = src.oclAsType(UML::Property) in
not p.oclIsUndefined() and
(not p.association.oclIsUndefined()
    and p.association.ownedEnd->includes(p)) and
(not p.association.memberEnd
->select( m | (not m.type.oclIsUndefined())
    and m.type.oclIsTypeOf(UML::UseCase))->notEmpty())
```

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Feature::ownedRelationship () : Relationship [0..*]

```
let qualifiers: Set(KerML::FeatureMembership) =
    from.qualifier
    ->collect(q | ElementFeatureMembership_Mapping.getMapped(q))->asSet() in
let typing: KerML::FeatureTyping =
    StructuralFeatureToFeatureTyping_Mapping.getMapped(from) in
let subsetting: Set(KerML::Subsetting) =
    from.subsettedProperty
    ->collect(p | PropertySubsetting_Mapping.getMapped(from, p))->asSet() in
let subsettingMultiplicityTyping: Set(KerML::Relationship) =
    subsetting->union(if typing.oclIsUndefined() then
                          Set{MultiplicityMembership_Mapping.getMapped(from)}
                      else
```

```
                              Set{MultiplicityMembership_Mapping.getMapped(from), typing}
                          endif)->asSet() in
let relationships: Set(KerML::Relationship) = qualifiers->union(
    if from.defaultValue.oclIsTypeOf(UML::OpaqueExpression) then
        subsettingMultiplicityTyping
        ->including(ElementOwningMembership_Mapping.getMapped(from.defaultValue))
    else
        subsettingMultiplicityTyping
    endif) in

if from.defaultValue.oclIsUndefined() then
    relationships
else
    relationships->including(
        if from.defaultValue.oclIsTypeOf(UML::OpaqueExpression) then
            DefaultValueOpaqueExpression_Mapping.getMapped(from.defaultValue)
        else
            DefaultValue_Mapping.getMapped(from.defaultValue)
        endif)
endif
```

7.7.12.2.35 OwnedEndMembership_Mapping

Description

Creates a membership relationship for *memberElement()*.

General Mappings

EndMembership_Mapping

Mapping Source

Property

Mapping Target

EndFeatureMembership

Owned Mappings

(none)

Applicable filters

This mapping applies only if the following (OCL) condition implemented by the operation *filter(src : Element) : Boolean* is verified:

```
src.oclIsKindOf(UML::Property)
    and not src.oclAsType(UML::Property).association.oclIsUndefined()
    and src.oclAsType(UML::Property).association.ownedEnd->includes(src)
```

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- EndFeatureMembership::ownedMemberFeature () : Feature [1]

  ```
  OwnedEnd_Mapping.getMapped(from)
  ```

7.7.12.2.36 Port_Mapping

Description

A UML4SysML::Port that is typed by an interface block is mapped to a SysML v2 PortUsage.

The following shows an example of what the textual SysML v2 syntax of the result of the transformation may look like.

```
port sysMLv1Port : SysMLv1InterfaceBlock;
port def SysMLv1InterfaceBlock
```

General Mappings

PropertyCommon_Mapping
NamedElementMain_Mapping

Mapping Source

Port

Mapping Target

PortUsage

Owned Mappings

(none)

Applicable filters

This mapping applies only if the following (OCL) condition implemented by the operation *filter(src : Element) : Boolean* is verified:

```
if src.oclIsTypeOf(UML::Port) and
not Helper.hasStereotypeApplied(src.owner,
'SysML::ConstraintBlocks::ConstraintBlock' ) then
    let p: UML::Port = src.oclAsType(UML::Port) in
    if p.type.oclIsUndefined() then
        false
    else
        true
    endif
else
    false
endif
```

Mapping rules

The mapping class only has inherited rules. See the mapping classes in the general mapping section for details.

7.7.12.2.37 PortUntyped_Mapping

Description

A UML4SysML::Port that is untyped is mapped to a SysML v2 PortUsage.

The following shows an example of what the textual SysML v2 syntax of the result of the transformation may look like.

```
port sysMLv1Port;
```

General Mappings

PropertyUntyped_Mapping

Mapping Source

Port

Mapping Target

PortUsage

Owned Mappings

(none)

Applicable filters

(none)

7.7.12.2.38 PropertyToFeatureChaining_Mapping

Description

The mapping class creates the SysML v2 FeatureChaining for the UML4SysML::Property mapping.

General Mappings

ToRelationship_Init
Mapping

Mapping Source

Property

Mapping Target

FeatureChaining

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureChaining::chainingFeature () : Feature [1]

  ```
  ElementMain_Mapping.getMapped(from)
  ```

7.7.12.2.39 QualifierMembership_Mapping

Description

Creates a membership relationship for *memberElement()*.

General Mappings

StructuralFeatureMembership_Mapping

Mapping Source

StructuralFeature

Mapping Target

FeatureMembership

Owned Mappings

(none)

Applicable filters

(none)

7.7.13 UseCases

SYSML21-542: Transformation: overview tables shall be revised

7.7.13.1 Overview

SYSML21-542: Transformation: overview tables shall be revised

Table 12. List of High-Level UseCases Mappings

SysML v1 Abstract Syntax/Stereotype	SysML v2 Abstract Syntax
Actor	PartDefinition
Extend	The semantics of the UML4SysML::Extend relationship is not supported by SysML v2.
ExtensionPoint	The semantics of the UML4SysML::Extend relationship is not supported by SysML v2 Therefore, UML4SysML::ExtensionPoint is also not covered by the transformation.
Include	No specific mapping specified yet.
UseCase	UseCaseDefinition

7.7.13.2 Mapping Specifications

7.7.13.2.1 Actor_Mapping

Description

A UML4SysML::Actor is mapped to a SysML v2 PartDefinition. The following shows an example of what the textual SysML v2 syntax of the result of the transformation may look like.

```
part def SysMLv1Actor;
```

General Mappings

ElementMain_Mapping
BehavioredClassifier_Mapping

Mapping Source

Actor

Mapping Target

PartDefinition

Owned Mappings

(none)

Applicable filters

(none)

7.7.13.2.2 Include_Mapping

Description

A UML4SysML::Include is mapped to a SysML v2 IncludeUseCaseUsage. The following shows an example of what the textual SysML v2 syntax of the result of the transformation may look like.

```
use case def SysMLv1UseCase1 {
        include use case : SysMLv1UseCase2;
}
use case def SysMLv1UseCase2;
```

General Mappings

ToOccurrenceUsage_Init
NamedElementMain_Mapping

Mapping Source

Include

Mapping Target

IncludeUseCaseUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- IncludeUseCaseUsage::ownedRelationship () : Relationship [0..*]

```
Set{IncludeFeatureTyping_Mapping.getMapped(from),
ReturnParameterFeatureMembership_Factory.create(),
EmptySubjectMembership_Factory.create()}
->union(self.oclAsType(ElementMain_Mapping).ownedRelationship())
```

7.7.13.2.3 IncludeFeatureTyping_Mapping

Description

Creates a feature typing relationship owned by the element *typedFeature()*.

General Mappings

ToFeatureTyping_Init
Mapping

Mapping Source

Include

Mapping Target

FeatureTyping

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureTyping::type () : Type [1]

```
from.addition
```

7.7.13.2.4 UseCase_Mapping

Description

A UML4SysML::UseCase is mapped to a SysML v2 UseCaseDefinition. The expected SysML v2 textual syntax of a mapped UML4SysML::UseCase with a defined subject is as follows.

```
use case def SysMLv1UseCase {
  subject subject_SysMLv1Block : SysMLv1Block;
}
part def SysMLv1Block;
```

Currently, only one use case subject is supported by the mapping class. Since the UML4SysML::Extend relationship is not considered by the SysML v1 to SysML v2 transformation, the extension points of a use case are also not mapped.

General Mappings

BehavioredClassifier_Mapping
NamedElementMain_Mapping

Mapping Source

UseCase

Mapping Target

UseCaseDefinition

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- UseCaseDefinition::ownedRelationship () : Relationship [0..*]

```
let properties : Set(UML::Element) =
    from.ownedElement->select(e | e.oclIsKindOf(UML::Property) and
        e.oclAsType(UML::Property).association.oclIsUndefined()) in
let actors : Set(UML::Property) =
    UML::Association.allInstances()
        ->collect(m | m.memberEnd)
        ->flatten()
        ->select( m | m.type = from)->collect(a | a.owningAssociation)
        ->collect( p | p.memberEnd->select( m | not (m.type = from) ))->flatten() in
let extensionPoints : Sequence(UML::Element) =
    from.ownedElement->select(e | e.oclIsKindOf(UML::ExtensionPoint)) in
let extend : Sequence(UML::Element) =
    from.ownedElement->select(e | e.oclIsKindOf(UML::Extend)) in
let include : Sequence(UML::Element) =
    from.ownedElement->select(e | e.oclIsKindOf(UML::Include)) in
let elements : Set(UML::Element) =
    (((from.ownedElement-properties) - extensionPoints) - extend) - include) in
```

```
let relationships : Sequence(KerML::Relationship) =
elements->collect(e | ElementOwningMembership_Mapping.getMapped(e))
->union(properties->collect(e | PropertyMembership_Mapping.getMapped(e)))
->including(UseCaseSubjectMembership_Mapping.getMapped(from))
->including(UseCaseObjectiveMembership_Mapping.getMapped(from))
->including(CommonReturnParameterReferenceUsageMembership_Mapping.getMapped(from))
->union(actors->collect(e | UseCaseActorMembership_Mapping.getMapped(e))) in
if from.classifierBehavior.oclIsUndefined() then
    relationships
else
    relationships
    ->including(BehavioredClassifierFeatureMembership_Mapping.getMapped(from))
endif
```

7.7.13.2.5 UseCaseActor_Mapping

Description

The mapping class creates the PartUsage representing an actor of the use case.

General Mappings

ToPartUsage_Init
Mapping

Mapping Source

Property

Mapping Target

PartUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- PartUsage::declaredName () : String [0..1]

  ```
  from.name
  ```

- PartUsage::ownedRelationship () : Relationship [0..*]

  ```
  Set{UseCaseActorFeatureTyping_Mapping.getMapped(from)}
  ```

7.7.13.2.6 UseCaseActorFeatureTyping_Mapping

Description

Creates a feature typing relationship owned by the element *typedFeature()*.

General Mappings

ToFeatureTyping_Init
Mapping

Mapping Source

Property

Mapping Target

FeatureTyping

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureTyping::type () : Type [1]

  ```
  from.type
  ```

7.7.13.2.7 UseCaseActorMembership_Mapping

Description

Creates a membership relationship for *memberElement()*.

General Mappings

ToActorMembership_Init
Mapping

Mapping Source

Property

Mapping Target

ActorMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ActorMembership::ownedMemberParameter () : Feature [1]

  ```
  UseCaseActor_Mapping.getMapped(from)
  ```

7.7.13.2.8 UseCaseEmptySubjectReferenceUsage_Mapping

Description

The mapping class creates an "empty" ReferenceUsage for the subject, if the subject is not given at the SysML v1 UseCase element.

General Mappings

ToReferenceUsage_Init
Mapping

Mapping Source

UseCase

Mapping Target

ReferenceUsage

Owned Mappings

(none)

Applicable filters

(none)

7.7.13.2.9 UseCaseObjectiveMembership_Mapping

Description

Creates a membership relationship for *memberElement()*.

General Mappings

ToObjectiveMembership_Init
Mapping

Mapping Source

UseCase

Mapping Target

ObjectiveMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ObjectiveMembership::ownedMemberFeature () : Feature [1]

  ```
  UseCaseObjectiveRequirementUsage_Mapping.getMapped(from)
  ```

7.7.13.2.10 UseCaseObjectiveRequirementUsage_Mapping

Description

The mapping class creates the RequirementUsage element for the use case objective. The element is not set by an element from the SysML v1 UseCase.

General Mappings

ToRequirementUsage_Init
Mapping

Mapping Source

UseCase

Mapping Target

RequirementUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- RequirementUsage::ownedRelationship () : Relationship [0..*]

  ```
  Set{UseCaseObjectiveSubjectMembership_Mapping.getMapped(from),
  CommonReturnParameterReferenceUsageMembership_Mapping.getMapped(from)}
  ```

7.7.13.2.11 UseCaseObjectiveSubjectMembership_Mapping

Description

Creates a membership relationship for *memberElement()*.

General Mappings

ToSubjectMembership_Init
Mapping

Mapping Source

UseCase

Mapping Target

SubjectMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- SubjectMembership::ownedMemberParameter () : Feature [1]

  ```
  UseCaseEmptySubjectReferenceUsage_Mapping.getMapped(from)
  ```

7.7.13.2.12 UseCaseSubjectFeatureTyping_Mapping

Description

Creates a feature typing relationship owned by the element *typedFeature()*.

General Mappings

ToFeatureTyping_Init
Mapping

Mapping Source

UseCase

Mapping Target

FeatureTyping

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureTyping::type () : Type [1]

```
if from.subject->size() > 0 then from.subject->get(0) else invalid endif
```

7.7.13.2.13 UseCaseSubjectMembership_Mapping

Description

Creates a membership relationship for *memberElement()*.

General Mappings

ToSubjectMembership_Init
Mapping

Mapping Source

UseCase

Mapping Target

SubjectMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- SubjectMembership::ownedMemberParameter () : Feature [1]

```
if from.subject->size() > 0 then
    UseCaseSubjectReferenceUsage_Mapping.getMapped(from)
else
    UseCaseEmptySubjectReferenceUsage_Mapping.getMapped(from)
endif
```

7.7.13.2.14 UseCaseSubjectReferenceUsage_Mapping

Description

The mapping class creates the ReferenceUsage element for the subject.

General Mappings

UseCaseEmptySubjectReferenceUsage_Mapping

Mapping Source

UseCase

Mapping Target

ReferenceUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ReferenceUsage::declaredName () : String [0..1]

  ```
  'subject_' + from.subject->get(0).name
  ```

- ReferenceUsage::ownedRelationship () : Relationship [0..*]

  ```
  Set{UseCaseSubjectFeatureTyping_Mapping.getMapped(from)}
  ```

7.7.14 Values

SYSML21-542: Transformation: overview tables shall be revised

7.7.14.1 Overview

SYSML21-542: Transformation: overview tables shall be revised

Table 13. List of High-Level Values Mappings

SysML v1 Abstract Syntax/Stereotype	SysML v2 Abstract Syntax
Duration	Expression
DurationConstraint	ConstraintDefinition
DurationInterval	Expression
DurationObservation	Mapping is not specified yet.
Expression	Expression, OperatorExpression
Interval	Expression
IntervalConstraint	ConstraintDefinition
LiteralBoolean	LiteralBoolean
LiteralInteger	LiteralInteger
LiteralNull	NullExpression
LiteralReal	LiteralRational

SysML v1 Abstract Syntax/Stereotype	SysML v2 Abstract Syntax
LiteralString	LiteralString
LiteralUnlimitedNatural	LiteralInfinity
OpaqueExpression	CalculationUsage
StringExpression	Expression, OperatorExpression
TimeConstraint	ConstraintDefinition
TimeExpression	TriggerInvocationExpression
TimeInterval	Expression
TimeObservation	Mapping is not specified yet.

7.7.14.2 Mapping Specifications

7.7.14.2.1 EqualOperatorExpressionFeature_Mapping

Description

The mapping class creates the feature element for the equal operator.

General Mappings

ToFeature_Init
Mapping

Mapping Source

TypedElement

Mapping Target

Feature

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Feature::ownedRelationship () : Relationship [0..*]

  ```
  Set{EqualOperatorExpressionFeatureValue_Mapping.getMapped(from)}
  ```

7.7.14.2.2 EqualOperatorExpressionFeatureValue_Mapping

Description

Creates a feature value relationship.

General Mappings

ToFeatureValue_Init
Mapping

Mapping Source

TypedElement

Mapping Target

FeatureValue

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureValue::value () : Expression [1]

  ```
  CommonFeatureReferenceExpression_Mapping.getMapped(from)
  ```

7.7.14.2.3 EqualOperatorExpressionOperandParameterMembership_Mapping

Description

Creates a membership relationship for *memberElement()*.

General Mappings

ToParameterMembership_Init
Mapping

Mapping Source

TypedElement

Mapping Target

ParameterMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ParameterMembership::visibility () : VisibilityKind [1]

  ```
  KerML::VisibilityKind::private
  ```

- ParameterMembership::ownedMemberParameter () : Feature [1]

  ```
  EqualOperatorExpressionFeature_Mapping.getMapped(from)
  ```

7.7.14.2.4 Expression_Mapping

Description

A UML4SysML::Expression element is mapped to a SysML v2 OperatorExpression element.

General Mappings

ToExpression_Init
NamedElementMain_Mapping

Mapping Source

Expression

Mapping Target

OperatorExpression

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- OperatorExpression::operator () : String [1]

  ```
  from.symbol
  ```

7.7.14.2.5 ExpressionElse_Mapping

Description

A UML4SysML::Expression element with operator "else" is mapped to a SysML v2 TextualRepresentation element with language set to "SysMLv1" and body set to "else".

General Mappings

Expression_Mapping

Mapping Source

Expression

Mapping Target

OperatorExpression

Owned Mappings

(none)

Applicable filters

This mapping applies only if the following (OCL) condition implemented by the operation *filter(src : Element) : Boolean* is verified:

```
src.symbol = 'else'
```

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- OperatorExpression::ownedRelationship () : Relationship [0..*]

```
self.oclAsType(ElementMain_Mapping).ownedRelationship()
  ->including(ExpressionElseMembership_Mapping.getMapped(from))
```

7.7.14.2.6 ExpressionElseMembership_Mapping

Description

Creates the membership relationship for the textual representation for the else guard condition specification.

General Mappings

ToOwningMembership_Init
Mapping

Mapping Source

Expression

Mapping Target

OwningMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- OwningMembership::ownedMemberElement () : Element [1]

  ```
  ExpressionElseSpecification_Mapping.getMapped(from)
  ```

7.7.14.2.7 ExpressionElseSpecification_Mapping

Description

Creates the textual representation for the else guard condition specification.

General Mappings

ToTextualRepresentation_Init
Mapping

Mapping Source

Expression

Mapping Target

TextualRepresentation

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- TextualRepresentation::body () : String [1]

  ```
  'else'
  ```

- TextualRepresentation::language () : String [1]

  ```
  'SysMLv1'
  ```

7.7.14.2.8 LiteralBoolean_Mapping

Description

The mapping class maps UML4SysML::LiteralBoolean to SysML v2 LiteralBoolean.

General Mappings

LiteralSpecificationCommon_Mapping

Mapping Source

LiteralBoolean

Mapping Target

LiteralBoolean

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- LiteralBoolean::value () : Boolean [1]

  `from.value`

7.7.14.2.9 LiteralInteger_Mapping

Description

The mapping class maps UML4SysML::LiteralInteger to SysML v2 LiteralInteger.

General Mappings

LiteralSpecificationCommon_Mapping

Mapping Source

LiteralInteger

Mapping Target

LiteralInteger

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- LiteralInteger::value () : Integer [1]

  ```
  from.value
  ```

7.7.14.2.10 LiteralNull_Mapping

Description

The mapping class maps UML4SysML::LiteralNull to SysML v2 NullExpression.

General Mappings

LiteralSpecificationCommon_Mapping

Mapping Source

LiteralNull

Mapping Target

NullExpression

Owned Mappings

(none)

Applicable filters

(none)

7.7.14.2.11 LiteralReal_Mapping

Description

The mapping class maps UML4SysML::LiteralReal to SysML v2 LiteralRational.

General Mappings

LiteralSpecificationCommon_Mapping

Mapping Source

LiteralReal

Mapping Target

LiteralRational

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- LiteralRational::value () : Real [1]

```
from.value
```

7.7.14.2.12 LiteralSpecificationCommon_Mapping

Description

The mapping class the is abstract base class for all concrete UML4SysML::LiteralSpecification mappings.

General Mappings

ValueSpecification_Mapping

Mapping Source

LiteralSpecification

Mapping Target

LiteralExpression

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- LiteralExpression::ownedRelationship () : Relationship [0..*]

```
let ownerships: Set(SYSML2::Relationship) =
    self.oclAsType(ElementMain_Mapping).ownedRelationship()
    ->including(CommonReturnParameterFeatureMembership_Mapping.getMapped(from)) in
if from.type.oclIsUndefined() then
    ownerships
else
    ownerships->including(LiteralSpecificationTyping_Mapping.getMapped(from))
endif
```

7.7.14.2.13 LiteralSpecificationFeatureTyping_Mapping

Description

Creates a feature typing relationship owned by the element *typedFeature()*.

General Mappings

TypedElementFeatureTyping_Mapping

Mapping Source

LiteralSpecification

Mapping Target

FeatureTyping

Owned Mappings

(none)

Applicable filters

(none)

7.7.14.2.14 LiteralString_Mapping

Description

The mapping class maps UML4SysML::LiteralString to the SysML v2 LiteralString.

General Mappings

LiteralSpecificationCommon_Mapping

Mapping Source

LiteralString

Mapping Target

LiteralString

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- LiteralString::value () : String [1]

```
if from.value.oclIsUndefined() then '' else from.value endif
```

7.7.14.2.15 LiteralUnlimitedUnbounded_Mapping

Description

The mapping class maps UML4SysML::LiteralUnlimited to SysML v2 LiteralInfinity if it is the unlimited value.

General Mappings

LiteralUnlimitedInteger_Mapping

Mapping Source

LiteralUnlimitedNatural

Mapping Target

LiteralInfinity

Owned Mappings

(none)

Applicable filters

This mapping applies only if the following (OCL) condition implemented by the operation *filter(src : Element) : Boolean* is verified:

```
(from.value = -1)
```

Mapping rules

The mapping class only has inherited rules. See the mapping classes in the general mapping section for details.

7.7.14.2.16 LiteralUnlimitedInteger_Mapping

Description

The mapping class maps UML4SysML::LiteralUnlimited to SysML v2 LiteralInteger if it is not the unlimited value.

General Mappings

LiteralSpecificationCommon_Mapping

Mapping Source

LiteralUnlimitedNatural

Mapping Target

LiteralInteger

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- LiteralInteger::value () : Integer [1]

  ```
  from.value
  ```

7.7.14.2.17 OpaqueExpressionAsValue_Mapping

Description

The mapping class maps a UML4SysML::OpaqueExpression if it is used as a value to a SysML v2 FeatureChainExpression.

General Mappings

ToExpression_Init
Mapping

Mapping Source

OpaqueExpression

Mapping Target

FeatureChainExpression

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureChainExpression::ownedRelationship () : Relationship [0..*]

  ```
  Set{OpaqueExpressionParameterMembership_Mapping.getMapped(from),
  CommonReturnParameterFeatureMembership_Mapping.getMapped(from)}
  ```

7.7.14.2.18 OpaqueExpression_Mapping

Description

A UML4SysML::OpaqueExpression element is mapped to a SysMLv2 CalculationUsage element.. The following shows an example of what the textual SysML v2 syntax of the result of the transformation may look like.

```
calc sysMLv1OpaqueExpression {
        return result : ScalarValues::Integer;
        language "Built-in Math"
        /*
```

```
          * result = 42 + 23;
         */
}
```

General Mappings

CommonAction_Mapping
ValueSpecification_Mapping

Mapping Source

OpaqueExpression

Mapping Target

CalculationUsage

Owned Mappings

(none)

Applicable filters

This mapping applies only if the following (OCL) condition implemented by the operation *filter(src : Element) : Boolean* is verified:

```
if src.owner.oclIsKindOf(UML::TimeExpression) then
  not src.owner.owner.oclIsKindOf(UML::TimeEvent)
else
  true
endif
```

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- CalculationUsage::ownedRelationship () : Relationship [0..*]

  ```
  Set{OpaqueExpressionMembership_Mapping.getMapped(from),
  OpaqueExpressionReferenceUsageReturnParameterMembership_Mapping.getMapped(from)}
  ->union(self.oclAsType(ElementMain_Mapping).ownedRelationship())
  ```

7.7.14.2.19 OpaqueExpressionFeature_Mapping

Description

The mapping class creates the feature of the FeatureChainExpression.

General Mappings

ToFeature_Init
Mapping

Mapping Source

OpaqueExpression

Mapping Target

Feature

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Feature::ownedRelationship () : Relationship [0..*]

```
Set{OpaqueExpressionFeatureValue_Mapping.getMapped(from),
OpaqueExpressionFeatureFeatureMembership_Mapping.getMapped(from)}
```

7.7.14.2.20 OpaqueExpressionFeatureFeature_Mapping

Description

The mapping class creates the Feature of the FeatureReferenceExpression.

General Mappings

ToFeature_Init
Mapping

Mapping Source

OpaqueExpression

Mapping Target

Feature

Owned Mappings

(none)

Applicable filters

(none)

7.7.14.2.21 OpaqueExpressionFeatureFeatureMembership_Mapping

Description

Creates a feature membership relationship for *ownedMemberFeature()*.

General Mappings

ToFeatureMembership_Init
Mapping

Mapping Source

OpaqueExpression

Mapping Target

FeatureMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureMembership::ownedMemberFeature () : Feature [1]

  `OpaqueExpressionFeatureFeature_Mapping.getMapped(from)`

7.7.14.2.22 OpaqueExpressionFeatureValue_Mapping

Description

Creates a feature value relationship.

General Mappings

ToFeatureValue_Init
Mapping

Mapping Source

OpaqueExpression

Mapping Target

FeatureValue

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureValue::value () : Expression [1]

  ```
  OpaqueExpressionFeatureValueExpression_Mapping.getMapped(from)
  ```

7.7.14.2.23 OpaqueExpressionFeatureValueExpression_Mapping

Description

The mapping class creates the value of the FeatureChainExpression that is a FeatureReferenceExpression.

General Mappings

ToExpression_Init
Mapping

Mapping Source

OpaqueExpression

Mapping Target

FeatureReferenceExpression

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureReferenceExpression::ownedRelationship () : Relationship [0..*]

  ```
  Set{OpaqueExpressionFeatureValueExpressionMembership_Mapping.getMapped(from),
  ReturnParameterFeatureMembership_Factory.create()}
  ```

7.7.14.2.24 OpaqueExpressionFeatureValueExpressionMembership_Mapping

Description

Creates a membership relationship for *memberElement()*.

General Mappings

ToMembership_Init
Mapping

Mapping Source

OpaqueExpression

Mapping Target

Membership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Membership::memberElement () : Element [1]

  ```
  from
  ```

7.7.14.2.25 OpaqueExpressionMembership_Mapping

Description

Creates a membership relationship for *memberElement()*.

General Mappings

ToOwningMembership_Init
Mapping

Mapping Source

OpaqueExpression

Mapping Target

OwningMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- OwningMembership::ownedMemberElement () : Element [1]

  ```
  OpaqueExpressionSpecification_Mapping.getMapped(from)
  ```

7.7.14.2.26 OpaqueExpressionParameterMembership_Mapping

Description

Creates a membership relationship for *memberElement()*.

General Mappings

ToParameterMembership_Init
Mapping

Mapping Source

OpaqueExpression

Mapping Target

ParameterMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ParameterMembership::ownedMemberParameter () : Feature [1]

  ```
  OpaqueExpressionFeature_Mapping.getMapped(from)
  ```

7.7.14.2.27 OpaqueExpressionReferenceUsageReturnParameterMembership_Mapping

Description

Creates a membership relationship for *memberElement()*.

General Mappings

ToReturnParameterMembership_Init
Mapping

Mapping Source

OpaqueExpression

Mapping Target

ReturnParameterMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ReturnParameterMembership::ownedMemberParameter () : Feature [1]

```
if from.type.oclIsUndefined() then
    OpaqueExpressionReferenceUsageUntyped_Mapping.getMapped(from)
else
    OpaqueExpressionReferenceUsage_Mapping.getMapped(from)
endif
```

7.7.14.2.28 OpaqueExpressionReferenceUsage_Mapping

Description

The mapping class creates the return parameter reference usage of the calculation usage.

General Mappings

ToReferenceUsage_Init
Mapping

Mapping Source

OpaqueExpression

Mapping Target

ReferenceUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ReferenceUsage::direction () : FeatureDirectionKind [0..1]

```
KerML::FeatureDirectionKind::_'out'
```

- ReferenceUsage::ownedRelationship () : Relationship [0..*]

```
Set{OpaqueExpressionReferenceUsageFeatureTyping_Mapping.getMapped(from)}
```

7.7.14.2.29 OpaqueExpressionReferenceUsageFeatureTyping_Mapping

Description

Creates a feature typing relationship owned by the element *typedFeature()*.

General Mappings

TypedElementFeatureTyping_Mapping

Mapping Source

OpaqueExpression

Mapping Target

FeatureTyping

Owned Mappings

(none)

Applicable filters

(none)

7.7.14.2.30 OpaqueExpressionReferenceUsageUntyped_Mapping

Description

The mapping class creates the return parameter reference usage of the calculation usage, if the UML4SysML::OpaqueExpression is untyped.

General Mappings

ToReferenceUsage_Init
Mapping

Mapping Source

OpaqueExpression

Mapping Target

ReferenceUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ReferenceUsage::direction () : FeatureDirectionKind [0..1]

  ```
  KerML::FeatureDirectionKind::_'out'
  ```

7.7.14.2.31 OpaqueExpressionSpecification_Mapping

Description

The mapping class creates the specification of the calculation usage based on the language and body of the UML4SysML::OpaqueExpression.

General Mappings

ToTextualRepresentation_Init
Mapping

Mapping Source

OpaqueExpression

Mapping Target

TextualRepresentation

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- TextualRepresentation::language () : String [1]

  ```
  if from.language->size() = 0 then invalid else from.language.get(0) endif
  ```

- TextualRepresentation::body () : String [1]

  ```
  if from.body->size() = 0 then invalid else from.body.get(0) endif
  ```

7.7.14.2.32 TimeExpression_Mapping

Description

A UML4SysML::TimeExpression is mapped to a SysML v2 Expression. The details of the mapping are not specified yet.

General Mappings

ValueSpecification_Mapping

Mapping Source

TimeExpression

Mapping Target

Expression

Owned Mappings

(none)

Applicable filters

This mapping applies only if the following (OCL) condition implemented by the operation *filter(src : Element) : Boolean* is verified:

```
not src.owner.oclIsKindOf(UML::TimeEvent)
```

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Expression::ownedRelationship () : Relationship [0..*]

```
let ownedComments : Set(KerML::Relationship) =
  from.ownedComment->reject(c | c.annotatedElement->includes(from))
  ->collect(c| CommentOwnership_Mapping.getMapped(c))->asSet() in
let expression : Set(KerML::Relationship) = if from.expr.oclIsUndefined() then
  Set{}
else
  Set{ElementOwningMembership_Mapping.getMapped(from.expr)}
endif in
(if from.type.oclIsUndefined() then
  Set{CommonReturnParameterFeatureMembership_Mapping.getMapped(from)}
else
  Set{LiteralSpecificationTyping_Mapping.getMapped(from),
    CommonReturnParameterFeatureMembership_Mapping.getMapped(from)}
endif)
->union(ownedComments)
->union(expression)
```

7.7.14.2.33 ValueSpecification_Mapping

Description

The mapping class is the abstract base class of all mapping classes for special value specifications.

General Mappings

NamedElementMain_Mapping
ToExpression_Init

Mapping Source

ValueSpecification

Mapping Target

Expression

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Expression::ownedRelationship () : Relationship [0..*]

```
(if from.type.oclIsUndefined() then
    Set{CommonReturnParameterFeatureMembership_Mapping.getMapped(from)}
else
    Set{LiteralSpecificationTyping_Mapping.getMapped(from),
    CommonReturnParameterFeatureMembership_Mapping.getMapped(from)}
endif)->union(self.oclAsType(ElementMain_Mapping).ownedRelationship())
```

7.8 Mappings from SysML v1.7 stereotypes

7.8.1 Overview

The following subclauses of Mappings from SysML v1.7 stereotypes are organized according to the main packages of SysML v1.

7.8.2 Activities

SYSML21-542: Transformation: overview tables shall be revised

7.8.2.1 Overview

SYSML21-542: Transformation: overview tables shall be revised

Table 14. List of High-Level Activities Mappings

SysML v1 Abstract Syntax/Stereotype	SysML v2 Abstract Syntax
Continuous	No specific mapping specified yet.
ControlOperator	The concept that an action can control other actions is not supported by SysML v2.
Discrete	No specific mapping specified yet.
NoBuffer	No specific mapping specified yet.
Optional	The stereotype states that the lower multiplicity of the parameter is 0. Since the multiplicity of the parameter is transformed, the additional statement that the parameter is optional is redundant. Therefore, the stereotype is not considered in the transformation.
Overwrite	No specific mapping specified yet.

SysML v1 Abstract Syntax/Stereotype	SysML v2 Abstract Syntax
Probability	No specific mapping specified yet.
Rate	No specific mapping specified yet.

7.8.2.2 Mapping Specifications

7.8.2.2.1 ProbabilityMetadataUsage_Mapping

Description

A SysML::Activities::Probability is mapped to a SysML v2 MetadataUsage owned by the appropriate target element of the UML4SysML::ActivityEdge or UML4SysML::ParameterSet.

The following shows an example of what the textual SysML v2 syntax of the result of the transformation may look like.

```
action def SysMLv1Activity {
    action sysMLv1Action1;
    succession sysMLv1ControlFlow1 first sysMLv1Action1 then sysMLv1Action2 {
        @SysMLv1Library::ProbabilityData {probability = 0.42;}
    }
    action sysMLv1Action2;
}
```

General Mappings

ToMetadataUsage_Init
Mapping

Mapping Source

Element

Mapping Target

MetadataUsage

Owned Mappings

(none)

Applicable filters

This mapping applies only if the following (OCL) condition implemented by the operation *filter(src : Element) : Boolean* is verified:

```
Helper.hasStereotypeApplied(src, 'SysML::Activities::Probability')
```

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- MetadataUsage::ownedRelationship () : Relationship [0..*]

```
Set{ProbabilityMetadataUsageFeatureTyping_Mapping.getMapped(from),
ProbabilityMetadataUsageFeatureMembership_Mapping.getMapped(from)}
```

7.8.2.2.2 ProbabilityMetadataUsageFeatureMembership_Mapping

Description

Creates a feature membership relationship for *ownedMemberFeature()*.

General Mappings

ToFeatureMembership_Init
Mapping

Mapping Source

Element

Mapping Target

FeatureMembership

Owned Mappings

(none)

Applicable filters

This mapping applies only if the following (OCL) condition implemented by the operation *filter(src : Element) : Boolean* is verified:

```
Helper.hasStereotypeApplied(src, 'SysML::Activities::Probability')
```

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureMembership::ownedMemberFeature () : Feature [1]

  ```
  ProbabilityMetadataUsageReferenceUsage_Mapping.getMapped(from)
  ```

7.8.2.2.3 ProbabilityMetadataUsageFeatureTyping_Mapping

Description

Creates a feature typing relationship owned by the element *typedFeature()*.

General Mappings

ToFeatureTyping_Init
Mapping

Mapping Source

Element

Mapping Target

FeatureTyping

Owned Mappings

(none)

Applicable filters

This mapping applies only if the following (OCL) condition implemented by the operation *filter(src : Element) : Boolean* is verified:

```
Helper.hasStereotypeApplied(src, 'SysML::Activities::Probability')
```

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureTyping::type () : Type [1]

  ```
  SYSML2::MetadataDefinition.allInstances()
  ->any(m | m.qualifiedName = 'SysMLv1Library::ProbabilityData')
  ```

7.8.2.2.4 ProbabilityMetadataUsageReferenceUsage_Mapping

Description

Creates a reference usage.

General Mappings

ToReferenceUsage_Init
Mapping

Mapping Source

Element

Mapping Target

ReferenceUsage

Owned Mappings

(none)

Applicable filters

This mapping applies only if the following (OCL) condition implemented by the operation *filter(src : Element) : Boolean* is verified:

```
Helper.hasStereotypeApplied(src, 'SysML::Activities::Probability')
```

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ReferenceUsage::ownedRelationship () : Relationship [0..*]

```
Set{ProbabilityMetadataUsageReferenceUsageRedefinition_Mapping.getMapped(from),
ProbabilityMetadataUsageReferenceUsageFeatureValue_Mapping.getMapped(from)}
```

7.8.2.2.5 ProbabilityMetadataUsageReferenceUsageFeatureValue_Mapping

Description

Creates a feature value relationship.

General Mappings

ToFeatureValue_Init
Mapping

Mapping Source

Element

Mapping Target

FeatureValue

Owned Mappings

(none)

Applicable filters

This mapping applies only if the following (OCL) condition implemented by the operation *filter(src : Element) : Boolean* is verified:

```
Helper.hasStereotypeApplied(src, 'SysML::Activities::Probability')
```

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureValue::value () : Expression [1]

```
let probability : OclAny =
Helper.getTagValue(from, 'SysML::Activities::Probability', 'probability') in
LiteralRational_Factory.create(probability)
```

7.8.2.2.6 ProbabilityMetadataUsageReferenceUsageRedefinition_Mapping

Description

Creates a redefinition relationship for the *redefiningFeature()* and the *redefinedFeature()*.

General Mappings

ToRedefinition_Init
Mapping

Mapping Source

Element

Mapping Target

Redefinition

Owned Mappings

(none)

Applicable filters

This mapping applies only if the following (OCL) condition implemented by the operation *filter(src : Element) : Boolean* is verified:

```
Helper.hasStereotypeApplied(src, 'SysML::Activities::Probability')
```

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Redefinition::redefinedFeature () : Feature [1]

```
SYSML2::AttributeUsage.allInstances()
->any(m | m.qualifiedName = 'SysMLv1Library::ProbabilityData::probability')
```

7.8.2.2.7 ProbabilityOwningMembership_Mapping

Description

Creates a owning membership relationship for *ownedMemberElement()*.

General Mappings

ToOwningMembership_Init
Mapping

Mapping Source

Element

Mapping Target

OwningMembership

Owned Mappings

(none)

Applicable filters

This mapping applies only if the following (OCL) condition implemented by the operation *filter(src : Element) : Boolean* is verified:

```
Helper.hasStereotypeApplied(src, 'SysML::Activities::Probability')
```

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- OwningMembership::ownedMemberElement () : Element [1]

  ```
  ProbabilityMetadataUsage_Mapping.getMapped(from)
  ```

7.8.2.2.8 RateMetadataUsage_Mapping

Description

A SysML::Activities::Rate and the specializations SysML::Activities::Discrete and SysML::Activities::Continuous are mapped to a SysML v2 MetadataUsage owned by the appropriate target element of the UML4SysML::ActivityEdge or UML4SysML::Parameter.

The following shows an example of what the textual SysML v2 syntax of the result of the transformation may look like.

```
succession flow sysMLv1ObjectFlow of SysMLv1Block
    from sysMLv1Action1.outputValue to sysMLv1Action1.inputValue {
        @SysMLv1Library::RateData {isDiscrete = true;}
}
```

The mapping of the rate instance value is not supported yet.

General Mappings

ToMetadataUsage_Init
Mapping

Mapping Source

Element

Mapping Target

MetadataUsage

Owned Mappings

(none)

Applicable filters

This mapping applies only if the following (OCL) condition implemented by the operation *filter(src : Element) : Boolean* is verified:

```
Helper.hasStereotypeApplied(src, 'SysML::Activities::Rate')
or Helper.hasStereotypeApplied(src, 'SysML::Activities::Continuous')
or Helper.hasStereotypeApplied(src, 'SysML::Activities::Discrete')
```

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- MetadataUsage::ownedRelationship () : Relationship [0..*]

```
let relationships : Set(KerML::Relationship) =
    Set{RateMetadataUsageFeatureTyping_Mapping.getMapped(from)} in
if Helper.hasStereotypeApplied(from, 'SysML::Activities::Discrete') then
    relationships
    ->including(
        RateMetadataUsageDiscreteFeatureMembership_Mapping.getMapped(from))
else if Helper.hasStereotypeApplied(from, 'SysML::Activities::Continuous') then
        relationships
        ->including(
            RateMetadataUsageContinuousFeatureMembership_Mapping.getMapped(from))
     else
        relationships
     endif
endif
```

7.8.2.2.9 RateMetadataUsageContinuousFeatureMembership_Mapping

Description

Creates a feature membership relationship for *ownedMemberFeature()*.

General Mappings

ToFeatureMembership_Init
Mapping

Mapping Source

Element

Mapping Target

FeatureMembership

Owned Mappings

(none)

Applicable filters

This mapping applies only if the following (OCL) condition implemented by the operation *filter(src : Element) : Boolean* is verified:

```
Helper.hasStereotypeApplied(src, 'SysML::Activities::Continuous')
```

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureMembership::ownedMemberFeature () : Feature [1]

```
RateMetadataUsageContinuousReferenceUsage_Mapping.getMapped(from)
```

7.8.2.2.10 RateMetadataUsageFeatureValue_Mapping

Description

Creates a feature value relationship.

General Mappings

ToFeatureValue_Init
Mapping

Mapping Source

Element

Mapping Target

FeatureValue

Owned Mappings

(none)

Applicable filters

This mapping applies only if the following (OCL) condition implemented by the operation *filter(src : Element) : Boolean* is verified:

```
Helper.hasStereotypeApplied(src, 'SysML::Activities::Rate')
or Helper.hasStereotypeApplied(src, 'SysML::Activities::Continuous')
or Helper.hasStereotypeApplied(src, 'SysML::Activities::Discrete')
```

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureValue::value () : Expression [1]

  ```
  LiteralBoolean_Factory.create(true)
  ```

7.8.2.2.11 RateMetadataUsageContinuousReferenceUsage_Mapping

Description

Creates a reference usage.

General Mappings

ToReferenceUsage_Init
Mapping

Mapping Source

Element

Mapping Target

ReferenceUsage

Owned Mappings

(none)

Applicable filters

This mapping applies only if the following (OCL) condition implemented by the operation *filter(src : Element) : Boolean* is verified:

```
Helper.hasStereotypeApplied(src, 'SysML::Activities::Continuous')
```

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ReferenceUsage::ownedRelationship () : Relationship [0..*]

  ```
  Set{RateMetadataUsageContinuousReferenceUsageRedefinition_Mapping.getMapped(from),
  RateMetadataUsageFeatureValue_Mapping.getMapped(from)}
  ```

7.8.2.2.12 RateMetadataUsageContinuousReferenceUsageRedefinition_Mapping

Description

Creates a redefinition relationship for the *redefiningFeature()* and the *redefinedFeature()*.

General Mappings

ToRedefinition_Init
Mapping

Mapping Source

Element

Mapping Target

Redefinition

Owned Mappings

(none)

Applicable filters

This mapping applies only if the following (OCL) condition implemented by the operation *filter(src : Element) : Boolean* is verified:

```
Helper.hasStereotypeApplied(src, 'SysML::Activities::Continuous')
```

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Redefinition::redefinedFeature () : Feature [1]

```
SYSML2::AttributeUsage.allInstances()
->any(m | m.qualifiedName = 'SysMLv1Library::RateData::isContinuous')
```

7.8.2.2.13 RateMetadataUsageDiscreteFeatureMembership_Mapping

Description

Creates a feature membership relationship for *ownedMemberFeature()*.

General Mappings

ToFeatureMembership_Init
Mapping

Mapping Source

Element

Mapping Target

FeatureMembership

Owned Mappings

(none)

Applicable filters

This mapping applies only if the following (OCL) condition implemented by the operation *filter(src : Element) : Boolean* is verified:

```
Helper.hasStereotypeApplied(src, 'SysML::Activities::Discrete')
```

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureMembership::ownedMemberFeature () : Feature [1]

```
RateMetadataUsageDiscreteReferenceUsage_Mapping.getMapped(from)
```

7.8.2.2.14 RateMetadataUsageDiscreteReferenceUsage_Mapping

Description

Creates a reference usage.

General Mappings

ToReferenceUsage_Init
Mapping

Mapping Source

Element

Mapping Target

ReferenceUsage

Owned Mappings

(none)

Applicable filters

This mapping applies only if the following (OCL) condition implemented by the operation *filter(src : Element) : Boolean* is verified:

```
Helper.hasStereotypeApplied(src, 'SysML::Activities::Discrete')
```

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ReferenceUsage::ownedRelationship () : Relationship [0..*]

  ```
  Set{RateMetadataUsageDiscreteReferenceUsageRedefinition_Mapping.getMapped(from),
  RateMetadataUsageFeatureValue_Mapping.getMapped(from)}
  ```

7.8.2.2.15 RateMetadataUsageDiscreteReferenceUsageRedefinition_Mapping

Description

Creates a redefinition relationship for the *redefiningFeature()* and the *redefinedFeature()*.

General Mappings

ToRedefinition_Init
Mapping

Mapping Source

Element

Mapping Target

Redefinition

Owned Mappings

(none)

Applicable filters

This mapping applies only if the following (OCL) condition implemented by the operation *filter(src : Element) : Boolean* is verified:

```
Helper.hasStereotypeApplied(src, 'SysML::Activities::Discrete')
```

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Redefinition::redefinedFeature () : Feature [1]

  ```
  SYSML2::AttributeUsage.allInstances()
  ->any(m | m.qualifiedName = 'SysMLv1Library::RateData::isDiscrete')
  ```

7.8.2.2.16 RateMetadataUsageFeatureTyping_Mapping

Description

Creates a feature typing relationship owned by the element *typedFeature()*.

General Mappings

ToFeatureTyping_Init
Mapping

Mapping Source

Element

Mapping Target

FeatureTyping

Owned Mappings

(none)

Applicable filters

This mapping applies only if the following (OCL) condition implemented by the operation *filter(src : Element) : Boolean* is verified:

```
Helper.hasStereotypeApplied(src, 'SysML::Activities::Rate')
or Helper.hasStereotypeApplied(src, 'SysML::Activities::Continuous')
or Helper.hasStereotypeApplied(src, 'SysML::Activities::Discrete')
```

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureTyping::type () : Type [1]

  ```
  SYSML2::MetadataDefinition.allInstances()
  ->any(m | m.qualifiedName = 'SysMLv1Library::RateData')
  ```

7.8.2.2.17 RateOwningMembership_Mapping

Description

Creates a owning membership relationship for *ownedMemberElement()*.

General Mappings

ToOwningMembership_Init
Mapping

Mapping Source

Element

Mapping Target

OwningMembership

Owned Mappings

(none)

Applicable filters

This mapping applies only if the following (OCL) condition implemented by the operation *filter(src : Element) : Boolean* is verified:

```
Helper.hasStereotypeApplied(src, 'SysML::Activities::Rate')
or Helper.hasStereotypeApplied(src, 'SysML::Activities::Continuous')
or Helper.hasStereotypeApplied(src, 'SysML::Activities::Discrete')
```

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- OwningMembership::ownedMemberElement () : Element [1]

  ```
  RateMetadataUsage_Mapping.getMapped(from)
  ```

7.8.2.2.18 Model Libraries

7.8.2.2.18.1 ControlValues

7.8.2.2.18.1.1 ControlValueKind

The enumeration ControlValueKind is mapped to the SysML v2 enumeration definition SysMLv1Library::Enumerations::ControlValueKind (see 7.3.2).

7.8.3 Allocations

SYSML21-542: Transformation: overview tables shall be revised

7.8.3.1 Overview

SYSML21-542: Transformation: overview tables shall be revised

Table 15. List of High-Level Allocations Mappings

SysML v1 Abstract Syntax/Stereotype	SysML v2 Abstract Syntax
Allocate	No specific mapping specified yet.
AllocationUsage	No specific mapping specified yet.
AllocateActivityPartition	No specific mapping specified yet.

7.8.3.2 Mapping Specifications

7.8.3.2.1 Allocation_Mapping

Description

A SysML::Allocations::Allocate is mapped to a SysML v2 AllocationDefinition if it is an allocation between definition elements.

The following shows an example of what the textual SysML v2 syntax of the result of the transformation may look like.

```
action def SysMLv1Activity {
        action sysMLv1Action;
}
part def SysMLv1Block {
        part sysMLv1PartProperty : AnotherSysMLv1Block;
}
part def AnotherSysMLv1Block;

// Allocation of definition
allocation def SysMLv1Allocation {
        end :>> source : SysMLv1Activity;
        end :>> target : SysMLv1Block;
}

// Allocation of usage
allocation def {
        end :>> source : SysMLv1Activity;
        end :>> target : SysMLv1Block;
        allocate source.sysMLv1Action to target.sysMLv1PartProperty;
}
// Allocation of usage to definition
allocation def {
        end :>> source : SysMLv1Activity;
        end :>> target : SysMLv1Block;
        allocate source.sysMLv1Action to target;
}
```

General Mappings

Abstraction_Mapping

Mapping Source

Abstraction

Mapping Target

AllocationDefinition

Owned Mappings

(none)

Applicable filters

This mapping applies only if the following (OCL) condition implemented by the operation *filter(src : Element) : Boolean* is verified:

```
(Helper.hasStereotypeApplied(src, 'SysML::Allocations::Allocate'))
```

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- AllocationDefinition::ownedRelationship () : Relationship [0..*]

```
let relationships : Set(KerML::Relationship) =
    Set{AllocationSourceFeatureMembership_Mapping.getMapped(from.client.get(0)),
    AllocationTargetFeatureMembership_Mapping.getMapped(from.supplier.get(0))}
    ->union(self.oclAsType(ElementMain_Mapping).ownedRelationship()) in
if from.client.get(0).oclIsKindOf(UML::Type) then
    relationships
else
    relationships->including(AllocationUsageFeatureMembership_Mapping.getMapped(from))
endif
```

7.8.3.2.2 AllocationFeatureMembership_Mapping

Description

Creates a feature membership relationship for *ownedMemberFeature()*.

General Mappings

ToFeatureMembership_Init
Mapping

Mapping Source

NamedElement

Mapping Target

FeatureMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureMembership::ownedMemberFeature () : Feature [1]

```
AllocationSourceReferenceUsage_Mapping.getMapped(from)
```

7.8.3.2.3 AllocationFeatureTyping_Mapping

Description

Creates a feature typing relationship owned by the element *typedFeature()*.

General Mappings

ToFeatureTyping_Init
Mapping

Mapping Source

NamedElement

Mapping Target

FeatureTyping

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureTyping::type () : Type [1]

```
if from.oclIsKindOf(UML::Type) then
    from
else
    from.owner
endif
```

7.8.3.2.4 AllocationReferenceUsage_Mapping

Description

Creates a reference usage.

General Mappings

ToReferenceUsage_Init
UniqueMapping

Mapping Source

NamedElement

Mapping Target

ReferenceUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ReferenceUsage::ownedRelationship () : Relationship [0..*]

```
Set{AllocationFeatureTyping_Mapping.getMapped(from),
AllocationSourceReferenceUsageRedefinition_Mapping.getMapped(from)}
```

- ReferenceUsage::isEnd () : Boolean [1]

```
true
```

7.8.3.2.5 AllocationSourceReferenceUsageRedefinition_Mapping

Description

Creates a redefinition relationship for the *redefiningFeature()* and the *redefinedFeature()*.

General Mappings

ToRedefinition_Init

Mapping Source

NamedElement

Mapping Target

Redefinition

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Redefinition::redefinedFeature () : Feature [1]

```
SYSML2::ReferenceUsage.allInstances()
->any(m | m.qualifiedName = 'Allocations::Allocation::source')
```

7.8.3.2.6 AllocationTargetFeatureMembership_Mapping

Description

Creates a feature membership relationship for *ownedMemberFeature()*.

General Mappings

ToFeatureMembership_Init

Mapping Source

NamedElement

Mapping Target

FeatureMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureMembership::ownedMemberFeature () : Feature [1]

```
AllocationTargetReferenceUsage_Mapping.getMapped(from)
```

7.8.3.2.7 AllocationTargetReferenceUsage_Mapping

Description

Creates a reference usage.

General Mappings

ToReferenceUsage_Init
UniqueMapping

Mapping Source

NamedElement

Mapping Target

ReferenceUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ReferenceUsage::ownedRelationship () : Relationship [0..*]

  ```
  Set{AllocationFeatureTyping_Mapping.getMapped(from),
  AllocationTargetReferenceUsageRedefinition_Mapping.getMapped(from)}
  ```

- ReferenceUsage::isEnd () : Boolean [1]

  ```
  true
  ```

7.8.3.2.8 AllocationTargetReferenceUsageRedefinition_Mapping

Description

Creates a redefinition relationship for the *redefiningFeature()* and the *redefinedFeature()*.

General Mappings

ToRedefinition_Init

Mapping Source

NamedElement

Mapping Target

Redefinition

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Redefinition::redefinedFeature () : Feature [1]

```
SYSML2::ReferenceUsage.allInstances()
->any(m | m.qualifiedName = 'Allocations::Allocation::target')
```

7.8.3.2.9 AllocationUsage_Mapping

Description

A SysML::Allocations::Allocate is mapped to a SysML v2 AllocationUsage owned by a AllocationDefinition if a usage element is source or target of the allocation relationship.

General Mappings

ToUsage_Init
Mapping

Mapping Source

Abstraction

Mapping Target

AllocationUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- AllocationUsage::ownedRelationship () : Relationship [0..*]

```
Set{AllocationUsageSourceEndFeatureMembership_Mapping.getMapped(from.client.get(0)),
AllocationUsageTargetEndFeatureMembership_Mapping.getMapped(from.target.get(0))}
```

7.8.3.2.10 AllocationUsageEndFeatureMembership_Mapping

Description

Creates a feature membership relationship for *ownedMemberFeature()*.

General Mappings

ToEndFeatureMembership_Init
Mapping

Mapping Source

NamedElement

Mapping Target

EndFeatureMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- EndFeatureMembership::ownedMemberFeature () : Feature [1]

  ```
  AllocationUsageSourceFeature_Mapping.getMapped(from)
  ```

7.8.3.2.11 AllocationUsageFeature_Mapping

Description

Creates a feature element as an end of the allocation usage relationship.

General Mappings

ToFeature_Init
Mapping

Mapping Source

NamedElement

Mapping Target

Feature

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Feature::ownedRelationship () : Relationship [0..*]

  ```
  Set{AllocationUsageSourceFeatureSubsetting_Mapping.getMapped(from)}
  ```

7.8.3.2.12 AllocationUsageFeatureChaining_Mapping

Description

Creates the first feature chaining element for the subsetting feature for the feature element which represents an end of the allocation usage relationship.

General Mappings

ToFeatureChaining_Init
Mapping

Mapping Source

NamedElement

Mapping Target

FeatureChaining

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureChaining::chainingFeature () : Feature [1]

  ```
  AllocationSourceReferenceUsage_Mapping.getMapped(from)
  ```

7.8.3.2.13 AllocationUsageFeatureChainingChainedFeature_Mapping

Description

Creates the second feature chaining element for the subsetting feature for the feature element which represents an end of the allocation usage relationship.

General Mappings

ToFeatureChaining_Init
Mapping

Mapping Source

NamedElement

Mapping Target

FeatureChaining

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureChaining::chainingFeature () : Feature [1]

  ```
  from
  ```

7.8.3.2.14 AllocationUsageFeatureMembership_Mapping

Description

Creates a feature membership relationship for *ownedMemberFeature()*.

General Mappings

ToFeatureMembership_Init
Mapping

Mapping Source

Abstraction

Mapping Target

FeatureMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureMembership::ownedMemberFeature () : Feature [1]

  ```
  AllocationUsage_Mapping.getMapped(from)
  ```

7.8.3.2.15 AllocationUsageFeatureSubsetting_Mapping

Description

Creates a subsetting relationship.

General Mappings

ToReferenceSubsetting_Init
Mapping

Mapping Source

NamedElement

Mapping Target

ReferenceSubsetting

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ReferenceSubsetting::ownedRelatedElement () : Element [0..*]

```
if from.oclIsKindOf(UML::Type) then
    Set{}
else
    Set{AllocationUsageSourceFeatureSubsettingFeature_Mapping.getMapped(from)}
endif
```

7.8.3.2.16 AllocationUsageFeatureSubsettingFeature_Mapping

Description

Creates the subsetting feature for the feature element which represents an end of the allocation usage relationship.

General Mappings

ToFeature_Init
Mapping

Mapping Source

NamedElement

Mapping Target

Feature

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Feature::ownedRelationship () : Relationship [0..*]

```
Set{AllocationUsageSourceFeatureChaining_Mapping.getMapped(from),
AllocationUsageFeatureChainingChainedFeature_Mapping.getMapped(from)}
```

7.8.3.2.17 AllocationUsageTargetEndFeatureMembership_Mapping

Description

Creates a feature membership relationship for *ownedMemberFeature()*.

General Mappings

ToEndFeatureMembership_Init

Mapping Source

NamedElement

Mapping Target

EndFeatureMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- EndFeatureMembership::ownedMemberFeature () : Feature [1]

```
AllocationUsageTargetFeature_Mapping.getMapped(from)
```

7.8.3.2.18 AllocationUsageTargetFeature_Mapping

Description

Creates a feature element as an end of the allocation usage relationship.

General Mappings

ToFeature_Init

Mapping Source

NamedElement

Mapping Target

Feature

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Feature::ownedRelationship () : Relationship [0..*]

  ```
  Set{AllocationUsageTargetFeatureSubsetting_Mapping.getMapped(from)}
  ```

7.8.3.2.19 AllocationUsageTargetFeatureChaining_Mapping

Description

Creates the first feature chaining element for the subsetting feature for the feature element which represents an end of the allocation usage relationship.

General Mappings

ToFeatureChaining_Init

Mapping Source

NamedElement

Mapping Target

FeatureChaining

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureChaining::chainingFeature () : Feature [1]

```
AllocationTargetReferenceUsage_Mapping.getMapped(from)
```

7.8.3.2.20 AllocationUsageTargetFeatureSubsetting_Mapping

Description

Creates a subsetting relationship.

General Mappings

ToReferenceSubsetting_Init

Mapping Source

NamedElement

Mapping Target

ReferenceSubsetting

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ReferenceSubsetting::ownedRelatedElement () : Element [0..*]

```
if from.oclIsKindOf(UML::Type) then
    Set{}
else
    Set{AllocationUsageTargetFeatureSubsettingFeature_Mapping.getMapped(from)}
endif
```

7.8.3.2.21 AllocationUsageTargetFeatureSubsettingFeature_Mapping

Description

Creates the subsetting feature for the feature element which represents an end of the allocation usage relationship.

General Mappings

ToFeature_Init

Mapping Source

NamedElement

Mapping Target

Feature

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Feature::ownedRelationship () : Relationship [0..*]

```
Set{AllocationUsageTargetFeatureChaining_Mapping.getMapped(from),
AllocationUsageFeatureChainingChainedFeature_Mapping.getMapped(from)}
```

7.8.4 Blocks

SYSML21-542: Transformation: overview tables shall be revised

7.8.4.1 Overview

SYSML21-542: Transformation: overview tables shall be revised

Table 16. List of High-Level Allocations Mappings

SysML v1 Abstract Syntax/Stereotype	SysML v2 Abstract Syntax
AdjunctProperty	The concept of adjunct properties is not needed in SysML v2, where the principal of the adjunct property can be used directly in the appropriate place.
BindingConnector	BindingConnectorAsUsage
Block	PartDefinition
BoundReference	No specific mapping specified yet.
ClassifierBehaviorProperty	The classifier behavior is already mapped to a property which also plays the role of the classifier behavior property. Therefore, there is no explicit mapping of a classifier behavior property.
ConnectorProperty	The connector property is a special case of an adjunct property and is not mapped, just like the adjunct property.
DirectedRelationshipPropertyPath	The stereotype is abstract is therefore not mapped. The concept of the DirectedRelationshipPropertyPath is included in the SysML v2 language.
DistributedProperty	No specific mapping specified yet.
ElementPropertyPath	The stereotype is abstract is therefore not mapped. The concept of the ElementPropertyPath is included in the SysML v2 language.

SysML v1 Abstract Syntax/Stereotype	SysML v2 Abstract Syntax
EndPathMultiplicity	No specific mapping specified yet.
NestedConnectorEnd	The concept of NestedConnectorEnd is already included in the SysML v2 language. It is not required to do an explicit mapping.
ParticipantProperty	No specific mapping specified yet.
PropertySpecificType	No specific mapping specified yet.
ValueType	AttributeDefinition

7.8.4.2 Mapping Specifications

7.8.4.2.1 AssociationBlock_Mapping

Description

An AssociationBlock is mapped to a SysML v2 ConnectionDefinition.

The SysML::Blocks::ParticipantProperties transformation is not defined yet. Therefore, the mapping is currently identical with the mapping of UML4SysML::AssociationClass.

General Mappings

AssociationClass_Mapping

Mapping Source

AssociationClass

Mapping Target

ConnectionDefinition

Owned Mappings

(none)

Applicable filters

This mapping applies only if the following (OCL) condition implemented by the operation *filter(src : Element) : Boolean* is verified:

```
Helper.hasStereotypeApplied(src, 'SysML::Blocks::Block')
```

Mapping rules

The mapping class only has inherited rules. See the mapping classes in the general mapping section for details.

7.8.4.2.2 BindingConnector_Mapping

Description

A SysML::Blocks::BindingConnector is mapped to a SysML v2 BindingConnectorAsUsage.

The following shows an example of what the textual SysML v2 syntax of the result of the transformation may look like.

```
part def SysMLv1Block1 {
        part sysMLv1PartProperty1 : SysMLv1Block2;
        part sysMLv1PartProperty2 : SysMLv1Block2;

        binding sysMLv1BindingConnector
            bind sysMLv1PartProperty1 = sysMLv1PartProperty2;
}
part def SysMLv1Block2;
```

General Mappings

Connector_Mapping

Mapping Source

Connector

Mapping Target

BindingConnectorAsUsage

Owned Mappings

(none)

Applicable filters

This mapping applies only if the following (OCL) condition implemented by the operation *filter(src : Element) : Boolean* is verified:

```
Helper.hasStereotypeApplied(src, 'SysML::Blocks::BindingConnector')
```

Mapping rules

The mapping class only has inherited rules. See the mapping classes in the general mapping section for details.

7.8.4.2.3 Block_Mapping

Description

A SysML::Blocks::Block is mapped to a SysML v2 PartDefinition.

The following shows an example of what the textual SysML v2 syntax of the result of the transformation may look like.

```
part definition SysMLv1Block;
```

General Mappings

Class_Mapping

Mapping Source

Class

Mapping Target

PartDefinition

Owned Mappings

(none)

Applicable filters

This mapping applies only if the following (OCL) condition implemented by the operation *filter(src : Element) : Boolean* is verified:

```
not src.oclIsTypeOf(UML::AssociationClass)
   and Helper.hasStereotypeApplied(src, 'SysML::Blocks::Block')
   and not Helper.hasStereotypeApplied(src, 'SysML::ConstraintBlocks::ConstraintBlock')
   and not Helper.hasStereotypeApplied(src, 'SysML::Ports&Flows::InterfaceBlock')
```

Mapping rules

The mapping class only has inherited rules. See the mapping classes in the general mapping section for details.

7.8.4.2.4 EncapsulatedBlock_Mapping

Description

A SysML::Block with *isEncapsulated=true* is mapped to a SysML v2 PartDefinition, and, additionally, gets a metadata feature defined by the SysML v1 library which represents the SysML v1 isEncapsulated property.

The following shows an example of what the textual SysML v2 syntax of the result of the transformation may look like.

```
part def SysMLv1EncapsulatedBlock {
  @SysMLv1Library::BlockData {isEncapsulated = true;}
}
```

General Mappings

Block_Mapping

Mapping Source

Class

Mapping Target

PartDefinition

Owned Mappings

(none)

Applicable filters

This mapping applies only if the following (OCL) condition implemented by the operation *filter(src : Element) : Boolean* is verified:

```
not src.oclIsTypeOf(UML::AssociationClass) and
   Helper.hasStereotypeApplied(src, 'SysML::Blocks::Block') and
   not Helper.hasStereotypeApplied(src, 'SysML::ConstraintBlocks::ConstraintBlock') and
   not Helper.hasStereotypeApplied(src, 'SysML::Ports&Flows::InterfaceBlock') and
   Helper.getTagValue(src, 'SysML::Blocks::Block', 'isEncapsulated')
```

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- PartDefinition::ownedRelationship () : Relationship [0..*]

```
let toElementFMS: Set(UML::Element) =
    from.ownedElement->select(e | e.oclIsKindOf(UML::Property) and
    (e.oclAsType(UML::Property).redefinedProperty->size() = 0)) in
let redefinedAttributes: Set(UML::Element) =
    from.ownedElement->select(e | from.oclIsKindOf(UML::DataType) and
    (e.oclAsType(UML::Property).redefinedProperty->size() > 0)) in
let generalizations : Set(UML::Generalization) =
    from.ownedElement->select(e | e.oclIsKindOf(UML::Generalization)) in
let toElementOMS: Set(UML::Element) =
    (((from.ownedElement - toElementFMS) - redefinedAttributes) -
    generalizations)  in
let relationships: Sequence(UML::Element) =
toElementOMS->collect(e | ElementOwningMembership_Mapping.getMapped(e))
->union(toElementFMS
    ->collect(e | ElementFeatureMembership_Mapping.getMapped(e)))
->union(redefinedAttributes
    ->collect(e | AttributeRedefinedMembership_Mapping.getMapped(e)))
->union(generalizations->collect(e | Generalization_Mapping.getMapped(e)))
->including(EncapsulatedBlockMetadataMembership_Mapping.getMapped(from)) in
if from.classifierBehavior.oclIsUndefined() then
    relationships
else
    relationships
    ->append(BehavioredClassifierFeatureMembership_Mapping.getMapped(from))
endif
```

7.8.4.2.5 EncapsulatedBlockMetadataMembership_Mapping

Description

Creates a membership relationship for *memberElement()*.

General Mappings

ToOwningMembership_Init
Mapping

Mapping Source

Class

Mapping Target

OwningMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- OwningMembership::ownedMemberElement () : Element [1]

```
EncapsulatedBlockMetadata_Mapping.getMapped(from)
```

7.8.4.2.6 EncapsulatedBlockMetadata_Mapping

Description

The mapping class creates the metadata for the property SysML::Blocks::Block::isEncapsulated.

General Mappings

ToMetadataUsage_Init
Mapping

Mapping Source

Class

Mapping Target

MetadataUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- MetadataUsage::ownedRelationship () : Relationship [0..*]

```
Set{EncapsulatedBlockMetadataFeatureTyping_Mapping.getMapped(from),
EncapsulatedBlockMetadataFeatureMembership_Mapping.getMapped(from)}
```

7.8.4.2.7 EncapsulatedBlockMetadataFeatureMembership_Mapping

Description

Creates a feature membership relationship for *ownedMemberFeature()*.

General Mappings

ToFeatureMembership_Init
Mapping

Mapping Source

Class

Mapping Target

FeatureMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureMembership::ownedMemberFeature () : Feature [0..1]

  ```
  EncapsulatedBlockMetadataReferenceUsage_Mapping.getMapped(from)
  ```

7.8.4.2.8 EncapsulatedBlockMetadataFeatureTyping_Mapping

Description

Creates a feature typing relationship owned by the element *typedFeature()*.

General Mappings

ToFeatureTyping_Init
Mapping

Mapping Source

Class

Mapping Target

FeatureTyping

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureTyping::type () : Type [1]

```
SYSML2::MetadataDefinition.allInstances()
->any(m | m.qualifiedName = 'SysMLv1Library::BlockData')
```

7.8.4.2.9 EncapsulatedBlockMetadataReferenceUsage_Mapping

Description

Creates a reference usage.

General Mappings

ToReferenceUsage_Init
Mapping

Mapping Source

Class

Mapping Target

ReferenceUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ReferenceUsage::ownedRelationship () : Relationship [0..*]

```
Set{EncapsulatedBlockMetadataRedefinition_Mapping.getMapped(from),
EncapsulatedBlockMetadataFeatureValue_Mapping.getMapped(from)}
```

7.8.4.2.10 EncapsulatedBlockMetadataFeatureValue_Mapping

Description

Creates a feature value relationship.

General Mappings

ToFeatureValue_Init
Mapping

Mapping Source

Class

Mapping Target

FeatureValue

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureValue::value () : Expression [1]

  ```
  LiteralBoolean_Factory.create(true)
  ```

7.8.4.2.11 EncapsulatedBlockMetadataRedefinition_Mapping

Description

Creates a redefinition relationship for the *redefiningFeature()* and the *redefinedFeature()*.

General Mappings

ToRedefinition_Init
Mapping

Mapping Source

Class

Mapping Target

Redefinition

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Redefinition::redefinedFeature () : Feature [1]

  ```
  SYSML2::AttributeUsage.allInstances()
  ->any(m | m.qualifiedName = 'SysMLv1Library::BlockData::isEncapsulated')
  ```

7.8.4.2.12 FlowPropertyPart_Mapping

Description

A UML4SysML::Property which is typed by a block and to which the SysML v1 stereotype FlowProperty has been applied is mapped as in the general mapping class PartProperty_Mapping but the target feature is always referential and the flow direction specified in the stereotype FlowProperty is considered.

General Mappings

PartProperty_Mapping

Mapping Source

Property

Mapping Target

PartUsage

Owned Mappings

(none)

Applicable filters

This mapping applies only if the following (OCL) condition implemented by the operation *filter(src : Element) : Boolean* is verified:

```
Helper.hasStereotypeApplied(src, 'SysML::Ports&Flows::FlowProperty')
    and not src.type.oclIsUndefined()
    and src.type.oclIsKindOf(UML::Class)
    and Helper.hasStereotypeApplied(src.type, 'SysML::Blocks::Block')
```

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- PartUsage::direction () : FeatureDirectionKind [0..1]

  ```
  Helper.getFlowDirectionKind(

  Helper.getTagValue(from,

  'SysML::Ports&Flows::FlowProperty', 'direction'))
  ```

- PartUsage::isComposite () : Boolean [1]

  ```
  false
  ```

7.8.4.2.13 PartProperty_Mapping

Description

A UML4SysML::Property which is typed by a block is mapped to a SysML::PartUsage. The derived property Property::isComposite is directly mapped to PartUsage::isComposite.

The following shows an example of what the textual SysML v2 syntax of the result of the transformation may look like.

```
part def SysMLv1Block1 {
        part sysMLv1PartProperty1 : SysMLv1Block2;
        ref part sysMLv1ReferencedPartProperty2 : SysMLv1Block2;
}
part def SysMLv1Block2;
```

General Mappings

PropertyTypedByClassInterface_Mapping

Mapping Source

Property

Mapping Target

PartUsage

Owned Mappings

(none)

Applicable filters

This mapping applies only if the following (OCL) condition implemented by the operation *filter(src : Element) : Boolean* is verified:

```
if src.oclIsKindOf(UML::Property) and not src.oclIsKindOf(UML::Port) then
    let p: UML::Property = src.oclAsType(UML::Property) in
    not p.type.oclIsUndefined() and
    Helper.hasStereotypeApplied(p.type, 'SysML::Blocks::Block') and
    (p.association.oclIsUndefined() or p.association.ownedEnd->excludes(p))
else
    false
endif
```

Mapping rules

The mapping class only has inherited rules. See the mapping classes in the general mapping section for details.

7.8.4.2.14 Model Libraries

7.8.4.2.14.1 PrimitiveValueTypes

7.8.4.2.14.1.1 Boolean

7.8.4.2.14.1.2 Complex

7.8.4.2.14.1.3 Integer

7.8.4.2.14.1.4 Number

7.8.4.2.14.1.5 Real

7.8.4.2.14.1.6 String

7.8.4.2.14.2 UnitAndQuantityKind

7.8.4.2.14.2.1 QuantityKind

7.8.4.2.14.2.2 Unit

7.8.4.2.15 ValueType_Mapping

Description

A SysML::Blocks::ValueType is mapped to a SysML v2 AttributeDefinition.

The following shows an example of what the textual SysML v2 syntax of the result of the transformation may look like.

```
attribute definition SysMLv1ValueType;
```

General Mappings

DataType_Mapping

Mapping Source

DataType

Mapping Target

AttributeDefinition

Owned Mappings

(none)

Applicable filters

This mapping applies only if the following (OCL) condition implemented by the operation *filter(src : Element) : Boolean* is verified:

```
Helper.hasStereotypeApplied(from, 'SysML::Blocks::ValueType')
```

Mapping rules

The mapping class only has inherited rules. See the mapping classes in the general mapping section for details.

7.8.5 ConstraintBlocks

7.8.5.1 Overview

SYSML21-542: Transformation: overview tables shall be revised

Table 17. List of High-Level SysML v1 ConstraintBlocks Mappings

SysML v1 Abstract Syntax/Stereotype	SysML v2 Abstract Syntax
ConstraintBlock	ConstraintDefinition

7.8.5.2 Mapping Specifications

7.8.5.2.1 ConstraintBlock_Mapping

Description

A SysML::ConstraintBlocks::ConstraintBlock is mapped to a SysML v2 ConstraintDefinition.

The following shows an example of what the textual SysML v2 syntax of the result of the transformation may look like.

```
onstraint def SysMLv1ConstraintBlock {
        in attribute a : ScalarValues::Integer;
        in attribute b : ScalarValues::Integer;
        in attribute c : ScalarValues::Integer;

        constraint constraintExpression {
                language "OCL2.0"
                /*
                 * c == a + b
                 */
         }
}
```

General Mappings

Class_Mapping

Mapping Source

Class

Mapping Target

ConstraintDefinition

Owned Mappings

(none)

Applicable filters

This mapping applies only if the following (OCL) condition implemented by the operation *filter(src : Element) : Boolean* is verified:

```
Helper.hasStereotypeApplied(src, 'SysML::ConstraintBlocks::ConstraintBlock')
```

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ConstraintDefinition::ownedRelationship () : Relationship [0..*]

```
let generalizations : Set(UML::Generalization) =
    from.ownedElement->select(e | e.oclIsKindOf(UML::Generalization)) in
let toElementFMS : Set(UML::Element) =
    from.ownedElement
    ->select(e | e.oclIsKindOf(UML::Property) or e.oclIsKindOf(UML::Constraint)) in
let toElementOMS: Set(UML::Element) =
    (from.ownedElement - generalizations) - toElementFMS  in
toElementOMS->collect(e | ElementOwningMembership_Mapping.getMapped(e))
->union(toElementFMS->collect(e | ElementFeatureMembership_Mapping.getMapped(e)))
->union(generalizations->collect(e | Generalization_Mapping.getMapped(e)))
->including(CommonReturnParameterReferenceUsageMembership_Mapping.getMapped(from))
```

7.8.5.2.2 ConstraintParameter_Mapping

Description

The mapping class maps SysML v1 constraint parameter to SysML v2 attribute usages.

General Mappings

PropertyCommon_Mapping
NamedElementMain_Mapping

Mapping Source

Property

Mapping Target

AttributeUsage

Owned Mappings

(none)

Applicable filters

This mapping applies only if the following (OCL) condition implemented by the operation *filter(src : Element) : Boolean* is verified:

```
if src.oclIsKindOf(UML::Property) and
Helper.hasStereotypeApplied(src.owner, 'SysML::ConstraintBlocks::ConstraintBlock') then
    let p: UML::Property = src.oclAsType(UML::Property) in
    if p.type.oclIsUndefined() then
        false
    else
        true
    endif
else
    false
endif
```

Mapping rules

The mapping class only has inherited rules. See the mapping classes in the general mapping section for details.

7.8.6 Model Elements

SYSML21-542: Transformation: overview tables shall be revised

7.8.6.1 Overview

Table 18. List of High-Level SysML v1 ModelElements Mappings

SysML v1 Abstract Syntax/Stereotype	SysML v2 Abstract Syntax
Conform	No specific mapping specified yet.
ElementGroup	Package
Expose	No specific mapping specified yet.
Problem	Comment
Rationale	Comment
Stakeholder	ItemDefinition
View	No specific mapping specified yet.
Viewpoint	No specific mapping specified yet.

7.8.6.2 Mapping Specifications

7.8.6.2.1 ProblemRationaleMetadataFeatureMembership_Mapping

Description

Creates a feature membership relationship for *ownedMemberFeature()*.

General Mappings

ToFeatureMembership_Init
Mapping

Mapping Source

Comment

Mapping Target

FeatureMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureMembership::ownedMemberFeature () : Feature [0..1]

```
ProblemRationaleMetadataReferenceUsage_Mapping.getMapped(from)
```

7.8.6.2.2 ProblemRationaleMetadataFeatureTyping_Mapping

Description

Creates a feature typing relationship owned by the element *typedFeature()*.

General Mappings

ToFeatureTyping_Init
Mapping

Mapping Source

Comment

Mapping Target

FeatureTyping

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureTyping::type () : Type [1]

```
if Helper.hasStereotypeApplied(from, 'SysML::ModelElements::Problem') then
  SYSML2::MetadataDefinition.allInstances()
    ->any(m | m.qualifiedName = 'ModelingMetadata::Issue')
else if Helper.hasStereotypeApplied(from, 'SysML::ModelElements::Rationale') then
  SYSML2::MetadataDefinition.allInstances()
    ->any(m | m.qualifiedName = 'ModelingMetadata::Rationale')
else invalid endif endif
```

7.8.6.2.3 ProblemRationaleMetadataReferenceUsage_Mapping

Description

Creates a reference usage.

General Mappings

ToReferenceUsage_Init
Mapping

Mapping Source

Comment

Mapping Target

ReferenceUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ReferenceUsage::ownedRelationship () : Relationship [0..*]

```
Set{ProblemRationaleMetadataRedefinition_Mapping.getMapped(from),
ProblemRationaleMetadataFeatureValue_Mapping.getMapped(from)}
```

7.8.6.2.4 ProblemRationaleMetadataFeatureValue_Mapping

Description

Creates a feature value relationship.

General Mappings

ToFeatureValue_Init
Mapping

Mapping Source

Comment

Mapping Target

FeatureValue

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureValue::value () : Expression [1]

  ```
  LiteralString_Factory.create(from.body)
  ```

7.8.6.2.5 ProblemRationaleMetadataMembership_Mapping

Description

Creates a membership relationship for *memberElement()*.

General Mappings

ToOwningMembership_Init
Mapping

Mapping Source

Comment

Mapping Target

OwningMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- OwningMembership::ownedMemberElement () : Element [1]

  ```
  ProblemRationaleMetadataUsage_Mapping.getMapped(from)
  ```

7.8.6.2.6 Concern_Mapping

Description

The concern comments of a SysML::ModelElements::Stakeholder or a SysML::ModelElements::Viewpoint are mapped to SysML v2 ConcernUsages. The concern comments of the stakeholder are mapped to ConcernUsages which reference the stakeholder item definition.

The following shows an example of what the textual SysML v2 syntax of the result of the transformation may look like.

```
item def SysMLv1Stakeholder {
        @SysMLv1Library::StakeholderData {isStakeholder = true;}
}
concern concernCommentXMI_ID {
```

```
        doc /* concern string */
        stakeholder : SysMLv1Stakeholder;
}
```

General Mappings

Comment_Mapping

Mapping Source

Comment

Mapping Target

ConcernUsage

Owned Mappings

(none)

Applicable filters

This mapping applies only if the following (OCL) condition implemented by the operation *filter(src : Element) : Boolean* is verified:

```
(not Helper.hasStereotypeApplied(src, 'SysML::ModelElements::ElementGroup')) and
((UML::Classifier.allInstances()
->select(s |
    Helper.hasStereotypeApplied(s, 'SysML::ModelElements::Stakeholder'))
->collect(c |
    Helper.getTagValue(c, 'SysML::ModelElements::Stakeholder', 'concernList'))
    ->flatten()
->includes(src)) or
(UML::Classifier.allInstances()
->select(s |
    Helper.hasStereotypeApplied(s, 'SysML::ModelElements::Viewpoint'))
->collect(c |
    Helper.getTagValue(c, 'SysML::ModelElements::Viewpoint', 'concernList'))
->flatten()->includes(src)))
```

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ConcernUsage::ownedRelationship () : Relationship [0..*]

```
let toStakeholderMS : Set(UML::Classifier) =
    UML::Classifier.allInstances()
    ->select(s |
        Helper.hasStereotypeApplied(s, 'SysML::ModelElements::Stakeholder'))
    ->select(s |
        Helper.getTagValue(s, 'SysML::ModelElements::Stakeholder', 'concernList')
    ->flatten()->includes(from))->asSet() in
toStakeholderMS
```

```
->including(
    CommonReturnParameterReferenceUsageMembership_Mapping.getMapped(from))
->including(EmptySubjectMembership_Factory.create())
->union(self.oclAsType(Comment_Mapping).ownedRelationship())
```

7.8.6.2.7 ConcernDocumentation_Mapping

Description

The mapping class creates the documentation element with the body string of the UML4SysML::Comment model element representing a concern.

General Mappings

ToDocumentation_Init
Mapping

Mapping Source

Comment

Mapping Target

Documentation

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Documentation::body () : String [1]

```
from.body
```

7.8.6.2.8 ConcernOwningMembership_Mapping

Description

Creates a owning membership relationship for *ownedMemberElement()*.

General Mappings

ToOwningMembership_Init
Mapping

Mapping Source

Comment

Mapping Target

OwningMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- OwningMembership::ownedMemberElement () : Element [1]

  ```
  ConcernDocumentation_Mapping.getMapped(from)
  ```

7.8.6.2.9 ConcernStakeholderMembership_Mapping

Description

Creates a membership relationship for *memberElement()*.

General Mappings

ToParameterMembership_Init
Mapping

Mapping Source

Classifier

Mapping Target

StakeholderMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- StakeholderMembership::ownedMemberParameter () : Feature [1]

```
ConcernStakeholderPartUsage_Mapping.getMapped(from)
```

7.8.6.2.10 ConcernStakeholderPartUsage_Mapping

Description

In SysML v1, the stakeholder element has concerns. In SysML v2, the Concern element has stakeholders. This mapping class creates a PartUsage of the type of the stakeholder for the concern element.

General Mappings

ToPartUsage_Init
Mapping

Mapping Source

Classifier

Mapping Target

PartUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- PartUsage::ownedRelationship () : Relationship [0..*]

```
Set{ConcernStakeholderPartUsageFeatureTyping_Mapping.getMapped(from),
ConcernStakeholderPartUsageOwningMembership_Mapping.getMapped(from)}
```

7.8.6.2.11 ConcernStakeholderPartUsageFeatureTyping_Mapping

Description

Creates a feature typing relationship owned by the element *typedFeature()*.

General Mappings

ToFeatureTyping_Init
Mapping

Mapping Source

Classifier

Mapping Target

FeatureTyping

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureTyping::type () : Type [1]

  `from`

7.8.6.2.12 ConcernStakeholderPartUsageOwningMembership_Mapping

Description

Creates a owning membership relationship for *ownedMemberElement()*.

General Mappings

ToOwningMembership_Init
Mapping

Mapping Source

Classifier

Mapping Target

OwningMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- OwningMembership::ownedMemberElement () : Element [1]

  `ConcernStakeholderPartUsageFeature_Mapping.getMapped(from)`

7.8.6.2.13 ConcernStakeholderPartUsageFeature_Mapping

Description

The mapping class creates a feature element for the concern stakeholder part usage.

General Mappings

ToFeature_Init
Mapping

Mapping Source

Classifier

Mapping Target

Multiplicity

Owned Mappings

(none)

Applicable filters

(none)

7.8.6.2.14 ElementGroup_Mapping

Description

A SysML::ModelElements::ElementGroup element is mapped to a SysML v2 Package with membership import relationships representing the grouping.

The following shows an example of what the textual SysML v2 syntax of the result of the transformation may look like.

```
package ElementGroupModel {
    part def SysMLv1Block1;
    attribute def SysMLv1ValueType;
    part def SysMLv1Block2 {
        part sysMLv1PartProperty:SysMLv1Block1;
    }
}

package SysMLv1ElementGroup {
    import ElementGroupModel::SysMLv1Block1;
    import ElementGroupModel::SysMLv1ValueType;
    import ElementGroupModel::SysMLv1Block2::sysMLv1PartProperty;

    @SysMLv1Library::ElementGroupData {criterion = "criterion string";}
}
```

General Mappings

Comment_Mapping

Mapping Source

Comment

Mapping Target

Package

Owned Mappings

(none)

Applicable filters

This mapping applies only if the following (OCL) condition implemented by the operation *filter(src : Element) : Boolean* is verified:

```
Helper.hasStereotypeApplied(src, 'SysML::ModelElements::ElementGroup')
```

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Package::ownedRelationship () : Relationship [0..*]

```
let elements : Set(KerML::Relationahip) =
    Helper.getTagValueAsElementColl(from,
        'SysML::ModelElements::ElementGroup', 'member')
    ->collect(e | CommonElementImport_Mapping.getMapped(e)) in
elements->including(ElementGroupMetadaMembership_Mapping.getMapped(from))
->union(self.oclAsType(ElementMain_Mapping).ownedRelationship())
```

- Package::declaredName () : String [0..1]

```
Helper.getTagValueAsString(from, 'SysML::ModelElements::ElementGroup', 'name')
```

7.8.6.2.15 ElementGroupMetadaMembership_Mapping

Description

Creates a membership relationship for *memberElement()*.

General Mappings

ToOwningMembership_Init
Mapping

Mapping Source

Comment

Mapping Target

OwningMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- OwningMembership::ownedMemberElement () : Element [1]

  ```
  ElementGroupMetadataUsage_Mapping.getMapped(from)
  ```

7.8.6.2.16 ElementGroupMetadataFeatureMembership_Mapping

Description

Creates a feature membership relationship for *ownedMemberFeature()*.

General Mappings

ToFeatureMembership_Init
Mapping

Mapping Source

Comment

Mapping Target

FeatureMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureMembership::ownedMemberFeature () : Feature [1]

  ```
  ElementGroupMetadataReferenceUsage_Mapping.getMapped(from)
  ```

7.8.6.2.17 ElementGroupMetadataFeatureTyping_Mapping

Description

Creates a feature typing relationship owned by the element *typedFeature()*.

General Mappings

ToFeatureTyping_Init
Mapping

Mapping Source

Comment

Mapping Target

FeatureTyping

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureTyping::type () : Type [1]

```
SYSML2::MetadataDefinition.allInstances()
->any(m | m.qualifiedName = 'SysMLv1Library::ElementGroupData')
```

7.8.6.2.18 ElementGroupMetadataFeatureValue_Mapping

Description

Creates a feature value relationship.

General Mappings

ToFeatureValue_Init
Mapping

Mapping Source

Comment

Mapping Target

FeatureValue

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureValue::value () : Expression [1]

```
let criterion: String =
  Helper.getTagValueAsString(
    from,
    'SysML::ModelElements::ElementGroup', 'criterion') in
LiteralString_Factory.create(criterion)
```

7.8.6.2.19 ElementGroupMetadataRedefinition_Mapping

Description

Creates a redefinition relationship for the *redefiningFeature()* and the *redefinedFeature()*.

General Mappings

ToRedefinition_Init
Mapping

Mapping Source

Comment

Mapping Target

Redefinition

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Redefinition::redefinedFeature () : Feature [1]

```
let m : SYSML2::Membership =
    SYSML2::AttributeUsage.allInstances()
    ->collect(dt | dt.owningRelationship)
    ->select(r | r.oclIsKindOf(SYSML2::Membership))
    ->any(m | m.memberName = 'criterion') in
if (m.oclIsUndefined()) then
        invalid
else
    m.memberElement
endif
```

7.8.6.2.20 ElementGroupMetadataReferenceUsage_Mapping

Description

Creates a reference usage.

General Mappings

ToReferenceUsage_Init
Mapping

Mapping Source

Comment

Mapping Target

ReferenceUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ReferenceUsage::ownedRelationship () : Relationship [0..*]

  ```
  Set{ElementGroupMetadataRedefinition_Mapping.getMapped(from),
  ElementGroupMetadataFeatureValue_Mapping.getMapped(from)}
  ```

7.8.6.2.21 ElementGroupMetadataUsage_Mapping

Description

The mapping class creates the metadata usage element for the SysML::ModelElements::ElementGroup mapping.

General Mappings

ToMetadataUsage_Init
Mapping

Mapping Source

Comment

Mapping Target

MetadataUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- MetadataUsage::ownedRelationship () : Relationship [0..*]

```
Set{ElementGroupMetadataFeatureTyping_Mapping.getMapped(from),
ElementGroupMetadataFeatureMembership_Mapping.getMapped(from)}
```

7.8.6.2.22 ProblemRationale_Mapping

Description

The mapping class combines the mapping of SysML::ModelElements::Problem and SysML::ModelElements::Rationale. The SysML::ModelElements::Problem is mapped to the library element ModelingMetadata::Issue and the SysML::ModelElements::Rationale is mapped to ModelingMetadata::Rationale.

The expected SysML v2 textual syntax of the mapping is as follows.

```
@ModelingMetadata::Issue {text = "This is a problem statement";}

@ModelingMetadata::Rationale {text = "This is a rationale statement";}
```

General Mappings

Comment_Mapping

Mapping Source

Comment

Mapping Target

Comment

Owned Mappings

(none)

Applicable filters

This mapping applies only if the following (OCL) condition implemented by the operation *filter(src : Element) : Boolean* is verified:

```
(not Helper.hasStereotypeApplied(src, 'SysML::ModelElements::ElementGroup')) and
(Helper.hasStereotypeApplied(src, 'SysML::ModelElements::Problem') or
Helper.hasStereotypeApplied(src, 'SysML::ModelElements::Rationale'))
```

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Comment::ownedRelationship () : Relationship [0..*]

```
self.oclAsType(ElementMain_Mapping).ownedRelationship()
->including(ProblemRationaleMetadataMembership_Mapping.getMapped(from))
```

7.8.6.2.23 ProblemRationaleMetadataRedefinition_Mapping

Description

Creates a redefinition relationship for the *redefiningFeature()* and the *redefinedFeature()*.

General Mappings

ToRedefinition_Init
Mapping

Mapping Source

Comment

Mapping Target

Redefinition

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Redefinition::redefinedFeature () : Feature [1]

```
if Helper.hasStereotypeApplied(from, 'SysML::ModelElements::Problem') then
  SYSML2::AttributeUsage.allInstances()
  ->any(m | m.qualifiedName = 'ModelingMetadata::Issue::text')
else if Helper.hasStereotypeApplied(from, 'SysML::ModelElements::Rationale') then
  SYSML2::AttributeUsage.allInstances()
  ->any(m | m.qualifiedName = 'ModelingMetadata::Rationale::text')
else
  invalid
endif
endif
```

7.8.6.2.24 ProblemRationaleMetadataUsage_Mapping

Description

The mapping class creates the metadata usage element for the SysML::ModelElements::Problem and SysML::ModelElements::Rationale transformation target.

General Mappings

ToMetadataUsage_Init
Mapping

Mapping Source

Comment

Mapping Target

MetadataUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- MetadataUsage::ownedRelationship () : Relationship [0..*]

```
Set{ProblemRationaleMetadataFeatureTyping_Mapping.getMapped(from),
ProblemRationaleMetadataFeatureMembership_Mapping.getMapped(from)}
```

7.8.6.2.25 Stakeholder_Mapping

Description

A SysML::ModelElements::Stakeholder is mapped to a SysML v2 ItemDefinition with metadata to tag it as a stakeholder. The concern comments of the stakeholder are mapped to ConcernUsages which reference the stakeholder item definition.

The following shows an example of what the textual SysML v2 syntax of the result of the transformation may look like.

```
item def SysMLv1Stakeholder {@SysMLv1Library::StakeholderData {isStakeholder = true;}}
concern concernCommentXMI_ID {
        doc /* concern string */
        stakeholder : SysMLv1Stakeholder;
}
```

General Mappings

Class_Mapping

Mapping Source

Class

Mapping Target

ItemDefinition

Owned Mappings

(none)

Applicable filters

This mapping applies only if the following (OCL) condition implemented by the operation *filter(src : Element) : Boolean* is verified:

```
Helper.hasStereotypeApplied(src, 'SysML::ModelElements::Stakeholder')
```

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ItemDefinition::ownedRelationship () : Relationship [0..*]

```
let toElementFMS: Set(UML::Element) =
  from.ownedElement
  ->select(e | (e.oclIsKindOf(UML::Property) and
    (e.oclAsType(UML::Property).redefinedProperty->size()
      = 0)) or e.oclIsKindOf(UML::Operation)) in
let redefinedAttributes: Set(UML::Element) =
  from.ownedElement
  ->select(e | from.oclIsKindOf(UML::DataType) and
   (e.oclAsType(UML::Property).redefinedProperty->size() > 0)) in
let generalizations : Set(UML::Generalization) =
  from.ownedElement
  ->select(e | e.oclIsKindOf(UML::Generalization)) in
let constraints : Set(UML::Constraint) =
  UML::Constraint.allInstances()
  ->select( c | c.constrainedElement->includes(from)) in
let toElementOMS: Set(UML::Element) =
  (((from.ownedElement - toElementFMS) - redefinedAttributes) -
  generalizations)  in
let relationships: Sequence(KerML::Relationship) =
toElementOMS->collect(e | ElementOwningMembership_Mapping.getMapped(e))
->union(toElementFMS->collect(e | ElementFeatureMembership_Mapping.getMapped(e)))
->union(constraints
  ->collect(e | ConstrainedElementFeatureMembership_Mapping.getMapped(e)))
->union(redefinedAttributes
  ->collect(e | AttributeRedefinedMembership_Mapping.getMapped(e)))
->union(generalizations->collect(e | Generalization_Mapping.getMapped(e)))
->including(StakeholderMetadataOwningMembership_Mapping.getMapped(from)) in
if from.classifierBehavior.oclIsUndefined() then
  relationships
else
  relationships
    ->append(BehavioredClassifierFeatureMembership_Mapping.getMapped(from))
endif
```

7.8.6.2.26 StakeholderMetadataUsage_Mapping

Description

The mapping class creates the metadata usage element for the SysML::ModelElements::Stakeholder mapping.

General Mappings

ToMetadataUsage_Init
Mapping

Mapping Source

Classifier

Mapping Target

MetadataUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- MetadataUsage::ownedRelationship () : Relationship [0..*]

```
Set{StakeholderMetadataFeatureTyping_Mapping.getMapped(from),
StakeholderMetadataFeatureMembership_Mapping.getMapped(from)}
```

7.8.6.2.27 StakeholderMetadataFeatureMembership_Mapping

Description

Creates a feature membership relationship for *ownedMemberFeature()*.

General Mappings

ToFeatureMembership_Init
Mapping

Mapping Source

Classifier

Mapping Target

FeatureMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureMembership::ownedMemberFeature () : Feature [1]

```
StakeholderMetadataReferenceUsage_Mapping.getMapped(from)
```

7.8.6.2.28 StakeholderMetadataFeatureTyping_Mapping

Description

Creates a feature typing relationship owned by the element *typedFeature()*.

General Mappings

ToFeatureTyping_Init
Mapping

Mapping Source

Classifier

Mapping Target

FeatureTyping

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureTyping::type () : Type [1]

```
SYSML2::MetadataDefinition.allInstances()
->any(m | m.qualifiedName = 'SysMLv1Library::StakeholderData')
```

7.8.6.2.29 StakeholderMetadataOwningMembership

Description

Creates a owning membership relationship for *ownedMemberElement()*.

General Mappings

ToOwningMembership_Init
Mapping

Mapping Source

Classifier

Mapping Target

OwningMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- OwningMembership::ownedMemberElement () : Element [1]

```
StakeholderMetadataUsage_Mapping.getMapped(from)
```

7.8.6.2.30 StakeholderMetadataReferenceUsage_Mapping

Description

Creates a reference usage.

General Mappings

ToReferenceUsage_Init
Mapping

Mapping Source

Classifier

Mapping Target

ReferenceUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ReferenceUsage::ownedRelationship () : Relationship [0..*]

```
Set{StakeholderMetadataReferenceUsageRedefinition_Mapping.getMapped(from),
StakeholderMetadataReferenceUsageFeatureValue_Mapping.getMapped(from)}
```

7.8.6.2.31 StakeholderMetadataReferenceUsageFeatureValue_Mapping

Description

Creates a feature value relationship.

General Mappings

ToFeatureValue_Init
Mapping

Mapping Source

Classifier

Mapping Target

FeatureValue

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureValue::value () : Expression [1]

  ```
  LiteralBoolean_Factory.create(true)
  ```

7.8.6.2.32 StakeholderMetadataReferenceUsageRedefinition_Mapping

Description

Creates a redefinition relationship for the *redefiningFeature()* and the *redefinedFeature()*.

General Mappings

ToRedefinition_Init
Mapping

Mapping Source

Classifier

Mapping Target

Redefinition

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Redefinition::redefinedFeature () : Feature [1]

```
SYSML2::AttributeUsage.allInstances()
->any(m | m.qualifiedName = 'SysMLv1Library::StakeholderData::isStakeholder')
```

7.8.6.2.33 Viewpoint_Mapping

Description

A SysML::ModelElements::Viewpoint is mapped to a SysML v2 ViewDefinition with an owned SysML v2 ViewpointUsage. In SysML v1, the viewpoint combines the purpose and stakeholder concerns as well as presentation information. This is covered by a SysML v2 ViewDefinition with owned SysML v2 ViewpointUsage.

The following shows an example of what the textual SysML v2 syntax of the result of the transformation may look like.

```
view def SysMLv1Viewpoint {
	viewpoint sysMLv1Viewpoint {
		frame concern1XmiID1;
		frame concern2XmiID2;
		metadata SysMLv1Library::ViewpointData {
			languages = ("language1","language2");
			presentations = ("presentation1", "presentation2");
		}
		require constraint {
			doc /* thisIsThePurpose */
		}
	}
	satisfy sysMLv1Viewpoint;
	rendering {
		action : SysMLv1ViewpointMethodBehavior1;
		action : SysMLv1ViewpointMethodBehavior2;
	}
}
action def SysMLv1ViewpointMethodBehavior1;
action def SysMLv1ViewpointMethodBehavior2;

item def SysMLv1Stakeholder {@SysMLv1Library::StakeholderData {isStakeholder = true;}}

concern concern1XmiID1 {
	doc /* Concern1 */
	stakeholder : SysMLv1Stakeholder;
}
concern concern2XmiID2 {
	doc /* Concern2 */
	stakeholder : SysMLv1Stakeholder;
}
```

General Mappings

Class_Mapping

Mapping Source

Class

Mapping Target

ViewDefinition

Owned Mappings

(none)

Applicable filters

This mapping applies only if the following (OCL) condition implemented by the operation *filter(src : Element) : Boolean* is verified:

```
Helper.hasStereotypeApplied(src, 'SysML::ModelElements::Viewpoint')
```

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ViewDefinition::ownedRelationship () : Relationship [0..*]

```
let toElementFMS: Set(UML::Element) =
    from.ownedElement->select(e | (e.oclIsKindOf(UML::Property) and
        (e.oclAsType(UML::Property).redefinedProperty->size() = 0)) or
        e.oclIsKindOf(UML::Comment)) in
let redefinedAttributes: Set(UML::Element) =
    from.ownedElement->select(e | from.oclIsKindOf(UML::DataType) and
        (e.oclAsType(UML::Property).redefinedProperty->size() > 0)) in
let generalizations : Set(UML::Generalization) =
    from.ownedElement->select(e | e.oclIsKindOf(UML::Generalization)) in
let toElementOMS: Set(UML::Element) =
    (((from.ownedElement - toElementFMS) - redefinedAttributes) -
    generalizations)  in
let relationships: Sequence(UML::Element) =
toElementOMS->collect(e | ElementOwningMembership_Mapping.getMapped(e))
->union(toElementFMS->collect(e | ElementFeatureMembership_Mapping.getMapped(e)))
->union(redefinedAttributes
    ->collect(e | AttributeRedefinedMembership_Mapping.getMapped(e)))
->union(generalizations->collect(e | Generalization_Mapping.getMapped(e)))
->including(ViewpointViewpointUsageFeatureMembership_Mapping.getMapped(from))
->including(ViewpointSatisfyFeatureMembership_Mapping.getMapped(from))
->including(ViewpointRenderingFeatureMembership_Mapping.getMapped(from))
->including(
    CommonReturnParameterReferenceUsageMembership_Mapping.getMapped(from)) in
if from.classifierBehavior.oclIsUndefined() then
    relationships
else
    relationships
```

```
        ->append(BehavioredClassifierFeatureMembership_Mapping.getMapped(from))
    endif
```

7.8.6.2.34 ViewpointConcernReferenceSubsetting_Mapping

Description

Creates a subsetting relationship.

General Mappings

ToReferenceSubsetting_Init
Mapping

Mapping Source

Comment

Mapping Target

ReferenceSubsetting

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ReferenceSubsetting::referencedFeature () : Feature [1]

  ```
  from
  ```

7.8.6.2.35 ViewpointConcernUsage_Mapping

Description

The mapping class creates the concern usage element for the SysML::ModelElements::Viewpoint mapping.

General Mappings

ToRequirementUsage_Init
Mapping

Mapping Source

Comment

Mapping Target

ConcernUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ConcernUsage::ownedRelationship () : Relationship [0..*]

```
Set{ViewpointConcernReferenceSubsetting_Mapping.getMapped(from),
EmptySubjectMembership_Factory.create(),
CommonReturnParameterReferenceUsageMembership_Mapping.getMapped(from)}
```

7.8.6.2.36 ViewpointConstraintUsage_Mapping

Description

The mapping class creates the constraint usage element for the SysML::ModelElements::Viewpoint mapping.

General Mappings

ToConstraintUsage_Init
Mapping

Mapping Source

Class

Mapping Target

ConstraintUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ConstraintUsage::ownedRelationship () : Relationship [0..*]

```
Set{ViewpointConstraintUsageOwningMembership_Mapping.getMapped(from),
ReturnParameterFeatureMembership_Factory.create()}
```

7.8.6.2.37 ViewpointConstraintUsageDocumentation_Mapping

Description

The mapping class creates the documentation element for the SysML::ModelElements::Viewpoint mapping.

General Mappings

ToDocumentation_Init
Mapping

Mapping Source

Class

Mapping Target

Documentation

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Documentation::body () : String [1]

```
Helper.getTagValueAsString(from, 'SysML::ModelElements::Viewpoint', 'purpose')
```

7.8.6.2.38 ViewpointConstraintUsageOwningMembership_Mapping

Description

Creates a owning membership relationship for *ownedMemberElement()*.

General Mappings

ToOwningMembership_Init
Mapping

Mapping Source

Class

Mapping Target

OwningMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- OwningMembership::ownedMemberElement () : Element [1]

  ```
  ViewpointConstraintUsageDocumentation_Mapping.getMapped(from)
  ```

7.8.6.2.39 ViewpointFramedConcernMembership_Mapping

Description

Creates a membership relationship for *memberElement()*.

General Mappings

ToFeatureMembership_Init
Mapping

Mapping Source

Comment

Mapping Target

FramedConcernMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FramedConcernMembership::ownedMemberFeature () : Feature [1]

  ```
  ViewpointConcernUsage_Mapping.getMapped(from)
  ```

7.8.6.2.40 ViewpointLanguagesMetadataFeatureMembership_Mapping

Description

Creates a feature membership relationship for *ownedMemberFeature()*.

General Mappings

ToFeatureMembership_Init
Mapping

Mapping Source

Class

Mapping Target

FeatureMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureMembership::ownedMemberFeature () : Feature [1]

  ```
  ViewpointLanguagesMetadataReferenceUsage_Mapping.getMapped(from)
  ```

7.8.6.2.41 ViewpointLanguagesMetadataFeatureValue_Mapping

Description

Creates a feature value relationship.

General Mappings

ToFeatureValue_Init
Mapping

Mapping Source

Class

Mapping Target

FeatureValue

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureValue::value () : Expression [1]

```
ViewpointLanguagesMetadataOperatorExpression_Mapping.getMapped(from)
```

7.8.6.2.42 ViewpointLanguagesMetadataRedefinition_Mapping

Description

Creates a redefinition relationship for the *redefiningFeature()* and the *redefinedFeature()*.

General Mappings

ToRedefinition_Init
Mapping

Mapping Source

Class

Mapping Target

Redefinition

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Redefinition::redefinedFeature () : Feature [1]

```
SYSML2::AttributeUsage.allInstances()
->any(m | m.qualifiedName = 'SysMLv1Library::ViewpointData::languages')
```

7.8.6.2.43 ViewpointLanguagesMetadataReferenceUsage_Mapping

Description

Creates a reference usage.

General Mappings

ToReferenceUsage_Init
Mapping

Mapping Source

Class

Mapping Target

ReferenceUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ReferenceUsage::ownedRelationship () : Relationship [0..*]

```
Set{ViewpointLanguagesMetadataRedefinition_Mapping.getMapped(from),
ViewpointLanguagesMetadataFeatureValue_Mapping.getMapped(from)}
```

7.8.6.2.44 ViewpointMetadataFeatureTyping_Mapping

Description

Creates a feature typing relationship owned by the element *typedFeature()*.

General Mappings

ToFeatureTyping_Init
Mapping

Mapping Source

Class

Mapping Target

FeatureTyping

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureTyping::type () : Type [1]

```
SYSML2::MetadataDefinition.allInstances()
->any(m | m.qualifiedName = 'SysMLv1Library::ViewpointData')
```

7.8.6.2.45 ViewpointLanguagesMetadataOperatorExpression_Mapping

Description

The mapping class creates the operator expression for the list of languages of the SysML::ModelElements::Viewpoint mapping.

General Mappings

ToOperatorExpression_Init
Mapping

Mapping Source

Class

Mapping Target

OperatorExpression

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- OperatorExpression::ownedRelationship () : Relationship [0..*]

  ```
  Helper.getTagValueAsStringColl(from, 'SysML::ModelElements::Viewpoint', 'language')
  ->collect(e | StringParameterMembership_Factory.create(e))
  ```

- OperatorExpression::operator () : String [1]

  ```
  ','
  ```

7.8.6.2.46 ViewpointMetadataOwningMembership_Mapping

Description

Creates a owning membership relationship for *ownedMemberElement()*.

General Mappings

ToOwningMembership_Init
Mapping

Mapping Source

Class

Mapping Target

OwningMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- OwningMembership::ownedMemberElement () : Element [1]

  ```
  ViewpointMetadataUsage_Mapping.getMapped(from)
  ```

7.8.6.2.47 ViewpointMetadataUsage_Mapping

Description

The mapping class creates the metadata usage element for the SysML::ModelElements::Viewpoint mapping.

General Mappings

ToMetadataUsage_Init
Mapping

Mapping Source

Class

Mapping Target

MetadataUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- MetadataUsage::ownedRelationship () : Relationship [0..*]

```
Set{ViewpointMetadataFeatureTyping_Mapping.getMapped(from),
ViewpointLanguagesMetadataFeatureMembership_Mapping.getMapped(from),
ViewpointPresentationsMetadataFeatureMembership_Mapping.getMapped(from)}
```

7.8.6.2.48 ViewpointPresentationsMetadataFeatureMembership_Mapping

Description

Creates a feature membership relationship for *ownedMemberFeature()*.

General Mappings

ToFeatureMembership_Init
Mapping

Mapping Source

Class

Mapping Target

FeatureMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureMembership::ownedMemberFeature () : Feature [1]

  ```
  ViewpointPresentationsMetadataReferenceUsage_Mapping.getMapped(from)
  ```

7.8.6.2.49 ViewpointPresentationsMetadataFeatureValue_Mapping

Description

Creates a feature value relationship.

General Mappings

ToFeatureValue_Init
Mapping

Mapping Source

Class

Mapping Target

FeatureValue

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureValue::value () : Expression [1]

  ```
  ViewpointPresentationsMetadataOperatorExpression_Mapping.getMapped(from)
  ```

7.8.6.2.50 ViewpointPresentationsMetadataOperatorExpression_Mapping

Description

The mapping class creates the operator expression for the list of presentations of the SysML::ModelElements::Viewpoint mapping.

General Mappings

ToOperatorExpression_Init
Mapping

Mapping Source

Class

Mapping Target

OperatorExpression

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- OperatorExpression::operator () : String [1]

  ```
  ','
  ```

- OperatorExpression::ownedRelationship () : Relationship [0..*]

```
Helper.getTagValueAsStringColl(from,
    'SysML::ModelElements::Viewpoint', 'presentation')
    ->collect(e | StringParameterMembership_Factory.create(e))
```

7.8.6.2.51 ViewpointPresentationsMetadataRedefinition_Mapping

Description

Creates a redefinition relationship for the *redefiningFeature()* and the *redefinedFeature()*.

General Mappings

ToRedefinition_Init
Mapping

Mapping Source

Class

Mapping Target

Redefinition

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Redefinition::redefinedFeature () : Feature [1]

```
SYSML2::AttributeUsage.allInstances()
->any(m | m.qualifiedName = 'SysMLv1Library::ViewpointData::presentations')
```

7.8.6.2.52 ViewpointPresentationsMetadataReferenceUsage_Mapping

Description

Creates a reference usage.

General Mappings

ToReferenceUsage_Init
Mapping

Mapping Source

Class

Mapping Target

ReferenceUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ReferenceUsage::ownedRelationship () : Relationship [0..*]

```
Set{ViewpointPresentationsMetadataRedefinition_Mapping.getMapped(from),
ViewpointPresentationsMetadataFeatureValue_Mapping.getMapped(from)}
```

7.8.6.2.53 ViewpointRenderingFeatureMembership_Mapping

Description

Creates a feature membership relationship for *ownedMemberFeature()*.

General Mappings

ToFeatureMembership_Init
Mapping

Mapping Source

Class

Mapping Target

FeatureMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureMembership::ownedMemberFeature () : Feature [1]

```
ViewpointRenderingUsage_Mapping.getMapped(from)
```

7.8.6.2.54 ViewpointRenderingUsage_Mapping

Description

The mapping class creates the rendering usage element for the SysML::ModelElements::Viewpoint mapping class.

General Mappings

ToPartUsage_Init
Mapping

Mapping Source

Class

Mapping Target

RenderingUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- RenderingUsage::ownedRelationship () : Relationship [0..*]

```
from.ownedOperation
->select( o | Helper.hasStereotypeApplied(o, 'Create') )
->collect( e |
    ViewpointRenderingUsageActionUsageFeatureMembership_Mapping.getMapped(e))
```

7.8.6.2.55 ViewpointRenderingUsageActionUsage_Mapping

Description

The mapping class creates the action usage element for the rendering usage element for the SysML::ModelElements::Viewpoint mapping class.

General Mappings

ToActionUsage_Init
Mapping

Mapping Source

Class

Mapping Target

ActionUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ActionUsage::ownedRelationship () : Relationship [0..*]

  `Set{ViewpointRenderingUsageActionUsageFeatureTyping_Mapping.getMapped(from)}`

7.8.6.2.56 ViewpointRenderingUsageActionUsageFeatureMembership_Mapping

Description

Creates a feature membership relationship for *ownedMemberFeature()*.

General Mappings

ToFeatureMembership_Init
Mapping

Mapping Source

Class

Mapping Target

FeatureMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureMembership::ownedMemberFeature () : Feature [1]

  `ViewpointRenderingUsageActionUsage_Mapping.getMapped(from)`

7.8.6.2.57 ViewpointRenderingUsageActionUsageFeatureTyping_Mapping

Description

Creates a feature typing relationship owned by the element *typedFeature()*.

General Mappings

ToFeatureTyping_Init
Mapping

Mapping Source

Class

Mapping Target

FeatureTyping

Owned Mappings

(none)

Applicable filters

(none)

7.8.6.2.58 ViewpointRequirementConstraintMembership_Mapping

Description

Creates a membership relationship for *memberElement()*.

General Mappings

ToFeatureMembership_Init
Mapping

Mapping Source

Class

Mapping Target

RequirementConstraintMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- RequirementConstraintMembership::ownedMemberFeature () : Feature [1]

  ```
  ViewpointConstraintUsage_Mapping.getMapped(from)
  ```

7.8.6.2.59 ViewpointSatisfyFeatureMembership_Mapping

Description

Creates a feature membership relationship for *ownedMemberFeature()*.

General Mappings

ToFeatureMembership_Init
Mapping

Mapping Source

Class

Mapping Target

FeatureMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureMembership::ownedMemberFeature () : Feature [1]

  ```
  ViewpointSatisfyRequirementUsage_Mapping.getMapped(from)
  ```

7.8.6.2.60 ViewpointSatisfyRequirementUsage_Mapping

Description

The mapping class creates the satisfy requirement usage element for the SysML::ModelElements::Viewpoint mapping.

General Mappings

ToRequirementUsage_Init
Mapping

Mapping Source

Class

Mapping Target

SatisfyRequirementUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- SatisfyRequirementUsage::ownedRelationship () : Relationship [0..*]

```
Set{ViewpointSatisfyRequirementUsageReferenceSubsetting_Mapping.getMapped(from),
EmptySubjectMembership_Factory.create(),
ReturnParameterFeatureMembership_Factory.create()}
```

7.8.6.2.61 ViewpointSatisfyRequirementUsageReferenceSubsetting_Mapping

Description

Creates a subsetting relationship.

General Mappings

ToReferenceSubsetting_Init
Mapping

Mapping Source

Class

Mapping Target

ReferenceSubsetting

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ReferenceSubsetting::referencedFeature () : Feature [1]

```
ViewpointViewpointUsage_Mapping.getMapped(from)
```

7.8.6.2.62 ViewpointViewpointUsage_Mapping

Description

The mapping class creates the embedded viewpoint usage for the SysML::ModelElements::Viewpoint mapping.

General Mappings

ToUsage_Init
Mapping

Mapping Source

Class

Mapping Target

ViewpointUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ViewpointUsage::declaredName () : String [0..1]

  ```
  from.name.substring(1,1).toLowerCase() + from.name.substring(2, from.name.size())
  ```

- ViewpointUsage::ownedRelationship () : Relationship [0..*]

  ```
  Helper.getTagValueAsElementColl(
      from, 'SysML::ModelElements::Viewpoint', 'concernList')
  ->collect(e | ViewpointFramedConcernMembership_Mapping.getMapped(e))
  ->including(ViewpointMetadataOwningMembership_Mapping.getMapped(from))
  ->including(EmptySubjectMembership_Factory.create())
  ->including(ViewpointRequirementConstraintMembership_Mapping.getMapped(from))
  ```

7.8.6.2.63 ViewpointViewpointUsageFeatureMembership_Mapping

Description

Creates a feature membership relationship for *ownedMemberFeature()*.

General Mappings

ToFeatureMembership_Init
Mapping

Mapping Source

Class

Mapping Target

FeatureMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureMembership::ownedMemberFeature () : Feature [1]

  ```
  ViewpointViewpointUsage_Mapping.getMapped(from)
  ```

7.8.7 PortsAndFlows

SYSML21-542: Transformation: overview tables shall be revised

7.8.7.1 Overview

SYSML21-542: Transformation: overview tables shall be revised

Table 19. List of High-Level Ports & Flows Mappings

SysML v1 Abstract Syntax/Stereotype	SysML v2 Abstract Syntax
AcceptChangeStructuralFeatureEventAction	AcceptActionUsage
AddFlowPropertyValueOnNestedPortAction	No specific mapping specified yet.
ChangeStructuralFeatureEvent	No specific mapping specified yet.
DirectedFeature	PerformActionUsage
FlowProperty	AttributeUsage, OccurrenceUsage, ReferenceUsage
FullPort	PortUsage
InterfaceBlock	PortDefinition
InvocationOnNestedPortAction	ItemFlow
ProxyPort	
TriggerOnNestedPort	No specific mapping specified yet.
~InterfaceBlock	PortDefinition

7.8.7.2 Mapping Specifications

7.8.7.2.1 AcceptChangeStructuralFeatureEventAction_Mapping

Description

The SysML::PortsAndFlows::AcceptChangeStructuralFeatureEventAction element is mapped to SysML v2 AcceptActionUsage. The details of the mapping are not defined yet.

General Mappings

AcceptEventAction_Mapping

Mapping Source

AcceptEventAction

Mapping Target

AcceptActionUsage

Owned Mappings

(none)

Applicable filters

This mapping applies only if the following (OCL) condition implemented by the operation *filter(src : Element) : Boolean* is verified:

```
 Helper.hasStereotypeApplied(src,
'SysML::Ports&Flows::AcceptChangeStructuralFeatureEventAction')
```

Mapping rules

The mapping class only has inherited rules. See the mapping classes in the general mapping section for details.

7.8.7.2.2 CommonFullPort_Mapping

Description

The abstract mapping class is the base class of the mapping classes for the SysML::Ports&Flows::FullPort mappings.

General Mappings

PropertyCommon_Mapping

Mapping Source

Port

Mapping Target

PartUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- PartUsage::ownedRelationship () : Relationship [0..*]

```
let typings: Set(KerML::FeatureTyping) = if from.type.oclIsUndefined() then
    Set{}
else
    Set{StructuralFeatureToFeatureTyping_Mapping.getMapped(from)}
endif in
let subsettings: Set(KerML::Subsetting) = from.subsettedProperty
    ->collect(p | PropertySubsetting_Mapping.getMapped(from, p))->asSet() in
let defaultValue: Set(KerML::OwningMembership) =
if from.defaultValue.oclIsUndefined() then
    Set{}
else
    Set{DefaultValue_Mapping.getMapped(from)}
endif in
typings->union(subsettings)->union(defaultValue)
->including(MultiplicityMembership_Mapping.getMapped(from))->asSet()
->including(FullPortMetadataOwningMembership_Mapping.getMapped(from))
```

7.8.7.2.3 ConjugatedPortDefinition_Mapping

Description

A SysML::Ports&Flows::InterfaceBlock element is mapped to a SysML v2 ConjugatedPortDefinition owned by the PortDefinition that is the target element of the main mapping of the SysML::Ports&Flows::InterfaceBlock.

General Mappings

ToClassifier_Init
Mapping

Mapping Source

Class

Mapping Target

ConjugatedPortDefinition

Owned Mappings

- portConjugation : PortConjugation_Mapping

Applicable filters

This mapping applies only if the following (OCL) condition implemented by the operation *filter(src : Element) : Boolean* is verified:

```
Helper.hasStereotypeApplied(src, 'SysML::Ports&Flows::InterfaceBlock')
```

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ConjugatedPortDefinition::ownedRelationship () : Relationship [0..*]

  ```
  Set{portConjugation.to}
  ```

7.8.7.2.4 FlowProperty_Mapping

Description

A UML4SysML::Property which satisfies the filter condition of PropertyTypedByClassInterface_Mapping and to which the SysML v1 stereotype FlowProperty has been applied is mapped as in the general mapping class PropertyTypedByClassInterface_Mapping but the target feature is always referential and the flow direction specified in the stereotype FlowProperty is considered.

General Mappings

PropertyTypedByClassInterface_Mapping

Mapping Source

Property

Mapping Target

OccurrenceUsage

Owned Mappings

(none)

Applicable filters

This mapping applies only if the following (OCL) condition implemented by the operation *filter(src : Element) : Boolean* is verified:

```
Helper.hasStereotypeApplied(src, 'SysML::Ports&Flows::FlowProperty')
    and ((not src.type.oclIsUndefined())
        and (src.type.oclIsKindOf(UML::Class)
        or src.type.oclIsKindOf(UML::Interface)))
```

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- OccurrenceUsage::isComposite () : Boolean [1]

  ```
  false
  ```

- OccurrenceUsage::direction () : FeatureDirectionKind [0..1]

  ```
  Helper.getFlowDirectionKind(Helper.getTagValue(from,
  'SysML::Ports&Flows::FlowProperty', 'direction'))
  ```

7.8.7.2.5 FlowPropertyAttribute_Mapping

Description

A UML4SysML::Property which satisfies the filter condition of Attribute_Mapping and to which the SysML v1 stereotype FlowProperty has been applied is mapped as in the general mapping class Attribute_Mapping with consideration of the flow direction specified in the stereotype FlowProperty.

General Mappings

Attribute_Mapping

Mapping Source

Property

Mapping Target

AttributeUsage

Owned Mappings

(none)

Applicable filters

This mapping applies only if the following (OCL) condition implemented by the operation *filter(src : Element) : Boolean* is verified:

```
Helper.hasStereotypeApplied(src, 'SysML::Ports&Flows::FlowProperty')
    and (not src.type.oclIsUndefined() and src.type.oclIsKindOf(UML::DataType))
```

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- AttributeUsage::direction () : FeatureDirectionKind [0..1]

  ```
  Helper.getFlowDirectionKind(Helper.getTagValue(from,
  'SysML::Ports&Flows::FlowProperty', 'direction'))
  ```

7.8.7.2.6 FlowPropertyUntyped_Mapping

Description

A UML4SysML::Property which satisfies the filter condition of PropertyUntyped_Mapping and to which the SysML v1 stereotype FlowProperty has been applied is mapped as in the general mapping class PropertyUntyped_Mapping but the target feature is always referential and the flow direction specified in the stereotype FlowProperty is considered.

General Mappings

PropertyUntyped_Mapping

Mapping Source

Property

Mapping Target

ReferenceUsage

Owned Mappings

(none)

Applicable filters

This mapping applies only if the following (OCL) condition implemented by the operation *filter(src : Element) : Boolean* is verified:

```
Helper.hasStereotypeApplied(src, 'SysML::Ports&Flows::FlowProperty')
  and src.type.oclIsUndefined()
  and not Helper.hasStereotypeApplied(src.owner,
    'SysML::ConstraintBlocks::ConstraintBlock')
```

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ReferenceUsage::isComposite () : Boolean [1]

  ```
  false
  ```

- ReferenceUsage::direction () : FeatureDirectionKind [0..1]

  ```
  Helper.getFlowDirectionKind(Helper.getTagValue(from,
  'SysML::Ports&Flows::FlowProperty', 'direction'))
  ```

7.8.7.2.7 FullPort_Mapping

Description

A SysML::Ports&Flows::FullPort element is mapped to a part usage in SysML v2 with metadata that marks the part usage as a full port. The metadata is defined in the SysML v1 library for SysML v2.

The mapping class FullPortUntyped_Mapping does the same for full ports that have no type.

The following shows an example of what the textual SysML v2 syntax of the result of the transformation may look like.

```
part sysMLv1FullPort : SysMLv1Block {SysMLv1Library::PortData {isFullPort = true;}}
```

General Mappings

Port_Mapping
CommonFullPort_Mapping

Mapping Source

Port

Mapping Target

PartUsage

Owned Mappings

(none)

Applicable filters

This mapping applies only if the following (OCL) condition implemented by the operation *filter(src : Element) : Boolean* is verified:

```
(not src.type.oclIsUndefined()) and
Helper.hasStereotypeApplied(src, 'SysML::Ports&Flows::FullPort')
```

Mapping rules

The mapping class only has inherited rules. See the mapping classes in the general mapping section for details.

7.8.7.2.8 FullPortMetadata_Mapping

Description

Create the metadata usage element to annotate a port with the information that its SysML v1 mapping source element is a SysML v1 full port element.

General Mappings

ToMetadataUsage_Init
Mapping

Mapping Source

Port

Mapping Target

MetadataUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- MetadataUsage::ownedRelationship () : Relationship [0..*]

  ```
  Set{FullPortMetadataFeatureTyping_Mapping.getMapped(from),
  FullPortMetadataFeatureMembership_Mapping.getMapped(from)}
  ```

7.8.7.2.9 FullPortMetadataFeatureMembership_Mapping

Description

Creates a feature membership relationship for *ownedMemberFeature()*.

General Mappings

ToFeatureMembership_Init
Mapping

Mapping Source

Port

Mapping Target

FeatureMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureMembership::ownedMemberFeature () : Feature [1]

  ```
  FullPortMetadataReferenceUsage_Mapping.getMapped(from)
  ```

7.8.7.2.10 FullPortMetadataFeatureTyping_Mapping

Description

Creates a feature typing relationship owned by the element *typedFeature()*.

General Mappings

ToFeatureTyping_Init
Mapping

Mapping Source

Port

Mapping Target

FeatureTyping

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureTyping::type () : Type [1]

```
SYSML2::MetadataDefinition.allInstances()
->any(m | m.qualifiedName = 'SysMLv1Library::PortData')
```

7.8.7.2.11 FullPortMetadataOwningMembership_Mapping

Description

Creates a owning membership relationship for *ownedMemberElement()*.

General Mappings

ToOwningMembership_Init
Mapping

Mapping Source

Port

Mapping Target

OwningMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- OwningMembership::ownedMemberElement () : Element [1]

```
FullPortMetadata_Mapping.getMapped(from)
```

7.8.7.2.12 FullPortMetadataReferenceUsage_Mapping

Description

Creates a reference usage.

General Mappings

ToReferenceUsage_Init
Mapping

Mapping Source

Port

Mapping Target

ReferenceUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ReferenceUsage::ownedRelationship () : Relationship [0..*]

```
Set{FullPortMetadataReferenceUsageRedefinition_Mapping.getMapped(from),
FullPortMetadataReferenceUsageFeatureValue_Mapping.getMapped(from)}
```

7.8.7.2.13 FullPortMetadataReferenceUsageFeatureValue_Mapping

Description

Creates a feature value relationship.

General Mappings

ToFeatureValue_Init
Mapping

Mapping Source

Port

Mapping Target

FeatureValue

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureValue::value () : Expression [1]

  ```
  LiteralBoolean_Factory.create(true)
  ```

7.8.7.2.14 FullPortMetadataReferenceUsageRedefinition_Mapping

Description

Creates a redefinition relationship for the *redefiningFeature()* and the *redefinedFeature()*.

General Mappings

ToRedefinition_Init
Mapping

Mapping Source

Port

Mapping Target

Redefinition

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Redefinition::redefinedFeature () : Feature [1]

  ```
  SYSML2::AttributeUsage.allInstances()
  ->any(m | m.qualifiedName = 'SysMLv1Library::PortData::isFullPort')
  ```

7.8.7.2.15 FullPortUntyped_Mapping

Description

A SysML::Ports&Flows::FullPort element is mapped to a part usage in SysML v2 with metadata that marks the part usage as a full port. The metadata is defined in the SysML v1 library for SysML v2.

The mapping class FullPort_Mapping does the same for full ports with a type.

The following shows an example of what the textual SysML v2 syntax of the result of the transformation may look like.

```
part sysMLv1FullPort {SysMLv1Library::PortData {isFullPort = true;}}
```

General Mappings

PortUntyped_Mapping
CommonFullPort_Mapping

Mapping Source

Port

Mapping Target

PartUsage

Owned Mappings

(none)

Applicable filters

This mapping applies only if the following (OCL) condition implemented by the operation *filter(src : Element) : Boolean* is verified:

```
src.type.oclIsUndefined() and
Helper.hasStereotypeApplied(src, 'SysML::Ports&Flows::FullPort')
```

Mapping rules

The mapping class only has inherited rules. See the mapping classes in the general mapping section for details.

7.8.7.2.16 InterfaceBlock_Mapping

Description

A SysML::Ports&Flows::InterfaceBlock element is mapped to a SysML v2 PortDefinition.

The following shows an example of what the textual SysML v2 syntax of the result of the transformation may look like.

```
port def SysMLv1InterfaceBlock;
```

General Mappings

Block_Mapping

Mapping Source

Class

Mapping Target

PortDefinition

Owned Mappings

(none)

Applicable filters

This mapping applies only if the following (OCL) condition implemented by the operation *filter(src : Element) : Boolean* is verified:

```
Helper.hasStereotypeApplied(src, 'SysML::Ports&Flows::InterfaceBlock')
```

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- PortDefinition::ownedRelationship () : Relationship [0..*]

```
self.oclAsType(Block_Mapping).ownedRelationship()
  ->including(InterfaceBlockOwningMembership_Mapping.getMapped(from))
```

7.8.7.2.17 InterfaceBlockConjugated_Mapping

Description

A SysML::Ports&Flows::~InterfaceBlock element is mapped to a SysML v2 PortDefinition. The SysML v1 constraints ensure that the port definition is compatible with the appropriate port definition, which is the target of the mapping of the original interface block. Instead of the special tilde symbol, the port definition name gets a "c" symbol as a prefix. The following shows an example of what the textual SysML v2 syntax of the result of the transformation may look like.

```
port def cSysMLv1InterfaceBlock;
```

General Mappings

InterfaceBlock_Mapping

Mapping Source

Class

Mapping Target

PortDefinition

Owned Mappings

(none)

Applicable filters

This mapping applies only if the following (OCL) condition implemented by the operation *filter(src : Element) : Boolean* is verified:

```
Helper.hasStereotypeApplied(src, 'SysML::Ports&Flows::~InterfaceBlock')
```

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- PortDefinition::declaredName () : String [0..1]

  ```
  'c' + from.name.substring(2,from.name.size())
  ```

7.8.7.2.18 InterfaceBlockOwningMembership_Mapping

Description

Creates a owning membership relationship for *ownedMemberElement()*.

General Mappings

ToOwningMembership_Init
Mapping

Mapping Source

Class

Mapping Target

OwningMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- OwningMembership::ownedMemberElement () : Element [1]

  ```
  ConjugatedPortDefinition_Mapping.getMapped(from)
  ```

7.8.7.2.19 OperationDirectedFeature_Mapping

Description

The mapping class sets the direction of the perform action usage if the SysML v1 mapping source operation has the stereotype SysML::Ports&Flows::DirectedFeature applied.

General Mappings

Operation_Mapping

Mapping Source

Operation

Mapping Target

PerformActionUsage

Owned Mappings

(none)

Applicable filters

This mapping applies only if the following (OCL) condition implemented by the operation *filter(src : Element) : Boolean* is verified:

```
Helper.hasStereotypeApplied(src, 'SysML::Ports&Flows::DirectedFeature')
```

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- PerformActionUsage::direction () : FeatureDirectionKind [0..1]

```
Helper.getKerMLFeatureDirectionKind(
Helper.getTagValueAsElement(
from,'SysML::Ports&Flows::DirectedFeature', 'featureDirection'
))
```

7.8.7.2.20 PortConjugation_Mapping

Description

Creates a PortConjugation between a PortDefinition and a ConjugatedPortDefinition element.

General Mappings

ToConjugation_Init
Mapping

Mapping Source

Class

Mapping Target

PortConjugation

Owned Mappings

- conjugatedPortDefinition : ConjugatedPortDefinition_Mapping

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- PortConjugation::originalPortDefinition () : Type [1]

  ```
  from
  ```

- PortConjugation::conjugatedType () : Type [1]

  ```
  conjugatedPortDefinition.to
  ```

7.8.8 Requirements

SYSML21-542: Transformation: overview tables shall be revised

7.8.8.1 Overview

SYSML21-542: Transformation: overview tables shall be revised

Table 20. List of High-Level Requirements Mappings

SysML v1 Abstract Syntax/Stereotype	SysML v2 Abstract Syntax
Copy	The copy relationship is not covered by SysML v2.
DeriveReqt	ConnectionUsage
Refine	Dependency
Requirement	RequirementUsage
Satisfy	SatisfyRequirementUsage
TestCase	VerificationCaseDefinition
Trace	Dependency
Verify	RequirementVerificationMembership

7.8.8.2 Mapping Specifications

7.8.8.2.1 DeriveReqt_Mapping

Description

A SysML::Requirements::DeriveReqt relationship is mapped to a SysML v2 DerivationConnections::Derivation model library element.

The following shows an example of what the textual SysML v2 syntax of the result of the transformation may look like.

```
requirement <'id1'> SysMLv1Requirement {
        doc /*
               * requirement text
               */
}
requirement <'id2'> SysMLv1RequirementDerived {
        doc /*
               * requirement text
               */
}
```

```
connection : DerivationConnections::Derivation
    connect SysMLv1RequirementDerived to SysMLv1Requirement;
```

General Mappings

Abstraction_Mapping
ToConnectionUsage_Init

Mapping Source

Abstraction

Mapping Target

ConnectionUsage

Owned Mappings

(none)

Applicable filters

This mapping applies only if the following (OCL) condition implemented by the operation *filter(src : Element) : Boolean* is verified:

```
Helper.hasStereotypeApplied(src, 'SysML::Requirements::DeriveReqt')
```

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ConnectionUsage::ownedRelationship () : Relationship [0..*]

  ```
  Set{DeriveReqtFeatureTyping_Mapping.getMapped(from),
  DeriveReqtSourceEndFeatureMembership_Mapping.getMapped(from),
  DeriveReqtTargetEndFeatureMembership_Mapping.getMapped(from)}
  ->union(self.oclAsType(ElementMain_Mapping).ownedRelationship())
  ```

7.8.8.2.2 DeriveReqtFeatureTyping_Mapping

Description

Creates a feature typing relationship owned by the element *typedFeature()*.

General Mappings

ToFeatureTyping_Init
Mapping

Mapping Source

Dependency

Mapping Target

FeatureTyping

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureTyping::type () : Type [1]

```
SYSML2::ConnectionDefinition.allInstances()
->any(m | m.qualifiedName = 'DerivationConnections::Derivation')
```

7.8.8.2.3 DeriveReqtSourceEndFeatureMembership_Mapping

Description

Creates a feature membership relationship for *ownedMemberFeature()*.

General Mappings

ToEndFeatureMembership_Init
Mapping

Mapping Source

Dependency

Mapping Target

EndFeatureMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- EndFeatureMembership::ownedMemberFeature () : Feature [1]

```
DeriveReqtSourceFeature_Mapping.getMapped(from)
```

7.8.8.2.4 DeriveReqtSourceFeature_Mapping

Description

The mapping class creates the source feature of the ConnectionUsage relationship for the mapping of the SysML v1 deriveReqt relationship.

General Mappings

ToFeature_Init
Mapping

Mapping Source

Dependency

Mapping Target

Feature

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Feature::ownedRelationship () : Relationship [0..*]

  ```
  Set{DeriveReqtSourceFeatureReferenceSubsetting_Mapping.getMapped(from)}
  ```

7.8.8.2.5 DeriveReqtSourceFeatureReferenceSubsetting_Mapping

Description

Creates a subsetting relationship.

General Mappings

ToReferenceSubsetting_Init
Mapping

Mapping Source

Dependency

Mapping Target

ReferenceSubsetting

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ReferenceSubsetting::referencedFeature () : Feature [1]

```
from.client->any(c | true)
```

7.8.8.2.6 DeriveReqtTargetEndFeatureMembership_Mapping

Description

Creates a feature membership relationship for *ownedMemberFeature()*.

General Mappings

ToEndFeatureMembership_Init
Mapping

Mapping Source

Dependency

Mapping Target

EndFeatureMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- EndFeatureMembership::ownedMemberFeature () : Feature [1]

```
DeriveReqtTargetFeature_Mapping.getMapped(from)
```

7.8.8.2.7 DeriveReqtTargetFeature_Mapping

Description

The mapping class creates the target feature of the ConnectionUsage relationship for the mapping of the SysML v1 deriveReqt relationship.

General Mappings

ToFeature_Init
Mapping

Mapping Source

Dependency

Mapping Target

Feature

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Feature::ownedRelationship () : Relationship [0..*]

  ```
  Set{DeriveReqtTargetFeatureReferenceSubsetting_Mapping.getMapped(from)}
  ```

7.8.8.2.8 DeriveReqtTargetFeatureReferenceSubsetting_Mapping

Description

Creates a subsetting relationship.

General Mappings

ToReferenceSubsetting_Init
Mapping

Mapping Source

Dependency

Mapping Target

ReferenceSubsetting

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ReferenceSubsetting::referencedFeature () : Feature [1]

```
from.supplier->any(c | true)
```

7.8.8.2.9 Refine_Mapping

Description

A SysML::Requirements::Refine relationship is mapped to a SysML v2 Dependency relationship annotated with a metadata usage tagging it as a former SysML v1 refine relationship.

The following shows an example of what the textual SysML v2 syntax of the result of the transformation may look like.

```
requirement <'id1'> SysMLv1Requirement {
        doc /*
              * requirement text
              */
}
use case def SysMLv1UseCase;

dependency from SysMLv1UseCase to SysMLv1Requirement {
        @SysMLv1Library::RefineData {isRefine = true;}
}
```

General Mappings

Abstraction_Mapping

Mapping Source

Abstraction

Mapping Target

Dependency

Owned Mappings

(none)

Applicable filters

This mapping applies only if the following (OCL) condition implemented by the operation *filter(src : Element) : Boolean* is verified:

```
Helper.hasStereotypeApplied(src, 'SysML::Requirements::Refine')
```

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Dependency::ownedRelationship () : Relationship [0..*]

```
self.oclAsType(ElementMain_Mapping).ownedRelationship()
->including(RefineAnnotation_Mapping.getMapped(from))
```

7.8.8.2.10 RefineAnnotation_Mapping

Description

The mapping class creates the annotation relationship for the SysML::Requirements::Refine mapping.

General Mappings

ToAnnotation_Init
Mapping

Mapping Source

Abstraction

Mapping Target

Annotation

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Annotation::annotatingElement () : AnnotatingElement [1]

  ```
  RefineMetadataUsage_Mapping.getMapped(from)
  ```

7.8.8.2.11 RefineMetadataFeatureMembership_Mapping

Description

Creates a feature membership relationship for *ownedMemberFeature()*.

General Mappings

ToFeatureMembership_Init
Mapping

Mapping Source

Abstraction

Mapping Target

FeatureMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureMembership::ownedMemberFeature () : Feature [1]

  ```
  RefineMetadataReferenceUsage_Mapping.getMapped(from)
  ```

7.8.8.2.12 RefineMetadataReferenceUsage_Mapping

Description

Creates a reference usage.

General Mappings

ToReferenceUsage_Init
Mapping

Mapping Source

Abstraction

Mapping Target

ReferenceUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ReferenceUsage::ownedRelationship () : Relationship [0..*]

  ```
  Set{RefineMetadataReferenceUsageRedefinition_Mapping.getMapped(from),
  RefineMetadataReferenceUsageFeatureValue_Mapping.getMapped(from)}
  ```

7.8.8.2.13 RefineMetadataReferenceUsageFeatureValue_Mapping

Description

Creates a feature value relationship.

General Mappings

ToFeatureValue_Init
Mapping

Mapping Source

Abstraction

Mapping Target

FeatureValue

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureValue::value () : Expression [1]

  ```
  LiteralBoolean_Factory.create(true)
  ```

7.8.8.2.14 RefineMetadataReferenceUsageRedefinition_Mapping

Description

Creates a redefinition relationship for the *redefiningFeature()* and the *redefinedFeature()*.

General Mappings

ToRedefinition_Init
Mapping

Mapping Source

Abstraction

Mapping Target

Redefinition

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Redefinition::redefinedFeature () : Feature [1]

```
SYSML2::AttributeUsage.allInstances()
->any(m | m.qualifiedName = 'SysMLv1Library::RefineData::isRefine')
```

7.8.8.2.15 RefineMetadataUsage_Mapping

Description

Create the metadata usage element to annotate a dependency relationship with the information that its SysML v1 mapping source element is a SysML v1 refine relationship.

General Mappings

ToMetadataUsage_Init
Mapping

Mapping Source

Abstraction

Mapping Target

MetadataUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- MetadataUsage::ownedRelationship () : Relationship [0..*]

```
Set{RefineMetadataUsageFeatureTyping_Mapping.getMapped(from),
RefineMetadataFeatureMembership_Mapping.getMapped(from)}
```

7.8.8.2.16 RefineMetadataUsageFeatureTyping_Mapping

Description

Creates a feature typing relationship owned by the element *typedFeature()*.

General Mappings

ToFeatureTyping_Init
Mapping

Mapping Source

Abstraction

Mapping Target

FeatureTyping

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureTyping::type () : Type [1]

  ```
  SYSML2::MetadataDefinition.allInstances()
  ->any(m | m.qualifiedName = 'SysMLv1Library::RefineData')
  ```

7.8.8.2.17 Requirement_Mapping

Description

A SysML::Requirement is mapped to a SysML v2 RequirementUsage.

The following shows an example of what the textual SysML v2 syntax of the result of the transformation may look like.

```
requirement <'id1'> SysMLv1Requirement {
        doc /*
                       * requirement text
               */

        requirement <'id2'> SysMLv1NestedRequirement {
                doc /*
                              * requirement text
                                 */
        }
}
```

General Mappings

NamedElementMain_Mapping
ToRequirementUsage_Init

Mapping Source

Class

Mapping Target

RequirementUsage

Owned Mappings

(none)

Applicable filters

This mapping applies only if the following (OCL) condition implemented by the operation *filter(src : Element) : Boolean* is verified:

```
Helper.isRequirement(src)
```

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- RequirementUsage::reqId () : String [1]

  ```
  let stereotype: UML::Stereotype = Helper.getRequirementStereotype(from) in
  Helper.getTagValueAsString(from, stereotype.qualifiedName, 'id')
  ```

- RequirementUsage::ownedRelationship () : Relationship [0..*]

  ```
  self.oclAsType(ElementMain_Mapping).ownedRelationship()
  ->including(CommonReturnParameterReferenceUsageMembership_Mapping.getMapped(from))
  ->including(RequirementDocumentationMembership_Mapping.getMapped(from))
  ->including(RequirementSubjectMembership_Mapping.getMapped(from))
  ```

7.8.8.2.18 RequirementDocumentation_Mapping

Description

The mapping class creates a Comment contained in a Requirement which contains the SysML::Requirements::AbstractRequirement::text property.

General Mappings

ToDocumentation_Init
Mapping

Mapping Source

Class

Mapping Target

Documentation

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Documentation::body () : String [1]

```
let stereotype: UML::Stereotype = Helper.getRequirementStereotype(from) in
Helper.getTagValueAsString(from, stereotype.qualifiedName, 'text')
```

7.8.8.2.19 RequirementDocumentationMembership_Mapping

Description

Creates a membership relationship for *memberElement()*.

General Mappings

ToOwningMembership_Init
Mapping

Mapping Source

Class

Mapping Target

OwningMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- OwningMembership::ownedMemberElement () : Element [1]

```
RequirementDocumentation_Mapping.getMapped(from)
```

7.8.8.2.20 RequirementSubject_Mapping

Description

The mapping class creates the subject reference usage element of the requirement. It is not used since the concept does not exist SysML v1.

General Mappings

ToReferenceUsage_Init
Mapping

Mapping Source

Class

Mapping Target

ReferenceUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ReferenceUsage::direction () : FeatureDirectionKind [0..1]

  `KerML::FeatureDirectionKind::_'in'`

7.8.8.2.21 RequirementSubjectMembership_Mapping

Description

The subject is not used, because it is not a SysML v1 concept, but must be created for a SysML v2 requirement.

General Mappings

ToParameterMembership_Init
Mapping

Mapping Source

Class

Mapping Target

SubjectMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- SubjectMembership::ownedMemberParameter () : Feature [0..1]

  ```
  RequirementSubject_Mapping.getMapped(from)
  ```

7.8.8.2.22 Satisfy_Mapping

Description

A SysML::Requirements::Satisfy relationship is mapped to a SysML v2 SatisfyRequirementUsage.

The following shows an example of what the textual SysML v2 syntax of the result of the transformation may look like.

```
// satisfy relationship from a block
part def SysMLv1Block {
        part sysMLv1PartProperty;
}
requirement <'ReqId1'> SysMLv1Requirement { doc /* requirement text */ }

ref :SysMLv1Block = all SysMLv1Block {
        satisfy requirement SysMLv1Requirement by self;
}

// satisfy relationship from a part property
satisfy SysMLv1Requirement by sysML1BlockUsage.sysMLv1PartProperty {
        sysMLv1BlockUsage : SysMLv1Block;
}
```

General Mappings

ToOccurrenceUsage_Init
Abstraction_Mapping

Mapping Source

Abstraction

Mapping Target

SatisfyRequirementUsage

Owned Mappings

(none)

Applicable filters

This mapping applies only if the following (OCL) condition implemented by the operation *filter(src : Element) : Boolean* is verified:

```
let satisfy: UML::Abstraction = src.oclAsType(UML::Abstraction) in
    if satisfy.oclIsUndefined() then
        false
    else
```

```
        Helper.hasStereotypeApplied(satisfy, 'SysML::Requirements::Satisfy')
    endif
```

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- SatisfyRequirementUsage::ownedRelationship () : Relationship [0..*]

```
let relationships : Set(KerML::Relationship) =
    self.oclAsType(ElementMain_Mapping).ownedRelationship()
->including(SatisfyFeatureTyping_Mapping.getMapped(from))
->including(SatisfySubjectSubjectMembership_Mapping.getMapped(from))
->including(CommonReturnParameterReferenceUsageMembership_Mapping.getMapped(from)) in
if from.client->any(c | true).oclIsKindOf(UML::Property) then
    relationships
    ->including(SatisfyReferenceUsageFeatureMembership_Mapping.getMapped(from))
else
    relationships
endif
```

7.8.8.2.23 SatisfyReferenceUsage_Mapping

Description

Creates a reference usage.

General Mappings

ToReferenceUsage_Init
Mapping

Mapping Source

Abstraction

Mapping Target

ReferenceUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ReferenceUsage::ownedRelationship () : Relationship [0..*]

```
Set{SatisfyReferenceUsageFeatureTyping_Mapping.getMapped(from)}
```

- ReferenceUsage::declaredName () : String [0..1]

```
from.client
->any(c | true).owner.name.substring(1,1).toLowerCase()
+ from.client
->any(c | true).owner.name.
substring(2,from.client->any(c | true).owner.name.size())
+ 'SatisfyClientUsage'
```

7.8.8.2.24 SatisfyReferenceUsageFeatureMembership_Mapping

Description

Creates a feature membership relationship for *ownedMemberFeature()*.

General Mappings

ToFeatureMembership_Init
Mapping

Mapping Source

Abstraction

Mapping Target

FeatureMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureMembership::ownedMemberFeature () : Feature [1]

```
SatisfyReferenceUsage_Mapping.getMapped(from)
```

7.8.8.2.25 SatisfySubjectReferenceUsage_Mapping

Description

Creates a reference usage.

General Mappings

ToReferenceUsage_Init
Mapping

Mapping Source

Abstraction

Mapping Target

ReferenceUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ReferenceUsage::ownedRelationship () : Relationship [0..*]

  ```
  Set{SatisfySubjectReferenceUsageFeatureValue_Mapping.getMapped(from)}
  ```

- ReferenceUsage::direction () : FeatureDirectionKind [0..1]

  ```
  KerML::FeatureDirectionKind::_'in'
  ```

7.8.8.2.26 SatisfySubjectReferenceUsageValue_Mapping

Description

The mapping class create the feature reference expression for the subject of the SatisfyRequirementUsage element.

General Mappings

ToFeatureReferenceExpression_Init
Mapping

Mapping Source

Abstraction

Mapping Target

FeatureReferenceExpression

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureReferenceExpression::ownedRelationship () : Relationship [0..*]

```
Set{SatisfySubjectReferenceUsageValueOwningMembership_Mapping.getMapped(from),
ReturnParameterFeatureMembership_Factory.create()}
```

7.8.8.2.27 SatisfySubjectReferenceUsageValueFeature_Mapping

Description

The mapping class creates the feature element for the feature reference expression of the subject of the SatisRequirementUsage element.

General Mappings

ToFeature_Init
Mapping

Mapping Source

Abstraction

Mapping Target

Feature

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Feature::ownedRelationship () : Relationship [0..*]

```
Set{SatisfySubjectReferenceUsageFeatureChaining_Mapping.getMapped(from),
SatisfySubjectReferenceUsageValueFeatureChainingProperty_Mapping.getMapped(from)}
```

7.8.8.2.28 SatisfySubjectReferenceUsageFeatureChaining_Mapping

Description

The mapping class creates the feature chaining element from SysML v2 SatisfyRequirementUsage's reference usage element.

General Mappings

ToFeatureChaining_Init
Mapping

Mapping Source

Abstraction

Mapping Target

FeatureChaining

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureChaining::chainingFeature () : Feature [1]

  ```
  SatisfyReferenceUsage_Mapping.getMapped(from)
  ```

7.8.8.2.29 SatisfySubjectReferenceUsageValueFeatureChainingProperty_Mapping

Description

The mapping class creates the feature chaining element from the source element of the SysML v1 satisfy relationship.

General Mappings

ToFeatureChaining_Init
Mapping

Mapping Source

Abstraction

Mapping Target

FeatureChaining

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureChaining::chainingFeature () : Feature [1]

```
from.client->any(c | true)
```

7.8.8.2.30 SatisfySubjectReferenceUsageFeatureValue_Mapping

Description

Creates a feature value relationship.

General Mappings

ToFeatureValue_Init
Mapping

Mapping Source

Abstraction

Mapping Target

FeatureValue

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureValue::value () : Expression [1]

```
SatisfySubjectReferenceUsageValue_Mapping.getMapped(from)
```

7.8.8.2.31 SatisfySubjectReferenceUsageValueOwningMembership_Mapping

Description

Creates a owning membership relationship for *ownedMemberElement()*.

General Mappings

ToOwningMembership_Init
Mapping

Mapping Source

Abstraction

Mapping Target

OwningMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- OwningMembership::ownedMemberElement () : Element [1]

  ```
  SatisfySubjectReferenceUsageValueFeature_Mapping.getMapped(from)
  ```

7.8.8.2.32 SatisfySubjectSubjectMembership_Mapping

Description

Creates a membership relationship for *memberElement()*.

General Mappings

ToSubjectMembership_Init
Mapping

Mapping Source

Abstraction

Mapping Target

SubjectMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- SubjectMembership::ownedMemberParameter () : Feature [1]

  ```
  SatisfySubjectReferenceUsage_Mapping.getMapped(from)
  ```

7.8.8.2.33 SatisfyFeatureTyping_Mapping

Description

Creates a feature typing relationship owned by the element *typedFeature()*.

General Mappings

ToFeatureTyping_Init
Mapping

Mapping Source

Abstraction

Mapping Target

FeatureTyping

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureTyping::type () : Type [1]

  ```
  from.supplier->any(s | true)
  ```

7.8.8.2.34 SatisfyReferenceUsageFeatureTyping_Mapping

Description

Creates a feature typing relationship owned by the element *typedFeature()*.

General Mappings

ToFeatureTyping_Init
Mapping

Mapping Source

Abstraction

Mapping Target

FeatureTyping

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureTyping::type () : Type [1]

  ```
  from.client->any(c | true).owner
  ```

7.8.8.2.35 TestCaseActivity_Mapping

Description

A SysML::Requirements::TestCase applied to an activity is mapped to a SysML v2 VerificationCaseDefinition element.

The following shows an example of what the textual SysML v2 syntax of the result of the transformation may look like.

```
verification def SysMLv1ActivityTestCase {
        return verdict : VerificationCases::VerdictKind;
}
```

General Mappings

ActivityAsDefinition_Mapping

Mapping Source

Activity

Mapping Target

VerificationCaseDefinition

Owned Mappings

(none)

Applicable filters

This mapping applies only if the following (OCL) condition implemented by the operation *filter(src : Element) : Boolean* is verified:

```
Helper.hasStereotypeApplied(src, 'SysML::Requirements::TestCase')
```

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- VerificationCaseDefinition::ownedRelationship () : Relationship [0..*]

```
let relationships : Set(KerML::Relationship) =
    Helper.activityOwnedRelationship(from) in
let verdictParameter : Set(UML::Parameter) =
    from.ownedElement->select(e | e.oclIsKindOf(UML::Parameter) and
    (e.oclAsType(UML::Parameter).type.name = 'VerdictKind')) in
let parameters : Set(UML::Paramter) =
    ((from.ownedElement->select(e | e.oclIsKindOf(UML::Parameter))) -
    verdictParameter) in
let verifyRelationships : Set(UML::Abstraction) =
    from.clientDependency
    ->select( v |
        Helper.hasStereotypeApplied(v, 'SysML::Requirements::Verify')) in
relationships
->union(parameters->collect(p | ParameterMembership_Mapping.getMapped(p)))
->union(verdictParameter
    ->collect(vp |
        TestCaseActivityReturnParameterMembership_Mapping.getMapped(vp)))
->including(EmptySubjectMembership_Factory.create())
->including(EmptyObjectiveMembership_Factory.create())
->union(verifyRelationships->collect(v | Verify_Mapping.getMapped(v)))
```

7.8.8.2.36 TestCaseActivityReturnParameterMembership_Mapping

Description

Creates a membership relationship for *memberElement()*.

General Mappings

ParameterMembership_Mapping

Mapping Source

Parameter

Mapping Target

ReturnParameterMembership

Owned Mappings

(none)

Applicable filters

(none)

7.8.8.2.37 TestCaseVerifyObjectiveMembership_Mapping

Description

Creates a the objective membership relationship.

General Mappings

UniqueMapping
ToFeatureMembership_Init

Mapping Source

Abstraction

Mapping Target

ObjectiveMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ObjectiveMembership::ownedMemberFeature () : Feature [1]

  ```
  TestCaseVerifyObjectiveRequirementUsage_Mapping.getMapped(from)
  ```

7.8.8.2.38 TestCaseVerifyObjectiveRequirementUsage_Mapping

Description

The mapping class creates the objective requirements usage of the SysML v2 verification case.

General Mappings

ToRequirementUsage_Init
UniqueMapping

Mapping Source

Abstraction

Mapping Target

RequirementUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- RequirementUsage::ownedRelationship () : Relationship [0..*]

  ```
  Set{Verify_Mapping.getMapped(from)}
  ```

7.8.8.2.39 TestCaseVerifyRequirementUsageReferenceSubsetting_Mapping

Description

Creates a subsetting relationship.

General Mappings

ToSubsetting_Init
Mapping

Mapping Source

Abstraction

Mapping Target

ReferenceSubsetting

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ReferenceSubsetting::referencedFeature () : Feature [1]

  ```
  from.supplier->get(0)
  ```

7.8.8.2.40 TestCaseVerifyRequirementUsage_Mapping

Description

The mapping class creates the requirements usage of the SysML v2 test case for the verify relationship.

General Mappings

ToUsage_Init
Mapping

Mapping Source

Abstraction

Mapping Target

RequirementUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- RequirementUsage::ownedRelationship () : Relationship [0..*]

  ```
  Set{TestCaseVerifyRequirementUsageReferenceSubsetting_Mapping.getMapped(from),
  EmptySubjectMembership_Factory.create(),
  CommonReturnParameterReferenceUsageMembership_Mapping.getMapped(from)}
  ```

7.8.8.2.41 Trace_Mapping

Description

A SysML::Requirements::Trace relationship is mapped to a SysML v2 Dependency relationship annotated with a metadata usage tagging it as a former SysML v1 trace relationship.

The following shows an example of what the textual SysML v2 syntax of the result of the transformation may look like.

```
requirement <'id1'> SysMLv1Requirement1 {
        doc /*
               * requirement text
               */
}
requirement <'id2'> SysMLv1Requirement2 {
        doc /*
               * requirement text
               */
}
dependency from SysMLv1Requirement1 to SysMLv1Requirement2 {
        @SysMLv1Library::TraceData {isTrace = true;}
}
```

General Mappings

Abstraction_Mapping

Mapping Source

Abstraction

Mapping Target

Dependency

Owned Mappings

(none)

Applicable filters

This mapping applies only if the following (OCL) condition implemented by the operation *filter(src : Element) : Boolean* is verified:

```
Helper.hasStereotypeApplied(src, 'SysML::Requirements::Trace')
```

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Dependency::ownedRelationship () : Relationship [0..*]

  ```
  self.oclAsType(ElementMain_Mapping).ownedRelationship()
  ->including(TraceAnnotation_Mapping.getMapped(from))
  ```

7.8.8.2.42 TraceAnnotation_Mapping

Description

The mapping class creates the annotation relationship for the SysML::Requirements::Trace mapping.

General Mappings

ToAnnotation_Init
Mapping

Mapping Source

Abstraction

Mapping Target

Annotation

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Annotation::annotatingElement () : AnnotatingElement [1]

  ```
  TraceMetadataUsage_Mapping.getMapped(from)
  ```

7.8.8.2.43 TraceMetadataFeatureMembership_Mapping

Description

Creates a feature membership relationship for *ownedMemberFeature()*.

General Mappings

ToFeatureMembership_Init
Mapping

Mapping Source

Abstraction

Mapping Target

FeatureMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureMembership::ownedMemberFeature () : Feature [1]

  ```
  TraceMetadataReferenceUsage_Mapping.getMapped(from)
  ```

7.8.8.2.44 TraceMetadataReferenceUsage_Mapping

Description

Creates a reference usage.

General Mappings

ToReferenceUsage_Init
Mapping

Mapping Source

Abstraction

Mapping Target

ReferenceUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- ReferenceUsage::ownedRelationship () : Relationship [0..*]

```
Set{TraceMetadataReferenceUsageRedefinition_Mapping.getMapped(from),
TraceMetadataReferenceUsageFeatureValue_Mapping.getMapped(from)}
```

7.8.8.2.45 TraceMetadataReferenceUsageFeatureValue_Mapping

Description

Creates a feature value relationship.

General Mappings

ToFeatureValue_Init
Mapping

Mapping Source

Abstraction

Mapping Target

FeatureValue

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureValue::value () : Expression [1]

```
LiteralBoolean_Factory.create(true)
```

7.8.8.2.46 TraceMetadataReferenceUsageRedefinition_Mapping

Description

Creates a redefinition relationship for the *redefiningFeature()* and the *redefinedFeature()*.

General Mappings

ToRedefinition_Init
Mapping

Mapping Source

Abstraction

Mapping Target

Redefinition

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- Redefinition::redefinedFeature () : Feature [1]

```
SYSML2::AttributeUsage.allInstances()
->any(m | m.qualifiedName = 'SysMLv1Library::TraceData::isTrace')
```

7.8.8.2.47 TraceMetadataUsage_Mapping

Description

Create the metadata usage element to annotate a dependency relationship with the information that its SysML v1 mapping source element is a SysML v1 trace relationship.

General Mappings

ToMetadataUsage_Init
Mapping

Mapping Source

Abstraction

Mapping Target

MetadataUsage

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- MetadataUsage::ownedRelationship () : Relationship [0..*]

```
Set{TraceMetadataUsageFeatureTyping_Mapping.getMapped(from),
TraceMetadataFeatureMembership_Mapping.getMapped(from)}
```

7.8.8.2.48 TraceMetadataUsageFeatureTyping_Mapping

Description

Creates a feature typing relationship owned by the element *typedFeature()*.

General Mappings

ToFeatureTyping_Init
Mapping

Mapping Source

Abstraction

Mapping Target

FeatureTyping

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- FeatureTyping::type () : Type [1]

```
SYSML2::MetadataDefinition.allInstances()
->any(m | m.qualifiedName = 'SysMLv1Library::TraceData')
```

7.8.8.2.49 Verify_Mapping

Description

A SysML::Requirements::Verify relationship is mapped to a SysML v2 RequirementVerificationMembership relationship.

The following shows an example of what the textual SysML v2 syntax of the result of the transformation may look like.

```
requirement <'id1'> SysMLv1Requirement {
        doc /*
               * requirement text
               */
}
verification def SysMLv1TestCase {
        objective objective_SysMLv1TestCase {
                verify SysMLv1Requirement;
        }
        return verdict : VerificationCases::VerdictKind;
}
```

General Mappings

ToRelationship_Init
Mapping

Mapping Source

Abstraction

Mapping Target

RequirementVerificationMembership

Owned Mappings

(none)

Applicable filters

(none)

Mapping rules

In addition to the inherited rules, the following lists the mapping class specific mapping rules for the target element properties.

- RequirementVerificationMembership::ownedRelatedElement () : Element [0..*]

  `Set{TestCaseVerifyRequirementUsage_Mapping.getMapped(from)}`

7.8.8.2.50 Model Libraries

7.8.8.2.50.1 Verdicts

7.8.8.2.50.1.1 VerdictKind

The enumeration VerdictKind is mapped to the SysML v2 VerificationCases::VerdictKind model library element.